DEPARTMENT OF WAR
Small Business Innovation Research (SBIR) Program

2026 SBIR Broad Agency Announcement (BAA)

IMPORTANT DATES

	Pre-Release	Open	Close
Release 1	Apr 13, 2026	May 06, 2026	Jun 03, 2026
Release 2	May 06, 2026	May 27, 2026	Jun 24, 2026
Release 3	Jun 03, 2026	Jun 24, 2026	Jul 22, 2026
Release 4	Jul 01, 2026	Jul 22, 2026	Aug 19, 2026
Release 5	Aug 05, 2026	Aug 26, 2026	Sep 23, 2026
Release 6	Sep 02, 2026	Sep 23, 2026	Oct 21, 2026

NOTE: DoW Services and Components may elect to release topics under this BAA once a month throughout fiscal year 2026. New research topics will pre-release on the first Wednesday of each month, based on the schedule above. Each release will have corresponding open and close dates applicable to the topics within that release. Any topic released that deviates from the open/close schedule in the matrix above will be noted within the Component-specific instructions and on DSIP.

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1.0 PROGRAM DESCRIPTION

1.1 Introduction

The Department of War (DoW) Small Business Innovation Research (SBIR) Program's objectives include stimulating technological innovation by strengthening the role of small businesses in meeting DoW research and development (R&D) needs, delivering innovative technology to the Warfighter, and growing the defense industrial base through commercialization.

The Small Business Administration (SBA), through its SBIR/STTR Policy Directive, purposely departs from traditional government solicitation formats and requirements, simplifying the SBIR/STTR award process and minimizing the regulatory burden on small businesses. Consistent with the SBA SBIR/STTR Policy Directive, DoW is soliciting proposals as a Broad Agency Announcement (BAA). DoW invites Small Business Concerns (SBCs) with the capability to conduct R&D and commercialize the results in any of the defense-related topic areas described in this SBIR Program BAA to submit proposals.

1.1.1 General BAA Procedures

Federal Acquisition Regulation (FAR) 2.101 defines a BAA as a general announcement of an agency's research interest including criteria for selecting proposals and soliciting the participation of all offerors capable of satisfying the Government's needs. FAR 35.016 prescribes the procedures for the use of BAAs as a competitive method for acquiring basic and applied research, and in some cases, advanced technology development. BAAs are intended to solicit scientific study and experimentation directed toward advancing knowledge, understanding and capability in areas of interest to the Government, rather than to acquire specific systems or hardware.

This BAA provides the framework for DoW to pursue innovative research concepts that address defense-related scientific and engineering challenges. By leveraging the flexibility of the SBA SBIR/STTR Policy Directive, this BAA encourages scientific and technical proposals most likely to yield significant results for DoW and the private sector.

This BAA may include broadly defined topics as well as more specific, individual program requirements. Topics are intentionally non-prescriptive and structured to accommodate varied technical and scientific approaches, encouraging novel solutions that align with DoW modernization goals. Proposals should be responsive to the specific problem statement(s) and area(s) of interest outlined in the release topics.

1.1.2 Acquisition Framework

This BAA is for research topics accepting Phase I or Direct to Phase II proposals. A separate BAA will not be issued requesting Phase II proposals. All proposing SBCs that receive a Phase I award from this BAA will be eligible to participate in Phase II competitions and potential Phase III awards. DoW Services/Components will notify Phase I awardees of the Phase II proposal submission requirements.

Depending on the nature of the proposed effort, and the statutory authorities available, the Government retains the flexibility to award a FAR-based procurement contract, purchase order, or Other Transaction (OT) under this solicitation, as appropriate. FAR-based contract types, including, Firm-Fixed-Price, Firm-Fixed-Price Level of Effort, Labor Hour, Time & Material, or Cost-Plus-Fixed-Fee type contracts may be negotiated at the discretion of the Component Contracting Officer. Additionally, the Government may award OTs pursuant to 10 U.S.C. 4021 (Research Projects) or 10 U.S.C. 4022 (Prototype Projects). Follow-on production OTs may also be awarded in accordance with 10 U.S.C. 4022(f) if the statutory conditions are met. Separately, awards may also be made under 10 U.S.C. 4023 (Procurement for Experimental Purposes), when appropriate.

Multiple procurements are planned and anticipated to be awarded for each topic. Each proposal will be

evaluated independently, and funding decisions will be made without regard to other submissions. DoW Services/Components may elect to fund several, one or none of the proposals received in response to a given topic. All awards are subject to both a risk-based due diligence security review and availability of funds, and the DoW is under no obligation to make any awards under this BAA. Offerors are solely responsible for all costs incurred in the preparation and submission of proposals. The DoW will not reimburse any expenses related to proposal development or pre-award activities. This BAA is not a mechanism for submitting unsolicited proposals, and unsolicited proposals will not be considered nor receive a response. Only proposals that are responsive to a posted topic release will be considered.

The instructions contained in this BAA are applicable to all scheduled topic releases. Each release will have corresponding open and close dates applicable only to those topics contained within the release. Additional instructions specific to an individual topic release will be contained within the component-specific instructions. All proposals submitted in response to any topic posted under this solicitation will be governed by this BAA. Proposals must conform to this announcement's terms.

1.2 A Three Phased Program

The SBIR Program has three phases, Phases I, II, and III. Phase I determines, to the extent possible, an idea's scientific, technical, and commercial merit and feasibility within the SBIR program. Phase I and II awards are made adhering to current SBA Policy Directive guidelines.

The Phase I period of performance is generally between six to twelve months. Proposals should focus on Research or Research & Development (R/R&D)to prove the proposed effort's scientific and technical feasibility, and commercialization potential, the successful completion of which is a prerequisite for further DoW support in Phase II. Proposing SBCs are encouraged to consider whether the research or R&D being proposed to DoW Services/Components also has private sector potential, either for the proposed application or as a base for other applications.

Phase II awards will be made to proposing SBCs based on results of Phase I awards and the Phase II proposal's scientific merit, technical merit, and commercialization potential. The period of performance is generally 24 months. The objective of Phase II is to continue and further develop the R/R&D effort from the completed Phase I award.

Phase III refers to work that derives from, extends, or completes an effort made under prior SBIR funding agreements, but is funded by sources other than the SBIR Program. Under Phase III, the SBC should focus on commercializing previously SBIR-funded technology and is required to obtain funding from either the private sector, a non-SBIR federal source, or both, to develop the prototype into a viable product or non-R&D service for sale in military or private sector markets.

1.3 OUSW(R&E) Critical Technology Areas

Although each DoW Service/Component develops SBIR and STTR topics tailored to their mission needs, topics may also align with the Office of the Under Secretary of Defense, Research & Engineering (OUSW(R&E)) Critical Technology Areas (CTAs). While many technologies may cross between these categories, these areas represent the broad and different approaches required to advance technologies crucial to the Department, with a focus on accelerating key capabilities' transitions to the Military Services and Combatant Commands.

OUSW(R&E) CTAs include:

- Applied Artificial Intelligence (AAI)
- Biomanufacturing (BIO)
- Contested Logistics Technologies (LOG)
- Quantum & Battlefield Information Dominance (QBID)
- Scaled Hypersonics (SHY)
- Scaled Directed Energy (SCADE)

Full descriptions of the OUSW (R&E) CTAs can be viewed here.

Below are additional Component Technology Priority Areas (TPAs) supporting DoW Services/Component-specific mission-critical areas:

- FutureG
- Trusted AI and Autonomy
- Biotechnology
- Advanced Computing and Software
- Integrated Sensing and Cyber
- Directed Energy (DE)
- Hypersonics
- Microelectronics
- Integrated Network Systems-of-Systems
- Quantum Science
- Space Technology
- Renewable Energy Generation and Storage
- Advanced Materials
- Human-Machine Interfaces
- Advanced Infrastructure & Advanced Manufacturing
- Combat Casualty Care
- Emerging Threat Reduction
- Military Infectious Diseases
- Military Operational Medicine
- Mission Readiness & Disaster Preparedness
- Nuclear
- Sustainment & Logistics

Full descriptions of the above TPAs can be reviewed here.

1.4 Eligibility and Performance Requirements

Each proposing entity must qualify as an SBC as defined in 13 C.F.R §§ 701-705 at time of award and certify to this in the proposal's cover sheet. The eligibility requirements for the SBIR/STTR programs are unique and do not correspond to those of other small business programs.

a. Proposing SBC must meet eligibility requirements for Small Business Ownership and Control (see 13 CFR § 121.702).
b. The proposing SBC must conduct a minimum of two-thirds of the Phase I research and/or analytical work. For Phase II, the proposing SBC must perform no less than 50 percent of the research and/or analytical work. The work percentage is measured via direct and indirect costs. Occasionally, deviations from these SBIR requirements may occur with the Funding Agreement officer's written approval after consultation with the agency SBIR/STTR program manager/coordinator. For more information on the percentage of work calculation during proposal submission, refer to section 3.7.
c. For both Phase I and II, the principal investigator's primary employment must be with the proposing SBC at the time of the award and during the conduct of the proposed effort. Primary employment means that more than one-half of the principal investigator's time is spent with the small business (based on a 40-hour work week). Primary employment with an SBC precludes full-time employment at another organization. Deviations from this requirement or changes to the principal investigator are subject to the Funding Agreement officer approval.
d. For both Phase I and Phase II, the SBC and its subcontractors must perform all research or R&D work in the United States.

e. Joint ventures and limited partnerships are permitted, provided that the entity qualifies as small business in accordance with the ownership requirements in 13 CFR 121.702(a)(1)(iii) and the size requirements in 13 CFR 121.702(c)(6). Proposing SBC must disclose joint ventures with existing (or planned) relationships/partnerships with any foreign entity or any foreign government-controlled companies. See sections 2.6 and 3.7 for more detail.

1.5 Majority Ownership in Part by Multiple Venture Capital, Hedge Fund, and Private Equity Firms

Unless otherwise noted in the participating Service/Component instructions, proposing SBCs that are multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds majority owned are **ineligible** to submit applications to or receive awards under this solicitation. If a Service/Component authorizes such participation, any proposing VCOC, hedge fund, and/or private equity fund-owned SBC, whether in part or in whole, must identify each foreign national, foreign entity, or foreign government holding or controlling greater than a 5 percent, either directly or indirectly held, equity stake in the proposing SBC. The proposing SBC must also identify any ultimate parent owner(s) and other entities and/or individuals owning more than a 5 percent equity stake in its ownership chain.

In accordance with the requirements of 13 CFR 121.702(a)(2), no single venture capital operating company, hedge fund, or private equity firm may own more than 50 percent of the concern unless that single venture capital operating company, hedge fund, or private equity firm qualifies as an SBC that is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States.

1.6 Performance Benchmark Requirements/Increased Minimum Performance Standards for Experienced Firms

Proposing SBCs with multiple prior SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The Phase I to Phase II Transition Rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The Commercialization Benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' application for more experienced firms. Find detailed information on benchmark calculations, increased performance standards for more experienced firms and consequence of failure to meet benchmarks here. SBA will notify companies failing either benchmark and the relevant officials at the participating agencies.

The SBIR/STTR Policy Directive defines the Departments of the Army, Navy, and Air Force each as its own federal agency, and the remaining DoW Components as an executive agency of the Department of War. Therefore, companies that fail to meet either of the benchmarks under the Increased Performance Standards for More Experienced Firms may not receive more than an overall total of 80 awards from DoW, as detailed in the breakdown below:

Army – 20 total Phase I and Direct to Phase II awards
Navy – 20 total Phase I and Direct to Phase II awards
Air Force – 20 total Phase I and Direct to Phase II awards
All other DoW Components combined – 20 total Phase I and Direct to Phase II awards

1.7 Direct to Phase II Program

15 U.S.C. §638 (cc), as amended by NDAA FY2012, Sec. 5106, and further amended by NDAA FY2019,Sec. 854, PILOT TO ALLOW PHASE FLEXIBILITY, allows DoW to make a SBIR Phase II award to an SBC with respect to a project, without regard to whether the SBC was provided a SBIR

program Phase I award with respect to such project. DoW does not guarantee Direct to Phase II opportunities will be offered in future solicitations.

Each eligible topic requires proposing SBCs provide documentation to demonstrate feasibility described in the Phase I section of the topic has been met. **Feasibility documentation cannot be based solely upon or logically extend from any prior or ongoing federally funded SBIR or STTR work.** The proposing SBC and/or the principal investigator must have substantially performed the work submitted in the feasibility documentation. The proposing SBC must demonstrate ownership or licensure of any intellectual property (IP) associated with such technology prior to proposal submission to enable it and its subcontractors to legally carry out the proposed work.

If the proposing SBC fails to demonstrate technical merit and feasibility equivalent to the Phase I level as described in the associated topic, the related Phase II proposal will not be accepted or evaluated, in accordance with the Service/Component-specific Direct to Phase II instructions.

Refer to the Service/Component-specific Direct to Phase II instructions for full details regarding Service/Component Direct to Phase II processes and proposal preparation requirements.

1.8 Program on Innovation Open Topics

15 U.S.C. §638(ww) requires DoW establish innovation open topic activities to:

a. increase the transition of commercial technology to the DoW;
b. expand the small business nontraditional industrial base;
c. increase commercialization derived from DoW investments; and
d. expand the ability for qualifying SBCs to propose technology solutions to meet DoW needs.

Unlike conventional topics, which specify the desired technical objective and output, open topics use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

Open topics released under this solicitation will be clearly identified as such in the title and topic objective. Proposal preparation instructions for open topics may vary significantly across DoW Services/Components. Proposing SBCs must carefully read and follow all instructions from the DoW Service/Component for the open topic of interest. Unless specifically noted in the Service/Component instructions, all requirements outlined in this solicitation remain in effect for open topics.

An SBC may only submit one proposal to each open topic. If an SBC submits more than one proposal for a single open topic, only the most recent certified proposal submitted prior to the submission deadline will receive an evaluation. All previously submitted proposals for the same open topic will be marked nonresponsive and will not receive an evaluation. Refer to the Service/Component-specific instructions for any additional requirements specific to open topics.

1.9 Discretionary Technical and Business Assistance (TABA)

DoW has not mandated the use of discretionary technical and business assistance (TABA). The proposing SBCs should review individual Service/Component-specific instructions to determine if TABA is offered by the Service/Component and follow instructions for requesting TABA funding.

1.10 Phase II Enhancement Policy

To further encourage the transition of SBIR research into both DoW acquisition programs and the private sector, certain DoW Services/Components developed their own Phase II Enhancement policies. Under

this policy, the Service/Component will provide a Phase II awardee with additional Phase II SBIR funding if the proposing SBC can match the additional SBIR funds with non-SBIR funds from DoW acquisition programs or the private sector.

See Service/Component instructions for more details on Phase II Enhancement opportunities.

1.11 Commercialization Readiness Program (CRP)

The SBIR/STTR Reauthorization Act of 2011 established the Commercialization Pilot Program (CPP) as a long-term program called the Commercialization Readiness Program (CRP). Each Military Department (Army, Navy, and Air Force) has a CRP; refer to the Service/Component instructions for further information.

The DoW SBIR/STTR Program has established the OSW Accelerated Research for Transition (ART) for purposes of technology maturation and transfer. For more information contact osd.ncr.ousd-r-e.mbx.sbir-sttr-tech-transition@mail.mil.

1.12 State and Other Available Assistance

Many states have established programs to provide services to those proposing SBCs and individuals wishing to participate in the Federal SBIR Program. These services vary from state to state, but may include:

a. Information and technical assistance;
b. Matching funds to SBIR recipients;
c. Assistance in obtaining Phase III funding.

Contact your State SBIR/STTR Support office for further information. SBCs may seek general administrative guidance from small and disadvantaged business utilization specialists located in various defense contract management activities throughout the continental United States.

1.13 Fraud and Fraud Reporting

Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. Sec 1001), punishable by a fine of up to $10,000, up to five years in prison, or both.

The DoW Office of Inspector General Hotline ("Defense Hotline") is an important avenue for reporting fraud, waste, abuse, and mismanagement within the DoW. The Office of Inspector General operates this hotline to receive and investigate complaints or information from contractor employees, DoW civilians, Service members, and the public. Individuals who wish to report fraud, waste or abuse may contact the Defense Hotline at (800) 424-9098 between 8:00 a.m. and 5:00 p.m. Eastern Time or visit their website to submit a complaint. Mailed correspondence should be addressed to the Defense Hotline, The Pentagon, Washington, DC 20301-1900, or email addressed to hotline@dodig.mil.

2.0 CERTIFICATIONS AND REGISTRATIONS

2.1 System for Award Management (SAM) Registration

The System for Award Management (SAM) allows proposing SBCs to provide basic information on business structure, capabilities, and financial and payment information with the Federal Government. Proposing SBCs must register in SAM here. Registration in SAM will generate the Unique Entity ID (UEI) number and the Commercial and Government Entry (CAGE) code. The UEI is required for registration in the U.S. Small Business Administration's (SBA) Company Registry. A proposing SBC that is already registered in SAM should verify the registration is active, and its representations and certifications are current to avoid award delay.

2.2 SBA Company Registry

Proposing SBCs must be registered in the SBA Company Registry. SBCs will be required to verify registration by providing the SBC Control ID and Proof of Registration/Certification during proposal submission.

2.3 Defense SBIR/STTR Innovation Portal (DSIP) Registration

Individuals from proposing SBCs must be registered in the DSIP to prepare and submit proposals. Proposing SBCs submitting through this site for the first time will be asked to register. All users are required to have an individual user account to access DSIP. It is recommended proposing SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process.

DSIP user accounts are authenticated by Login.gov. Users who do not already have a Login.gov account will be required to create one. Users who already have a Login.gov account can link their existing Login.gov account with their DSIP account. Job Aids and Help Videos to walk you through the process are in the Learning & Support section of DSIP.

Be advised the sharing of accounts and passwords is a violation of the Terms of Use for Login.gov and DoW policy.

Registered SBCs will have a designated DSIP Firm Admin responsible for creating the Firm PIN, controlling access for other users in the SBC and completing and maintaining the Firm-level forms, which must be completed before any proposals can be submitted.

Users should complete their account registrations as soon as possible to avoid any delays in proposal submissions.

NOTE: The DSIP application is only accessible from within the United States, which is defined as the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

2.3.1 DSIP Assistance and Support

For assistance with the DSIP application, please visit the Learning & Support section of DSIP. Email DSIP Support at DoDSBIRSupport@reisystems.com only for further assistance with issues pertaining directly to the DSIP application. Questions submitted to DSIP Support will be addressed in the order received during normal operating hours (Monday through Friday, 9:00 a.m. to 5:00 p.m. ET). Please include information on your SBC, a proposal number (if applicable), and screenshots of any pertinent errors or issues encountered.

DSIP Support cannot provide updates to proposal status after submission, such as proposal selection/non-selection status or contract award status. Contact the DoW Service/Component that originated the topic following the Service/Component-specific instructions given at the beginning of that Service/Component-specific instructions.

2.4 Required Certifications

At the time of proposal submission, each SBC must certify via the Cover Sheet of the proposal that it meets the size, ownership, and other requirements of the SBIR Program. In addition, the Policy Directive includes certification requirements set forth in the SBIR and STTR Extension Act of 2022 (Public Law 117-183). SBCs are required to certify that they are meeting the Program's requirements during the life

cycle of the funding agreement: at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the total Phase II award amount, and prior to final payment on the Phase II award.

2.5 Due Diligence Program to Assess Security Risks

15 U.S.C. §638(vv) requires the DoW, in coordination with the SBA, to establish and implement a due diligence program to assess and, when possible, mitigate national security risks from SBCs seeking a federally funded award. The Department will use the proposal and information in response to the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (proposal submission Volume 7) to conduct a risk-based due diligence review of the following areas: cybersecurity practices; patent analysis; employee analysis and foreign ownership, including the SBC's financial ties and obligations (which shall include surety, equity, and debt obligations); and SBC employees' ties to a foreign country, foreign person, or foreign entity. Disclosure of foreign affiliations or relationships on this form does not mean denial. The Department will also assess proposals utilizing open-source analysis and analytical tools, for the purpose of confirming the accuracy of the information provided as well as determining if the proposing SBC failed to disclose the information set forth in 15 U.S.C. 638(g)(13).

After reviewing the proposing SBC's responses to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries, if DoW determines it appropriate the Department may ask the SBC to provide true copies of any contractual or financial obligation or other agreement specific to a business arrangement or joint-venture like arrangement with an enterprise owned by a foreign state or any foreign entity in effect during the five-year period preceding the SBC's proposal submission.

The DoW may not make awards that pose unacceptable security risks pursuant to 15 U.S.C. §638(vv). If the risk-based due diligence review concludes that an SBC should not be eligible for the specific SBIR or STTR award due to a national security risk that cannot be adequately mitigated, the proposal will not receive consideration for possible award regardless of the results of the technical review of the proposal. Attachment 2: Defense SBIR and STTR Due Diligence Program Common Risk Matrix of the memo titled "Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program", dated 13 May 2024, provides details on the factors for assessing SBC risk during the due diligence review.

2.6 Cybersecurity Maturity Model Certification (CMMC) Program

DoW has established the Cybersecurity Maturity Model Certification (CMMC) Program in order to verify contractors have implemented required security measures necessary to safeguard Federal Contract Information (FCI) and Controlled Unclassified Information (CUI). CMMC requirements will apply to all DoW solicitations and contracts for which a defense contractor or subcontractor will process, store, or transmit FCI or CUI on its unclassified contractor information systems. It is important to note that subcontractors are subject to flow-down requirements if the subcontractor will process, store, or transmit FCI or CUI on its unclassified contractor information systems.

The CMMC Program provides assessments at three levels, each incorporating security requirements from existing regulations and guidelines.

Level 1: Basic Safeguarding of FCI

- Requirements: Annual self-assessment and annual affirmation of compliance with the 15 security requirements in FAR clause 52.204-21.

Level 2: Broad Protection of CUI

- Requirements:
    1. Either a self-assessment or an independent assessment by an authorized CMMC Third-Party Assessment Organization (C3PAO) every three years, as specified in the solicitation.

- Determined by the type of information processed, transmitted, or stored on the contractor or subcontractor information systems.

2. Annual affirmation, verify compliance with the 110 security requirements in NIST SP 800-171 Revision 2.

Level 3: Higher-Level Protection of CUI Against Advanced Persistent Threats

- Requirements:
  1. Achieve CMMC Status of Final Level 2.
  2. Undergo an assessment every three years by the Defense Contract Management Agency's (DCMA) Defense Industrial Base Cybersecurity Assessment Center (DIBCAC).
  3. Provide an annual affirmation verifying compliance with the 24 identified requirements from NIST SP 800-172.

Projected CMMC level requirements will be identified within the respective topic. Proposers should carefully review and consider these updated requirements when preparing their proposals, as compliance may impact proposed costs and technical approaches. For additional CMMC information and resources visit https://dodcio.defense.gov/cmmc/About/.

2.6.1 Resources for Understanding FOCI & Cybersecurity Readiness

DoW has partnered with Project Spectrum to provide online resources and courses on Understanding Foreign Ownership, Control, or Influence (FOCI) and improving cybersecurity readiness, resiliency, and compliance as it relates to CMMC. SBCs can register and access these courses and resources at projectspectrum.io.

2.7 Joint Ventures

A small business joint venture entity must submit, with its proposal, the representation required in paragraph (c) of FAR solicitation provision 52.212-3, Offeror Representations and Certifications-Commercial Products and Commercial Services, and paragraph (c) of FAR provision 52.219-1, Small Business Program Representations, in accordance with 52.204-8(d) and 52.212-3(b) for the following categories:

a. Small business;
b. Service-disabled veteran-owned small business;
c. Women-owned small business (WOSB) under the WOSB Program;
d. Economically disadvantaged women-owned small business under the WOSB Program; or
e. Historically underutilized business zone small business.

These representations can be found here and must be uploaded to Volume 5, Supporting Documents of the proposal submission in DSIP, if applicable.

2.8 Conflicts of Interest

Contract awards to an SBC owned by or employing current, or previous, Federal Government employees could create conflicts of interest for those employees, which may be a violation of federal law.

2.9 Organizational Conflicts of Interest (OCI)

FAR 9.5 Requirements

In accordance with FAR 9.5, proposing SBCs are required to identify and disclose all facts relevant to potential organizational conflicts of interest (OCIs) involving the proposing SBC's organization and any proposed team member (sub-awardee, consultant). The proposing SBC is responsible for providing this disclosure with each submitted proposal. The disclosure must include the proposing SBC's, and as applicable, proposed team member's OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposing SBC has taken, or intends to take, to prevent the existence of

conflicting roles that might bias the proposing SBC's judgment, and to prevent the proposing SBC from having unfair competitive advantage. The OCI mitigation plan will specifically discuss the disclosed OCI in the context of each of the OCI limitations outlined in FAR 9.505-1 through FAR 9.505-4.

Agency Supplemental OCI Policy

DoW Services/Components also may have a supplemental OCI policy prohibiting contractors/performers from concurrently providing scientific engineering technical assistance (SETA), advisory and assistance services (A&AS), or similar support services, and being a technical performer. As part of the FAR 9.5 disclosure requirement, a proposing SBC must affirm whether the proposing SBC or any proposed team member (sub-awardee, consultant) is providing SETA, A&AS, or similar support to any DoW Service/Component office(s) under: (a) a current award or sub-award; or (b) a past award or sub-award that ended within one calendar year prior to the proposal's submission date. If SETA, A&AS, or similar support is or was provided to any DoW Service/Component office(s), the proposal must include:

a. The name of the DoW Service/Component office receiving the support;
b. The prime contract number;
c. Identification of proposed team member (sub-awardee, consultant) providing the support; and
d. An OCI mitigation plan in accordance with FAR 9.5.

Government Procedures

In accordance with FAR 9.503, 9.504 and 9.506, the Government will evaluate OCI mitigation plans to avoid, neutralize, or mitigate potential OCI issues before award and determine whether it is in the government's interest to grant a waiver. The U.S. Government will only evaluate OCI mitigation plans for proposals determined selectable under the solicitation evaluation criteria and funding availability.

The government may require proposing SBCs provide additional information to support evaluation of the proposing SBC's OCI mitigation plan.

If the government determines a proposer failed to fully disclose an OCI; or failed to provide a government waiver as described above; or failed to reasonably provide additional information the government requested when evaluating the proposer's OCI mitigation plan, the government may reject the proposal and withdraw it from consideration for award.

2.10 Research Involving Human Subjects/Human Subject Research (RIHS/HSR)

All research involving human subjects, to include use of human biological specimens and human data, shall comply with the applicable federal and state laws, and agency policy/guidelines for human subject protection (see Section 5.2 and Appendix B).

Institutions receiving funding for research involving human subjects must provide documentation of a current federal assurance of compliance with federal regulations for human subject protection; for example a Department of Health and Human Services, Office for Human Research Protections federal-wide assurance (http://www.hhs.gov/ohrp). The awarding DoW Service/Component may also request additional federal assurance documentation. All institutions engaged in human subject research, to include subcontractors, must also have a valid assurance. In addition, personnel involved in human subject research must provide documentation of completed appropriate training for the protection of human subjects. Institutions proposing to conduct human subject research that meets one of the exemption criteria in 32 CFR 219.101 are not required to have a federal assurance of compliance. Proposing SBCs should clearly segregate research activities involving human subjects from other R&D activities in their proposal.

If selected, institutions must also provide documentation of institutional review board (IRB) approval, or a determination from an appropriate official in the institution, that the work meets one of the exemption criteria with 32 CFR 219. As part of the IRB review process, evidence that all investigators are appropriately trained should accompany the protocol. The protocol, separate from the proposal, must include a detailed description of the research plan, study population, risks and benefits of study participation, recruitment and consent process, and data collection and analysis.

The amount of time required for the IRB to review and approve the protocol will vary based on the IRB's procedures, the complexity of the research, the level of risk to study participants and the responsiveness of the investigator. The average IRB approval process can last between one and three months. Once the IRB has approved the research, the awarding DoW Service/Component will review the protocol and the IRB's determination to ensure that the research will be conducted in compliance with both DoW and Service/Component policies. The DoW review process can last between three to six months. Ample time should be allotted to complete both the IRB and DoW approval processes prior to recruiting subjects. **No funding can be used towards human subject research until ALL approvals are granted. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.11 Research Involving Animal Subjects

All research, development, testing, experimentation, education, or training involving the use of animals shall comply with the applicable federal and agency rules on animal acquisition, transport, care, handling, and use (see Section 5.2 and Appendix B).

For submissions containing animal use, proposals should briefly describe plans for their institutional animal care and use committee (IACUC) review and approval.

All recipients must receive their IACUC's approval, as well as secondary or headquarters-level approval from a DoW veterinarian trained or experienced in laboratory animal medicine and science. **No animal research may be conducted using DoW funding until all appropriate DoW office(s) grant approval. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.12 Research Involving Recombinant DNA Molecules

All research involving recombinant DNA molecules shall comply with the applicable federal and state law, regulation, and additional agency guidance. An institutional biosafety committee must approve the research.

In addition to the standard federal and DoW procurement certifications, the SBA SBIR Policy Directive requires the proposing business concerns provide certain information at time of award and during the award life cycle. Each proposing SBC must provide this additional information at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the Phase II total award amount, and prior to final payment on the Phase II award.

2.13 Federal Acquisition Supply Chain Security Act Orders

FAR 52.204-29 Federal Acquisition Supply Chain Security Act (FASCA) Orders—Representation and Disclosures and FAR 52.204-30 FASCA Orders—Prohibition are included in this solicitation. In accordance with FAR 52.204-29 and FAR 52.204-30, proposing SBCs must review FASCSA orders here for covered articles, or any products or services produced or provided by a source, that an applicable FASCSA order prohibits.

During contract performance, the contractor shall review SAM.gov at least once every three months, or as

the contracting officer advises, to check for covered articles, or products or services produced subject as part of any new FASCSA order(s) that could impact their supply chain, and report to the contracting officer any covered article, or product or service produced or provided by a source provided to the government or used during the contract performance.

The proposing SBC represents that, via proposal submission under this solicitation, it conducted a reasonable inquiry, and it does not propose to provide or use any covered article, or any products or services produced or provided by a source, if an applicable FASCSA prohibited the covered article or the source effective this solicitation's issue date.

2.14 BlueUAS Certification

SecWar *Unleashing U.S. Military Drone Dominance* memorandum dated 10 July 2025 provides a goal to eliminate barriers to purchase and train with emerging UAS technology. As of December 2025, the Department's BlueUAS Certification process has transitioned from the Defense Innovation Unit (DIU) to the Defense Contract Management Agency (DCMA). Companies who are seeking BlueUAS certification can review the process and submit their product(s) for assessment via DCMA's Blue List portal: https://bluelist.appsplatformportals.us/Blue-UAS-Portal/.

IMPORTANT: Unless specifically required per the topic information, companies are not required to have BlueUAS certification to compete for awards as included in this solicitation, nor are DoW SBIR/STTR program offices responsible for reimbursement of costs incurred by companies pursuing BlueUAS certification.

3.0 PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS

3.1 Introduction

The proposal must provide sufficient information to demonstrate to the evaluator(s) that the proposed work represents an innovative approach to an important scientific or engineering problem and is worthy of support under the stated criteria. The proposed research or R&D must be responsive to the chosen topic, although it does not need to use the exact approach specified in the topic, if applicable. SBCs should consider the following:

a. Does the technical approach have a reasonable chance of meeting the topic objective (if applicable)?
b. Is this approach innovative, not routine, with potential for commercialization?
c. Does the proposing SBC have the capability to implement the technical approach or can it obtain the appropriate people and equipment for the task?

DSIP provides a structure for providing the following proposal volumes:

a. Volume 1: Proposal Cover Sheet
b. Volume 2: Technical Volume
c. Volume 3: Cost Volume
d. Volume 4: Company Commercialization Report
e. Volume 5: Supporting Documents
f. Volume 6: Fraud, Waste and Abuse Training
g. Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

Each Service/Component guidance on allowable proposal content may vary. A completed proposal submission in DSIP does NOT indicate that each proposal volume has been completed in accordance with the Service/Component-specific instructions. Accordingly, it is the proposing SBC's responsibility to consult the Service/Component-specific instructions for detailed guidance,

including required proposal documentation and structure, cost and duration limitations, budget structure, TABA allowance and proposal page limits.

3.2 Export-Controlled Topic Requirements

Export of all unclassified technical data with military or space application in the possession of, or under the control of, a DoW Component information, which includes, in some circumstances, release to foreign nationals within the United States, without first obtaining approval, authorization, or license from the Department of State for items controlled by the International Traffic in Arms Regulations (ITAR), or the Department of Commerce for items controlled by the Export Administration Regulations (EAR), may constitute a violation of law.

Pursuant to Defense Federal Acquisition Regulation Supplement (DFARS) Procedures Guidance and Information 225.7901-2, your firm should direct its attention to the clause at DFARS 252.225-7048, Export-Controlled Items for questions concerning compliance with ITAR/EAR.

Further, in accordance with Department of Defense Directive 5230.25, Withholding of Unclassified Technical Data from Public Disclosure, contractors, or subcontractors that will handle technical data that might have military or space applications, must certify that they will comply with all applicable U.S. laws that control the export of sensitive data, as follows:

> If any portion of the proposed SBIR effort is subject to ITAR an SBC must complete a fully certified DD Form 2345, Military Critical Technical Data Agreement, and provide a copy or evidence of application submission within Volume 5. The DD Form 2345, Military Critical Technical Data Agreement, instructions, and Frequently Asked Questions (FAQs) may be found at the United States/Canada Joint Certification Program (JCP) website, JCP Portal.
>
> If any portion of the proposed SBIR effort is subject to EAR, your firm must submit for and obtain the proper export licenses through the Department of Commerce's Bureau of Industry and Security on-line system, SNAP-R. Failure to obtain the proper export licenses in a timely manner will significantly delay contract award, become grounds for cancellation of the contract action, or become grounds for termination of an existing contract.

Topics under this announcement may be subject to ITAR/EAR and may be identified as such. However, export control compliance statements found in this document are not meant to be all inclusive. They do not remove any liability from the applicant to comply with applicable ITAR or EAR export control restrictions. For additional guidance on export control considerations, SBCs are encouraged to review the SBIR|STTR ITAR tutorial, which provides an overview of ITAR and EAR and their applicability to federally funded research.

3.3 Classified Proposals

Classified proposals will not be accepted under the DoW SBIR Program. If topics require classified work during Phase II, the proposing SBC must have a facility clearance to perform the work. For more information on facility and personnel clearance procedures and requirements, please visit the Defense Counterintelligence and Security Agency (DCSA) website at: https://www.dcsa.mil/mc/ctp/fc/.

3.4 Promotional Materials

Promotional and non-project related discussion is discouraged, and additional information provided via website links or on computer disks, CDs, DVDs, video tapes or any other medium will not be accepted or considered in the proposal evaluation.

3.5 Prior, Current, or Pending Support of Similar Proposals or Awards

While it is permissible, with proposal notification, to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program solicitations, it is unlawful to enter negotiation for contracts requiring essentially equivalent effort. If there is any question concerning prior, current, or pending support of similar proposals or awards, it must be disclosed to the soliciting agency or agencies as early as possible and declared on the proposal cover sheet.

3.6 Marking Proprietary Proposal Information

Proposing SBCs that include data in their proposals they do not want disclosed to the public for any purpose, or only used for government evaluation purposes, shall:

a. Mark the first page of each volume of the proposal submission with the following legend:

   "This proposal includes data that shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this proposing SBC as a result of-or in connection with-the submission of this data, the Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]"; and

b. Mark each sheet of data it wishes to restrict with the following legend:

   "Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

The DoW assumes no liability for disclosure or use of unmarked data and may use or disclose such data for any purpose.

Restrictive notices notwithstanding, support contractors may handle proposals and final reports submitted through DSIP for administrative purposes only; they are required to adhere to appropriate non-disclosure agreements.

3.7 Phase I Proposal Instructions

a. **Proposal Cover Sheet (Volume 1)**

   The proposal cover sheet is prepared on DSIP. The cover sheet must include a brief technical abstract that describes the proposed R&D project and an anticipated benefits and potential commercial applications discussion. Each section should be no more than 3,000 characters. **Do not include proprietary or classified information in the proposal cover sheet**. If your proposal is selected for negotiation and possible award, the technical abstract and anticipated benefits discussion may be publicly released online. DSIP will assign a proposal number once the cover sheet is saved. You may modify the cover sheet as needed until the topic closes.

   NOTE: the amounts listed in the percentage of work (POW) certification question on the proposal cover sheet are derived from SBC-entered information in the cost volume (Volume 3). Details on the calculation can be viewed in DSIP during proposal submission.

If the POW calculations fall below eligibility requirements, the funding agreement officer must upload either an explanatory letter or approval to the certification question to complete the submission. Some DoW Services/Components will not accept any deviations from the POW minimum requirements. Please refer to the Service/Component instructions regarding deviations acceptance to the POW requirements.

b. **Technical Volume Format (Volume 2)**

1. **File Type.** The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the technical volume file. If a virus is detected, the proposal may be rejected. **Do not lock, password protect or encrypt the uploaded file. Do not include or embed active graphics, such as videos, moving pictures, or other similar media, in the document.**

2. **Length.** It is the proposing SBC's responsibility to verify that the technical volume does not exceed the page limit after upload to DSIP. Please refer to Service/Component-specific instructions for how a technical volume is handled if the stated page count is exceeded. Some Services/Components will reject the entire technical proposal if the proposal exceeds the stated page count.

3. **Layout.** Number all proposal pages consecutively. Submit a direct, concise, and informative research or R&D proposal (no type smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins, including the header). Each header on each page in the technical volume should contain the proposing SBC's name, topic number, and the DSIP-assigned proposal number from the cover sheet.

c. **Technical Volume Content (Volume 2)**

The Technical Volume should cover the following items <u>in the order given below</u>:

1. Identification and Significance of the Problem or Opportunity
2. Phase I Technical Objectives
3. Phase I Statement of Work
4. Related Work
5. Relationship with Future Research or Research and Development
6. Commercialization Strategy
7. Key Personnel
8. Foreign Citizens
9. Facilities/Equipment
10. Subcontractors/Consultants
11. Prior, Current, or Pending Support of Similar Proposals or Awards
12. Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software

A Phase I technical volume template is available in Appendix A to provide details and helpful guidelines for completing each section of your Phase I technical proposal.

Refer to the Service/Component-specific Direct to Phase II instructions for details on proposal preparation and technical volume content requirements.

d. **Cost Volume Content (Volume 3)**

Complete the cost volume using the DSIP cost volume form. Some items in the cost breakdown may not apply to the proposed project. There is no need to provide information on each individual item; make sure to provide enough information for evaluators to understand the requested funds' planned use if a contract is awarded.

1. List all key personnels' names and include their individual hours dedicated to the project as direct labor.
2. While special tooling and test equipment and material cost may be included under Phase I, equipment and material inclusion will be carefully reviewed relative to need and appropriateness for the work proposed. Special tooling and test equipment purchases must, in the Service/Component contracting officer's opinion, be advantageous to the U.S. Government and should relate directly to the specific topic. These may include such items as innovative instrumentation or automatic test equipment. Title to property the U.S. Government furnished or acquired with government funds will be vested with the DoW Service/Component, unless it is determined that title transfer to the contractor would be more cost effective than the DoW Service/Component equipment recovery.
3. Cost for travel funds must be justified and related to the project needs.
4. Cost sharing is permitted for proposals under this solicitation; cost sharing is not required, nor will it be an evaluation factor in the Phase I proposal consideration.
5. A Phase I option (if applicable) should be fully costed separately from the Phase I (base) approach.
6. All subcontractor costs and consultant costs, such as labor, travel, equipment, materials, must be detailed at the same level as prime contractor costs. Provide detailed subcontractor costs substantiation in your cost proposal. Supporting Documents (Volume 5) may be used if additional space is needed.

If a proposal is selected for negotiation and possible award, you must be prepared to submit further documentation to the Service/Component contracting officer to substantiate costs (e.g., a cost estimates explanation for equipment, materials, and consultants or subcontractors). For more information about cost proposals and accounting standards, see visit DCAA's website.

e. **Company Commercialization Report (Volume 4)**

The company commercialization report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. SBA requires SBIR and STTR awardees to update and maintain their organization's CCR on SBIR.gov. Commercialization information is required upon the last deliverable's completion under the funding agreement. Thereafter, SBIR and STTR awardees are requested to voluntarily update the information in the database annually for a minimum 5-year period.

If the proposing SBC has prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards, regardless of whether the project has any commercialization to date, the firm admin must download the CCR's PDF copy from SBIR.gov and upload it to DSIP's "Firm Forms" section. The DSIP firm admin completes the firm forms are completed and are applies them to all proposals the proposing SBC submits. To fulfill the DSIP CCR requirement complete the following:

1. Log into the firm account at https://www.sbir.gov/.
2. Navigate to My Dashboard > My Documents to view or print the information currently contained in the "Company Registry Commercialization Report."

3. Create or update the commercialization record, from the company dashboard, scroll to the "My Commercialization" section and click the "Create/Update Commercialization" tab under "Current Report Version." Please refer to the "Instructions" and "Guide" documents contained on dashboard for more detail on completing and updating the CCR. **Ensure the report is certified and submitted.**
4. Click the "Company Commercialization Report" PDF under the dashboard's "My Documents" section to download the CCR PDF.
5. Upload the CCR PDF (downloaded from SBIR.gov in previous step) to the "Company Commercialization Report" in DSIP's "Firm Forms" section. The firm admin must complete this upload action.

In Volume 4 of the DSIP proposal submission, the proposing SBC will be prompted to answer: "Do you have a new or revised Company Commercialization Report to upload?" There are three possible courses of action:

a. If the proposing SBC has prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards and **has a new or revised CCR from SBIR.gov to upload to DSIP**, select YES.
   1. If the user is the firm admin, they can upload the CCR PDF from SBIR.gov directly on this page. It will also be updated in the "Firm Forms" and be associated with all new or in-progress proposals the proposing SBC submitted. If the user is not the firm admin, they will receive a message that they do not have access and must contact the firm admin to complete this action.
   2. **WARNING:** Uploading a new CCR under the DSIP "Firm Forms" section or clicking "Save" or "Submit" in one proposal's Volume 4 is considered a change for ALL proposals under any open solicitations. If a proposing SBC has previously certified and submitted any Phase I or Direct to Phase II proposals under ***any*** solicitation ***still open***, those proposals will be automatically reopened. Proposing SBCs will have to recertify and resubmit affected proposals. If a proposing SBC does not recertify or resubmit affected proposals, they will not be considered fully submitted and will not be evaluated.

b. If the proposing SBC has prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards, and **no new or revised CCR from SBIR.gov to upload to DSIP**, select NO.
   1. If a prior CCR was uploaded to the "Firm Forms", the proposing SBC will see a file dialog box at the bottom of the page and can view the previously uploaded CCR. This read-only access allows the proposing SBC to confirm the firm admin uploaded the CCR.
   2. If no file dialog box appears at the bottom of the page **there is no previously uploaded CCR in the DSIP "Firm Forms**." To fulfill the DSIP CCR requirement the firm admin must follow steps 1-5 listed above to download a CCR PDF from SBIR.gov and upload it to the DSIP "Firm Forms" to be included with all proposal submissions.

c. If the proposing SBC has **NO** prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards, the CCR upload from SBIR.gov is not required and SBC will select NO. The proposal's CCR section will be marked complete.

Please refer to the Service/Component-specific instructions for details on how the CCR information will be considered during proposal evaluations.

f. **Supporting Documents (Volume 5)**

Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3).

The following documents may be included in Volume 5, if applicable to the proposal. Refer to Service/Component-specific instructions for additional Volume 5 requirements. **Reminder: A completed proposal submission in DSIP does NOT indicate the mandatory supporting documents have been uploaded in accordance with the Service/Component-specific instructions.**

1. Letters of support
2. Additional cost information
3. Funding agreement certification
4. Technical data rights (assertions)
5. Lifecycle certification
6. Allocation of rights
7. Verification of Eligibility of Small Business Joint Ventures, if applicable
8. DD Form 2345, Militarily Critical Technical Data Agreement, if applicable (see section 3.2)

g. **Fraud, Waste, and Abuse Training (Volume 6)**

The fraud, waste, and abuse (FWA) training is **required** for DoW SBIR/STTR proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA. The training currently consists of a 3-page PDF, consistent with the tutorial provided by the SBA. This training material must be thoroughly reviewed once per year and can be found here and in the DSIP proposal submission module for Volume 6. Plan time to review the tutorial during completion of Volume 6, prior to the proposal submission deadline. The individual designated as the Proposal Owner must complete this training on behalf of the proposing small business.

h. **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**

In accordance with 15 U.S.C. §638(vv) and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this solicitation to assess security risks of SBCs seeking a federally funded award. SBCs must complete the DSIP Volume 7 webform "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (NOTE: PDF uploads are no longer accepted). The corporate official cannot certify and submit the full proposal until the Volume 7 webform is fully completed and submitted.

Please be aware that the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" form WILL NOT be accepted as a supporting document in DSIP's Volume 5 proposal submission. Do not upload any previous versions of this form to Volume 5.

For additional details, please refer to Section 2.5. The disclosure questions are below:

1. Is any owner or covered individual of the applicant or awardee party to any malign foreign talent recruitment program? If yes, disclose the first and last name of each owner or covered individual, identify their role (i.e., owner or covered individual), and the malign foreign talent recruitment program.

2. Is there a parent company, joint venture, or subsidiary, of the applicant or awardee that is based in or receives funding from, any foreign country of concern? If yes, disclose the name, full address, applicant or awardee relationships (i.e., parent company, joint venture, or subsidiary) of each entity based in, or funded by, any foreign country of concern.
3. Does the applicant or awardee have any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an enterprise owned by a foreign state or any foreign entity? If yes, disclose the name of each enterprise or foreign entity, type of obligation, agreement, or arrangement (*i.e.,* contractual, financial, or other), description of obligation, agreement, or arrangement, and the foreign state(s) and/or the country of the foreign entity (or entities).
4. Is the applicant or awardee wholly owned in a foreign country? If yes, disclose the foreign country.
5. Does the applicant or awardee have any venture capital or institutional investment? If yes, proceed to question 5a. If no, proceed to question 6.

   5a. Does the investing entity have a general partner or any other individual holding a leadership role who has a foreign affiliation with any foreign country of concern? If yes or unable to determine, disclose the venture capital or institutional investing entity's name, the percentage of ownership obtained by the investing entity, and the type of investment (i.e., equity, debt, or combination of equity and debt).
6. During the previous 5-year period, did the applicant or awardee have any technology licensing or intellectual property sales or transfers, to a foreign country of concern? If yes, disclose the name, address, and country, of the institution or entity that licensed, purchased, or received the technology or intellectual property.
7. Is there any foreign business entity, offshore entity, or entity outside the United States related to the applicant or awardee? If yes, disclose the entity name, relationship type (i.e., foreign business entity, offshore entity, entity outside the United States), description of the relationship to the applicant or awardee, and entity address and country.
8. Does the applicant or awardee have an owner, officer, or covered individual that has a foreign affiliation with a research institution located in a foreign country of concern? If yes, disclose the first and last name of each owner, officer, or covered individual that has a foreign affiliation with a foreign country of concern, identify their role (i.e., owner, officer, or covered individual), and the name of the foreign research institution and the foreign country of concern where it is located.

3.8 Phase II Proposal Information

Only Phase I awardees may submit Phase II proposals. Phase II proposals submission must follow individual Service/Component instructions. Awarding DoW Services/Component, either in the Phase I award or via subsequent notification, will provide details on Phase II proposal due date, content, and submission requirements. If a proposing SBC submits their Phase II proposal prior to the individual Service/Component's dates, it may be rejected without evaluation.

Due to specific limitations on the amount of funding and number of awards awarded to a particular proposing SBC per topic using SBIR/STTR program funds, head of agency determinations are required before a different agency may make an award using another agency's topic. This limitation does not

apply to Phase III funding. Please contact your original sponsoring agency before submitting a Phase II proposal to an agency other than the one who sponsored the original topic.

SBIR/STTR Policy Directive Section 4(b)(1)(i) allows that, at the agency's discretion, projects awarded a Phase I under a solicitation for SBIR may transition in Phase II to STTR and vice versa. A proposing SBC wishing to transfer from one program to another must contact their designated technical monitor to discuss the reasons for the request and the agency's ability to it. The transition may be proposed prior to award or during the Phase II effort performance. Agency disapproval of a request to change programs shall not be grounds for granting relief from any contractual performance requirement. All approved transitions between programs must be noted in the contracting officer-signed Phase II award or award modification that indicates the removal or addition of the research institution and the revised percentage of work requirements.

3.8.1 Phase II Commercialization Strategy

At a minimum, the commercialization strategy must address the following five questions:

a. What will be the first product to use this technology?
b. Who will be the customers, and what is the estimated market size?
c. How much money is needed to bring the technology to market, and how will that money be raised?
d. Does the proposing SBC contain marketing expertise and, if not, how will the SBC acquire that expertise?
e. Who are the proposing SBC's competitors, and what is the price and/or quality advantage over those competitors?

The commercialization strategy must also include a schedule showing the anticipated quantitative commercialization results at 1) one year after the Phase II project starts, 2) at Phase II completion, and 3) after Phase II completion (i.e., additional investment amount, sales revenue, etc.). After Phase II award, the proposing SBC is required to report actual sales and investment data in its SBA company commercialization report via SBIR.gov's "My Dashboard" on a minimum annual basis. Please refer to the Service/Component-specific instructions for guidance on formatting, page count and other details.

3.8.2 Phase II Adequate Accounting System

To reduce the small business's risk and avoid potential contracting delays, companies interested in pursuing Phase II SBIR contracts and other contracts of similar size with the DoW, may require an adequate accounting system in place per General Accepted Accounting Principles, Generally Accepted Government Auditing Standards, Federal Acquisition Regulation (FAR) and Cost Accounting Standards. The Defense Contract Audit Agency (DCAA) will audit the accounting system. See DCAA's website for requirements and standards, the audit process overview, and a pre-award system adequacy checklist.

4.0 METHOD OF SELECTION AND EVALUATION CRITERIA

4.1 Evaluation Process

All proposals will be evaluated and judged on a competitive basis in terms of technical capability and technical value. Proposals will be initially screened to determine responsiveness to the topic objective. Proposals passing this initial screening will be technically evaluated by engineers, scientists, or subject matter experts to determine the most promising technical and scientific approaches. As a common statement of work does not exist, each proposal will be assessed on the merit of the approach in achieving the technical objectives established in the topic.

4.2 Evaluation Criteria

Proposals will be evaluated based on the criteria outlined below, unless otherwise specified in the Service/Component-specific instructions. Selections will be based on a determination of the overall technical value of each proposal and an evaluation of the cost volume, with the appropriate method of analysis given the contract type to be awarded, for selection of the proposal(s) most advantageous to the Government, considering the following factors which are listed in descending order of importance:

a. The soundness, technical merit, and innovation of the proposed approach and its incremental progress toward topic or subtopic solution.
b. The qualifications of the proposed principal/key investigators, supporting staff, and consultants. Qualifications include not only the ability to perform the R&D but also the ability to commercialize the results.
c. The potential for commercial (government or private sector) application and the benefits expected to accrue from this commercialization.

Cost or budget data submitted with the proposals will be considered during evaluation.

Technical reviewers will base their conclusions only on information contained in the proposal. Do not assume reviewers are acquainted with the proposing SBC, key individuals, or any referenced experiments. Relevant supporting data such as journal articles, literature, including Government publications, etc., should be included based on requirements listed in Service/Component-specific instructions.

4.3 Proposal Status & Feedback

Proposing SBCs will be notified of selection or non-selection status for an award by the DoW Service/Component that originated the topic <u>no later than 90 days</u> of the closing date for this solicitation. Please refer to the Service/Component-specific instructions for details.

After final selection decisions have been announced, the SBC may be provided proposal feedback in the form of a written debrief. This debriefing process varies across the DoW Services/Components. Please refer to the Service/Component-specific instructions for details on the debriefing processes.

4.4 Award Denials

The DoW will not make an award under the SBIR program if it determines:

a. The SBC submitting the proposal
    i. has an owner or covered individual that is party to a malign foreign talent recruitment program;
    ii. has a business entity, parent company, or subsidiary located in the People's Republic of China or another foreign country of concern; or
    iii. has an owner or covered individual that has a foreign affiliation with a foreign entity located in the People's Republic of China or another foreign country of concern; and
b. The relationships and commitments described in clauses (i) through (iii) of subparagraph (A)

i. interfere with the capacity for DoW-supported activities to be carried out;
ii. create duplication with DoW-supported activities;
iii. present concerns about conflicts of interest;
iv. were not appropriately disclosed to the DoW;
v. violate Federal law or terms and conditions of DoW-awarded contracts or other agreements; or
vi. pose a risk to national security.

4.5 Pre-Award and Post Award Solicitation Protests

Interested parties have the right to protest via procedures in FAR Subpart 33.1; protests exclusively related to this solicitation's terms must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil

For protests filed with the Government Accountability Office (GAO), a copy of the protest shall be submitted to the email address listed above (pre-award ONLY) or DoW Service/Component POC (post-selection/award decision ONLY) within one day of filing with the GAO. Protests of small business status of a selected proposing SBC may also be made to the SBA via the procedures in FAR § 19.302.

For the purposes of a protest related to a particular topic selection, non-selection or award decision, protests should be served to the point-of-contact (POC) listed in the instructions of the DoW Component that authored the topic.

5.0 ADDITIONAL CONSIDERATIONS

5.1 Award Information

The number of awards will be consistent with the Component's RDT&E budget. No awards will be made until evaluation of all qualified proposals for a specific topic is complete.

Each proposal selected for negotiation and possible award will be funded under a negotiated instrument and may include a reasonable fee or profit consistent with normal profit margins consistent with normal profit margins provided to profit-making proposing SBCs for R/R&D work.

Contract value varies among the DoW Services/Components; it is important for proposing SBCs to review Service/Component-specific instructions regarding award size.

The SBA SBIR/STTR Policy Directive, Section 7(c)(1)(ii), states agencies should issue the award no more than 180 days after the closing date of the topic.

5.2 Contract Requirements

Upon contract award, the contractor will be required to make certain legal commitments through acceptance of U.S. Government contract clauses in the Phase I contract. The examples below are illustrative of the types of provisions the Federal Acquisition Regulation requires in the Phase I contract. This is not an exhaustive provisions list that could be included in Phase I contracts, nor does it contain specific clause wording. Appendix C of this solicitation contains additional potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Copies of complete general provisions will be made available prior to award.

While this solicitation does not address all potential clauses for resultant funding agreements, proposers are specifically directed to the Defense Federal Acquisition Regulation Supplement (DFARS) clause 252.204-7021, Cybersecurity Maturity Model Certification (CMMC) Requirements. This clause update reflects significant changes to cybersecurity requirements. Additional information provided in section 2.13 of this solicitation.

Examples of general provisions:

a. **Standards of Work**. Work performed under the Funding Agreement must conform to high professional standards.
b. **Inspection**. Work performed under the Funding Agreement is subject to Government inspection and evaluation at all times.
c. **Examination of Records**. The Comptroller General (or a duly authorized representative) must have the right to examine any pertinent records of the Awardee involving transactions related to this Funding Agreement.
d. **Default**. The Federal Government may terminate the Funding Agreement if the contractor fails to perform the work contracted.
e. **Termination for Convenience**. The Funding Agreement may be terminated at any time by the Federal Government if it deems termination to be in its best interest, in which case the Awardee will be compensated for work performed and for reasonable termination costs.
f. **Disputes.** Any dispute concerning the Funding Agreement that cannot be resolved by agreement must be decided by the contracting officer with right of appeal.
g. **Contract Work Hours**. The Awardee may not require an employee to work more than 8 hours a day or 40 hours a week unless the employee is compensated accordingly (for example, overtime pay).
h. **Equal Opportunity**. The Awardee will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin.
i. **Equal Opportunity for Veterans**. The Awardee will not discriminate against any employee or application for employment because he or she is a disabled veteran or veteran of the Vietnam era.
j. **Equal Opportunity for People with Disabilities**. The Awardee will not discriminate against any employee or applicant for employment because he or she is physically or intellectually disabled.
k. **Officials Not to Benefit**. No Federal Government official may benefit personally from the SBIR/STTR Funding Agreement.
l. **Covenant Against Contingent Fees**. No person or agency has been employed to solicit or secure the Funding Agreement upon an understanding for compensation except bona fide employees or commercial agencies maintained by the Awardee for the purpose of securing business.
m. **Gratuities**. The Funding Agreement may be terminated by the Federal Government if any gratuities have been offered to any representative of the Government to secure the award.
n. **Patent Infringement**. The Awardee must report each notice or claim of patent infringement based on the performance of the Funding Agreement.
o. **American Made Equipment and Products**. When purchasing equipment or a product under the SBIR/STTR Funding Agreement, purchase only American-made items whenever possible.

5.3 Agency Recovery Authority and Ongoing Reporting

In accordance with Section 5 of the SBIR and STTR Extension Act of 2022, the DoW will:

a. require an SBC receiving an award under its SBIR program to repay all amounts received from the federal agency under the award if,
   1. the SBC makes a material misstatement that the federal agency determines poses a risk to national security; or
   2. there is a change in the SBC's ownership, entity structure, or other substantial change in circumstances that the federal agency determines poses a risk to national security; and

b. require an SBC receiving an award under its SBIR program to regularly report to the federal agency and the administration throughout the duration of the award on

1. any change to a disclosure required under the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form;
2. any material misstatement made under paragraph (A) above; and
3. any change described in paragraph (B) above.

5.4 Copyrights

With prior written permission of the contracting officer, the awardee may copyright (consistent with any appropriate national security considerations) material developed with DoW support. DoW receives a royalty-free license for the Federal Government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

5.5 Patents

SBCs normally may retain the principal worldwide patent rights to any invention developed with U.S. Government support. The government receives a royalty-free license for its use, reserves the right to require the patent holder to license others in certain limited circumstances, and requires that anyone exclusively licensed to sell the invention in the United States must normally manufacture it domestically. 35 U.S.C. § 205 authorizes that the government will not make public any information disclosing a government-supported invention for a period of five years to allow the awardee to pursue a patent. See also Section 6.8, Invention Reporting.

5.6 Invention Reporting

SBIR awardees must report inventions to the Service/Component within two months of the inventor's report to the awardee, via either paper documentation submission, including fax, or through the Edison Invention Reporting System at www.iedison.gov for participating agencies.

5.7 Technical Data Rights

Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this solicitation generally remain with the contractor, except that the U.S. Government obtains a royalty-free license to use such technical data only for government purposes during the period commencing with contract award and ending not less than twenty years after that date. This data should be marked with the restrictive legend specified in DFARS 252.227-7018 (JAN 2025). Upon expiration of the twenty-year restrictive license, the government has government purpose rights in the SBIR data. During the license period, the U.S. Government may not release or disclose SBIR data to any person other than its support services contractors except: (1) for evaluation purposes; (2) as expressly permitted by the contractor; or (3) a use, release, or disclosure that is necessary for emergency repair or overhaul government-operated items.

If a proposing SBC plans to submit assertions in accordance with DFARS 252.227-7017 (JAN 2025), those assertions must be identified and assertion of use, release, or disclosure restriction must be included with your proposal submission, at the end of the technical volume. The contract cannot be awarded until assertions are approved.

5.8 Final Technical Reports - Phase I through Phase III

a. **Content**: A final report is required for each project phase. The reports must contain in detail the project objectives, work performed, results obtained, and estimates of technical feasibility. A completed SF 298, "Report Documentation Page," will be used as the first page of the report. The DoW Service/Component also may require monthly status and/or progress reports.

b. **SF 298 Form "Report Documentation Page" Preparation**:
   a. If desirable, the proposing SBC may also use language from its Phase II proposal to cover Phase I progress in the final report.

b. For each unclassified report, the proposing SBC submitting the report should fill in Block 12 (Distribution/Availability Statement) of the SF 298, "Report Documentation Page," with the following statement: "Distribution authorized to U.S. Government only; Proprietary Information, (Date of Determination). Other requests for this document shall be referred to the Service/Component SBIR Program Office."

   Note: Data developed under a SBIR contract is subject to SBIR Data Rights, under which DFARS 252.227-7018 (JAN 2025) (see Section 5.7, Technical Data Rights) provides protection. The sponsoring DoW activity, after reviewing the proposing SBC's entry in Block 12, has final responsibility for assigning a distribution statement.

   For additional information on distribution statements see the following Defense Technical Information Center (DTIC) website.

c. Block 14 (Abstract) of the SF 298, "Report Documentation Page" must include as the first sentence, "Report developed under SBIR contract for topic [insert solicitation topic number. [Follow with the topic title, if possible.]" The abstract must identify the work's purpose and briefly describe the work conducted, the findings or results, and the effort's potential applications. **Since DoW will publish the abstract, it must not contain any proprietary or classified data, and type "UU" in Block 17.**

d. Block 15 (Subject Terms) of the SF 298 must include the term "SBIR Report".

c. **Submission**: In accordance with DFARS 252.235-7011, submit an electronic copy of the approved final scientific or technical report, not a summary, delivered under the contract to the Defense Technical Information Center (DTIC) through the web-based input system at https://discover.dtic.mil/submit-documents/ as required by DoD Instruction 3200.12, DoW Scientific and Technical Information Program (STIP). Include a completed Standard Form (SF) 298, Report Documentation Page, in the document, or complete the web-based SF 298. Additional submission resources are available here.

   Delivery will normally be within 30 days after completion of the Phase I technical effort.

   Other requirements regarding reports and/or other deliverables submission will be defined in each contract's contract data requirements list (CDRL). Special instructions for submitting CLASSIFIED reports will be defined in the contract's delivery schedule.

   DO NOT email classified or controlled unclassified reports, or reports containing SBIR Data Rights protected under DFARS 252.227-7018 (JAN 2025).

6.0 PROPOSAL SUBMISSION

6.1 Submission Details

DSIP is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed guidance on DSIP proposal submission is found here.

Deadline for Receipt: Complete proposals must be certified and submitted in DSIP no later than the close date of the release listed on the cover page of this solicitation. Proposals cannot be submitted in DSIP after the deadline is reached and will not be accepted or evaluated.

The final proposal submission includes successful completion of all firm level forms, all required proposal volumes, and electronic corporate official certification. Although signatures are not required on the electronic forms at the time of submission, the proposal must be certified electronically by the corporate official for it to be considered submitted. If the proposal is selected for negotiation and possible award, the DoW Component program will contact the proposing SBC for signatures prior to award. **Please plan to submit proposals as early as possible to allow time for troubleshooting any possible issues before the topic close. DSIP Support is unable to assist with submission issues once a deadline has passed and cannot provide submission extensions. DoW is not responsible for missed proposal submission due to system latency.**

If the proposal status is "In Progress" or "Ready to Certify" it will NOT be considered submitted, even if all volumes are added prior to the topic close date. The proposing SBC may modify all proposal volumes prior to the topic close date.

6.2 Technical Questions

a. **Direct Contact with Topic Authors**. During the pre-release period, the names of the topic authors, their phone numbers and/or email addresses are published with the topic on the DSIP Topics and Topic Q&A page. During this time, proposing SBCs can contact topic authors via telephone or email to ask technical questions about specific solicitation topics. Questions must be limited to specific information related to understanding a particular topic's requirements. Proposing SBCs may not ask for advice or guidance on solution approach and may not submit additional material to the topic author.

   If information provided during an exchange with the topic author is deemed necessary for proposal preparation, that information will be made available to all parties through DSIP Topic Q&A.

   After the pre-release period, questions must be asked through DSIP Topic Q&A as described below. No further direct contact is allowed between proposing SBCs and topic authors, unless the topic author is responding to a question submitted during the pre-release period.

b. **DSIP Topic Q&A.** Proposing SBCs may submit written questions through DSIP Topic Q&A here, where all questions and answers are posted on a non-attribution basis for public viewing. DSIP Topic Q&A opens on the pre-release date and closes two weeks prior to the topic close date.

   Proposing SBCs may use the topic search feature on DSIP to locate a topic of interest. Use the form at the bottom of the topic description, enter and submit the question. Answers are generally posted within seven business days of question submission and also e-mailed directly to the inquirer.

   Questions submitted through the DSIP Topic Q&A are limited to technical information focused on understanding a topic's requirements. Any other questions, such as asking for advice or guidance on solution approach, or administrative questions, such as SBIR or STTR program eligibility, technical proposal/cost proposal structure and page count, budget and

duration limitations, or proposal due date are not appropriate and will not receive a response; for administrative questions, refer to a topic's Service/Component-specific instructions.

Once the proposal submission closes, no communication of any kind is allowed either with the topic author or through topic Q&A regarding submitted proposals.

Throughout the topic solicitation period, proposing SBCs should frequently monitor DSIP for updates and amendments to the topics and DSIP Topic Q&A for questions and answers.

7.0 Participating Component Instructions & Research Topics

The following section contains all Component-specific proposal preparation instructions and research topics this BAA.

DoW 2026 SBIR CSO
Release 1

April 13, 2026: Topics Pre-Release
May 06, 2026: Topics Open; DoW begins accepting proposals in DSIP
May 20, 2026: DSIP Topic Q&A closes to new questions at **12:00 p.m. ET**
June 03, 2026: Topics Close; Deadline for receipt of proposals is **12:00 p.m. ET**

Participating Services/Components:

- Department of the Navy (Navy)
- Department of the Air Force (DAF)
- Defense Advanced Research Projects Agency (DARPA)
- Defense Health Agency (DHA)
- Office of the Secretary of War – Strategic Capabilities Office (OSW-SCO)
- United States Special Operations Command (SOCOM)

DEPARTMENT OF THE NAVY (DON)
Small Business Innovation Research (SBIR)
DoW 2026 SBIR BAA Release 1
Phase I Proposal Submission Instructions

IMPORTANT

- **The following instructions apply to topics:**
  - **DON26BZ01-NV001 through DON26BZ01-NV039**

- Submitting small business concerns (SBCs) are encouraged to thoroughly review the DoW SBIR/STTR Program Broad Agency Announcement (BAA) and register for the DSIP Listserv to remain apprised of important programmatic changes.
  - The DoW Program BAA is located at: https://www.dodsbirsttr.mil/submissions/login. Select the tab for the appropriate BAA cycle.
  - Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

- The information provided in the DON Proposal Submission Instructions takes precedence over the DoW Instructions posted for this BAA.

- **DON Phase I Technical Volume (Volume 2) page limit is not to exceed 10 pages.**

- Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF) or any combination of these are eligible to submit proposals in response to DON topics advertised in this BAA. Information on Majority Ownership in Part and certification requirements at time of submission for these proposing SBCs are detailed in the section titled ADDITIONAL SUBMISSION CONSIDERATIONS.

- Phase I Technical Volume (Volume 2) and Supporting Documents (Volume 5) templates, specific to DON topics, are available at https://www.navysbir.com/links_forms.htm.

- The DON may consider the following FAR and Non-FAR contract strategies when issuing Phase I awards: Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT). The DON may consider the following FAR and Non-FAR contracting strategies when issuing Phase II awards: Cost Plus Fixed Fee (CPFF), FFP, BOA, or Prototype OT.

- This BAA is issued under regulations set forth in Federal Acquisition Regulation (FAR) 35.016 and awards will be made under "other competitive procedures". The policies and procedures of FAR Subpart 15.3 shall not apply to this BAA, except as specifically referenced in it. All procedures are at the sole discretion of the Government as set forth in this BAA. Submission of a proposal in response to this BAA constitutes the express acknowledgement to that effect by the proposing SBC.

INTRODUCTION

The DON SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DON's Fleet through research and development (R&D) topics that have dual-use potential, but

primarily address the needs of the DON. More information on the programs can be found on the DON SBIR/STTR website at www.navysbir.com. Additional information on DON's mission can be found on the DON website at www.navy.mil.

For questions regarding this BAA, use the information in Table 1 to determine who to contact for what types of questions.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS BAA

Type of Question	When	Contact Information
Program and administrative	Always	Navy SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	BAA Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoW SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoW SBIR/STTR Program BAA for details.
	BAA Open	DoW SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoW SBIR/STTR Program BAA for details.
Electronic submission to the DoW SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific BAA instructions and forms	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DON SYSTEMS COMMANDS (SYSCOM) SBIR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
DON26BZ01-NV001	Ms. Tatiana Sears	Marine Corps Systems Command (MCSC)	smb_mcsc_sbir_admins@usmc.mil
DON26BZ01-NV002 to DON26BZ01-NV014	Ms. Kristi DePriest	Naval Air Systems Command (NAVAIR)	navair-sbir@us.navy.mil
DON26BZ01-NV015 to DON26BZ01-NV021	Mr. Jason Schroepfer	Naval Sea Systems Command (NAVSEA)	NSSC_SBIR.fct@navy.mil

DON26BZ01-NV022 to DON26BZ01-NV036	Ms. Lore-Anne Ponirakis	Office of Naval Research (ONR)	usn.pentagon.cnr-arlington-va.mbx.onr-sbir-sttr@us.navy.mil
DON26BZ01-NV037 to DON26BZ01-NV039	Mr. Scott Steward	Strategic Systems Programs (SSP)	ssp.sbir@ssp.navy.mil

PHASE I SUBMISSION INSTRUCTIONS

The following section details requirements for submitting a compliant Phase I proposal to the DoW SBIR/STTR Programs.

(NOTE: Proposing SBCs are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.)

DoW SBIR/STTR Innovation Portal (DSIP). Proposing SBCs are required to submit proposals via the DoW SBIR/STTR Innovation Portal (DSIP); and follow proposal submission instructions in the DoW SBIR/STTR Program BAA on the DSIP at https://www.dodsbirsttr.mil/submissions. Proposals submitted by any other means will be disregarded. Proposing SBCs submitting through DSIP for the first time will be asked to register. It is recommended that SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposals that are not successfully certified electronically in DSIP by the Corporate Official prior to BAA Close will NOT be considered submitted and will not be evaluated by DON. Proposals that are encrypted, password protected, or otherwise locked in any portion of the submission will be REJECTED unless specifically directed within the text of the topic to which you are submitting. Please refer to the DoW SBIR/STTR Program BAA for further information.

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoW SBIR/STTR Program BAA.

- **Technical Proposal (Volume 2)**
  - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
    - Not to exceed ten (10) pages, regardless of page content
    - Single column format, single-spaced typed lines
    - Standard 8 ½" x 11" paper
    - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
    - No font size smaller than 10-point
    - Include, within the ten-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified. Phase I Options are exercised upon selection for Phase II.
    - Work proposed for the Phase I Base must be exactly six (6) months.
    - Work proposed for the Phase I Option must be exactly six (6) months.

- Additional information:
  - A Phase I proposal template specific to DON to meet Phase I requirements is available at https://navysbir.com/links_forms.htm
  - A font size smaller than 10-point is allowable for headers, footers, imbedded tables, figures, images, or graphics that include text. However, proposing SBCs are cautioned that if the text is too small to be legible it will not be evaluated.

- **Cost Volume (Volume 3).**
  - Cost Volume (Volume 3) must meet the following requirements or the proposal will be REJECTED:
    - The Phase I Base amount must not exceed $140,000.
    - Phase I Option amount must not exceed $100,000.
    - Costs for the Base and Option must be separated and clearly identified in Volume 3.
    - For Phase I, a minimum of two-thirds of the work is performed by the proposing SBC. The two-thirds percentage of work requirement must be met in the Base costs as well as in the Option costs. DON will not accept deviations from the minimum percentage of work requirements for Phase I. The percentage of work is measured by both direct and indirect costs. To calculate the minimum percentage of work for the proposing SBC the sum of all direct and indirect costs attributable to the proposing SBC represent the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) is the denominator. The subcontractor percentage is calculated by taking the sum of all costs attributable to the subcontractor (Total Subcontractor Costs (TSC)) as the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) as the denominator.
      - Proposing SBC Costs (included in numerator for calculation of the SBC):
        - Total Direct Labor (TDL)
        - Total Direct Material Costs (TDM)
        - Total Direct Supplies Costs (TDS)
        - Total Direct Equipment Costs (TDE)
        - Total Direct Travel Costs (TDT)
        - Total Other Direct Costs (TODC)
        - General & Administrative Cost (G&A)

        NOTE: G&A, if proposed, will only be attributed to the proposing SBC.
      - Subcontractor Costs (numerator for subcontractor calculation):
        - Total Subcontractor Costs (TSC)
      - Total Cost (i.e., Total Cost before Profit Rate is applied, denominator for either calculation)
    - **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. A value above or below $0.00 entered in the Cost Sharing field will not be considered in the Phase I contract award.**

  - Additional information:
    - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
    - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
    - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3. When a proposal is selected for award, be

prepared to submit further documentation to the SYSCOM Contracting Officer to substantiate costs (e.g., an explanation of cost estimates for equipment, materials, and consultants or subcontractors).

- **Company Commercialization Report (Volume 4)**. DoW collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DON may or will require to process a proposal, if selected, for contract award.

  - Proposing SBCs must review and submit the following items, as applicable:
    - **Majority Ownership in Part.** Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, are eligible to submit proposals in response to DON topics advertised within this BAA. Complete the certification as detailed under ADDITIONAL SUBMISSION CONSIDERATIONS.

  - Additional information:
    - Proposing SBCs may include the following administrative materials in Supporting Documents (Volume 5); a template is available at https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing SBC may want to include in Volume 5:
      - Additional Cost Information to support the Cost Volume (Volume 3)
      - SBIR/STTR Funding Agreement Certification
      - Data Rights Assertion
      - Disclosure of Information (DFARS 252.204-7000)
      - Prior, Current, or Pending Support of Similar Proposals or Awards
      - Foreign Citizens
    - Details of Request for Discretionary Technical and Business Assistance (TABA), if proposed, is to be included under the Additional Cost Information section if using the DON Supporting Documents template.
    - Do not include documents or information to substantiate the Technical Volume (Volume 2) in Volume 5 (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.
    - A font size smaller than 10-point is allowable for documents in Volume 5; however, proposing SBCs are cautioned that the text may be unreadable.

- **Fraud, Waste and Abuse Training Certification (Volume 6)**. DoW requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this BAA to assess security risks presented by SBCs seeking a Federally funded award. SBCs must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details.

PHASE I EVALUATION AND SELECTION

The following section details how the DON SBIR/STTR Programs will evaluate Phase I proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DON SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoW and DON SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing SBC has met eligibility requirements and followed the instructions for the Proposal Cover Sheet as specified in the DoW SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).** The DON will evaluate and select Phase I proposals using the evaluation criteria specified in the Method of Selection and Evaluation Criteria section of the DoW SBIR/STTR Program BAA, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation and proposals will not be compared to one another. Cost is not an evaluation criterion and will not be considered during the evaluation process; the DON will only do a compliance review of Volume 3. Due to limited funding, the DON reserves the right to limit the number of awards under any topic.

  The Technical Volume (Volume 2) will undergo a compliance review (prior to evaluation) to verify the proposing SBC has met the following requirements or the proposal will be REJECTED:
  - Not to exceed ten (10) pages, regardless of page content
  - Single column format, single-spaced typed lines
  - Standard 8 ½" x 11" paper
  - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
  - No font size smaller than 10-point, except as permitted in the instructions above.
  - Include, within the 10-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified.
  - Work proposed for the Phase I Base must be exactly six (6) months.
  - Work proposed for the Phase I Option must be exactly six (6) months.

- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will only undergo a compliance review to verify the proposing SBC has met the following requirements or the proposal will be REJECTED:
  - Must not exceed values for the Base ($140,000) and Option ($100,000).
  - Must meet minimum percentage of work; a minimum of two-thirds of the work is performed by the proposing SBC. The two-thirds percentage of work requirement must be met in the Base costs as well as in the Option costs. DON will not accept deviations from the minimum percentage of work requirements for Phase I.
  - **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. A value above or below $0.00 entered in the Cost Sharing field will not be considered in the Phase I contract award.**

- **Company Commercialization Report (CCR) (Volume 4).** The CCR (Volume 4) will not be evaluated by the DON nor will it be considered in the award decision. However, all proposing SBCs

must refer to the DoW SBIR/STTR Program BAA to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing SBC has included items in accordance with the PHASE I SUBMISSION INSTRUCTIONS section above.

- **Fraud, Waste, and Abuse Training Certificate (Volume 6).** Not evaluated.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Due Diligence Program to Assess Security Risks. Refer to the DoW SBIR/STTR Program BAA to ensure compliance with Volume 7 requirements.

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing SBCs to consider during proposal preparation and submission process.

Due Diligence Program to Assess Security Risks. The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of War, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by SBCs seeking a Federally-funded award. Please review the Certifications and Registrations section of the DoW SBIR/STTR Program BAA for details on how DoW will assess security risks presented by SBCs. The Due Diligence Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The SBIR and STTR Policy Directive section 9(b) allows the DON to provide TABA (formerly referred to as DTA) to its awardees. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; and commercializing the SBIR/STTR product or process, including intellectual property protections. Proposing SBCs may request, in their Phase I Cost Volume (Volume 3) and Phase II Cost Volume, to contract these services themselves through one or more TABA providers in an amount not to exceed the values specified below. The Phase I TABA amount is up to $6,500 and is in addition to the award amount. The Phase II TABA amount is up to $25,000 per award, is to be included as part of the award amount, and is limited by the established award values for Phase II by the SYSCOM (i.e., within the $2,000,000 or lower limit specified by the SYSCOM). As with Phase I, the amount proposed for TABA cannot include any profit/fee by the proposing SBC and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the SBIR proposing SBC. A Phase II project may receive up to an additional $25,000 for TABA as part of one additional (sequential) Phase II award under the project for a total TABA award of up to $50,000 per project. An SBC receiving TABA will be required to submit a report detailing the results and benefits of the service received. This TABA report will be due at the time of submission of the final report.

Request for TABA funding will be reviewed by the DON SBIR/STTR Program Management Office.

If the TABA request does not include the following items the TABA request will be denied.

- TABA provider(s) (firm name)
- TABA provider(s) point of contact, email address, and phone number
- An explanation of why the TABA provider(s) is uniquely qualified to provide the service

- Tasks the TABA provider(s) will perform (to include the purpose and objective of the assistance)
- Total TABA provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA must NOT:
- be subject to any indirect costs, profit, or fee by the SBIR proposing SBC
- propose a TABA provider that is the SBIR proposing SBC
- propose a TABA provider that is an affiliate of the SBIR proposing SBC
- propose a TABA provider that is an investor of the SBIR proposing SBC
- propose a TABA provider that is a subcontractor or consultant of the requesting SBC otherwise required as part of the paid portion of the research effort (e.g., research partner, consultant, tester, or administrative service provider)

TABA requests must be included in the proposal as follows:
- Phase I:
  - Online DoW Cost Volume (Volume 3) – the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.
- Phase II:
  - DON Phase II Cost Volume (provided by the DON SYSCOM) - the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.

Proposed values for TABA must NOT exceed:
- Phase I: A total of $6,500
- Phase II: A total of $25,000 per award, not to exceed $50,000 per Phase II project

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing SBC shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design, production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). An SBC whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DON Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DON Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Majority Ownership in Part. Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, **are eligible** to submit proposals in response to DON topics advertised within this BAA.

For proposing SBCs that are a member of this ownership class the following must be satisfied for proposals to be accepted and evaluated:

a. Prior to submitting a proposal, SBCs must register with the SBA Company Registry Database.
b. The proposing SBC within its submission must submit the Majority-Owned VCOC, HF, and PEF Certification. A copy of the SBIR VC Certification can be found on https://navysbir.com/links_forms.htm. Include the SBIR VC Certification in the Supporting Documents (Volume 5).
c. Should a proposing SBC become a member of this ownership class after submitting its proposal and prior to any receipt of a funding agreement, the proposing SBC must immediately notify the Contracting Officer, register in the appropriate SBA database, and submit the required certification, which can be found on https://navysbir.com/links_forms.htm.

System for Award Management (SAM). It is strongly encouraged that proposing SBCs register in SAM, https://sam.gov, by the Close date of this BAA, or verify their registrations are still active and will not expire within 60 days of BAA Close. Additionally, proposing SBCs should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. An SBC selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Cybersecurity Maturity Model Certification (CMMC) Program. DoW has established the CMMC Program to verify that awardees have implemented required security measures necessary to safeguard Federal Contract Information (FCI) and Controlled Unclassified Information (CUI). CMMC Level requirements are identified within each topic. Proposing SBCs should anticipate that a Projected CMMC Level for Phase II award may be higher than the Projected CMMC Level advertised in the Phase I topic. Proposing SBCs should carefully review and consider the CMMC requirements as compliance may impact proposed costs and technical approach. Please review the DoW SBIR/STTR Program BAA for additional information on the CMMC Program.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.240-7997, in order to be considered for award, an SBC is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit https://www.sprs.csd.disa.mil/nistsp.htm. For in-depth tutorials on these items, please visit https://www.sprs.csd.disa.mil/webtrain.htm.

Human Subjects, Animal Testing, and Recombinant DNA. Due to the short timeframe associated with Phase I of the SBIR/STTR process, the DON does **not** recommend the submission of Phase I proposals that require the use of Human Subjects, Animal Testing, or Recombinant DNA. For example, the ability to obtain Institutional Review Board (IRB) approval for proposals that involve human subjects can take 6-12 months, and that lengthy process can be at odds with the Phase I goal for time-to-award. Before the DON makes any award that involves an IRB or similar approval requirement, the proposing SBC must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human, animal, or recombinant DNA protocols. It will not impact the DON's evaluation, but requiring IRB approval may delay the start time of the Phase I award and if approvals are not obtained within two months of notification of selection, the decision to award may be terminated. If the use of human, animal, and recombinant DNA is included under a Phase I or Phase II proposal, please carefully review the requirements at: https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-

and-protections/research-protections. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

Government Furnished Equipment (GFE). Due to the typical lengthy time for approval to obtain GFE, it is recommended that GFE is not proposed as part of the Phase I proposal. If GFE is proposed, and it is determined during the proposal evaluation process to be unavailable, proposed GFE may be considered a weakness in the technical merit of the proposal.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I BAA Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Debriefs. Requests for a debrief must be made within 15 calendar days of select/non-select notification via email as specified in the select/non-select notification. Please note debriefs are typically provided in writing via email to the Corporate Official identified in the proposal of the proposing SBC within 60 days of receipt of the request. Requests for oral debriefs may not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the debrief request.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Pre-award agency protests related to the terms of the BAA must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DON Topics may be obtained from the DON SYSCOM Program Managers listed in Table 2 above. For protests filed with the GAO, a copy of the protest must be submitted to the appropriate DON SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DON reserves the right to limit the number of awards under any topic. Any notification received from the DON that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct compliance review of Volume 3 to confirm eligibility of the proposing SBC, and to take other relevant steps necessary prior to making an award.

Contract Types. A Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT) may be used for Phase I awards.

Funding Limitations. In accordance with the SBIR and STTR Policy Directive section 4(b)(5), there is a limit of one sequential Phase II award per SBC per topic. The maximum Phase I proposal/award amount including all options is $240,000. The Phase I Base amount must not exceed $140,000 and the Phase I Option amount must not exceed $100,000. The maximum Phase II proposal/award amount including all options (including TABA) is $2,000,000 (unless non-SBIR/STTR funding is being added). Individual SYSCOMs may award amounts, including Base and all Options, of less than $2,000,000 based on available funding. The structure of the Phase II proposal/award, including maximum amounts as well as breakdown between Base and Option amounts will be provided to all Phase I awardees either in their Phase I award or a minimum of 30 days prior to the due date for submission of their Initial Phase II proposal.

Contract Deliverables. Contract deliverables for Phase I are typically a kick-off brief, progress reports, and a final report. Required contract deliverables (as stated in the contract) must be uploaded to https://www.navysbirprogram.com/navydeliverables/.

Payments. The DON makes three payments from the start of the Phase I Base period, and from the start of the Phase I Option period, if exercised. Payment amounts represent a set percentage of the Base or Option value as follows:

Days from Start of Base Award or Option	Payment Amount
15 Days	50% of Total Base or Option
90 Days	35% of Total Base or Option
180 Days	15% of Total Base or Option

Transfer Between SBIR and STTR Programs. Section 4(b)(1)(i) of the SBIR and STTR Policy Directive provides that, at the agency's discretion, projects awarded a Phase I under a BAA for SBIR may transition in Phase II to STTR and vice versa.

PHASE II GUIDELINES

Evaluation and Selection. All Phase I awardees may submit an **Initial** Phase II proposal for evaluation and selection. The evaluation criteria for Phase II is the same as Phase I (as stated in this BAA). The Phase I Final Report and Initial Phase II Proposal will be used to evaluate the SBC's potential to progress to a workable prototype in Phase II and transition the technology to Phase III. Details on the due date, content, and submission requirements of the Initial Phase II Proposal will be provided by the awarding SYSCOM either in the Phase I contract or by subsequent notification.

Awards. The DON will consider the following for Phase II award: Cost Plus Fixed Fee (CPFF), Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT). Phase II awards can be structured in a way that allows for increased funding levels based on the project's transition potential. To accelerate the transition of SBIR/STTR-funded technologies to Phase III, especially those that lead to Programs of Record and fielded systems, the Commercialization Readiness Program was authorized and created as part of section 5122 of the National Defense Authorization Act of Fiscal Year 2012. The statute set-aside is 1% of the available SBIR/STTR funding to be used for administrative support to accelerate transition of SBIR/STTR-developed technologies and provide non-financial resources for the SBCs (e.g., the Navy SBIR Transition Program ,STP).

Navy SBIR Transition Program (STP). All Phase II awardees have the opportunity to participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: https://navystp.com. Phase II awardees will be contacted separately regarding this program.

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm that developed the technology as a result of a Phase I or Phase II award. The DON will give Phase III status to any award that falls within the above-mentioned description. Consequently, DON will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DON protect the rights of the SBIR/STTR firm.

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DON26BZ01-NV001 TITLE: Amphibious Combat Vehicle (ACV) Maneuver Improvements

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Human-Machine Interfaces

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Improve the Human-Machine Interface for operator control in the transition from water to land and improve water mobility in the surf zone for Amphibious Combat Vehicles (ACVs).

DESCRIPTION: The ACV is an adaptation of an Italian Combat Vehicle with enough changes to weight and buoyancy that water mobility has been negatively impacted. Improvements are needed to the operator's controls and water propulsion hardware to reduce operator workload and improve maneuverability in the water and surf zone.

The new design shall make the transition between land and water operations easier for the operator by simplifying the human-machine interface (i.e., having to monitor and actuate fewer controls). The focus should be on determining innovative functional capability and controls which will reduce cognitive load on the operator when entering and exiting the surf zone and traversing through water. The new design should also make the vehicle more responsive to the operator's input. The new design needs to consider maintenance and corrosion control. The design needs to maintain or, if possible, improve water speed, bollard pull, and water operation fuel economy. The new design shall not result in degraded performance as baselined by the current propulsion system.

The ACV powerpack currently provides a maximum of 690hp at 1,800rpm (490kW) with maximum torque of 2,036 ft-lb (2,761 N-m) at 1,500rpm. The size of the area that an improved water propulsion device MUST FIT is 28 inches by 26 inches by 26 inches not including potential bracketry. The propulsion device must operate in shallow water where it will be exposed to sand, mud and small rocks in the water flow.

PHASE I: Design a new control system and concepts to improve steering in both the water and transition modes (land and fording). Modeling and simulation will be used to document the new design, and a control system model will be developed to simulate system operation with operator inputs, corresponding water propulsion system outputs, and system feedback to the operator. A bench top system may be built to show how the concept would work and allow users to comment on the design. The design will be documented in a performance specification and an architecture diagram to be used in Phase II to build a prototype for testing.

PHASE II: Develop the design concept from Phase I into a fully functional prototype. The prototype system will be lab tested once the design is mature. The test plan for the prototype will be prepared by the awardee, and reviewed by the government, to test the performance against the specification developed in Phase I. Lessons learned from the prototype system will be used to update the performance specification developed in Phase I.

Phase II Option (if exercised): Upon successful lab performance of the prototype system, the government will make an ACV available, and a refined design will be generated and integrated into the ACV. An updated computer model of the refined system will be generated to simulate operation of the refined design for the government prior to vehicle integration. The ACV with the refined design will be tested at a commercial or government facility to validate performance. The government will provide test support to include operators. The test plan for the refined design will be prepared by the awardee, and reviewed

by the government, to test the performance against the updated Phase II performance specification. The refined design and test results will be documented in the final report.

PHASE III DUAL USE APPLICATIONS: Upon successful completion of testing, the final design shall be provisioned and catalogued with the necessary logistical documents to enable ACV modification and employment of the system based on battalion desire to implement on their vehicles.
The main transition path will be to the ACV. Other military vehicles that require a fording or swim mode may also be interested, to include the Stryker and the Advanced Reconnaissance Vehicle (ARV). Commercial applications include amphibious vehicles used to deliver cargo to remote locations where ports are not available.

REFERENCES:

1. BAE ACV Spec Sheet: https://www.baesystems.com/dam/jcr:36eb6327-a224-4f84-bc43-ab0a1914c618
2. ACV Entering Water: https://www.youtube.com/shorts/g37R9OcQ1nE
3. ACV Propellers: https://www.youtube.com/shorts/UWBseEba99M
4. ACV Launch from Beach with Camera at Propeller: https://www.youtube.com/shorts/KVLL_keP_d4

KEYWORDS: Amphibious; Maneuver; Controls; Water Propulsion; Water Mobility; Propulsor; Steering; Amphibious Combat Vehicle; ACV

DON26BZ01-NV002 TITLE: Integrated Metal Ceramic Matrix for High Strength Steels

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop an integrated metal matrix for high strength steels.

DESCRIPTION: Landing gear components are limited to the use of high strength steels due to their harsh loading applications and various environmental conditions. Typically, high strength steels are used to survive the load requirements. The two technologies currently applied to most landing gear components are Hard Chrome and high velocity oxygen fuel (HVOF). Each has their disadvantages that affects landing gear components. A replacement for Hard Chrome and HVOF is required to improve the readiness and safety of landing gear components.

Hard Chrome's main disadvantage is that it hides corrosion underneath the chrome plating which can lead to stress corrosion cracking in high strength steels. This failure mode would cause the complete loss of a landing gear system as the landing gear essentially snaps into pieces due to high stresses of landing. If corrosion is found before stress corrosion cracking occurs it still leads to the complete scrapping of landing gear components. This is due to Hard Chrome having no repair method. The only option for Hard Chrome is to replace, remove, and then reapply which takes days of machining and post machining. In addition to the machining, the application requires hazardous chemicals and produces waste that creates a health and safety risk to the fleet and its manufacturing personnel. Lastly, another risk with Hard Chrome is the dimensional limitations it provides. If too little or too much Hard Chrome is applied, the coating will immediately delaminate and damage landing gear and hydraulic components due to the foreign object debris (FOD) inside the system.

HVOF comes with its disadvantages as well. HVOF requires extremely low surface roughness on the pistons which have poor tribology. The poor tribology causes the hydraulics seals to perform dry and wear the seals away extremely quickly. Hydraulic fluid cannot stick to the walls of the piston due to the low surface roughness.

On top of the hydraulic disadvantages, the surface roughness requires precision post machining for long durations to survive the landing gear environments. In the fleet, the main issue seen with HVOF is spalling when the landing gear experiences high strains. When this occurs, the landing gear components must be removed and replaced.

This topic seeks an innovative solution that provides an integrated metal matrix for high strength steels that boosts the performance of and extends a component's survivability and improves a system's operational readiness and lifecycle costs. Current technology for titanium uses waveform energy. The process generates a targeted physical reaction within a substrate, activating the substrate at an atomic level for precise placement and gradient depth control of an integrated infusion. This infusion results in a matrix composite material that leverages the strengths of both components. The chemical bonding between a ceramic and the titanium alloy involves a combination of covalent and ionic characteristics — sharing and exchanging of electrons. This combination enhances the mechanical properties of the composite material, such as properties and porosity mitigation for corrosion protection, hardness for wear resistance, thermal stability, and overall durability, resulting in a metal-matrix suitable for various high-performance applications. Current technology can tailor characteristics such as hardness, electrical conductivity, thermal and oxidation, and mechanical strength. These meticulous adjustments enable the creation of the matrix with specific, desired functionalities, enhancing their performance in various applications to defeat corrosion, wear, erosion, thermal, and other challenges. For instance, a metal matrix

composite gradient depth infusions of titanium nitride (TiN) achieved hardness ratings of 2800-3100HV (micro-Vickers). Currently, the process is limited to transition metals; however, there is a need to adapt and develop it for application to high strength steels. This innovative solution will provide the benefits of both Hard Chrome and HVOF while eliminating the current limitations of the respective coatings.

PHASE I: The Phase I Option focuses on identifying a potential coating by evaluating the compatibility of metal integration properties with the proposed high- strength steel. This includes determining whether a metal matrix can be successfully formed and sustained on the high strength steel surface. Attention will be given to identifying optimal surface characteristics—such as roughness, texture, patterning, and placement adjustments—to enhance oil retention and lubricity within landing gear components. Desired material properties and suitable tooling methods will be established to achieve the required metal integration. Sample coupons will be created as feasibility evidence for developing the coating process. This will be followed by analysis and characterization of the metal integration within high-strength steel substrates. Finally prototype plans will be developed to realize initial geometric characteristics for a titanium alloy component tailored to the project's specifications.

PHASE II: Develop prototyped landing gear components with internal components using the developed integrated metal matrix. Perform landing gear qualification testing to ensure prototyped integrated metal matrix components can withstand landing gear environments.
Establish wear patterns, production process, and related properties.

PHASE III DUAL USE APPLICATIONS: Integrate the landing gear components into fleet aircraft.
Metal matrix composites are employed in advanced industries due to their high modulus and strength, favorable wear and corrosion resistance, and other good properties at elevated temperatures.
Aerospace: High-temperature components like exhaust nozzles, heat shields, and other components.
Engine components: Turbine disks, impellers, and other engine parts requiring high strength-to-weight ratios.

Structural components: Structures where lightweight and high strength are crucial.
Automotive: Engine parts like Piston rings, brake discs, and rotors benefit due to their high strength, wear resistance, and thermal conductivity.
Lightweight construction components to reduce vehicle weight, improving fuel efficiency.
Electronics: Thermal management heat sinks and electronic packaging to dissipate heat and improve device performance.
Industrial: Cutting tools due to their high strength and wear resistance. Wear-resistant parts in industrial machinery and tools where high wear resistance is needed.

REFERENCES:

1. Chen, L-Y.; Qin, P.; Zhang, L. and Zhang, L-C. "An overview of additively manufactured metal matrix composites: preparation, performance, and challenge." International Journal of Extreme Manufacturing, June 20, 2024. https://iopscience.iop.org/article/10.1088/2631-7990/ad54a4
2. Chandrasekar, P.; Balusamy, V.; Chandran, K.S. Ravi and Kumar, Harish. "Laser surface hardening of titanium–titanium boride (Ti–TiB) metal matrix composite." Scripta Materialia, Volume 56, Issue 7, April 2007, pp. 641-644, ISSN 1359-6462. https://doi.org/10.1016/j.scriptamat.2006.11.035 (https://www.sciencedirect.com/science/article/pii/S135964620600844X)

KEYWORDS: Landing Gear; Coatings; Metal-matrix; Ceramics; High-Strength Steels; Hard Chrome

DON26BZ01-NV003 TITLE: High Clutter Virtual Objects for Modeling and Simulation

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a modeling and simulation (M&S) application for generating high-fidelity, thermally-attributed virtual electro-optic and infrared (EO/IR) object models—specifically weather-explicit 3D clouds and/or debris fields—for integration into real-time scene generation systems.

DESCRIPTION: Developmental and Operational Testing (DT/OT) of Missile Warning Systems (MWS), Infrared Countermeasures (IRCM), and Intelligence, Surveillance, and Reconnaissance (ISR) systems are currently limited to in-flight tests or the use of recorded flight video in digital system models (DSM). These methods do not adequately replicate the complexity of battlefield, industrial, and urban environments, especially under high-clutter, thermally dynamic conditions.
To enhance system survivability and test realism, validated synthetic 3D scene models are required to represent high-fidelity thermal environments unachievable through traditional Test & Evaluation (T&E) methods. These models enable more effective assessment of performance and operational effectiveness across a range of mission scenarios.

The Navy's EO/IR Direct Inject (EOIRDI) initiative employs the Synchronized Kilohertz Injection Projection (SKIP) scene generation system to support hardware-in-the-loop engagements.
SKIP is capable of operating across multiple formats:
• 2k x 2k at 60 Hz
• 512 x 512 at 500 Hz
• 320 x 320 at 1kHz

To fully utilize SKIP's capabilities, synthetic test engagements must be developed to match specific system-under-test (SUT) frame rates and resolution formats, incorporating unique geographic locations and weather conditions.

This topic seeks a M&S application for generating high-fidelity, thermally-attributed virtual EO/IR object models—specifically weather-explicit 3D clouds and/or debris fields—for integration into real-time scene generation systems.

The tool must enable validated six degrees of freedom (6-DOF) physics-based simulations to support live, virtual, and constructive (LVC)-based survivability assessments of modern threat engagement systems in cluttered battlefield environments involving air-to-air missiles (AAM) and surface-to-air (SAM) threats. The solution must support scene generation at real-time frame rates (60 Hz, 500 Hz, 1 kHz) using the EO/IR rendering framework built on OpenSceneGraph (OSG) and Virtual Planet Builder (VPB). Models must include optical, thermal, and physical attributes—such as spectral absorption, emissivity, and reflectivity—across the MWIR band (3.0–5.0 microns), with scalability to 0.2–20.0 microns. The system will enable creation, rendering, and validation of thermally accurate clutter (clouds/debris) varying by temperature, atmospheric composition, and precipitation to support enhanced DT/OT of threat detection and survivability systems.

PHASE I: Perform an analysis to design a comprehensive 3D cloud and/or debris field model demonstrating the development strategy and defining the M&S requirements for software plugins using the OSG 3D virtual framework environment for integration into DoW scene generation systems. Explore the technical method for model engineering, system integration, verification and validation testing. Write a final report for prototype plans in phase II.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Development of 3D Cloud and/or debris field models for integration into DoW scene generation using the OSG framework. The radiometrically accurate 3D models are initially for use with a Navy Scene Generator system. Each model will be delivered with metadata from its generation that is needed to be fully incorporated into the Scene Generator both geometrically and radiometrically.

PHASE III DUAL USE APPLICATIONS: Continue 3D Cloud model development with the addition of new cloud, debris field, and other hot entity model types for addition of high clutter objects to threat engagement scenarios for high fidelity T&E of aircraft installed MWS and FLIR systems. All new 3D model types of data need include calibration meta data.
High clutter 3D cloud and debris field models have the potential for fire fighters and pilot training for building virtual training environments.

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KEYWORDS: Development Testing (DT); Digital System Model (DSM); Electro-Optical (EO); Infrared (IR); Intelligence, Surveillance, and Reconnaissance (ISR); IR Countermeasures (IRCM); Infrared (IR); Midwave IR (MWIR); Missile Warning Systems (MWS); OpenSceneGraph (OSG); Operational Testing (OT); Virtual Planet Builder (VPB)

DON26BZ01-NV004 TITLE: Aircraft Formation Flight Control Technology for Heterogeneous Formation Flight

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an advanced flight control architecture to enable greater range and endurance through precise automatic station keeping while flying in formation and exploiting vortex-generated upwash from upstream aircraft.

DESCRIPTION: Wake surfing (i.e., flying trail in close formation within the upwash of one or several lead aircraft) has demonstrated significant fuel savings on the order of 10-20%. Researchers have conducted multiple studies and executed flight demonstrations in the past that validated performance gains. However, the adoption of an operational capability still faces challenges.

One key challenge is the technical approach for trailing aircraft to maintain precise relative position behind upstream aircraft in the optimal location to maximize efficiency. While this task can be performed through manual pilot station keeping, the task is workload intensive and is not practical for long missions. There is a need for an autopilot flight control capability to maintain the position for optimum fuel savings (i.e., the “sweet spot”), realizing this significant range/endurance benefit opportunity with minimal or zero pilot workload. Flight control architectures must be capable of precise station keeping in aircraft formations of similar/dissimilar and manned/unmanned fixed wing aircraft. Flight control architectures may include techniques to sense the location of the vortex/upwash effects both with and without explicit knowledge of aircraft relative positions.

The objective is to create robust flight control laws for trailing aircraft in similar or dissimilar formations to exploit the benefits of wake surfing. Unique aircraft hardware and modifications should be minimized to the greatest extent possible to achieve this objective. To achieve robust control law development for precision formation flight, the problem can be broken into coarse and precision tracking problems, with some interdependencies between the two. It is strongly desired that both problems be solved without additional hardware integration for participating vehicles and zero data-link demands.

For coarse acquisition and tracking, it is expected that the relative position between participating aircraft needs to be established and maintained in the general vicinity of the lead’s wing-tip vortex. Relative position must be maintained while sequencing waypoints or tracking a heading or ground track to accomplish ingress/egress mission segments. Consideration in the development of coarse acquisition and tracking capability should be given to Global Positioning System unavailability.

For precision position tracking and control, it is expected that aircraft sensors (e.g. air data, inertial, flight controls) affected by the influences of the wing tip vortex on the trail aircraft can be identified and exploited to locate optimal position. Control architecture gains and surface mixing influences necessary

for acquiring and tightly tracking the optimal location in the presence of the non-linear wing tip vortices and free stream turbulence must be considered.

PHASE I: Define and develop a control law approach that provides a robust coarse and precision tracking schemes for automated formation flight to improve range. Create a control law development plan detailing the approach, rationale, schedule, key evaluations, robustness analysis, and other milestones. The plan should clearly identify expected parameters to be used for both the coarse and precision tracking loops (e.g., engine fuel flow, pitch vs Angle of Attack relationship changes, trim impacts), requirements, and rationale for their selection/derivation. Expectations for parameters sources (such as existing hardware, datalinks, or derived parameters) should be clearly documented, and any new hardware requirements should be made explicit. Control law architecture for all axes, expected gain setting, expected surface mixing approaches shall be discussed. The plan shall identify key analyses and iteration cycles to be performed in the maturation of control law algorithms. Preliminary modeling and simulation results assessing feasibility of the concept, including an accurate representation of the trailing vortex effects, are desired but not required during this Phase.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and implement a prototype solution for wake surfing including control algorithms (i.e., relative positioning and optimized vortex benefit positioning) evaluated and tuned for use between multiple dissimilar aircraft. Software capabilities also include: formation management algorithms, mission optimization algorithms, formation entry and exit algorithms/procedures, and displays unique to wake surfing. Implement the prototype solution in a six degree-of-freedom (6DOF) simulation environment (including pilot-in-the-loop) to demonstrate and evaluate the algorithms and displays. Produce analysis and reports describing the prototype solution and results.

The 6DOF simulation environment should include a) multiple aircraft (at least two aircraft of different type) to determine technical feasibility of the relative positioning algorithm and b) representation of the concept formation with 3+ aircraft to determine effect of larger formation sizes. The simulation must include an accurate representation of the systems that affect flight dynamic performance (actuation, latencies, hardware/sensors, etc.). Control algorithms should be able to handle different lead aircraft and aircraft sequencing inside the formation, handle a variety of maneuvers (e.g., turns and descents), enter and exit formation including failure contingency management, manage keep out zones for safety, and manage formation stability.

Navy aircraft simulation environments may be available for use and control law evaluation during Phase II.

PHASE III DUAL USE APPLICATIONS: Integrate the Phase II-developed algorithms and displays into future manned and unmanned platforms, including Collaborative Combat Aircraft Programs of Record. Dual use applications include relative navigation without GPS aiding, UAS swarming, and robust flight control systems.

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KEYWORDS: Wake Surfing; Formation Flight; Precise Relative Navigation; Vortex; Upwash; Long Range

DON26BZ01-NV005 TITLE: Low-Cost Multi-Mission Dip Sonar for Mine Warfare and Anti-submarine Warfare (Inner/Middle Zone)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Integrated Sensing and Cyber;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop initial designs for a reduced cost, next generation helicopter dip sonar system utilizing multi-frequency band capabilities for traditional and enhanced anti-submarine warfare (ASW) capabilities for both inner and middle zone coverage (broadening to wide area search) as well as introducing aviation (naval) mine countermeasure (AMCM) capabilities.

DESCRIPTION: The United States Navy has long utilized dipping sonar systems on aircraft for Air ASW. The most recent sonar systems continue to show dominance in the Air ASW role with the ability to cover larger and larger areas of ocean. Simultaneously, various configurations of acoustic, electro-optic and electromagnetic sensor systems have been used in AMCM operations, with the newest remaining fielded systems offering limited mission coverage. As the Navy looks to new maritime strike future vertical lift capabilities, there will be an increased effort to combine capabilities into fewer unique aircraft platforms. To facilitate the merger of missions into fewer aircraft, it will become crucial to also combine more mission capabilities into individual mission systems. The resultant design from this effort is expected to provide increased capabilities across more aircraft of a singular configuration with the combination of improved Air ASW capability and added AMCM capability into a singular mission system, which in turn also will reduce the expected training and logistics costs with fewer variants of equipment to cover. Additionally, with continued retirements of existing mine-countermeasures systems, the Fleet will have an urgent need for other air-based AMCM capabilities/coverage and may want to consider implementing capabilities on other naval helicopters using existing, modified, or new sensors of acoustic, electro-optic, magnetic, and radio-frequency types.

Traditionally, the Navy developed and fielded acoustic ASW and AMCM systems independently while the physics of the underwater acoustic environment is a shared problem with differing targets and typical frequency bands of interest as a result. Additionally, acoustic ASW systems (i.e., sonobuoys and helicopter dip sonars) are of compact size and can be utilized on a medium lift helicopter or smaller, while acoustic AMCM systems have typically targeted installation on heavy-lift helicopters. Incorporation of a secondary frequency band capability into a helicopter dip sonar transducer assembly would quickly bring AMCM capability to a typically large number of traditionally ASW helicopters and bring air-based AMCM capability to the Navy's air-capable ships, simultaneously with ASW capability. The multi-mission capability of such a sonar transducer assembly would also allow one aircraft, without reconfiguring, cover both ASW and AMCM mission sets for reduced maintenance and reducing the equipment needed to be stored while afloat in space-constrained ships.

The objective is to develop initial designs for a reduced cost, next generation helicopter dip sonar system utilizing multi-frequency band capabilities for traditional and enhanced anti-submarine warfare (ASW) capabilities for both inner and middle zone coverage (broadening to wide area search) as well as introducing aviation (naval) mine countermeasure (AMCM) capabilities.

The system would also be utilized either in its full capability configuration or at a reduced capability configuration as a retrofit into the multi-mission helicopter as a replacement for the existing dipping sonar system transducer, while at a decreased unit and sustainment cost (below Class A mishap thresholds if lost in flight, with a goal of below a Class C threshold).

Minimally funded Science and Technology efforts have previously been performed to assess USN dipping sonar capability to detect naval mines using the system, acoustic pulses/frequencies, and processing in its existing ASW configuration and have shown success in detecting nearly every naval mine based on post-flight data analysis. Enhancing that capability with a secondary frequency band and associated beam steering, as well as uniquely developed pulses and processing across both frequency bands, is expected to provide a significant AMCM capability while retaining both traditional ASW superiority and enhanced ASW detection and classification capabilities for certain scenarios.

In addition to introducing AMCM capabilities into a traditional ASW sensor system, no significant improvements in the traditional ASW sonar transducer assemblies available from industry have been introduced since the last dipping sonar system competitive source selection conducted in the late 1980s. Increasing costs of the existing USN sonar systems continue to drive concerns regarding the long term affordability of the existing fielded systems and any future variants thereof, and continue to pose a risk of generating an equipment cost loss equivalent to a Class A mishap record if the transducer is lost from the aircraft. As such, decreasing the recurring production costs of a future transducer assembly are of significant concern and ensuring improved supportability. Noting that sonobuoys are similar advanced acoustic sensor systems made in large quantities for production unit costs of less than $15k/each indicates that a highly capable sonar transducer design would be capable of being generated with a much more reasonable forecast production cost well below $500k/each.

Additionally, the ability for the new sonar transducer to be retrofit in place of existing USN fielded sonar transducers (form/fit/function compatible) used on the existing USN aircraft while utilizing existing sonar processing (~3-5 kHz frequency band) and bringing AMCM capability and new added ASW capabilities to the traditionally ASW-focused helicopters is of interest utilizing a higher frequency band in the same unit.

Lastly, it would be a significant advancement in helicopter-based ASW capabilities if a tertiary frequency band below 2 kHz was also added to expand mission capabilities to broach wide area search and explore advantages of convergence zone type capabilities, while retaining the inherent existing direct path detection coverage of the mid-frequency 3-5kHz band, for full spectrum coverage of the surrounding areas.

The new multi-frequency band sonar transducer would be desired to have at least the following characteristics:
- Primary transmit array would be omnidirectional for ASW in the horizontal plane
- Primary acoustic transmit band for ASW: 3-5 kHz.
- Primary receive array would be capable of supporting 24 beams for primary ASW capabilities
- Consider using Single Crystal transducer technology or other new technology to reduce the weight and improve bandwidth.
- Overall weight must be less than 180 lbs.

- Primary electronics power and transmission signal power for the unit must be provided from an external transmitter/amplifier.
- Primary acoustic processing must occur offboard (not within unit)
- Secondary higher frequency band must be selected for AMCM mission optimization
- Secondary transmit and receive array functionality could reuse the primary arrays, utilizing electronic or physical manipulation as needed/possible to optimize AMCM. Alternatively, integrating other transmit and/or receive arrays within the same assembly may be acceptable.
- The secondary array capabilities would consider abilities to steer beams both horizontally and vertically depending on both mine and submarine targets of interest.
- As allowable, a tertiary capability of covering lower frequencies for longer range area searches and overlap with current other low frequency system operational frequencies (below 2 kHz) is preferred, broaching wide area search capabilities, with a system not requiring logistical complexities of storing large quantities of sonobuoys while associated to an aircraft deployed on a ship, and taking advantage of long range convergence zone detections
- Mechanically extended and retracted arrays are acceptable, as these are traditionally used in the most recent ASW sonar transducers.
- Will be capable of storage within an aircraft body for forward flight, ideally with an overall stowed diameter of no greater than 210 mm for the primary body and an overall length no greater than 1275 mm (some extensions for stabilizing features may be permissible).
- The CG of the sonar transducer assembly body will be designed to be as low as possible for stability in lowering operations, with an upper limit of no greater than 35% of the length of the overall unit as measured from the bottom.
- The final fielded unit would incorporate a water thermocouple for measuring the water column temperature during lowering operations, a method for bottom proximity detection, a capability to protect itself during electrostatic discharge when lowered from a helicopter into the sea water, redundant depth sensing capabilities, angular orientation reporting relative to vertical, and a method for determining bearing orientation of the array (e.g., magnetic compass, field sensors, other).
- Acoustic elements would be physically or electronically steerable in the vertical plane, providing enhanced sea bottom scanning for bottomed targets, and ideally, ability to determine target depth for setting of weapon depth deployments for improved success in target engagement
- The unit design would be able to withstand operating depths to at least 2500 ft.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Design a concept for a low-cost multi-frequency band sonar transducer assembly capable of supporting both ASW and AMCM mission sets. Create scale electronic models of the concept showing the integration of both capabilities within a single assembly, illustrate and explain conceptually how the assembly would be utilized for both ASW and AMCM missions in relation to the two (or more) primary frequency band capabilities.

Demonstrate via analysis and simulation of the ability to use the mid-frequency band to cover traditional both inner and middle zone ASW through mono-static and multi-static methods combined with new technology and unique operating/processing methods.

Via analysis and simulation, demonstrate the effectivity of the mid-frequency band and the higher frequency band for rapid detection of mines at various points in the water column including floating, moored, and bottomed types.
Prepare a Phase II plan.

PHASE II: Develop a scale prototype of the sonar transducer assembly designed in Phase I for demonstrating physical sizing, fit, weight, CG, and mechanical functionality (not necessarily operable except for manual manipulation) to allow for demonstration of fit into a USN ASW-capable helicopter. Develop and demonstrate an acoustically functional and watertight representative prototype array in water, including development of an initial design specification for the sonar transducer. Initial demonstration and validation via computer modeling in full, or in part, will be considered if funding availability limits full hardware/software prototype construction.
Prepare a Phase III commercialization/transition plan.

It is probable that the work under this effort will be classified under Phase II (see the Description for details).

PHASE III DUAL USE APPLICATIONS: Develop, build, and deliver a flight-worthy and fully functional sonar transducer assembly. Conduct full box level functional and environmental qualification (test and/or analysis), support field testing conducted by the USN, and support flight test operations on a USN MH-60R in both anti-submarine and mine-countermeasure scenarios. Create a system specification and drawing set for the final product. Verify through test compliance to specification and verification of modeling performed in Phase II. Develop and deliver high level concept of employment and operation. Assess and report on viability of using for mine-classification in addition to detection capabilities. Enhancements in underwater sonar systems could be applied to improved sonar systems used for offshore geographical exploration (mining, oil, etc.), marine surveys, and additionally could be beneficial in accelerating methods for search/rescue/recovery of personnel and equipment associated with ships and aircraft lost at sea. Perform analyses and modeling on effectivity in these alternate roles and any other applications. Provide a final report of this assessment.

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KEYWORDS: Sonar; anti-submarine warfare; ASW; undersea warfare; USW; aviation mine countermeasure; AMCM; Mine detection; Acoustic

DON26BZ01-NV006 TITLE: High-Gain Directional Low-Frequency Sonobuoy Array

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Advanced Materials;Integrated Network Systems-of-Systems

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a high-gain, low-frequency vertical line array of vector sensors capable of long-range passive detection and enhanced signal processing, deployable in an A-size form factor.

DESCRIPTION: To enhance anti-submarine warfare (ASW) detection and directional sensitivity in deep waters, the U.S. Navy requires a high-gain directional, low-frequency (< 500Hz) sonobuoy array, housed in an A-size form factor. It is expected that array gain will be achieved by using modelled and measured vertical noise profiles. The system must leverage novel sensor configurations and array geometries to maximize low-frequency Signal-to-Noise Ratio (SNR) while remaining compatible with existing sonobuoy communication and processing platforms. Advanced beamforming, signal processing, and robust hardware integration are crucial for extended detection ranges and minimized false alarms. Environmental factors like multipath interference, ambient noise, self-noise, and sensor stability must be addressed to ensure reliable performance in contested environments. This sonobuoy will bolster fleet ASW capabilities by delivering superior signal clarity, longer-range detection, and a decisive operational advantage through improved sensor capabilities and operational durations.

The objective is to develop a high-gain, low-frequency vertical line array of vector sensors capable of long-range passive detection and enhanced signal processing, deployable in an A-size form factor.
The system will be deployed from Navy Maritime Patrol and Reconnaissance Aircraft, have capability across multiple operational environments, and will utilize the necessarily varied hardware configurations, passive processing, and frequency characteristics to consistently achieve critical ASW metrics.
The sonobuoy must support deep-water tactical operations. Deployment depths up to 1000’ and 8 hours of life is required. The array design will provide 17 dB of gain at the design frequency in a three-dimensional isotropic ambient noise field as a minimum. The maximum saturation level will be 128 dB/µPa at 100 Hz with a total dynamic range of 96 dB. The sensor solution must be low power and fit within an "A" size sonobuoy (4.875-inch diameter x 36-inch length, weight under 40 pounds). Acoustic data sent to the aircraft from each vector sensor shall consist of Omni, Sine, and Cosine data. The communications link must comply with NATO's STANAG 4718. Long term plans include using the array in a persistent sonobuoy.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow

contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Establish the baseline sensor requirements working in conjunction with the Navy Technical Point of Contact (TPOC). Perform comprehensive analytical and numerical modeling to define the optimal sensor and array design for achieving the necessary gain using low frequency vertical noise profiles. Concurrently, develop a preliminary system architecture that addresses array geometry, deployment mechanisms, and communication protocols. Conduct trade studies on various sensor technologies, including velocity sensors, to select the most effective array configuration. Environmental noise factors will also be evaluated to determine their impact on overall system performance. Conduct trade studies on passive processing enhancements and adaptive beamforming to maximize detection range at these low frequencies.

A proof-of-concept simulation for a tactical sonobuoy will be developed to demonstrate the feasibility of the proposed approach, guiding both design decisions and risk mitigation. The Phase I effort will conclude with the generation of a high-level prototype design to be implemented during Phase II, ensuring a clear path from concept to operational capability.

Demonstrate materials/software/hardware required for prototype development can be sourced, produced, or obtained within a reasonable timeframe to allow for prototype development without requiring design changes due to unavailability, backorders, or extended lead times.

Demonstrate that preliminary testing/modelling has been performed on the materials/software/hardware showing that they possess the characteristics used for modeling or that at least the component has the characteristics and performance required for the system to achieve the objectives.

Phase I Requirements:
• Test design concepts(s) that have been demonstrated and validated to meet the objectives and can be scaled down to fit within the A-size constraints.
• Awardee must provide a full report outlining where their models have been performed with sufficient accuracy and depth to produce meaningful results to meet the objective.
• Demonstrate that preliminary testing/modelling has been performed on the materials/software/hardware showing that they possess the characteristics used for modeling or that at least the component has the characteristics and performance required for the system to achieve the objective.
• Awardee must also prove the models and results provided in the report originated from the SBIR effort and provide evidence that the raw materials/software/hardware required for prototype development can be sourced, produced, or obtained within a reasonable timeframe to allow for prototype development without requiring design changes due to unavailability, backorders, or extended lead times.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Design and fabricate an over-the-side deployable sonobuoy prototype rooted in Phase I findings and conduct over-the-side testing in both controlled facilities and actual aquatic environments to validate performance.

Integrate beamforming and signal processing algorithms optimized for the low-frequency range, ensuring robust detection and adaptation under varied ambient noise conditions.

Provide a thorough evaluation of the system's capabilities in deep-water scenarios, refining hardware and software elements in collaboration with domain experts.
Finalize the design concept by detailing a comprehensive roadmap for Phase III transition, including manufacturability considerations and integration with existing ASW platforms to maximize system synergy.

Conduct a study to determine the feasibility of extending the concept to a persistent capability with an operational life of 24 hours or greater.
Work in Phase II may become classified. Please see note in Description section.

PHASE III DUAL USE APPLICATIONS: Develop a production-ready design and specification for the Phase II solution and its accompanying algorithms, then proceed with integrated engineering and operational testing of the air-deployed system to verify full operational functionality in Navy-supported scenarios.

Demonstrate the system's adaptability and resilience in diverse maritime environments, ensuring seamless integration with Navy airborne ASW platforms and laying the groundwork for operational deployment. Upon successful qualification, transition to the Fleet and refine operational parameters through at-sea trials, leveraging real-world performance data.

Lastly, explore commercial applications, including marine mammal detection, underwater resource exploration, and environmental monitoring, to extend the solution's utility beyond Navy missions. Technology developed under this SBIR topic could be leveraged to achieve smaller and lighter systems capable of attaining key ASW measurements. This type of system capability may be of interest to the undersea mapping, exploration, seismology, and weather communities and used for monitoring marine mammals or icebergs. Government agencies such as the National Oceanographic and Atmospheric Administration (NOAA) and the Department of Commerce are continually trying to upgrade their measurement and data collection capability.

These sensors could fulfill a need to provide in-situ measurements for low rfequency acoustics. By developing reliable, low-cost sensor components, more capability and performance can be achieved.

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KEYWORDS: Anti-Submarine Warfare; ASW; Sonobuoy; Low-Frequency Acoustics; Directional Arrays; Passive Detection; Beamforming; Underwater Sensing

DON26BZ01-NV007 TITLE: Test and Alert System for Type 1 Encryption Device Hold-up Batteries (HUB)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a common Type 1 encryption device Hold-Up Battery (HUB) tester and accompanying low battery alert device.

DESCRIPTION: Develop a universal Hold-Up Battery (HUB) tester and integrated low-voltage alert system for Type 1 encryption devices. These devices rely on HUB batteries to retain mission-critical software. Failure to replace depleted batteries within specified intervals often renders them inoperable, necessitating costly returns to depots or vendors for software recovery.

The proposed solution must provide:
• A non-invasive HUB battery tester compatible across multiple device types
• A low-battery alert mechanism to signal impending voltage failure
• A streamlined method for monitoring and managing battery replacement intervals
This capability will significantly reduce lifecycle costs, improve operational readiness, and mitigate the risks associated with device storage in long-term vault conditions.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Investigate and propose design approaches for a universal HUB testing device compatible with a range of Type 1 encryption devices Emphasis will be placed on a non-invasive testing methodology to assess battery health without compromising device security or data integrity.
Evaluate the technical feasibility of developing a compact, attachable low-battery alert module capable of operating within storage conditions and security constraints typical of Type 1 encryption devices.
Identify common HUB battery characteristics across platforms and establish baseline voltage thresholds for end-of-life alerts.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Based on Phase I modeling, design and develop a prototype HUB testing device for Type 1 encryption devices. Execute Developmental Test and Evaluation (DT&E) activities to identify the system's capabilities, limitations and deficiencies. Provide DT&E data for cost, performance and schedule tradeoffs.
It is probable that the work under this effort will be classified under Phase II (see the Description for details).

PHASE III DUAL USE APPLICATIONS: Further develop the prototype(s) generated in Phase II for use in COMSEC facilities and Fleet activities that store and maintain Type 1 encryption devices with HUBs. By identifying and developing devices that test and alert to HUB status, this research benefits the private sector by enabling reliable and robust commercial solutions for testing, tracking and replacing batteries that may be failing. This could potentially lead to improved performance, reduced downtimes and replacement/recovery cost associated with these devices when the batteries fail.

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KEYWORDS: Type 1 Encryptor; Hold-Up Battery; HUB; COMSEC; Vault Storage; Bricked Encryptor; Battery Testing

DON26BZ01-NV008 TITLE: Automated Expeditionary Airfield Assembly

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Sustainment;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Research, develop, and evaluate robotic system methodologies for automating or augmenting the assembly of Expeditionary Airfield (EAF) matting to enhance operational efficiency.

DESCRIPTION: EAFs serve as vital shore-based aviation support systems that enable the rapid deployment and recovery of military aircraft in environments lacking established infrastructure. Currently, assembling EAF matting is a manual process carried out by Marines—a task that is physically demanding, labor-intensive, and exposes personnel to potential hazards.
Developing a robotic system capable of assisting with or fully automating this assembly process would offer significant operational benefits: increasing efficiency, reducing risk to personnel, and enabling Marines to focus on higher-priority mission objectives. The level of autonomy should allow for the robots to navigate and control without human assistance, which includes obstacle avoidance, path planning, and grasping. Such a solution would improve overall force readiness and effectiveness in austere and time-critical operational scenarios.

The approach includes defining and developing a viable system concept, while investigating various robotic configurations—such as mobile manipulators and assistive technologies—for their effectiveness in EAF mat handling, alignment, and interconnection across diverse and austere terrains.
The research will evaluate the proposed system's capacity to:
• Traverse and operate on uneven or unstable surfaces
• Manipulate and position heavy EAF mat sections with precision
• Endure harsh environmental and operational conditions
• Integrate seamlessly with current EAF deployment procedures

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor and/or subcontractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Demonstrate the technical feasibility of a robotic system capable of automating or augmenting the assembly of EAF prefabricated surfaced aluminum (PSA) Flat Top-Nested (Top-N) Trackway mats. This research will lay the groundwork for future development of a deployable robotic solution, enhancing safety, speed, and efficiency in EAF setup operations.

Focus on designing and modeling key system components to evaluate performance across critical metrics, including payload capacity, reach, manipulation precision, power consumption, and operational endurance. Under the Phase I Option, if exercised, simulations will be conducted to assess system behavior in representative virtual EAF deployment environments and to identify key technical risks and milestones. The Phase I effort will culminate in a comprehensive final report detailing the proposed system architecture, design specifications, and simulation-based performance evaluations, along with a clearly defined Phase II roadmap for prototype development, field testing, and integration planning.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and demonstrate a functional prototype of the robotic system for automated or semi-automated assembly of EAF mats. Refine the selected robotic system concept and fabricate a fully functional prototype incorporating the chosen locomotion system, end-effectors, sensors, and control architecture.

Rigorous testing will be conducted in both laboratory and field settings, culminating in a demonstration of the prototype's capabilities in a representative EAF deployment scenario. Testing data and operator feedback will be used to refine the prototype's design and functionality.
Finally, a comprehensive transition plan will be developed, outlining the steps required to move the robotic system towards technology demonstration and eventual deployment, ultimately enhancing the capabilities and effectiveness of expeditionary forces.
The robot shall be able to handle the PSA mats in some manner to aid in the assemble of the airfield, be a closed system, and able be able to operate in a realistic environment.
The system will be judged on feasibility, time to assemble, ease of use, and overall size and mass.

Deliverables include a prototype; the open interface specification; software design documents; the uncompiled, human-readable source code; associated comments and documentation; and any tuned parameters and weights; schematics of the robot.
Work in Phase II may become classified. Please see note in the Description section.

PHASE III DUAL USE APPLICATIONS: Leverage Phase II findings to develop a robust and deployable robotic system for EAF mat assembly, optimized for real-world operational scenarios. Refine the design and engineering of the locomotion system, end-effectors, sensors, deployment mechanisms, power and charging strategies, and control architecture using collected test date and operator feedback.
Crucially, the system will undergo hardening against electrical, environmental, and cyber threats. The resulting system must demonstrate sustained operation in deployed environments, achieving significant reductions in manning requirements, operational costs, and/or deployment time. Conduct rigorous field testing that culminates in a full-scale demonstration of EAF deployment, showcasing the system's deployment method and power management capabilities.

Collected test data and operator feedback will drive further system refinement and optimization of operational use cases. Evaluation criteria will include feasibility, assembly time, deployment method efficiency, power solution effectiveness, ease of use, and overall size and mass. Deliverables include a full-scale EAF assembly demonstration, open interface specifications, software design documents, uncompiled and documented source code, tuned parameters and weights, robot schematics, and a comprehensive deployment and delivery plan.

The technology developed for this SBIR topic will have dual use in construction allowing for the rapid deployment of flooring and laying of other interlocking material. Other technologies, such as the development of man-unmanned teaming, perception modeling, and enhanced understanding of unobservable environmental conditions, will drive advancements in robotics, computer vision, and autonomy, with broad implications across multiple domains.

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KEYWORDS: Robotics; Artificial Intelligence; AI; Navigation; Manipulation; Automation; Navigation

DON26BZ01-NV009 TITLE: Open Architecture for a Low Volume Software Defined Radio (SDR) for Navy Aircraft

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Integrated Network Systems-of-Systems

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design, develop, and demonstrate an innovative airborne radio system with a reduction compared to current airborne radios. The solution will incorporate a Modular Open Systems Approach (MOSA) and Model-Based Systems Engineering (MBSE) methodologies to ensure seamless integration across Navy and Marine Corps platforms including fixed wing, rotary wing and UAV aircraft.

DESCRIPTION: The Navy seeks an innovative, open-architecture airborne radio system optimized for a minimal Size, Weight, and Power (SWaP) to ensure seamless integration across a wide range of NAVAIR platforms, such as the SH-60, F/A-18, E-2D, and MQ-4C.This system will leverage a MOSA to ensure future adaptability and significantly reduce the cost and complexity of radio upgrades. The goal is to provide a pathway for future modifications without impacting existing platform infrastructure. Developing aircraft radio systems presents significant challenges due to stringent SWaP constraints, harsh environmental conditions, and demanding Electromagnetic Compatibility (EMC) standards. Equally critical is robust cybersecurity, requiring adherence to standards like NIST SP 800-53 and the integration of security measures throughout the system design lifecycle.

The objective of this SBIR topic is to design, develop, and demonstrate an innovative airborne radio system optimized for SWaP efficiency. The system must satisfy current security and operational demands, while providing a modular, scalable architecture that accommodates future technology upgrades and supports evolving communication waveforms.

An open architecture is also critical to sustain radio systems through their lifecycle. The MOSA leverages a robust ecosystem of established standards, including Sensor Open Systems Architecture (SOSA) and Modular Open RF Architecture (MORA) that enable modularity and interoperability. Additionally, applying an MBSE to radio system design will enhance system understanding, enable early defect detection and improve documentation.

Additionally, the resulting radio system architecture should adhere to the following technical goals:
• Fit within the tight size constraints of two VNX+ standard cards (78 mm x 89 mm x 19 mm each). Note that a VNX+ power supply, backplane and I/O connectors will be external to the solution.
• Support two separate Transmit and Receive RF channels. One RF channel capable of 30MHz to 6HGz operating frequency and the other capable of supporting 30MHz to 31GHz
• Support at least 60MHz instantaneous bandwidth
• Support transmit power amplifier capable of reliably delivering an average 25 Watts of RF power on transmit channel 1 and 1 Watt of RF power on transmit channel 2

• Interoperability with MORA devices for control and I/Q data sharing
• Capable of Digital Pre Distortion (DPD)
• Capable of programmable RF waveforms including VHF/UHF communications waveforms including AM/FM, Air Traffic Control (ATC), Public Safety, Have Quick II, SATURN, SINCGARS, DAMA, MUOS, JPALS, and Automatic Direction Finding (ADF), Link-16
• Capable of 1024-QAM OFDM modulation with 1000 subcarriers

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop an initial design for a novel SwaP-optimized airborne radio system utilizing MOSA and MBSE principles that is readily integrable across Navy and Marine Corps platforms, encompassing fixed wing, rotary wing and UAV aircraft. Provide analysis to determine the feasibility of the design by meeting the technical goals defined in the Description.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop a prototype that includes the high-risk technology elements previously identified. Continue to refine the MBSE design developed in Phase I and demonstrate prototype functionality in a laboratory environment.
Work in Phase II may become classified. Please see note in Description section.

PHASE III DUAL USE APPLICATIONS: Further develop/refine the prototype(s) generated in Phase II for inclusion in a tactical radio for Navy and Marine aircraft that includes qualification and flight testing. By identifying radio technologies adaptable to harsh Navy and Marine aviation environments, this research benefits the private sector by enabling more reliable and robust commercial solutions. For example, technologies proven resilient in demanding military aircraft environments can be applied to industries such as mining, oil and gas exploration, or even emergency services communication, leading to improved performance and reduced downtime in these challenging conditions.

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KEYWORDS: Radio; Modular; Communications; Open; Signal

DON26BZ01-NV010 TITLE: E-2D Large Language Model Entity (ELLMENT)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and implement a traceable, explainable, referenced, and reasoned Large Language Model (LLM) that functions as an on-demand Natural Language Processing (NLP) decision-support assistant for Naval Flight Officers (NFOs) and mission crew aboard a carrier-based, all weather, tactical battle management, airborne early warning, and command and control aircraft.

DESCRIPTION: Artificial Intelligence/Machine Learning (AI/ML) technologies are transforming how complex data is understood and acted upon in operational environments. This SBIR topic seeks to explore the development of a domain-specific LLM system to support rapid insight generation from structured and unstructured documents (e.g., Tactics, Techniques, and Procedures [TTPs]), mission logs, communications, and other high-volume data sources relevant to tactical operations.

The goal is to deliver a modular, self-contained AI/NLP solution that can assist NFOs and mission crew by summarizing, reasoning over, and extracting meaning from dense operational material in real time. This LLM must be specifically designed to operate in a stand-alone configuration in accordance with information assurance policies, with mechanisms for traceability, where the information came from and how is it connecting to the goal, source attribution, and model transparency. The system must also support future extensibility to multi-modal data ingestion.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Define and develop the foundational architecture and baseline capability for implementing Large Language Model Operations (LLMOps) in support of mission decision-aid tools for the E-2D platform, as outlined by Gallagher et al. [Ref 2].
1. Security, Ethics, and Data Governance Planning

• The small business will collaborate with relevant Navy civilian representatives—such as TPOCs and PMA-231 S&T leads—to:
• Establish appropriate data classification levels for training and deployment environments
• Define a cybersecurity framework aligned with DOW and platform-specific requirements
• Incorporate an ethical AI governance structure, including bias mitigation and auditability provisions

2. LLM Selection and Mission Alignment
• An appropriate LLM architecture will be selected based on mission-specific demands of the aircraft operator, with consideration for:
• Performance in tactical and technical language domains
• Model transparency and explainability
• Compatibility with in-theater deployment constraints

3. Corpus Curation and Model Training
• The selected LLM will be trained on an aircraft relevant corpus, including—but not limited to—mission-specific Tactics, Techniques, and Procedures (TTPs), doctrine documents, and communication logs. Training methodologies will include:
• Prompt engineering
• Fine-tuning with Navy-specific linguistic patterns and use cases
• Retrieval-Augmented Generation (RAG) to support on-demand referencing of large knowledge bases

4. Evaluation and Output Validation
• Model performance will be assessed using a comprehensive metrics suite, as recommended by Diaz-de-Arcaya et al. (2024) and Gallagher et al. (2023), including:
• Response accuracy and relevance
• Appropriateness and alignment with operational context
• Bias detection and mitigation
• Trustworthiness
• Independent Subject Matter Expert (SME) evaluation

5. Deployment Pathways and Phase II Readiness
• As part of final Phase I efforts, the small business will:
• Evaluate and down-select hardware and software deployment options (e.g., computer architecture, human-machine interface designs)
• Develop a baseline implementation roadmap for transitioning to Phase II prototype construction and TRL advancement

PHASE II: The developed LLM will be deployed to a stand-alone laboratory environment, for rigorous evaluation in an Operator-in-the-Loop (OITL) configuration. In this setup, NFOs and mission operators will engage with the LLM across representative command and control mission scenarios to assess its efficacy as a real-time natural language decision-support system.
Subject Matter Experts (SMEs) in specific operations and AI/ML will conduct structured evaluations using predefined metrics identified in Phase I —including response accuracy, contextual relevance, trustworthiness, and bias sensitivity. Iterative testing cycles will drive continuous refinement of the model's behavior and performance . Performance referring to the system's suggestions compared to SME suggestions.

To support future scale-up, candidate computing architectures will be assessed, including emerging platforms such as quantum-accelerated processing (e.g., D-Wave). These evaluations will focus on increasing operational capacity, expanding conversational memory (buffer length) and handling of larger mission datasets in constrained compute environments.

A lifecycle monitoring framework will also be established to operate the LLM Ops strategy introduced in Phase I. This includes procedures for tracking long-term model performance, retraining triggers, audit logs, output traceability, and alignment with evolving mission requirements.
Work in Phase II may become classified. Please see note in the Description section.

PHASE III DUAL USE APPLICATIONS: Upon successful completion of final verification and validation (V&V) testing, the developed system will be authorized for transition to designated operational platforms and associated industry partners, in alignment with established Navy acquisition and technology transition procedures.

In parallel, the capability has garnered interest from additional mission-critical stakeholders—specifically ONR Code 32, in connection with Anti-Submarine Warfare (ASW) mission domains. This cross-domain interest highlights the system's adaptability and potential for broader operational utility beyond its original use case, further enhancing the value and return on investment for the Department of the Navy. The development and refinement of such an LLM pushes the boundaries of AI and NLP, contributing to the overall advancement of these technologies. The need for traceable, referenced data management promotes innovation in data governance, lineage tracking, and knowledge management, which are valuable for private sector organizations dealing with large datasets.

Examples of Dual-Use Applications include:

- Predictive Maintenance: Predicting equipment failures and optimizing maintenance schedules
- Supply Chain Optimization: Optimizing supply chain logistics
- Threat Detection: Identifying and responding to cyber threats
- Security Auditing: Automating security audits

Overall, this approach has a more focused and specialized domain than current commercial applications. While LLMs like Gemini and ChatGPT focused on cloud-based approach, the proposed LLM suggests that a targeted local network approach can be a forward design to target specific problems. Some examples of alternative cloud-based approaches could include Neuromorphic computing, Local LLMs, LLMs on edge devices, and Small Language Models (SLMs).

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KEYWORDS: Large language model; LLMs; Natural Language Processing; NLP; Multi-modal approaches

DON26BZ01-NV011 TITLE: Compact Battery Operated Mid-wave Infrared (MWIR) Hyperspectral, High-Definition, Real-Time Video Camera Integrated with Photonic Crystal

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Microelectronics;Quantum Science

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and demonstrate a compact battery-operated mid-wave infrared (MWIR) hyperspectral imaging (HSI) photonic chip video camera for integration into mobile network enabled small sensor platforms.

DESCRIPTION: Hyperspectral imaging allows quantitative evaluation of material composition and spatial distribution and finds numerous applications in areas such as remote sensing and military reconnaissance. In particular, the operational utility of HSI for detection, recognition and identification of hard-to-detect targets in environments cluttered with background noise is especially critical. Spectral imaging can aid the detection, acquisition and tracking of a potentially camouflaged, low-signature target, with significantly improved accuracy that cannot otherwise be detected using more conventional imaging means.

Conventional HSI systems [Refs 1, 2] tend to use large, bulky optical elements, such as a Michelson interferometer or other tunable optical filter components to spectrally resolve the input optical signals, and therefore usually have the characteristics of significant size, weight, and power (SWaP) consumption, mechanical complexity, as well as non-compliance with military specifications. More importantly, the mechanical mechanism of the conventional tunable filtering system gives rise to extremely slow spectral scanning speed and thus, slow imaging speed at that rate of one hyperspectral image per approximately one to two minutes. Traditionally, hyperspectral algorithms have considered only static images, and existing algorithms process single frames without regard for sequential similarities or correlations. The difficulty in capturing and processing hyperspectral video sequences in real-time is correlated directly to the high dimensionality of the data. As a result, conventional HSI systems cannot be deployed to the more demanding field applications that require images that can be captured and analyzed on a real-time basis at a much higher frame rates due to HSI's inherent image acquisition speed bottleneck.

A typical hyperspectral image consists of a high-resolution 3-dimensional (3-D) data cube, with two dimensions in space and a third dimension in wavelength. A focal plane array (FPA) can only acquire a 2D data set in one exposure. In the conventional approach, spectral scanning is thus often used to attain the third dimension of wavelength for a 3-D data cube. As stated earlier, this process makes HSI operation very slow because wavelength scanning requires multiple exposures over a specific spectral range. In addition to the very slow scan speed, it also suffers from a low signal-to-noise ratio (SNR) resulted from a high level of noise in infrared detectors and a low light throughput caused by narrow-band filters used in spectral scanning. The use of narrow-band filters also limits the number of spectral bands. Infrared spectroscopy routinely uses spectral multiplexing to overcome the challenge of detector noise. This is known as the Fellgett's multiplexing advantage [Ref 3]). The best example is Fourier-transform infrared spectroscopy (FTIR). Instead of spectral scanning, it projects an unknown spectrum onto a serial of sinusoidal functions constructed by a Michelson interferometer and thus greatly improves light throughput. However, it is difficult to integrate FTIR with FPA because of their bulky size and single channel operation. Recently, on-chip multiplexing has emerged as a new approach for hyperspectral sensing. It uses the spectral response of judiciously designed nanostructures to construct the multiplexing basis. Exploiting optical interference and resonance effects at the nanoscale, these nanostructures can provide a highly complex and diverse range of response functions suitable for efficient multiplexing [Ref 2]. They can be directly integrated into FPA to have an ultra-compact form factor. Multiplexing can be

performed in both spectral and spatial domain. Advanced data-driven optimization such as machine learning can be used together with compressive sensing to reconstruct 3D data cube in single-shot operation [Ref 4].

It is therefore the objective of this SBIR topic to develop a battery-operated, compact, high-performance MWIR HSI camera system capable of capturing HSI video at real-time or higher frame rates in the room temperature thermal infrared region. One of the key challenges in on-chip photonic multiplexing of a photonic crystal-integrated FPA is the computational design. Constructing the multiplexing basis is a delicate balance between the physical limit of on-chip photonic structures and the imposed requirement from demultiplexing algorithms. The former requires solving multi-scale Maxwell's equations, and the latter requires large-scale data-driven optimization of demultiplexing algorithms. The coupled design process needs to be iterated efficiently to reach any useful design. It is expected that this challenge can be addressed by using massively parallel simulation of electrodynamics paired with efficient optimization algorithms such as adjoint method.

The project should demonstrate a systematic design method that leverages large-scale simulation, machine learning, and data-driven design to realize real-time hyperspectral video imaging. The final goal of this project is to experimentally demonstrate a battery operated MWIR HSI video camera with the following specifications.
System required parameters include:

1. Wavelength range: 3-5 microns
2. Array size: Threshold -- 1280 x 1024 pixels; Objective -- 2048 x 1536pixels
3. Spectral resolution: below 5 nm
4. Pixel pitch: Threshold – 12 microns; Objective – 8 microns
5. Real-time hyperspectral video imaging Programmable; 0.0015 Hz to 125 Hz frames per second
6. Acquisition time of hyperspectral image with 500 spectral bands: < 40 ms or minimum 25 video data cubes (each with 500 spectral bands) per second.
7. Size and Weight: 7.5 grams and < 4.9 cm^3
8. Battery Type: Lithium-ion battery enhanced by using carbon-based nanostructures with a specific energy > 600 Wh/kg at 0.5C discharge rate, and specific capacity of > 600 Ah/kg.
9. Low power consumption, starting at 600 mW
10. Humidity Non-condensing between 5% - 95%
11. Non-Operating Temperature Range -57 °C to +80 °C (-65 °F to +176 °F)
12. Operating Temperature Range -40 °C to +71 °C (-40 °F to +160 °F)
13. Operational Altitude 40,000 ft (~12km)
14. Shock 40g w/ 11ms half-sine pulse, 3-axis
15. Vibration 5.8 grms 3-axis, 1hr each

Responding to the 21.2 AC1 S&T Domain comments: Surface Optics produces multi-spectral camera that can only provide multispectral images with about 10 spectral bands. Also, their multi-spectral camera is in the SWIR band. This current topic is for the first time a topic that can produce MWIR hyperspectral images at better than real-time video frame rate (24 frames per second or higher) with up to 500 (not 10 in the multispectral camera situation) high-resolution hyperspectral images per frame. This current proposed technology can produce up to 50 times more spectral information than the current multispectral camera in the market. Hence, there is zero overlap in terms of technology innovation between what Surface Optics and other commercial concerns market as multi-spectral or hyperspectral cameras and this current topic. In fact, the current proposed topic's performance and SWaP are far superior to any commercially available hyperspectral images by a 10 to 1 to 50 to 1 wide margin.

PHASE I: Demonstrate the feasibility of using massive parallel computing to design hyperspectral photonic chips. The design should shorten the acquisition time by 100X, i.e., reducing the time from seconds in traditional HSI to a few milliseconds. It should improve the spectral resolution by 10 X, i.e., going from tens of nanometers to a few nanometers, which translates into over 500 spectral bands in the MWIR band (3 to 5 um). The spatial resolution should be consistent with today's FPA resolution to reach 3.1 -mega pixel for high-definition images. Lastly, the design could be realized in ultra-compact form factor, reducing traditional HSI's size and weight by 10 – 20 times.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Optimize the unit cell design and lattice lengths to allow for (i) polarization insensitivity, (ii) WFOV performance, and (iii) broadband performance. Demonstrate a fully coupled design process using machine learning and physical simulation. Perform experimental verification of the generated design by demonstrating a real time MWIR hyperspectral imager with 5 nm spectral resolution, 1.3-million-pixel count, at 30 Hz frame rate. Demonstrate tunable focal lengths using lens-embedded photonic crystals.

PHASE III DUAL USE APPLICATIONS: Outputs from Phase II are anticipated to be TRL 7 but may require additional effort to refine to a more manufacturable design.
Concentrate on the manufacturability as well as the fabrication process itself to prepare for commercial offerings of a fully functional product. Demonstrate and market the final product to DoW and the Navy transition partner. Perform additional Phase III planning once a design is established and manufacturing requirements and manufacturability become more apparent.
The commercial potential includes new handheld and portable instruments for chemical, photometric, and biological sensing. Photonic crystal cameras can be integrated into compact form factors that enable in situ measurement for manufacturing process analysis and in-process feedback control. Applications include solid-state lighting characterization and testing, emissions control, portable sensing, and personal health care.

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KEYWORDS: Photonic crystal; Hyperspectral; Camera; Focal plane array; On-chip; Filter

DON26BZ01-NV012 TITLE: Optical Power Limiters Countering Frequency Agile Lasers and Dazzlers

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Directed Energy (DE);Space Technology

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop Mid-Wave Infrared/Long-Wave Infrared (MWIR/LWIR) nonlinear optical (NLO) dyes embedded in sol-gel glass operating as an Optical Power Limiter that protects optical sensors from damage caused by high-intensity light by reducing transmittance at high input power levels such as from frequency agile lasers and dazzlers.

DESCRIPTION: The proliferation of commercial, visible, and infrared wavelength laser systems is increasingly becoming a threat to our warfighters, which drives the need for further research and development for electro-optical/infrared (EO/IR) sensor. Current fielded sensor protection equipment is limited to fixed wavelength filters. However, broad band filters that are designed to circumvent multiwavelength laser threats are plagued by low transmittance, which degrades the sensitivity and performance of the sensor. Future warfighter threats include frequency agile lasers and dazzlers which have the potential of defeating fixed filters. Self-activating (passive) devices, where protection is activated by the incoming radiation (optical limiters), are the best approach to counter frequency agile and short pulse laser threats. The current state of the art of optical limiters are hampered by off-state low transmittance, low laser damage threshold, high activation laser threshold, and narrow field-of-view (FOV) and bandwidth. In addition, a sensor's size, weight, and complexity greatly affect the user's acceptance as a potential optical-limiting device. A sensor protection device is generally designed as an insert, an add-on, or replacement to the optical system. The optical limiter must be designed not to impact the sensor's FOV and optical transmission. Currently available systems are very bulky and narrow band in their protection.

This SBIR topic solicits new, innovative NLO dyes embedded in sol-gel glass to provide sensor protection from frequency-agile laser and dazzlers operating in the MWIR/LWIR spectrum. The proposed NLO dyes embedded in sol-gel glass should allow ample transmission of ambient MWIR/LWIR light and be of high optical quality so as not to significantly degrade sensor performance. It should have a fast response time when exposed to dangerous fluence levels, sufficient to react to and block incident laser pulses to a high optical density. The dyes should be capable of changing from a high transmission state to a very low transmission state within sufficiently short time to block nearly all of the light contained in a light pulse emitted from frequency agile lasers and dazzlers . When harmful radiation is no longer incident, it must recover to a high transmission state in a short amount of time so that the sensor's optics are not interrupted or significantly degraded after exposure. The proposal should discuss in detail the spectral transmittance in the attenuating state, activation threshold, response time, optical density in the attenuating state, and recovery time of the technology, the electric and other parameters of the excited state to be taken for measurements, excimer formation as well as any other important technical details. The NLO dyes embedded in sol-gel glass critical requirements are:

1) Wavelengths – threshold MWIR 3 to 5 micron goal MWIR/LWIR 3 to 12 microns;
2) Response time: < 1ns
3) Recovery time: < 1ms
4) Low-intensity transparency is > 50%
5) For light intensity or fluence above the limiting threshold (LT), the attenuation is > 20dB
6) The Damage threshold (DT) is at least 10 times larger than that of the nonlinear optical material used
7) The fluence limiting threshold (LT) is below 500 milli-joules/cm^2/pulse
8) Multiple use without performance degradation exceeds 10,000 pulses

9) Wide acceptance and protection angles
10) Testing should be performed using f-number optics no greater than f/10, unless a higher f-number is required by a specific application
11) Dynamic range (~120 dB)
12) Rapid response time (~20 us)
13) Optical limiting threshold of 6.5 W / cm2 at room temperature.

Use of government materials, equipment, data, or facilities will not be offered and will not be required. If the technology is capable of exceeding any of the above requirements, the proposal should note this as well. Likewise, the proposal should note any limitations inherent to the proposed technology.
New and innovative material solutions may be proposed to provide new options for sol-gel glass production. Potential candidates include but are not limited to vanadium dioxide, use of commercially available or novel silanes and solvents. Processing approaches could include methods to control the rate of curing of the glass and the type, material, and shape of container used for the cure, as well as the cure temperature.

The goal is to develop a process that can make larger optical elements more reliably. Well established materials and processes may be proposed with a focus on improving the manufacturability, producibility, and reliability for current and next generation optical elements. Increasing size, manufacturing yield, and reducing cost while at the same time reducing manufacturing variability is desired. Proposers must have experience in the production of dye containing sol-gel glasses.
A second requirement of the optical elements are dyes which have the required optical transmittance/absorbance properties while being compatible with the sol-gel materials and production methods and are reliably available from domestic sources. This is currently a challenge. The performer will be required to identify suitable dyes for the optical elements and to design synthetic approaches to any dyes that are not commercially available from reliable domestic sources. The performer will synthesize any required dyes not commercially available from domestic sources in amounts exceeding 10 grams by the end of Phase II and have the capability to produce the dye(s) at batch sizes of at least 10 grams going forward or to work with another domestic producer to do so, or both. Proposers should have documented experience in the design, synthesis, and production of novel and existing absorbing and fluorescing dyes in the infrared regions of the spectrum and must have demonstrated the ability to reliably and reproducibly synthesize, purify, and characterize light-absorbing dyes at greater than 10-gram batch size. The proposal should clearly identify the current state of the art of the sol-gel and dyes of interest including both technical and manufacturing readiness and how the proposed work will advance readiness for the proposed optical elements.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a NLO dye embedded in sol-gel glass protection concept designed to meet the (NLO) dyes embedded in sol-gel glass critical requirements stated. Identify critical fabrication processes for

realizing this concept. Conduct theoretical analysis and limited laboratory laser irradiation experiments on sample materials or devices to prove the feasibility of the concept.
Demonstrate a clear ability to prepare at least 1 inch diameter optically clear sol gel glass boules that are suitable for cutting and polishing. Demonstrate experience in putting organic or organic/inorganic dyes into the sol gel and preparing 1 inch diameter optically clear sol gel boules that could be cut and polished into optical flats. Provide description and photos of procedures utilized in "Phase I-like" effort that will carry into the Phase II proposal. The Phase I deliverables will also include prototype plans to be developed in Phase II, 3-dimensional model, Weight Budget, Trade-off analysis, and preliminary lab test data and supporting analysis.

All NLO dyes embedded in sol-gel glass protection prototype testing is expected to be performed at the awardee's site with the awardee's equipment (no Government supplied equipment). Any remaining concept refinements needed after a Phase I completion will be addressed early in the Phase II effort, ideally in parallel with the design efforts.

PHASE II: Develop and demonstrate a NLO dye embedded in sol-gel glass protection prototype system. Prototype optical limiting for mid-infrared transparent windows should be built in the form, fit and function of, or integrated for use in conjunction with, common Embedded Image Periscopes (EIPs) or embedded vision blocks on ground combat vehicles. This a NLO dye embedded in sol-gel glass prototype shall be jamming, damage, and device tested for critical requirements listed in the Topic Description, broadband laser protection performance, linear absorption, and degradation to optical system performance in a laboratory environment. Factors to be considered for Advanced Naval Technology Exercise demonstration include, but are not limited to, optical density upon laser illumination, response time, recovery time, linear optical properties under normal daylight illumination, manufacturability, and environmental stability. Phase II deliverables will include a prototype a NLO dye embedded in sol-gel glass laser protection system limiter satisfying the critical requirements as specified in the Topic Description, interim sample materials (if applicable), test data, monthly progress reports, semi-annual progress reviews, a final review, and a final report.

The design process should include planning for demonstration and testing/measurements. Fabrication and demonstration of a prototype is expected to require a substantial portion of the Phase II program due to component purchase lead times and several iterations of fabrication to refine the process. With proper planning for demonstration and testing, the final portion of the Phase II program should be relatively short and produce high quality data that indicates the NLO dyes embedded in sol-gel glass protection prototype function as a high-quality optical component near a virtual focal plane.
It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Pursue commercialization of the various technologies developed in Phase II for transitioning expanded mission capability to a broad range of potential government and civilian users and alternate mission applications. Phase III award may be for additional research and development or direct procurement of products and services developed in coordination with the program.

Commercial applications could include coatings on car windows to attenuate incoming headlights, and coatings on windows of buildings to reduce heating from the sun. This system could be applied to other military platforms as well as the commercial and private airline industries as a defense against real world terrorist threats.

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KEYWORDS: Sol-Gel Glasses; Laser protection; Frequency-agile laser; Dazzlers; Mid-Wave Infrared; Long-Wave Infrared

DON26BZ01-NV013 TITLE: AI-Assisted Modernization and Optimization of Theater Mission Planning Center (TMPC) Software

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an AI-driven toolset to automate the modernization of Theater Mission Planning Center (TMPC) software, focusing on refactoring legacy code, optimizing performance, integrating advanced cybersecurity, and ensuring compatibility with modern and next-generation mission systems.

DESCRIPTION: TMPC software is built on legacy code that presents challenges in maintainability, performance optimization, cybersecurity, and integration with evolving mission systems. Current modernization efforts rely on manual refactoring, which is time-consuming, error-prone, and costly. There is a critical need for an AI-driven capability to automate code refactoring, optimize computational efficiency, and integrate cybersecurity features without disrupting TMPC's core functions. This effort will enable seamless software upgrades while maintaining backward compatibility with existing operational platforms.

The proposed solution will leverage machine learning (ML) and natural language processing (NLP) to analyze, refactor, and optimize TMPC's codebase while preserving mission-critical functionalities. Additionally, AI-assisted software validation and security enhancements will ensure that modernized TMPC software meets the evolving requirements of Navy mission planning environments.
Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Conduct an analysis of TMPC's existing software architecture to identify modernization needs. Develop conceptual AI-based models for legacy code analysis, refactoring, and cybersecurity integration. Demonstrate proof-of-concept AI-driven refactoring on a representative TMPC software component. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop a prototype AI toolset for TMPC code modernization. Integrate cybersecurity enhancements and automated validation/testing. Conduct functional and performance testing and validation. Ensure compatibility with existing and next-generation TMPC hardware/software.
Work in Phase II may become classified. Please see note in the Description.

PHASE III DUAL USE APPLICATIONS: Bring the AI tool from prototype to full-scale, improving the speed, security, and cost-effectiveness of software upgrades.
The same technology that helps the Navy can also potentially help the private sector fix and secure older software more easily.

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KEYWORDS: AI-assisted; Legacy Code; Machine Learning; Natural Language Processing; Tomahawk; Mission Planning

DON26BZ01-NV014 TITLE: Non-Radio Frequency, Covert Maritime Transceiver

COMPONENT TECHNOLOGY PRIORITY AREA(S): FutureG;Integrated Network Systems-of-Systems;Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a reliable and covert transceiver for use in contested areas where the use of traditional radio frequencies are not permitted in order to remain concealed. The Navy is looking for new technologies that can transmit and receive wireless communications from distances of at least 5km. The signal medium may be, but not limited to, acoustic, infrared, or ultraviolet. The communications link must be highly resistant to interference, detection, and exploitation.

DESCRIPTION: Covert communications have continuously evolved during the history of warfare. Paradigm shifts in communication (in warfare) have enabled evolutionary tactical advantages that have lasted for finite periods of time until an adversary adjusts technology and tactics to detect, and in some cases monitor, seemingly covert communications. Various modalities are available to attempt to provide secure, covert communications including many Radio Frequency (RF) techniques, free-space optics (laser comm.) and others. Due to the United States's reliance on RF for communications and sensing (e.g., radar), various peer-adversaries have engineered around many of these modalities putting secure communications at risk. For this reason, it is necessary to go "out-of-band" to provide a modality of communication not commonly used and enabled by technology that is wholly new and therefore restricted by rarity. Another limitation to this application is the need to avoid bulky, power-hungry systems that may require a high degree of attention in order to operate properly.

Therefore, the Navy is looking for a low power, small communications transceiver that offers low probability of intercept (LPI) and low probability of detection (LPD). The new technology must be able to acquire, track, and maintain a secure communications link between rapidly moving vehicles (manned and unmanned). Emerging applications include cognitive operations with other autonomous systems for armed combat, Intelligence, Surveillance, Reconnaissance (ISR), casualty extraction, and field communications. Each of these applications have different objectives but all require uninterrupted, high bandwidth, and secure communications.
Attributes:

- Must be able to communicate between two or more points at least 5km away
- Low Size, Weight, and Power/Cost (SWaP-C)
- Reliable, continuous communication link
- Field Programmable
- LPI/LPD
- Flexible data rate requirement (up to 10MB/s)

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating

procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: The evaluation technical merit of Phase I proposals (evaluation criteria defined in the BAA instruction) will be based on design, system functionality, durability and feasibility. In the development of the system, assessment of the performance parameters and identification of the constraints and limitations is required, and a full rationale supporting the advantages and strengths of the design. Justification of the feasibility study will be based on research, part/component availability, sound engineering principles, and market research. Describe maximum transmission distance between points and the reliability of the wireless link.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: The end state goal of the proposed initiative is to develop and test a prototype system that implements the Phase I design. This prototype will have its ability to communicate wirelessly between two points tested in a relevant environment with input and direction from the Government Technical Point of Contact (TPOC. The system should demonstrate that it meets the objective and system attributes in this SBIR topic and will be evaluated based on the reliability and capability of the communications link.

Work in Phase II may become classified. Please see note in the Description section.

PHASE III DUAL USE APPLICATIONS: Complete final testing and perform necessary integration and transition for use in monitoring operations, remote surveillance and reconnaissance applications with appropriate platforms and agencies, and future combat systems under development.
Commercially, this product could be used to enable remote environmental and security monitoring or point to point secure communications.

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KEYWORDS: Covert transceiver communications; Field programmable; Wireless communications; interference resistant; Low power consumption; Cognitive applications; Radio Frequency; non-RF

DON26BZ01-NV015 TITLE: Virtual Reality Model Walkthrough

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop software for a commercially available Virtual Reality (VR) headset to view new ship construction models in an immersive environment.

DESCRIPTION: When constructing a DDG-51 Class Destroyer, Navy engineers regularly need to perform design reviews to verify and validate proposed ship changes. Currently, these design reviews are held using screenshots and model sharing of the ship's Computer Aided Design (CAD) models. However, 2D rendering of 3D spaces and objects can make it challenging to assess the actual layout and configuration of items. This can lead to errors in the ship design process, requiring costly rework later in the ship construction cycle.
The Navy seeks an innovative solution for VR software that allows Navy engineers to view the ship construction models as though they were standing in space. The proposed solution would allow the shipbuilder and the Navy to be better able to detect and correct errors early in the construction process. Additionally, such software could be used to train new engineers in the layout and navigation of the ship before they board it for the first time. There is currently no commercial technology that can meet this need.

The development of VR software faces several technical challenges. First, the shipyards use Computer Aided Three-Dimensional Interactive Application (CATIA) and Ship Constructor CAD models. The VR model must be capable of accurately using the outputs of both these CAD programs. The Navy understands this can be difficult and will require good knowledge of CAD file formats. Secondly, the user must be able to navigate virtual space and manipulate the environment. Many VR programs have some form of self-directed navigation. Destroyer spaces can have complex interior layouts and minimizing any motion sickness the user might experience while navigating VR can be a challenge. The solution should be able to load and view multiple CAD files, navigating between them with minimal lag and overlaying them to view discrepancies.

PHASE I: Develop a concept for a VR Model Walkthrough solution that meets the requirements listed in the Description. Demonstrate the feasibility of the concept in meeting Navy needs and establish that the concept can be developed into a useful product for the Navy. Feasibility will be established via a program demo or other means deemed appropriate. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Based on the results of Phase I, develop and deliver a prototype VR Model Walkthrough solution. The prototype software will be evaluated to determine capability in meeting the performance

goals defined in the Phase II Statement of Work. Product performance will be demonstrated through multiple evaluations over the development cycle. An extended test by Navy personnel will be used to refine the prototype into a design that meets Navy requirements. Prepare a Phase III manufacturing and development plan to transition the Virtual Reality Model Walkthrough to Navy use.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology to Navy use. Develop operations, maintenance, and technical manuals for the software to support the transition to the Navy.
There are many potential commercial applications for a VR Model to aid engineering design and training. Notable examples include commercial construction, commercial shipbuilding, architecture, and test and research reactors.

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2. Xie, Biao et al. "A Review on Virtual Reality Skill Training Applications." Frontiers in Virtual Reality, vol. 2, no. 1, 30 Apr. 2021. www.frontiersin.org/articles/10.3389/frvir.2021.645153/full, https://doi.org/10.3389/frvir.2021.645153

KEYWORDS: Virtual Reality (VR); Ship Design; 3D Software; CAD Models; Ship Construction; Self-Directed Navigation in VR

DON26BZ01-NV016 TITLE: Superconducting Magnetic Energy Storage (SMES) Power Interfaces

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Directed Energy (SCADE)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Directed Energy (DE);Renewable Energy Generation and Storage

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a Superconducting Magnetic Energy Storage (SMES) system to support intermittent pulsed power loads by providing a consistent load to the generation source during pulsed power duty cycle.

DESCRIPTION: A Navy ship's electric plant and the electrical load aboard the vessel mimics an electrical microgrid structure to distribute power. Conventional plant designs have separate mechanical propulsion and weapons systems with the electrical plant to support hotel and combat systems. Future all-electric naval ships will require all prime movers to have the functionality of distributed electrical generators to power a wide variety of loads ranging from conventional electronics, electric propulsion systems, and pulsed power systems to drive electric weaponry. The pulsed power systems will draw power from the ship's electrical distribution to enable continuous operation.
While large-scale energy storage may support operations, high-rate intermittent storage is necessary to ensure the electrical distribution and prime movers are provided with relatively consistent loading. During the charge process of the pulsed power system, a considerable amount of power will be drawn from the electrical grid for time durations on the order of seconds with a lapse in between charges. The large power drawn in an intermittent fashion is difficult to control and difficult for non-stiff electrical generators to supply. Enabling technologies to support a supplemental high-rate storage system is required for pulsed power loads to be effectively used on board the ship without disruption to other loads or damage to the distributed generators.

SMES systems are a relatively new technology that can charge and discharge energy at rates to support the various loads that new Navy ship designs are targeting. Innovative R&D is needed to model and validate novel high-rate, intermittent energy storage and control architectures that can rapidly accept high intermittent currents to load-level prime movers during the pulsed-power duty cycle. The architecture should be designed to minimize the impact this type of operation has on the electrical generators and support the pulsed load modules' operation. The energy storage must be able to accept rapid charge from the generation system within the constraints of the duty cycle of the pulsed power system and then provide this stored energy on the order of seconds to allow for cyclic capability in a continuous manner. New high-peak power energy storage technologies and designs are needed to accomplish this goal. Control system architectures and algorithms must also be developed to ensure load leveling in all modes of operations while ensuring safety and constant operation. These devices, with the requisite conversion schemes, are necessary in highly dense packages to allow for implementation in volumetrically

constrained environments. Proof of principle hardware tests and validated computer design models are desired.

The Navy seeks a full-scale pulsed power SMES system to store energy between 4-10 MJ at a 2-4 MW power level. The energy storage system developed is expected to charge at a rate of > 1 MW and to deliver power > 1 MW. The energy will be pulsed at a power duty cycle > 80% at a discharge/charge ratio of 1:1 and accept power at a sub-second response rate. The Navy desires the energy storage interface to withstand voltages > 1000 V.

PHASE I: Develop a concept for an intermediate storage approach that utilizes advanced high-rate components to continuously accept and provide power to operate on a load leveling basis. At a minimum, modeling and simulation should be performed to aid in proving the concepts are feasible. Small scale proof of concept experimentation may also be performed to demonstrate hardware's ability to drive high peak powers with compact and sensible architecture and package. Control algorithms that maintain load leveling should be developed and demonstrated on small-scale hardware systems. Provide objective quality evidence in the form of reports and briefings that their development will satisfy the technical specifications in the description. If modeling is performed the results of the modeling shall be included in the reports and briefings. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver a prototype that demonstrates the conceptual architecture and controls at a relevant scale that aligns to the requirements as provided within this topic for voltage and rates. Ensure that modes of continuous operation will be shown without degradation of the device and will support operations under elevated temperature regimes up to 122°F. (Note: Cooling and other interfaces shall be specified and demonstrated for performance.) Build additional intermediate storage devices to be tested at a facility by exposing them to a variety of pulsed power system concepts as well as abusive conditions. Cycle the modules for extended periods to fully characterize degradation and capacity loss with use under relevant conditions. Provide data and results from testing of the hardware, which shall be included in the reports and briefings to the Navy. Deliver any Phase II-developed hardware to the Navy for additional evaluation.

PHASE III DUAL USE APPLICATIONS: Assist the Navy in transitioning the technology to Navy use. Apply the knowledge gained in Phase II to build a multiple-MW scale system to support intermediate storage operations, ensuring that the system will be able to provide load leveling performance as defined within the topic and will be demonstrated as such.
In microgrid applications, additional areas of usage are high-rate charge/discharge applications including fast-dispatch frequency regulation, large power system load leveling and scheduling.
SMES has been implemented to stabilize power in the electrical grid in papermill factories in South Africa and the electrical power feed for a semiconductor manufacturing facility in Japan. It could be commercially adapted for other manufacturing uses.

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5. “2019 Naval Power and Energy System Technology Development Roadmap.” Naval Sea Systems Command (NAVSEA). https://www.navsea.navy.mil/Portals/103/Documents/2019_NPES_TDR_Distribution_A_Approved_Final.pdf

KEYWORDS: Superconducting Magnetic Energy Storage (SMES); High-Charge Rate; High-Discharge Rate; Power Dense Energy Storage; Pulsed-Power Delivery; High-Duty Cycle Energy Storage

DON26BZ01-NV017 TITLE: Bridge Environmental Light Pollution Mitigation and Control

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and deploy a safe, sustainable technology suited for controlling light pollution, thereby reducing ambient light levels across a bridge environment and providing adequate situational task lighting at select workstations across the bridge.

DESCRIPTION: The Navy seeks a light mitigation technology for adequate situational lighting compliant with the Bridge Light Pollution Mitigation and Control Program (BLPM & CP). A comprehensive review of collisions involving U.S. Navy ships cited bridge lighting conditions as a possible contributing factor, stating the need to adhere to military standards for light producing displays and equipment installed on the bridges of surface combatant ships. The principal BLPM & CP's objective is to resolve non-compliance of current bridge equipment and hardware with Military Standard MIL-STD-1472H, DOW Design Criteria Standard for Human Engineering [Ref 2]. Existing hardware often fails to satisfy requirements as outlined in the referenced standard (MIL-STD-1472H).

Light pollution mitigation efforts are necessary for all light producing technology installed on surface ship bridges/pilot houses. Reducing the undesirable effects of excessive or poorly designed lighting (i.e., light pollution) on night vision and bridge-watch stander performance will create greater situational awareness for crew members in a darkened bridge environment, therefore enhancing ship safety at sea.
The Navy seeks light mitigation technology for the bridge environment that complies with MIL-STD-1472H and enhances the effectiveness of all lights (e.g., screens, indicator lights, LED) during dark operations. This solution must also include a ruggedized work light that complies with free translation in three-dimensional space and free rotation on all three axes of rotation. There is currently no commercial technology that can meet this need.

The light should comply with all surface ship environmental standards regarding Electromagnetic Environmental Effects (E3), shock, vibration, and power quality and be able to produce light at the levels described in MIL-STD-1772H. The work light shall allow bridge watch standers to observe printed material at nighttime while still preserving night vision. A shipboard bridge work light must provide a focused beam of light with minimal glare, must be adjustable to direct light precisely where needed, and must offer the ability to control brightness levels with a cool color temperature to minimize eye strain while performing detailed tasks like reading or writing notes especially for crew members in a darkened bridge environment.

This scope of this effort includes all light emitting devices on the bridge, and is not limited to the following bridge systems:

- Navigation Radar
- Surface Search Radar
- Situational Awareness Radar
- Electronic Charting System
- Ship Control Consoles
- Voyage Data Recorder
- Bridge-to-bridge Radio
- Hull, Mechanical, and Electrical (HM&E) Systems

• Damage Control Equipment/Displays
• Command, Control, Communications, Computers, and Intelligence (C4I) Systems

Desired light mitigation solution parameters include but are not limited to:

• Overlay applications, easily applied to existing displays, requiring no special tools, equipment, hardware, fixtures, adhesives, tapes, or fasteners.
• Collapsible, foldable, stackable, and/or portable solutions to allow effective and easy storage when not in use.
• Various optical densities and sizes of Neutral Density filter material may be overlaid on displays.
• Solutions shall allow operator adjustment during application or installation.
• Temporary covers, fixtures, filters, shades, etcetera must not alter the original design characteristics nor interfere with normal operation of mitigated light emitting sources.
• Technology should not require external electrical power nor include additional electronic control systems or require any form of computer network connections.
• Solution shall not leave any adhesive residue behind on surfaces after removal.
• Mitigations may also include other formed caps to cover various instrumentations to reduce or eliminate light pollution associated with installed bridge equipment and other environmental light polluters.
• Solution must be able to withstand extreme environmental conditions (e.g., high humidity, persistent vibration, temperature below 40° degrees Fahrenheit, etc.).

PHASE I: Develop a concept for reducing ambient light levels across a bridge environment that meets the requirements outlined in the Description. Demonstrate the feasibility of the concept in meeting Navy needs and establish that the concept can be developed into a useful product for the Navy. Feasibility will be established via modeling or other means deemed appropriate. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver a prototype light pollution mitigation system. The prototype will be evaluated to determine capability in meeting the performance goals defined in the Phase I. Product performance will be demonstrated through evaluation, modeling, and demonstration over the required range of parameters. Prepare a Phase III manufacturing and development plan to transition technology for Navy use. Prototypes expected during Phase II must apply to indicators and systems. The Navy will measure the suitability and sustainability of prototype using the criteria outlined in the references.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the light pollution mitigation system to Navy use. Develop installation, maintenance, and operations manuals during deployment to support the transition to the fleet.

The potential for application beyond military use cases exists within the commercial shipping industries like commercial fishing, cruise lines, cargo transport, oceanographic exploration, and other seagoing operations involving the need for optimized crew performance on the bridges of large ships during darkened and nighttime conditions. There are many potential commercial applications for light pollution mitigation technology in inimical conditions.

REFERENCES:

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2. “MIL-STD-1472H, Department of Defense Design Criteria Standard: Human Engineering (15-SEP-2020).” https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36903
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KEYWORDS: Bridge Illumination; Light Pollution; Darkened Ship; Night Vision; Safety at Sea; Collision

DON26BZ01-NV018 TITLE: Lightweight Beaching Ramp for Ships

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a durable, lightweight, corrosion-resistant beach ramp for newly constructed Medium Landing Ships (LSMs).

DESCRIPTION: The Medium Landing Ship (LSM) is a new construction beachable vessel intended to perform ship-to-shore amphibious movement of cargo, equipment, and troops. To accomplish the loading and unloading of equipment, a large vehicle ramp is needed for Roll-On/Roll-Off (RO-RO) capability. Large RO-RO ramps are used on multiple ship classes and are commonly made of heavy steel, susceptible to corrosion and maintenance issues. On beaching vessels, these ramps are constantly subjected to saltwater immersion and are expected to traverse a long distance to provide a safe transfer of vehicles to the shore. Without a properly functioning ramp, the primary mission of these beaching vessels is compromised. There is currently no commercial technology that can meet this need.
The Navy seeks a reliable, low maintenance, corrosion resistant ramp system for beaching vessels. The legacy ramps are mostly made of steel and are often submerged in seawater while rolling stock compromises paint protection causing corrosion. Due to complex geometric challenges of beaching a large vessel, complex articulating beaching ramps tend to be very long and heavy (typically about 75' and 110 tons). These length and weight challenges of deployment systems tend to be unreliable and often have mission degrading failures. The solution should be at least 13 ft in width and 75 ft in length and support a maximum vehicle load of 70 tons (tire contact load of 32,100 lbs. over 24" x 25.5" patch area). The ramp should be articulated from a single hinge point using hydraulics on the ship. However, the solution can be divided into as many subsections as necessary such as employing the use of multiple folds to accommodate the length and weight requirements. The time it takes to deploy the ramp, however, shall not exceed 30 minutes.

The technology should utilize maintainable systems for deployment and retraction. The developed solution should reduce maintenance requirements while maximizing reliability of the deployment/retraction system. The durability of corrosion resistant material should last the life cycle of the ship (30 years). The solution should take extreme environmental conditions such as wind, humidity, and sea spray into consideration as such conditions can decrease the life cycle of a technology. The solution should have a nonskid surface that is able to withstand the demands of high traffic during loading and offloading of heavy equipment and vehicles.

PHASE I: Develop a concept for a lightweight beaching ramp for ships that meets the requirements in the Description. Demonstrate the feasibility of the concept in meeting Navy needs by material testing and analytical modeling. If the Phase I Option is exercised, include the initial layout and capabilities to build the prototype in a Phase II plan.

PHASE II: Develop and deliver a scaled prototype beaching ramp that meets the requirements of the Description. The prototype will be installed on an appropriate test platform in a simulated environment for durability and load testing. These evaluation results will be used to refine the prototype into a design that will meet the LSM specifications. Prepare a Phase III development plan and cost analysis to transition the technology into Navy use.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the lightweight beaching ramp for Navy use on the LSM program. Refine the design of the lightweight beaching ramp based on Phase II testing results and prepare for Low-Rate Initial Production (LRIP).
The durable lightweight beaching ramp will have private sector commercial potential for all types of RO-RO ships of this scale operating in the near-shore or on-shore environment, which currently use expensive to procure and maintain steel designs. Commercial applications include ferries, RO-RO ships, transport aircraft, the oil and mineral industry, and cold climate research and exploration.

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KEYWORDS: Medium Landing Ship (LSM); lightweight Ramp; Lightweight ferry ramps; Advanced Materials; Corrosion resistance; Roll-On/Roll-Off (RO-RO) ramps

DON26BZ01-NV019 TITLE: Improved Portable Underway Replenishment (UNREP) Tester/Trainer

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

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OBJECTIVE: Develop a portable Underway Replenishment (UNREP) Tester/Trainer to be used pier side to replace legacy in-port Shipboard Qualifications Testing that requires a supply ship and pier space for the two ships.

DESCRIPTION: Underway Replenishment (UNREP) systems are used to transfer cargo and liquid (i.e., fuel, water) at sea between U.S. Navy ships. A tensioned line is rigged between the two ships, and the cargo/liquid is transferred along the highline between the ships. Delivery ships (i.e., T-AKE, T-AO, and T-AOE) have the equipment used to transfer the cargo and fuel, while the receiving ships (i.e., combatants, carriers, amphibious) have a simple connection point to connect the line from the delivery ship and then receive the cargo/liquids.

The Navy requires in-port testing to verify the installation and training of the fleet on how to operate the UNREP systems. Currently this testing requires the delivery ship to connect their cargo and fuel systems to the receiving ship in addition to the pier space for the receiving and delivery ships. Delays in testing often occur due to limited pier space and delivery ships not being readily available. This has increasingly become an issue as the fleet increases their size, placing a greater demand for pier space availability. There is currently no commercial technology that can meet this need.

The Navy seeks a portable UNREP Tester/Trainer System that will test the current design and use innovative power and controls to meet the Navy's needs. The new trailer should be able to test the UNREP stations on a receiving ship and provide training opportunities for the fleet while in-port. The proposed solution can be self-propelled or towable to allow use in various locations. It must be self-powered for UNREP testing/training. The trainer/tester will need to be fixed to the pier to allow all required testing. The testing/training will be for both cargo and liquid systems. Currently, cargo transfer uses a trolley that is pulled back and forth on the tensioned highline, while the liquid transfer uses a hose hanging from the tensioned highline. No fuel would be transferred; all liquid training and testing will be dry. The tensioned highline testing will include pulling at different angles including above and below the horizon and fore and aft of the station. The Navy UNREP testing requirements are further defined in NWP 4-01.4 [Ref 1].

PHASE I: Develop a concept for an UNREP tester/trainer that meets the requirements outlined in the Description. Demonstrate the feasibility of the concept in meeting Navy requirements and establish that the concept can be developed into a useful product for the Navy. Feasibility will be established via

computer modeling or other means deemed appropriate. The Phase I Option, if exercised, include the initial design specifications and capabilities description to build a prototype solution in a Phase II plan.

PHASE II: Develop and deliver a prototype for an UNREP tester/trainer system. The prototype will be evaluated and tested to determine capability in meeting the defined performance goals. Product performance will be demonstrated through prototype evaluation, modeling, and demonstration over the required range of parameters. An extended pier side test with an active/available ship will be used to refine the prototypes into a design that will meet Navy requirements. Prepare a Phase III manufacturing and development plan to transition the UNREP tester/trainer system to Navy use.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the portable UNREP tester/trainer to Navy use. Develop installation, maintenance, and operations manuals for the UNREP tester/trainer to support transition to the fleet.
Additionally, the finished product has applications for the Military Sealift Command Fleet in qualifying mariners to operate with Navy Ships at sea. There is potential use as a test fixture for commercial companies that are trying to get into the UNREP business and are building their own UNREP systems that will need to meet the same requirements as current Navy design.

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KEYWORDS: Underway Replenishment (UNREP); Pier side Training; Pier side Testing; Maintenance; Ship-to-ship Cargo Transfer; Tensioned Highline

DON26BZ01-NV020 TITLE: Auto-Focus Detection Capability for SONAR Systems

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Human-Machine Interfaces;Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

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OBJECTIVE: Develop an auto-focus signal processing capability to optimize detection of quiet contacts by arrays of hydrophones.

DESCRIPTION: Arrays of hydrophones are used to detect, classify, and localize contacts in the ocean environment. Finding a contact, especially a quiet contact, is extremely challenging due to the large volume of data that needs to be searched as well as the large number of other noise sources (e.g., shipping, fishing, whales, etc.) that generate clutter on the displays.

Array signal processing, also known as beamforming, steers many beams to spatially filter the noise environment and generate a 3-D data volume that is a function of time, frequency, and bearing (i.e., steered beam) that are processed to generate several detection surfaces.

Several parameters can be adjusted to optimize the detection of a signal on an array. One of these parameters is focus range. (Other parameters are more sensitive and will be provided to Phase II awardees). However, only a limited number of display surfaces are typically generated due to processing constraints, and this may not provide the best opportunity to detect all signals. Furthermore, the operators typically have a large workload and are only able to search for a limited number of the available display surfaces.

Automation approaches have been developed for decades to help reduce operator workload. However, a well-trained operator can still detect lower Signal to Noise Ratio (SNR) signals than the state-of-the-art automation. The main reason for this is if the automation detection threshold is adjusted to detect lower SNR signals, it will cause an increase in the number of false alerts that detracts from the search process. Another approach that is used to reduce the operator workload is ORing, which combines multiple Passive Narrow Band (PNB) displays by taking the maximum value at each time/frequency bin and then combines all contacts found on any of the displays onto a single display; however, it also takes the maximum of the noise bins. This results in ORing loss by increasing the noise floor and reducing the overall SNR.

As a result, automation has not yet solved the operator workload problem and operators are still required to conduct manual search on a limited number of detection surfaces. This leads to system losses that can at times be significant and offers an opportunity to mitigate those losses with a new processing paradigm.

The objective of this SBIR topic is to develop a signal processing approach that will auto-focus on the signal processing (much like a digital camera does) with respect to parameters such as focus range. There is currently nothing available commercially.

The easiest example to understand is range focusing. Let's assume we are trying to track whales and there are several of them at different ranges. If we process a single far field (i.e., distant) focus range, then the close-range whales may barely be detected. Instead, if we process several focus ranges, let's say 10, from close to far, there will be one focus range where each of the whales displays the clearest signal with the highest SNR. Over time, the whales will swim closer and farther, and the best detection range will change. The problem is that the operator doesn't have time to look at the detection surfaces for all 10 focus ranges so instead we need to combine them into a single display that contains the higher SNR instance of each whale regardless of the range where they are.
Different whales will also have different broadband signatures and would be more detectable when averaging over different frequency bands. The optimal frequency band may also vary as the ambient noise environment (such as nearby shipping and weather conditions) changes. If the processing generates a large number of detections in multiple frequency bands, then a user will be able to find the most detectable instance of each whale over time.

Processing multiple focus ranges is relatively straightforward and is largely just brute force processing. The innovative part of this SBIR topic is the use of this larger data volume to build a combined display that contains the best representation of every available signal. This combined display would be the primary search space for the operators and would also be provided with other automation algorithms. One of the keys to success will be developing an alternative to standard ORing that takes the maximum value at each pixel across the beams being ORed. It is speculated that improvements are possible since the SNR of the signals will be well behaved across the ORing dimension. For example, if multiple focus ranges are combined, there will be one focus range where the signal is strongest, but the signal will gradually degrade as the difference between the focus range and the actual range increases. For pixels that contain noise instead of signal, it is expected that the levels will be more random and that this could be exploited to enhance the signal without increasing the background noise.

Overall, it is expected that this auto focus approach will allow system gains that are currently not being realized with the current signal processing and automation approach. This would significantly improve system performance by providing earlier detections and longer holding times of contact without increasing the operator workload or requiring a complete overhaul of the signal processing and automation framework. And although this does come at an increased computational cost, it would allow us to squeeze every dB out of the signal processing.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVSEA in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for a technical approach for implementing an auto -focus signal processing and automation capability that accomplishes the intent identified in the Description. Demonstrate this approach by using the focus range parameter as an example using an unclassified simulated dataset. Establish feasibility through analysis and modelling.

During the Phase I period of performance, the government team will provide the simulated dataset that was developed to stress test the various conditions. If the performer does not have experience with array signal processing, this dataset would be preprocessed display surfaces for different focus ranges. If the performer does have experience with array signal processing, then this dataset could be either preprocessed display surfaces or element level timeseries array data. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in a Phase II plan.

PHASE II: Develop and deliver a prototype auto-focus signal processing and automation capability using additional optimization parameters identified by the Government team. This will be implemented as a research code and tested against classified datasets provided by the Government to the Phase II awardee(s). Based on the results of the testing, this approach will be refined to improve the system's performance.

It is important to note that this is not intended to be a tracker development effort unless the awardee's solution requires it. If needed, the government will process the test data using an existing tracker solution and provide the track data to the awardee.
Performance and technical requirements to be evaluated include earlier detections and increased time holding of quiet contacts, improvements in the operator's ability to search the data space, and/ or improvements in operator workflow.

Deliver a prototype, software description document, a working copy of the development code, and test results from processing two or more classified datasets.
At the end of Phase II, the development code may be used by the government team for an independent evaluation using additional datasets not provided to the awardee to determine whether follow-on Phase III efforts are justified.

It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology to Navy use to allow for further experimentation and refinement. Develop production level code that is containerized and assist with integration efforts to incorporate the SBIR-developed code into the Government's Processing Testbed (PTB) which allows development software to be tested on real-time or playback data. Successful integration and testing in PTB would lead to further integration into production software. This technology does have significant dual-use applicability. The underlying concept is built upon array processing fundamentals that are applicable to SONAR, RADAR, communications, geophysical exploration, astrophysics, and biomedical imaging.

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KEYWORDS: SONAR; Array Signal Processing; Surveillance Automation; Acoustic Detection; Acoustic Target Classification; Sonar Operator Workload

DON26BZ01-NV021 TITLE: Robocasting Ceramic Sensors

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a low-cost, flexible manufacturing technique to produce large format ceramics for undersea sensor applications.

DESCRIPTION: Piezoelectric ceramic materials are essential materials to produce undersea sensors. Many existing undersea sensors rely on a dry press manufacturing process that produces the ceramic components used in many fielded sensors. Existing piezoelectric ceramic components are becoming increasingly difficult to source due to a shrinking supplier base and a desire by many private companies to stop manufacturing lead-based products. Additionally, these components have been largely unchanged since the 1960's with little to no performance enhancements to ships' critical systems.
The goal of this SBIR topic is to support the development of new agile manufacturing techniques to produce large format ceramics and that require less capital overhead and would be easier to stand up in new cottage businesses if the current supply base continues to degrade. The secondary goal is to improve the electrical and acoustic performance of these large format ceramic materials by utilizing textured ceramic technology.

Textured ceramic materials have an aligned microstructure that can exhibit enhanced properties compared to traditionally manufactured ceramics with randomly oriented gains. One documented benefit is an improved piezoelectric performance for sonar sensor applications (early prototypes have shown upwards of 12dB improvement in performance, enabling sensors to detect potential threats much farther out). Current manufacturing techniques to produce textured ceramics are costly, inefficient, and typically limited to smaller sensor geometries. There is currently no known commercial technology that solves these problems.

There is a need for the ability to produce textured ceramic materials in a larger format than is currently available through tape casting and existing additive manufacturing techniques. The process of robocasting or direct ink writing of a shear thinning ceramic paste shows great potential as a flexible manufacturing technique to produce ceramics for undersea sensors. The hardware requirements for the robocasting process are often affordable, relatively simplistic instruments that can be adapted to additively manufacture ceramics. There has been recent research demonstrating that extruding a ceramic paste through a high aspect ratio nozzle can align high aspect ratio particles within a material, allowing to produce textured piezoelectric ceramics through a robocasting process.

The primary focus of this SBIR topic would be to validate the feasibility to integrate a Navy piezoelectric ceramic with a robocasting or direct ink write slurry system. The system must demonstrate the ability to properly extrude a ceramic paste that will support the buildup of sequential layers and produce a prototype part. Key criteria for success will include the ability to consistently extrude a layer of ceramic paste, support proper adhesion between layers, and produce high percent solids loading of the paste; and the ability to sinter the materials to produce dense final parts.

The secondary focus will be to demonstrate the ability of the additive manufacturing hardware to properly align high aspect ratio platelets during the printing process. These platelets should be dispersed in the piezoelectric ceramic and aligned within each print layer. This technique should be flexible enough to

produce prototype samples of varying sizes. Common geometries include cylinders with 1in outer diameter as well as rings that are greater than 4in in outer diameter.

Prototype parts of multiple geometries will need to be produced and undergo binder burn off and sintering. Sintered prototypes will need to have electrodes applied and the parts will have to undergo a poling process. Prototype parts will be evaluated by Naval Surface Warfare Center Crane Division for density, surface finish, particle/grain alignment, texture fraction as well as electrical and acoustic properties. Textured prototype parts will be electrically tested for resonance frequency, capacitance, dielectric constants, and loss factors to be compared to traditionally manufactured non-textured materials. The awardee will aim to create a prototype that exceeds a capacitance of 200pf while minimizing the loss tangent. The awardee will then revisit particle alignment and binder composition as needed to improve acoustic and electrical performance.

PHASE I: Develop a concept for a ceramic paste suitable for additive manufacturing that utilizes Navy piezoelectric ceramics and can align high aspect ratio ceramic platelets within the constraints listed in the Description. Feasibility may be demonstrated by analysis, modelling, and simulation, the fabrication and testing of initial test geometries , or some combination of all three. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver prototype hardware based on Phase I work. Demonstrate the ability to construct a prototype ceramic that meets the constraints listed in the Description. The prototype hardware will be delivered at the end of Phase II ready to be tested by the Government.

PHASE III DUAL USE APPLICATIONS: Assist the Navy in transitioning the technology to Navy use. Scale/volume/speed of production will also be optimized in this phase. Finalized equipment and consumables needed to produce the parts will then be made available for Crane/Navy to purchase. This new technology will support Navy programs/platforms by providing advanced piezoelectric transducers with better performance and capability.

This added technology/capability will also assist in other projects that require advanced, textured ceramics including hypersonic radomes as well as various sensors in the commercial sector and the military. Potential commercial applications include medical imaging devices, civilian watercraft navigation and fishing devices, and infrastructure inspection equipment.

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KEYWORDS: Additive manufacturing; Robocasting; Direct Ink Writing; textured ceramics; shear alignment; piezoelectric

DON26BZ01-NV022 TITLE: Extremely Wide Band Digital Recording System for Artificial Intelligence/Machine Learning Development

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a small and dense data recorder that can store > = 8 Petabytes of information in < = 4 u of 19-inch rack space, will be scalable and flexible in nature, and will demonstrate the interfacing to > = two different interface protocols each supporting > 400 GB/sec data transfer rates for > = 30 seconds.

DESCRIPTION: In today's environment, emphasis is put on how Artificial Intelligence/Machine Learning (AI/ML) can solve most of the Department of War's (DOW) problems as long as the AI/ML algorithms are trained correctly. This training requires vast amounts of relevant data. Unlike commercial websites where the algorithm developers can have the public train them based on security selection images, the DOW does not have vast stores of relevant data sets much less a global community to train the algorithms. Unfortunately, very few to none of the fielded program of record (POR) systems have the ability to record (at the tactical edge) relevant data products in sufficient quantity to help algorithm developers.

This SBIR topic is intended to develop extremely deep sensor data recorders for implementation/fielding on tactical platforms for tactical sensors at the tactical edge. These recording devices must be able to be integrated easily into the platform's sensor suite and be able to record the relevant data products for use in future algorithm development and training.
These recorders must easily adapt to various networking infrastructures (e.g., InfiniBand, NVLINK, PCIe, and or Ethernet, etc.) and support the extreme streaming bandwidths for wideband (500Mhz and greater I/Q data) Radio Frequency (RF) digital data and high definition (4 K or greater) streaming video. These recording devices must be scalable in nature, at a minimum take up less than or equal to 4u of face plate volume in a 19-inch rack, and record greater than 8 petabytes of storage.
These devices must meet all NSA data at rest encryption requirements and be developed in a manner to easily acquire a volatility certification letter. References 7, 8, and 9 are provided for informational purposes, further information may be provided to Phase II awardee. These prototype devices will be installed on manned and unmanned platforms. With that in mind, they must be developed with remote and/or autonomous operations in mind. These prototype devices will deliver the hardware and the requisite software to perform recording, playback, librarying, and search functions for the data on the devices.

Key requirements:
- Less than or equal to 4u of 19-inch rack volume

- Must meet class B shipboard installation Environmental Qualification Testing (EQT)
- Greater than 8 petabytes of data storage
- Must meet data at rest security requirements
- Must meet non-volatility certification requirements
- Have networking architecture demonstrating ability to configure to multiple types of networks
- Have a minimum of two different networking options where each networking option can sustain > 400 GB/s data rate
- Compliance with shipboard installation environmental qualification requirements
- Ability to perform data at rest encryption and the ability to meet volatility requirements for system posture changes
- Ability to consume data from a defined sensor and parse/tag this data
- Ability to record and playback from both local and remote users

Conduct, at a minimum, two lab demonstrations at the developer's facility and one integration and demonstration at a government lab. (Note: The government lab will provide testing and validation of the capabilities and provide immediate feedback to the developer for further refinement of the prototype.) Work with the government lab to develop a shipboard installation and testing plan. If the Phase II Option is exercised, focus on getting the prototype system ready to be installed and tested at sea during a government-defined testing event.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop and provide a detailed schedule out through Phase II options, as well as a detailed technical description as to how they will achieve success. The initial deliverable of the Phase I award will be a kickoff meeting detailing how they will get to the final briefing.
The final briefing will show specifically to meet the following requirements:

- Less than or equal to 4u of 19inch rack volume
- Meeting class B shipboard installation Environmental Qualification Testing (EQT)
- Greater than 8 Petabytes of data storage
- Data at rest security requirements
- Non-Volatility certification requirements
- Networking architecture demonstrating ability to configure to multiple types of networks.
- Interface descriptions on how external systems will interface and operate the prototype device remotely and locally
- Examples of the user interfaces and the schema for formatting, recording, librarying, and playback
- Cost and schedule program management plan
- Phase I option will be showcasing software modules and fundamental breadboard designs and present the detailed plans for Phase II and Phase II option.

PHASE II: Hold a kickoff meeting with a detailed development plan including costing (recurring and non-recurring separated) development; and detailed security and testing plans
These plans will include detailed:

- technical plans
- security plans
- EQT plans
- lab testing plans (both at developers facility and at government labs) utilizing different types of networks.
- ship installation and at sea testing (Phase II Option, if exercised, will be integration and testing at sea)

After the kickoff meeting and with government concurrence of the plan, the awardee will focus on developing the solution meeting all the security, environmental qualifications, and performance requirements agreed to.
The system to be developed shall meet the following requirements:

- < = 4u of 19 inch rack space
- > = 8 Petabytes of storage
- Minimum of two different networking options where each networking option can sustain > 400 GB/s data rate
- The prototype system will show compliance with shipboard installation environmental qual requirements
- The prototype system shall show the ability to perform data at rest encryption and the ability to meet volatility requirements for system posture changes.
- The prototype system shall show the ability to consume data from a defined sensor and parse and tag this data.
- The prototype system shall demonstrate the ability to record and playback from both local and remote users

There will be at a minimum two lab demonstrations at the developers facility and one integration and demonstration at a government lab. The government lab will provide testing and validation of the capabilities and provide immediate feedback to the developer for further refinement of the prototype.

- The awardee will work with the government lab to develop a shipboard installation and testing plan.
- The Phase II option Option, if exercised, will focus on getting the prototype system ready to be installed and tested at sea during a government defined testing event.

It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: The awardee will clearly and in detail describe how this capability will transition to a Navy program of record (POR). This plan will also describe how it will be used in the POR and the initial concept of what data products will be recorded.
Any commercial industry looking for low cost optical processing on extremely large data sets will benefit from this technology.

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KEYWORDS: Digital Recorders: Radar Interfaces; Combat Control Interfaces; Electro Optic data; Continuous Digital Intermediate Frequency; CDIF; Burst Digital Intermediate Frequency; BDIF

DON26BZ01-NV023 TITLE: Risk-Aware Regenerative AI-based Multimodal Visual-Tactical (ISRT) (Observant-AI) – Monitor, Understand, Alert, and Assist

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Integrated Sensing and Cyber;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop risk-aware artificial intelligence (AI)-based computing methods motivated by three naval challenge problems that enable insightful active cross-domain (Sea-Space-Air-Land-Cyber) situational awareness and AI-assisted course of action and countermeasures in real-time conditions, namely, “LIVE” machine self-teaching (i.e., Regenerative AI); contextual machine exploitation; contextual networking to gain insights from accessible all-source-intelligence (ASI) and multimodal sensors; and proactive AI-assisted targeteer and decision support to manned and unmanned assets. The Observant-AI is envisioned as a distributed system of mission-focused AI agents that self-organize and share insights via ad hoc networking. The agents autonomously form mission-oriented collaborative teams to process and fuse multidomain anomalous events and activities for real-time AI-generated visual-tactical understanding, monitoring, alerts, and related operational risks. It applies natural language explanations for human-AI interactions, course of action assistance, and reasoning about risky engagements. For example, submersible X is tracking you, change course to southwest, speed up…; Cargo-Ship Y is armed, Container Marking is…, Departing Port XYZ; 20 UAVs are shadowing, armed, turn around go south; Littoral Zone X, Torpedo-Mines, Bottom-Mines, Deep Fencing, Actively Guarded, Speed Boats, Risk-High Navigation, Need Minesweeper, Check with CENTCOM; etc.

DESCRIPTION: Problem scope and capability concerns. First, over the past three decades, advancements in AI and machine learning (ML) for applications in hybrid networked teaming of manned and unmanned systems and sensors have unlocked new possibilities across a range of naval operations for novel missions. On the other hand, the defensive and offensive effectiveness of these technologies against near-peer adversaries remains a significant challenge.

Second, current Naval ISRT operations follow rigorous protocols supported by wide-ranging wargaming scenarios to plan tactics, techniques, and procedures (TTPs) with contingencies as operations unfold. TTPs focus on various situational details, such as adversary strength, leadership temperament, past and present operational performance, logistics, and exploitation opportunities for friendly cross-domain actions and effects. These plans are vital to be followed. However, they are extremely vulnerable to human biases and omissions that undermine the assessment of evidence, statistical analysis, and the understanding of cause and effect.

Third, generative AI methods are being integrated into the operational planning process and can enrich the development of a range of ISRT strategies. However, it must start all over again if “Unknown-

Unknown" events crash the ongoing TTPs. Also, generative AI needs high-quality training datasets; otherwise, it is prone to inaccuracies and biases.

This SBIR topic will develop Observant-AI agents as a class of regenerative AI that learn in real time, enables active visual and tactical monitoring of anomalous activities, and trigger I&W alerts in naval operations. The envisioned Observant-AI agents proactively enforce the fail-safe execution of approved ISRT operational plans. They exploit unexpected events in real-time by leveraging insights from all-source intelligence (ASI) and remote sensors (i.e., space assets). They generate and execute novel all-domain ISRT TTPs plans consistent with the approved plans to counter evolving adversarial intents and undesired events, LIVE. In other words, the Observant-AI agents enable fault-tolerant mission-focused reconfiguration by analyzing existing assets' capabilities through novel tactical teaming arrangements from approved deployable capabilities (sensors, manned and unmanned weapon platforms, intelligence data sources, etc.). Observant-AI will automatically alert the chain of command at all levels with emerging or mission-altering observables that may interfere with operational objectives.
The goal of the effort is to perform a combination of offline and online predictive engagement modeling to plan for trusted AI-enabled TTPs that will strategically adjust plans in real time to adapt to emerging events and conditions. It will use Monte Carlo simulation to model the probability of various outcomes under countless AI-generated Red vs. Blue engagement (action-reaction) scenarios for offline TTP planning and mission success assessment. Regenerative AI will ensure Observant-AI can quickly adapt the blue's creative ISRT strategies against near-peer adversaries (Red). Regenerative AI offers unique capabilities such as learning from sparse data and predicting complex interactions. It will achieve this objective by testing novel all-domain penetration strategies, including offensive cyber and information operations to find advantageous strategies, then running them against many emerging scenarios, identifying the vulnerability points and engagement risks, and modifying strategies to sustain their performance with acceptable risks.

Critical AI technology components and developments are as follows:

1. Contextual modeling: relational modeling, graph-based modeling, spatial modeling, logic-based modeling, uncertainty modeling, ontology-based modeling, hybrid context modeling.
2. Multidomain multimodal all-source intelligence data and signals: multi-level secure connectivity and access.
3. Data learning: decision tree classifier, multilayer perception classifier, collaborative filtering, frequent pattern mining, K-means, deep learning.
4. Data quality, data interoperability, data generation.
5. Data storage: signal-oriented database, graph-based database, associative database, text-oriented database.
6. Spatiotemporal synchronization methods for multimodal data across decentralized architectures.
7. Multimodal contextual signal processing and fusion.
8. Cross-domain contextual collaborative learning, inference, and recognition.
9. Contextual collaboration, adaptation, and teaming via ad-hoc networking.
10. Contextual reasoning, risk assessment, and risk reduction.
11. Contextual query, question-answering (Q&A), and natural language processing.
12. Contextual priority-based task management and balancing competing multifaceted ISRT operational objectives such as persistence, endurance, opportunistic collections, and targeting.
13. AI-risk escalation control methods that will not erode decisions across the integrated chain-of-command.
14. AI-assisted targeteer maneuvers and engagements.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating

procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Determine the technical feasibility of designing and developing the Observant-AI technology described in the Description section. Draw key distinctions for the proposed design approach compared to the current state-of-the-art naval ISRT information exploitation systems. Ensure that the proposed design reveals a clear set of steps that naval ISRT platforms and joint services can capitalize on now for a transformative AI-based ISRT operation with dominant applications within the next five years. Motivate the design with three compelling challenge problems supported by relevant datasets. Consider challenge problems corresponding to cross-domain littoral operations and navigational risks countering anti-access/denied-access enforcement scenarios. (Note: The scenarios need to consider seaside terrain, weather, the maritime order-of-battle, the movement; engagement rules and doctrine; logistics and supply demands, etc. Hostile enforcements may include hypersonic missiles, ballistic missiles attacks from land or undersea, sea-skimming cruise missile attacks, anti-submarine track and trail, torpedo attacks, anti-space jamming, littoral mining, swarming weapons, and decoy and deception tactics.) Perform testing and demonstrations may use a mix of OSINT datasets, synthetic datasets from DON, MCML, AIS maritime traffic, commercial satellite imagery, or similar sources. Conduct end-to-end Observant-AI system performance assessment, including:

1. Cross-domain contextual collaborative learning, inference, and recognition.
2. Multi-agent cross-domain contextual collaborative teaming and adaptation via ad hoc networking.
3. Cause and effect sensitivity analysis on contextual understanding.
4. Confidence rate on AI-generated proactive TTP enhancements on engagement plans, options, and risk reduction associated with the ups and downs of encounters.
5. Confidence rate on AI-risk escalation controls enabling trust decisions across the integrated chain-of-command.
6. Efficiency gains in human responsiveness through timely decision-making, chain-of-actions, and resources spent.

Performance criteria must include sensitivity (true-positive rate), specificity (true-negative rate), precision (positive predictive value), miss rate (false negative rate), false discovery rate, and false omission rate. Performance metrics (considering outcomes will depend on data quality):

1. Analytic Completeness: not just identifying and stopping hostile acts, but also how it occurred by synthesizing the entire chain of events what would have happened had it not been stopped < 95%
2. Uniqueness: Signature attributes definable and retrievable (who, what, why, where, when) < 95%
3. Validity: Supporting evidence < 95%
4. Consistency: Periodic signature updates of attributes from various sources that reinforce linkages < 95%
5. Accuracy: Overcoming noisy data < 95%
6. Accuracy metrics for ingesting and classifying multimodal data: structured data mining and interpretation - accuracy of 95% over 98% captured content; unstructured data mining and interpretation – accuracy of 90% over 95% captured content.

Deliverables include end-to-end initial prototype technology, T&E, demonstration, a plan for Phase II, and a final report.

PHASE II: Conduct proof-of-concept and prototype development incorporating the recommended candidate technology from Phase I. Demonstrate the operational effectiveness based on the following criteria: (a) prioritized sensor alerts, (b) prioritized threat escalation, (c) measured severity of events, and (d) measured analytic completeness. Test and demonstrate the improved capability based on the performance metrics detailed for Phase I with the following requirements: Analytic Completeness < 98%, Uniqueness < 98%, Validity < 98%, Consistency < 98%, and Accuracy < 98%. Provide the following deliverables: analytics, signal processing tools, models, prototypes, T&E and demonstration results, interface requirements, and final report. Final report will include a detailed design of the system and a plan for transition to the program of record in Phase III.
It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Advance these capabilities to TRL-7 and integrate the technology into the Maritime Tactical Command and Control POR, Marine Air-Ground Task Force Command and Control, or ISR processing platforms at the Marine Corps Information Operations Center. Once conceptually and technically validated, demonstrate dual-use applications of this technology in civilian law enforcement and security services.

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KEYWORDS: Risk-Aware, Regenerative, Artificial Intelligence, Machine Learning, Contextual, Multimodal, Cross-Domain, Visual-Tactical ISRT

DON26BZ01-NV024 TITLE: 3D-Heterogeneously Integrated Photonic (HIP) Imaging Sensor

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Integrated Sensing and Cyber;Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design, fabricate, and verify the performance of a 3D-heterogeneously integrated photonic (HIP) imaging sensor consisting of a detector array, read-out integrated circuit (ROIC), and photonic transmitter.

DESCRIPTION: Emerging military electro-optical and infrared (EO/IR) sensors enable high resolution through small pixels, wide field-of-view through large arrays, and high frame rate through high sensitivity and low latency. For the most advanced focal plane array (FPA) sensors, the data bandwidth dictated by the high pixel count and bit rate is reaching the limits of conventional copper wire interconnects. Datalinks using optical interconnects offer a unique and commercially mature solution that can obviate the copper bandwidth limitation, while offering additional advantages of lower power, lower cost, and on-chip integration. For large arrays, the high data rate can be further managed by tiling synchronized, independently addressed smaller arrays, which divides the serialized data stream into multiple parallel paths, while also improving foundry yield. However, existing FPA layouts place read-out electronics, including column analog-to-digital converters, serializers, and bias sources, along the periphery of the imaging chip. To enable tiling with sub-pixel gaps between tiles, the peripheral electronics must be moved below the detector layer. A photonic layer could also be added to create a 3D vertically integrated FPA stack, enabling large arrays to operate at exceptionally high data rates. 3D heterogeneous integration of the FPA stack can be accomplished using bump-bonding, direct-bond integration, or other techniques, but ultra-low capacitance connections are required for low-noise operation to permit the short photon integration times inherent to high-frame-rate imaging. As militarily relevant EO-IR imagers often operate at cold temperatures of 100K+/-20K, the 3D HIP FPA transmitter must also perform well under cryogenic conditions. When tiled in large arrays of small pixels, the 3D-HIP imaging sensor will provide concurrent wide-FOV, high-resolution, and ultra-high frame rate, circumventing conventional imaging sensor paradigms. Frame rate should use 1 KHz as the goal is to address high data rate challenges, however, since the pixel size and format are flexible for this effort, this is not a hard requirement. This SBIR topic’s intent is the development and maturation of 3D heterogeneous integration (3DHI) of electrical and optical/photonic layers that achieves high bandwidth interconnection.

PHASE I: Perform a trade study of design variables. Create a concept for a 3D-HIP imaging sensor design. All design features must be supported by quantitative modelling, simulations, or general trade analysis. The design should be adaptable to all EO/IR spectral bands, formats, and pixel sizes. Address detector, ROIC, and photonic layer designs and interconnections. Within the photonic layer, the laser type

and location (inside vs outside the cold space), the optical modulation scheme, optical and electronic bandwidth and compatibility, and energy efficiency must be addressed. The proposed design must include a detailed noise analysis, including component capacitance projections and variances, along with other noise sources. A full link analysis must be provided, including both energy efficiency and data bandwidth. Prepare a Phase II plan that includes the fabrication, integration, and testing strategy.

PHASE II: Fabricate a prototype 3D-HIP imaging sensor based on the Phase I design. It is expected, but not required, that detector, electronic, and photonic layers will be fabricated and integrated at separate foundries, potentially using different foundry nodes. While only a 3D-HIP transmitter is required, the output must be received and processed into imagery. Fabrication of a symmetrically designed transmitter and receiver pair is encouraged. Moreover, the transmitter chip should be compatible with formation of a tiled array. The transmitter-to-receiver connection should employ optical fiber of nominally 1 meter length. The transceiver performance will be thoroughly documented in the Phase II final report.

PHASE III DUAL USE APPLICATIONS: Support the transition to Navy use. High-resolution, wide-FOV, high-speed imagers will find wide use in many commercial and industrial applications such as computer vision, autonomous navigation, security and industrial facility surveillance monitoring.

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KEYWORDS: Sensors, read-out integrated circuit, ROIC, photonics, tiling, chiplet, heterogeneous integration

DON26BZ01-NV025 TITLE: Leveraging Machine Learning for Advanced Passive Sonar Tracking

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop advanced automation to detect, locate, classify, and correlate contacts across multiple sonar sensors and multiple display surfaces.

DESCRIPTION: Passive sonar systems employ a standardized signal processing pipeline to track, classify, and localize underwater contacts. This automated process, often referred to as "automation," begins after front-end processing generates visual displays for sonar operator analysis and automated processing. Existing algorithms that track energy signatures on these displays typically include Kalman filters, probabilistic multi-hypothesis trackers, and particle filters. However, these traditional tracking methods, as implemented in current operational systems, often fail to fully leverage the potential of modern machine learning techniques. This SBIR topic seeks to incorporate cutting-edge machine learning technologies into passive sonar processing to significantly improve tracking, classification, fusion, and localization of current anti-submarine warfare passive sonar systems. The specific threshold and goals for performance improvement are as indicated in the following table.

Targeted Improvement	**Metric**	**Threshold**	**Objective**
Tracking	Increase Hold Time Ratio	10%	20%
Tracking	Reduce Time to Detect	10&	20%
Classification	Increase Probability of Correct Classification	10%	15%
Classification	Reduce Probability of False Alerts	10%	15%
Track Fusion	Increase Probability of Correct Association	15%	20%
Localization	Reduce Area of Uncertainty	15%	20%

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and

Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop algorithms that improve sonar automation for tracking, localization, classification, and multi-sensor fusion. The approach will reduce the burden of operators to maintain and promote tracks and be supported by theory.

PHASE II: Implement the proposed approach in a simulated environment (e.g., MATLAB) and demonstrate stated performance using government-provided data from a Navy sonar system. Important metrics will be, but not limited to, probability of correct association, hold time ratio, time to track, and probability of correct classification.
It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Support transition to Navy use.
This effort is anticipated to have dual-use applications in commercial surveillance systems with towed arrays or ISR uncrewed aerial vehicles. The performer shall identify possible non-Navy applications for their technology.

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KEYWORDS: Multi-sensor data fusion, operator workload reduction, advanced automation

DON26BZ01-NV026 TITLE: Passive-Active Combo System for Unmanned Characterization of Littoral Environments

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a lightweight, integrated passive imaging and LiDAR system, deployable on an unmanned aerial platform for target detection, feature characterization, and bathymetry retrieval in littoral environments. The system should be light enough for deployment from a Group 2 (max. gross takeoff weight: 21 – 55 lbs.) unmanned aerial vehicle (UAV).

DESCRIPTION: Achieving and maintaining maritime dominance in the coastal battlespace requires the Navy and Marine Corps to have superior situational awareness. A key component of this dominance is the ability to rapidly characterize shallow, nearshore environments [Ref 1] in real-time using agile, unmanned aerial platforms. To this end, a system is needed that provides (1) bathymetry retrieval; (2) detection and discrimination of underwater targets; and (3) characterization of the land-ocean interface (i.e., surface type, topography, and shallow-water bathymetry).

Current UAV-based shallow water and littoral zone characterization relies on either (1) passive imagers alone or (2) bathymetric LiDAR systems deployed on larger airborne platforms or in separate missions. While passive imagers effectively characterize surface features, bathymetric LiDAR is necessary for bathymetry retrieval and underwater target detection. Simultaneous deployment of both a high-performance passive imager and a bathymetric LiDAR on a Group 2 UAV is challenging due to payload weight limitations. Systems that attempt this combination often compromise sensor performance or utilize topographic LiDAR [Ref 2], which uses near-infrared wavelengths unsuitable for bathymetry retrieval. One potential solution is a system that can accommodate a passive imager and a dual-wavelength LiDAR that operates at two wavelengths – one where light penetrates deep into the water column and another with very little to no penetration into the water column – which can be used to effectively discriminate between LiDAR returns from the water surface and the substrate. The heaviest part of a topo-bathy LiDAR is the scanning component. A non-scanning, nadir-viewing LiDAR system would be light enough for simultaneous deployment on a Group 2 UAV. The passive imager could be hyperspectral or multispectral but should provide sufficient spectral information to spectrally characterize the water column and the land-ocean interface and discriminate underwater objects and features. Single nadir lines of LiDAR returns from adjacent flight lines could be mapped onto corresponding spatially explicit imaging data to build three-dimensional profiles of bathymetry. Coincidental LiDAR and imaging data could also be used to train a regression-based machine learning (ML) model to estimate depths from the imaging data, similar to previous empirical approaches [Ref 3].

The system should provide rapid onboard processing of passive spectral and LiDAR data and real-time downlink of preliminary output to a ground station. The output should include a true-color composite of the target area, a topo-bathy map, a target detection map (showing locations of targets of interest, which could be new objects or objects with known properties pre-programmed into the system), and a terrain characterization map (showing information on the terrain type, concentration of optically significant constituents in the water column, and bottom type). Performers may use simple or sophistical techniques to retrieve information from spectral imaging data, such as simple band-ratio algorithms, spectral inversion based on radiative transfer modeling, spectral derivatives, or ML techniques. The system should provide the above information for coastal waters up to 20 meters depth in moderately turbid waters (diffuse attenuation coefficient at 490 nm, Kd(490) ˜ 2-4 m-1). Note: SBIR funds may be used to purchase a Group 2 UAV to serve as a platform for the imager + LiDAR combo system.

PHASE I: Develop a preliminary observing system simulation experiment to simulate optical and spectral models for the combined passive imaging and LiDAR system. Perform sensitivity analysis of system performance for a range of design configurations under varying conditions of turbidity and optical complexity of shallow water environments. Conduct a feasibility study for the proposed system. Provide a report of the feasibility study and an initial layout of the proposed system design.

PHASE II: Develop the prototype based on optimal design configurations determined from the Phase I feasibility study. Finalize the approach for exploiting spectral information from the passive imager and combining spectral information with LiDAR returns for retrieval of three-dimensional bathymetry and characterization of the water column and the nearshore terrain. Limited demonstrations of the prototype are also required.

PHASE III DUAL USE APPLICATIONS: Upon successful demonstration of the prototype in Phase II, the system shall be flight-tested, developed into a commercial product against existing requirements of the Navy's Airborne Littoral Mine Detection System (ALMDS) and the Marine Corps' Standoff Explosive Detection System for operational coastal characterization and mine detection for transition consideration.
Blue-green LiDAR serves environmental monitoring, underwater mapping, and marine ecology purposes. With effective and specific processing, data acquired from this technology can be tailored for either commercial or military applications.

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KEYWORDS: Passive + active imaging system; topo-bathy LiDAR; mine detection; coastal characterization; unmanned aerial systems; nearshore bathymetry

DON26BZ01-NV027 TITLE: Automated Ice Detection and Polar Navigation Tool (PolarNav)

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Integrated Sensing and Cyber;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a prototype for a system that integrates information on sea ice conditions from a diverse set of sources, including shipboard instruments, airborne and spaceborne sensors, and sea ice model output, to yield optimized route options as a planning aid for navigation through ice-infested waters in polar regions.

DESCRIPTION: Recent trends of warming in the Arctic have led to a steady decrease in the extent of multi-year sea ice, a corresponding increase in seasonal sea ice, and an overall lengthening of the navigable season [Refs 1, 2], thereby making the Arctic increasingly open to maritime traffic. Vessels operating in and near sea ice must make navigation decisions that balance the capabilities of the ship with the objectives of their voyage. Such route planning is complicated by the dynamic nature of sea ice, as it is subject to movements caused by a number of factors such as the Beaufort Gyre, transpolar drift, and weather events, which are even more pronounced on the thinner, seasonal ice. A system capable of aiding navigation teams in route planning based on ice observations and forecasts over time scales on the order of hours to days is essential for safe navigation through polar regions.
Currently, ice navigation relies heavily on manual processes. A majority of route planning information comes from satellite imagery, either optical or synthetic aperture radar (SAR), or from forecast information from entities like the U.S. National Ice Center. Due to dynamic weather conditions and rapid movement, the operational value of overhead imagery is sometimes temporally limited. These longer-range data sources are augmented by shipboard systems, such as onboard radar systems for icebergs, in-situ ice floe, and pack ice detection, which typically have detection ranges on the order of a few tens of kilometers. These close-range systems help inform tactical navigation decisions and near-term route planning.

Key aspects of ice analysis, whether conducted onboard or remotely, are ice edge definition, identification of ice types (e.g., seasonal ice, multi-year ice) and concentration, and detection of ice features such as ridges and icebergs. This analysis is then presented to the navigation team and command who assess the current and planned route and make course adjustments as necessary. Current ice forecasts do not always adequately account for projected ice movement over the next 12-96 hours, which is crucial for effective route planning. Moreover, the analysis and route planning are often separate functions, each conducted by distinct teams based on their own personal experience and knowledge. This separation can lead to suboptimal decisions and increased risk.

The goal of this SBIR topic is to develop a prototype tool that helps ships make safe navigation decisions in the Arctic. The tool should leverage established ice prediction models and incorporate other available sources to assimilate models and improve forecasts. These additional sources may include:

- Onboard sensors: Radar, thermal cameras (forward-looking infrared), and microwave sensors on the ship.
- Aircraft sensors: Sensors on airplanes and unmanned aerial systems (if available).
- Satellites: Optical and SAR data, dynamically updated with every new overpass.
- Iceberg records: Historical data on where icebergs have been seen/located (e.g., from the U.S. Coast Guard's International Ice Patrol).

The envisioned product is a geographic-information-system-based tool that uses artificial intelligence, first-principles algorithms, and automated data processing schemes to combine information from the above sources, update model-based predictions, provide 12–96-hour sea ice forecasts, and suggest potential navigation routes. Route options should consider vessel specifications, such as ice resistance characteristics and fuel consumption rate, and provide options for fastest route to destination, shortest route to destination, route with minimal wear/tear on vessel and crew, and maximum safe speed based on ship hull type/construction. Ultimate route decisions should be left to the vessel's navigation team.

PHASE I: Draft a conceptual framework for dynamic route planning based on sea ice characterization and forecasts from data fused and integrated from disparate sources. Define and develop in detail the concept and methodologies for extracting and combining data from diverse sources to provide nowcasts and forecasts of sea ice conditions and route planning options at tactically relevant spatial and temporal scales. Prepare a report containing preliminary results of retrieving sea ice characteristics using fused multi-modal satellite imagery, a framework for improving predictions through assimilation of data from diverse sources, and a framework for dynamic route planning which will be the basis for the proposed tool. If the Phase I Option is exercised, carry out a simple demonstration using multi-temporal and sequential datasets from multiple satellites and/or in situ measurements and modeled sea ice predictions for a specific region to test approaches for improving predictions of sea ice distribution and movement and use the combined information for route planning in hindcast and forecast modes.

PHASE II: Develop a prototype data analysis and route planning software tool that can be tested operationally on a vessel and is in the form of a standalone system with a display interface showing the latest satellite imagery of the ocean in the vicinity of the vessel, akin to the display on a Global Positioning System (GPS) based automobile navigation system. This prototype should be able to connect to data streams from instruments onboard the vessel, near-real-time satellite data that could be downloaded through the vessel's onboard communication system (e.g., satellite communications), and sea ice model output; produces nowcasts and 12-96-hour forecasts of sea ice conditions; and provides multiple route options for navigation, optimized for the fastest route, shortest route, the most fuel-efficient route, or the route with the least ice encounter.

PHASE III DUAL USE APPLICATIONS: Further develop the prototype into a commercial tool for integration onto a U.S. Coast Guard icebreaker or an ice-hardened Navy vessel. The tool, operating onboard the vessel in a place of the navigation team and command's choosing, shall provide real-time updates of sea ice conditions in the vicinity of the vessel and route options that the vessel navigation and command team can use to make informed decisions for sailing through ice-infested waters. The tool will also find its use in commercial industries such as shipping, fishing, and tourism in the polar regions.

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KEYWORDS: Polar navigation; artificial intelligence/machine learning; AI/ML; sea ice forecast; route planning; satellite imagery; data fusion; Arctic; Antarctic; ice identification; ice classification; ice

prediction; ice analyst; Meteorology and Oceanography; METOC; remote sensing; modeling; shipboard sensors; human-machine interface; big data

DON26BZ01-NV028 TITLE: Overlay/Bond Coatings that Resist Hot Corrosion in Navy Gas Turbines

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop overlay or bond coatings and a coating model that enables longer service and prediction of corrosion, oxidation and overall degradation when exposed to marine Naval environments as a function of corrosivity, stress, and various temperature combinations via integrated computational material engineering (ICME), which will foster creation of new coatings resistant to these degradation modes.

DESCRIPTION: Marine gas turbine engines serve as primary and auxiliary power sources for several current classes of ships in the U.S. Navy. It is desirable for marine gas turbine engines to have a mean time between removals of 20,000 hours. While some engines have approached this goal, others have fallen significantly short. The main reason for this shortfall is various forms of hot corrosion (Type I and Type II) damage in the hot section turbine hardware due to intrusion of salts from the marine air and/or from sulfur in the gas turbine combustion fuels.
The synergistic effect of stress- and deposit-induced high temperature corrosion can lead to other corrosion mechanisms. Corrosion fatigue as well as fatigue often initiates at stress risers. Metallurgical examination of several failed marine gas turbine blades that had operated between 5,000 and 10,000 hours was performed and compared to “unfailed” blades with 18,000 operating hours from a similar marine engine. Deposition occurring at sites under the platform of unfailed turbine blades revealed pitting at those sites.

Further examination revealed poor coating quality (i.e., high porosity and variable thickness) under the platform of first stage turbine blades that allowed salts to permeate through the coating to the alloy surface and initiate hot corrosion. Further coating examination under the platform showed highly variable coating thicknesses (0-40 µm) in the curved area of transition between the under platform and the blade stem. In a few cases, coatings were non-existent on the “unfailed” blades. The Cobalt Chromium Aluminum Yttrium (CoCrAlY) coating, when present, usually was porous or the available coating under the platform was highly contaminated due to lack of adequate spray deposition in these non-line-of-sight areas. CoCrAlY coating thicknesses at other sites along the blade stem were 35 µm to 105 µm (1.4 to 4.1 mils) and devoid of porosity. The corrosion that was observed under the platform in all cases was caused by Type II, low-temperature hot corrosion, which occurs in the temperature range of 649°-732°C (1,200°-1,350°F). Corrosion penetrated the porous coating and caused further undercutting of the coating along the coating/alloy substrate interface, Type II hot corrosion caused pitting at these locations under the platform, which caused stress risers where corrosion fatigue cracks initiated. These pits advanced through the blade stems to varying degrees.

The synergistic effect of stress- and deposit-induced high temperature corrosion leads to the premature failure of aero turbine blades reportedly due to stress corrosion cracking. The lower shank of aero gas turbine blades, which operates below 600°C is susceptible to this mode of failure. Two important factors that lead to stress corrosion cracking of single crystal nickel-based superalloys are the type of deposits that form on components (these include alkali chlorides and sulfates which are introduced through the environment) and the concentration of SOx in the environment. Therefore, it is important to understand the synergistic role of deposits and sulfur containing gases on the stress corrosion cracking susceptibility of single crystal nickel-based superalloys below 600°C.

PHASE I: Demonstrate an understanding of what differences and influences exist between aviation and marine propulsion. Determine the mechanism for the observed corrosion at 500°-550°C. if stress corrosion cracking (SCC) is the prevalent corrosion mechanism, determine the interplay with NaCl, Na2SO4, SOx, and stress. For a given concentration of the chemical compounds, determine at what stress initiates SCC. Initiate correlations that should begin to formulate the ICME model framework to create a coating that would avoid reactions leading to SCC. (Note: For shipboard operations this would be either an overlay or diffusion coatings. For aero applications, this would lead to creation of a bond coat that would also promote long thermal barrier coating (TBC) life (goal: > 5K hours) and assist in maximizing corrosion and oxidation resistance by changes in coating chemistry and structure while not impacting fatigue, SCC, or substrate strength of the substrate alloys. It is suggested that the starting TBC be yttria-stabilized zirconia.) Lastly, perform a short-term (~200 hours) high temperature test to validate the feasibility of the ICME model.

PHASE II: The ICME framework shall be further expanded to include compatibility of the TBC to different bond coats as well as further development, modification, and maturation of the ICME model. Collaboration with coating and engine gas turbine original equipment manufacturers (OEMs) is encouraged for advice and direction for further developments of the ICME models and strategies. Correlate into the ICME-derived model the interaction of chromium and aluminum content in a coating that leads to the formation of chromia or alumina scales. Coatings on several alloys shall be tested to determine coating compatibility and assess the impact of coatings on alloy substrate properties in a burner-rig or similar test environment that includes salt ingestion. Coatings shall be applied onto alloy substrates by at least one recognized commercial coating process (line-of-sight and/or non-line-of-sight). The expected deliverables will be: (1) optimized coating corrosion resistance to SCC for a given set of alloys and (2) an ICME-derived model that would predict and assist in the development of future overlay or bond coats to minimize SCC in gas turbine that are compatible with multiple alloy substrates.

PHASE III DUAL USE APPLICATIONS: The ICME model will be further developed and matured through the expansion of bond coat/overlay coat chemistry and structure with the selected strategies to mitigate salt interaction that could lead to SCC. TBCs are permeable so the bond coat must form an impermeable barrier to avoid salt interaction with the alloy substrate that would tend to cause SCC. Engage with a gas turbine engine OEM to have an appropriate bond coat-TBC system applied on select static and/or rotating engine components of a current Navy engine and testing in cycling temperature test. The expected deliverables will be: (1) a TBC(s) compatible to corrosion and hot corrosion-resistant bond coat substrates, (2) TBC(s) resistant to SCC in the marine environment, and (3) an ICME-derived model that would predict and assist in the development of future TBC systems (i.e., alloy, bond coat, TBC, TBC strategy to minimize SCC with marine aero engine operational environment). Engage with a marine engine OEM to further develop the TBC technology for incorporation into current and future Navy ship engines.
Development of long-lived TBC systems able to withstand hot corrosion and subsequent SCC at temperatures below 600°C for U.S. Navy applications will also enable more efficient service for commercial applications that employ industrial gas turbines. Marine gas turbine engines are industrial gas turbines that are intended for Naval use. Successful development of better coatings for the current alloys, capable of extended service in the highly corrosive Naval operating environment, should enable subsequent use in commercial applications such as cargo ships, cruise ships, ferries, and tankers if the business case justifies the results.

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KEYWORDS: Hot Corrosion, Stress Corrosion Cracking, Environmental-Induced Cracking, Corrosion Fatigue, Gas Turbines, Superalloys

DON26BZ01-NV029 TITLE: Low Cost Malleable Metastructure Adherents for Maritime Environments

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Directed Energy (SCADE)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Microelectronics;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a material with the ability to rapidly and cost effectively produce metastructures or frequency selective surfaces which can be adhered to naval assets or similar systems (e.g., apertures, super-structures substructures, deployable, etc.).

DESCRIPTION: Several industries and Department of War (DOW) systems rely on Frequency Selective Surfaces (FSS), metastructures, or comparable materials to protect critical assets, including communications, radar, and Electromagnetic Warfare (EW) systems. Similar materials are also used as protective coatings for Electro-Optical/Infrared (EO/IR) systems—particularly in airborne and maritime applications—where they are consistently challenged by harsh maritime environments. These coatings, covers, and materials are susceptible to degradation from salt, ultraviolet (UV) radiation, and water intrusion due to their attachment to substructures, structures, or apertures.

Furthermore, the manufacturing and application of these materials are often considered expensive, time-consuming, and technically demanding due to platform-specific requirements. Recent constraints within the industrial base—such as the reduced availability of certain materials like CFC resins and polymers—have further exacerbated production challenges. These limitations have driven up costs, which have not benefited from economies of scale or broader adoption.

This SBIR topic seeks to develop alternative solutions that offer frequency selectivity, moldability (to conform to existing superstructures, substructures, or complex geometries), and resilience to maritime environments. In theory, such advancements would enable optimal dynamic performance across RF, microwave, or EO/IR domains, while maintaining durability in challenging conditions.

FSS remains the incumbent solution of choice, given its broadband frequency response, manufacturability, and superior durability in maritime conditions—advantages not matched by commercially available polymer-based fiberglass radomes, which typically lack frequency selectivity or the directive enhancements required by DOW systems. The reduction in availability and manufacturability of certain composites—due to regulatory restrictions or hazardous byproducts—has created an urgent need to pursue viable alternatives. Operating apertures across multiple frequency octaves remains a significant challenge for manufacturers and original equipment manufacturers (OEMs).

Addressing the outlined challenges while achieving required performance objectives will likely demand innovation across multiple technical disciplines, including:

- Frequency Response – such as L, S, C, X and Ku Band and/or EOIR: Optical, midwave, longwave, others
- Advanced high-performance materials (ceramics, polymers or superalloys)
- Novel manufacturing or machining techniques
- Advanced 3 D optimized material additive manufacturing
- 3D optimized structures, magnetics or similar (inductor/capacitive/parasitic imbedded circuits)
- Highly resilient coatings, or new coating application techniques to existing materials
- Highly flexible embedded thin film materials

While existing materials with modifications will be considered, alternative solutions are also welcomed. However, the potential impact of these alternative designs—relative to existing materials or coatings—will be a factor during the selection process. Proposers should clearly identify any necessary mitigation

considerations (e.g., storage, handling, disposal, etc.) required to support a credible path to qualification and approval for shipboard or airborne use.

The primary objective of this SBIR effort is to develop a material capable of broadband performance—defined here as the ability to provide frequency response across multiple octaves compared to existing materials. However, the proposed material must also be operationally viable and capable of meeting several critical performance objectives. Specifically, the solution should:

1. demonstrate through-performance (S21) in a near-field environment across multiple frequency octaves.
2. operate effectively across multiple bands of the EO/IR spectrum.
3. adhere to materials with sharp angles and varied geometries.
4. be capable of long-term storage without degradation after manufacturing or adherence to a structure.
5. withstand at least five years in a maritime environment without significant performance degradation (defined as <0.5 dB variance).
6. be rapidly applied to a surface with minimal preparation, achieving adherence in less than 24 hours.
7. demonstrate a reduction in abatement of signal return in multiple bands within the microwave and or the EO/IR energy regime radio frequency/midwave (RF/MW).
8. demonstrate that at scale the production cost can be lower than production of existing materials.

Acceptable solutions must also align with intended deployment scenarios, including shipboard/surface, Unmanned Aerial Systems (UAS), and land-based applications. For demonstration purposes, a commercial broadband antenna or a commercially available EO/IR camera may serve as the interface to evaluate proposed materials as radomes, covers, or adapters under defined boundary conditions. Demonstrations must show functional performance across at least two frequency bands—within the L-band to Ku-band range (e.g., S-band and C-band).

PHASE I:

1. Material Concept Evaluation

Investigate and identify novel materials or coatings capable of providing broadband frequency selectivity across RF, microwave, and EO/IR domains, with an emphasis on alternatives to restricted or environmentally hazardous substances (e.g., CFC resins, specific polymers).

2. Environmental Compatibility Assessment

Assess the proposed material's theoretical or lab-based resistance to maritime environmental stressors, including saltwater exposure, UV radiation, and water ingress.

3. Geometric and Structural Adaptability

Demonstrate initial feasibility for adherence or conformability of materials to complex substructures and geometries relevant to DOW platforms (i.e., airborne, shipboard, and ground-based).

4. Initial Performance Modeling

Develop simulation-based predictions or benchtop validations of frequency performance across multiple octaves in the RF and EO/IR spectrum (e.g., S21 transmission characteristics, optical transmission in multiple EO/IR bands).

5. Risk and Mitigation Planning

Identify potential risks (e.g., storage, degradation, application time) and propose mitigation strategies for eventual shipboard or airborne qualification.

PHASE II:
1. Prototype Fabrication
Design, manufacture, and deliver functional prototype(s) of the developed material or coating, tailored for maritime-relevant conditions and representative platform geometries.
2. Performance Validation Across Frequency Bands
Validate the prototype's frequency-selective behavior through laboratory and controlled-environment testing. Demonstrate multi-band performance (minimum two distinct bands, e.g., S- and C-band) from L-band to Ku-band.
3. EO/IR Performance Characterization
Conduct EO/IR transmission testing to confirm broadband optical performance through multiple EO/IR spectral bands, suitable for integration with commercial EO/IR sensors.
4. Environmental Endurance Testing
Evaluate long-term durability under simulated maritime conditions, including extended salt spray, UV exposure, and temperature/humidity cycling to validate 5+ year service life with minimal (<0.5 dB) performance degradation.
5. Rapid Application Demonstration
Demonstrate field-level application procedures confirming surface adherence with minimal preparation and application time under 24 hours.
6. Platform Integration Assessment
Assess integration potential with at least one DOW-relevant application (e.g., UAS radome, shipboard sensor cover), including initial qualification planning and boundary condition analysis.

PHASE III DUAL USE APPLICATIONS:
1. Qualification for Operational Platforms
Complete the qualification and certification process for use of the material on military platforms (i.e., shipboard, airborne, and land-based), including necessary safety, handling, and environmental compliance documentation.
2. Transition to DOW Programs of Record (PoRs)
Integrate the developed material into one or more PoRs or acquisition pathways (e.g., Navy UxS platforms, EW pods, surface combatant radar housings) through partnerships with prime contractors or system integrators.
3. Production Scale-Up and Cost Reduction
Establish a scalable manufacturing process that ensures material consistency, repeatability, and cost-efficiency, including options for low-rate initial production (LRIP) and full-rate production (FRP).
4. Commercial Dual-Use Expansion
Explore and initiate commercial applications of the developed material or coating, including broadband antennas, protective camera housings, or telecom equipment enclosures, leveraging interest from non-defense markets.
5. Sustainment and Lifecycle Support Plan
Develop a comprehensive sustainment strategy including repair, refurbishment, and replacement options, tailored for DOW logistics pipelines and long-term deployment in austere environments.

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KEYWORDS: Frequency Selective Surfaces, Metastrucutures, Engineered Materials, Coatings, Metamaterials, Phase Changing Materials

DON26BZ01-NV030 TITLE: Artificial Intelligence and Machine Learning (AI/ML) for Additive Manufacturing (AM)

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Automate additive manufacturing (AM) through advanced computational techniques (i.e., artificial intelligence and machine learning [AI/ML], digital twins, etc.) to select optimal materials and manufacturing parameters to meet mission requirements in terms of component performance.

DESCRIPTION: AM has enabled new designs and rapid fabrication. However, there are no automatic tools available to computationally link across build platform to part performance. This SBIR topic seeks to leverage AI/ML, digital twins, and process simulation to select optimal materials and manufacturing parameters to meet rapidly changing mission requirements. A user should be able to input material type, part geometry, and AM system details into the prototype tools to automatically generate optimized build parameters along with accurate mechanical performance predictions.
While some tools in the current market can address part of this need, none are known which can integrate across the entire material lifecycle from pre-build to performance in a single ready-to-use package. The focus of this effort will be investigating legacy parts (i.e., obsolete castings and forgings) which need rapid production to avoid long lead times. Leveraging physics-informed AI/ML technologies and digital twins to optimize printing based on geometry and material properties will mitigate build defects and reduce post-processing while enabling performance prediction.
From a technical standpoint, the prototype tool(s) developed under this topic should seamlessly integrate across the component lifecycle, from initial design (or reverse engineering) to build parameter optimization to mechanical performance prediction in structural metals, to enable the user to accurately fabricate mission-critical components. The tool(s) must be part and AM build system agnostic to ensure scalability to multiple locations across the Navy's manufacturing enterprise with various materials, systems, and performance requirements.

PHASE I: Define and develop a concept which leverages AI/ML, digital twins, and process simulation to select optimal materials and manufacturing parameters to meet rapidly changing mission requirements. Perform modeling and simulation with pointed physical testing for validation on a single component to demonstrate feasibility of the proposed concept. Required Phase I deliverables (in addition to the Contract Deliverables listed in the DON BAA instruction) will include a report on how the proposed concept will be expanded should the proposer be awarded a Phase II contract.

PHASE II: Expand the concept into full prototype tool development and validation using at least two additional components of different material classes and AM build systems. Demonstrate reduction in material fabrication time through automatic parameter generation while also reducing defect rates and material waste. Required Phase II deliverables will include:

a) A report on how the proposed concept can be expanded to other materials and systems not demonstrated in the Phase I and II taskings
b) Production of prototype tool(s) ready for delivery and demonstration at two U.S. Navy affiliated facilities.

PHASE III DUAL USE APPLICATIONS: Delivery of the final prototype tool(s) to U.S. Navy facilities will demonstrate the feasibility of the proposed solutions. Follow-on demonstrations to non-Navy

participants will enable other DOW, DoE, government, and industry partners to ability to view the solution and continue transition to other facilities. The expectation is that the tool(s) will be leveraged by any organization in need of efficient digital tools to predict component performance based on manufacturing details.

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KEYWORDS: Additive Manufacturing; AM; Artificial Intelligence; AI; Machine Learning; ML; AI/ML; Digital Twin

DON26BZ01-NV031 TITLE: Tele-Operated/Autonomous Mechatronic Vehicle Kits for Use In Mixed Disaster Environments

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Human-Machine Interfaces;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Create a mechatronic kit (roughly 3 feet wide, 3 feet long, 3 feet high, and weighing 220 pounds) with parts that can be easily swapped and reconfigured – including motors, sensors, and software – so that small teams can quickly build and change the mechatronic system in the field to handle many different tasks.

DESCRIPTION: First responder teams are small and have many tasks. Mechatronic solutions could help, but current solutions suffer from key shortfalls:
- They are an overly special purpose, limited in broad utility
- They lack common architecture, and cannot be repurposed or decomposed/recomposed for easy transport and repair
- They cannot be physically combined for additional power, speed or endurance
- Many key prime mover components are not manufactured at United States

This topic aims to create mechatronic prototypes that first responders can build in the field from basic parts (like frames, motors, computers, and controls). These mechatronic systems will be easy to customize for different needs. Preconfigured applications could include:
- Simple logistics support (e.g., carrying, loading, unloading)
- Intermediate applications (e.g., inspection, search and recovery)
- Advanced capabilities (e.g., waterproof, fire resistant, cold tolerant)

This topic aims to develop an open-ended physical architecture like an “Erector Set” with potential for endless variation.

PHASE I: Identify and design a prime mover configuration (such as electric brushless DC hub motors with Electronic Speed Control (ESC)) suitable for powering a variety of mechatronic systems with the following characteristics:
- o Rapid installation and removal
- o “Stackable” or arrangeable for higher power applications
- o Adaptable to track drives, direct wheel drives, gear trains, power take-offs (PTOs), propulsion shafts or other motive configurations
- o Domestically manufactured

Define and develop a suitable standard, material and configuration for structural purposes with the following characteristics:
- o Wide discretion in final configuration
- o Ease of assembly/disassembly
- o Folds, collapses or breaks down into minimal space
- o Able to host sensors and/or payloads

Prototype desired

PHASE II: Develop a prototype kit that resembles an erector set with multiple options for field construction of useful small mechatronic systems for logistics, security, surveillance and force projection at remote installations by small groups of first responders.
Develop prototype configurable U/Is that controls mechatronic vehicles/devices and allow for easy customizability by the first responders. Threshold: deliver 10 kits; objective: deliver 32 kits.

PHASE III DUAL USE APPLICATIONS: Product has utility to a wider group of users in three ways:

- Joint users, law enforcement, security forces, homeland security and other users with similar mission sets
- The development of an industry-recognized construction standard for small and mid-sized customized mechatronic systems
- STEM, academic research activities, hobbyists and industrial partners who are transitioning from sub-scale to intermediate mechatronic systems

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KEYWORDS: Mechatronic; Automation; Robotics; Modularity; Flexible; Configurable

DON26BZ01-NV032 TITLE: Automated Assessment and Adaptive Training for Simulated Fire Support Coordination

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Design and develop training tools that assess individual performance in a scenario-based fire support simulator and adapt instruction/scenarios based on that assessment without the need of an instructor in the loop.

DESCRIPTION: Recent Marine Corps publications have emphasized that effective fires employment remains a critical element of Marine Corps lethality and readiness [Refs 2, 3]. Fire support coordination (FSC) is the complex process of planning, integrating, and synchronizing the delivery of indirect fires (e.g., artillery, mortars) and close air support (CAS) to assist maneuver forces on the battlefield. At the company level, fire support is executed by a Fire Support Team (FiST) composed of several members, such as a FiST lead, Forward Observer (FO), Fire Support Officer (FSO), Joint Forward Observer (JFO), and Joint Terminal Attack Controller (JTAC). Some of these roles have a prescribed training pipeline (e.g., JFO, JTAC), whereas others do not (e.g., FiST Lead). The focus of the training for these roles is on individual skill development rather than team-based, integrated execution of fire support in support of a maneuver element. Opportunities for Marines to train collectively in combined arms integration are limited currently. Simulation-based training is available at a few designated locations, but these events require substantial instructor support to simulate different roles across multiple fires agencies and platforms (e.g., Fire Direction Center, CAS aircraft, Ground Force Commander). Live-fire Integrated Training Exercises (ITX), such as Fire Support Coordination Exercises (FSCEX), are costly (e.g., munitions), time and manpower intensive, occur infrequently, and have safety and external agency constraints not present in virtual training (e.g., FAA, host-nation restrictions). Furthermore, both simulated and live training environments require instructors to observe and assess performance with no automated assessment tooling. Across live and simulated events, these assessments are often not standardized and are subjective in nature, limiting opportunities for systematic assessment at the team or individual level. The lack of systematic assessment also limits the ability for Marine Corps to diagnose and address performance impacting the lethality of FiSTs.

Marines need a capability to assess foundational skills in their individual roles (e.g., crawl phase) that is embedded within a realistic fires simulator without requiring instructor facilitation or a full complement of FiST members. Automated assessment within the simulator allows for more objective-based metrics of performance and diagnosis of strengths and weaknesses of individual trainees. That enables schoolhouses and units to track performance on standardized metrics, which could be helpful for readiness assessments. Furthermore, Training and Education 2030 [Ref 3] outlines a student-centered adaptive training solution that tailors the training to the individual Marine based on an assessment of performance and prior research has demonstrated improved student learning outcomes and decreasing time through adaptive training methods – e.g. remediation or adaptive scenario difficulty [Refs 1, 4].

The desired capability of the fire support training solution is to provide tailored training to the individual FiST trainee based on the system's assessment of performance, targeting skill areas where the trainee is weakest. Providing targeted reps and sets will maximize training time in the crawl phase in addition to improving preparation for the team-based virtual (e.g., walk phase) and live exercises (e.g., run phase). Providing a capability that allows FiST members to practice individual skills ahead of time ensures that trainees can focus on team skill development such as communication and coordination skills with other entities during the time and resource constrained team-based events. Simulation solutions must

communicate via standard federated simulation protocols (e.g., DIS6/7, HLA RPR FOM [Refs 5,6]). Preference is given, but not required for submission, to proposals that incorporate or interoperate with existing and/or approved DOW simulation platforms with existing Authority To Operate (ATO) documentation for USMC.

PHASE I: Define and develop a concept for a scenario-based fire support simulation for individual FiST members that incorporates embedded assessment and adaptive training capabilities without the need for an instructor present. The focus of Phase I is developing and demonstrating an individual training solution for a FiST lead while also developing the broader concept for the remaining FiST individual roles. The concept shall include: (1) specific plans for how fundamental skills will be assessed at an individual FiST member level within the simulation and how the simulation will adapt based on the embedded performance assessment (the FiST lead focus / use case must be used to demonstrate a prototype to showcase the maturity and innovation for the proposed solution); (2) an evaluation plan to determine the validity of the assessments and training effectiveness.

PHASE II: Develop a prototype for a scenario-based fire support simulation that incorporates embedded assessment and adaptive training capabilities without the need for an instructor or role players present for the FiST lead role. Develop a plan to expand the assessment and adaptive training capabilities for additional roles in the FiST. Conduct validation of assessments with appropriate end users with coordination assistance from ONR.
Produce the following deliverables: (1) a working prototype of the system with performance data output that can be readable by instructors; (2) supporting software documentation; (3) an assessment validation report that demonstrates the suitability and utility of the training capability in the USMC training pipeline; (4) plans for expanding the assessment and training capabilities to new roles within the FiST.

PHASE III DUAL USE APPLICATIONS: Support the Marine Corps in transitioning the technology for Marine Corps use. Develop the software for evaluation to determine its effectiveness in either a formal Marine Corps schoolhouse or other training setting. As appropriate, focus on broadening capabilities of the training prototype to support additional roles, mission types, and fire support capabilities (e.g., loitering munitions, UAS, etc.).
Applications for dual-use include simulation-based individual and team assessments for training, such as law enforcement and air traffic controllers. The requirement to train and assess large numbers of aircraft controllers to meet personnel shortages has made global headlines. For air traffic control, having objective assessments and simulation-based training is paramount for safety considerations. Other relevant civilian applications that require significant simulation-based resources to train and assess individual and team skills in simulation-based environments include esports, which is a multi-billion-dollar industry with expected continued growth (https://www.statista.com/outlook/amo/esports/worldwide).

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KEYWORDS: Assessment; embedded assessment; adaptive training; fire support; simulation-based training; scenario-based training

DON26BZ01-NV033 TITLE: Applied Neuroanalytics for Optimization of Naval Training and Operational Readiness

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Human-Machine Interfaces

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and demonstrate a neuro-enhanced artificial intelligence (AI) system that captures, analyzes, and operationalizes neurophysiological and behavioral data to provide near real-time, adaptive feedback for improved training efficiency, performance, and operational readiness of U.S. Navy personnel.

DESCRIPTION: The U.S. Navy Force Design 2045 (CNO NavPlan 2024) highlights the importance of the warfighter and human-machine teaming in the future fight, emphasizing the criticality of developing high-performing teams and leaders that are resilient, adaptable, and warrior tough while supporting an increasingly hybrid Fleet of manned assets augmented with thousands of unmanned assets. The future fight will likely require operators to 1) digest and synthesize large amounts of data from an extensive network of humans and machines, 2) make decisions more rapidly due to advances in AI, enhanced connectivity, and autonomous weaponry and 3) oversee a greater number and types of robotics, including swarms (RAND, 2024).

Critical features of this paradigm shift towards manned-unmanned teaming and emphasis on improving warfighter performance are how we train operators. Training is at the forefront of the modernization of Naval operations to enhance readiness and lethality, and this will depend heavily on the cognitive resilience and decision-making capacity of warfighters in these novel, high-stress environments. Traditional training paradigms typically neglect real-time measurement and integration of cognitive and physiological performance states (e.g., mental effort, task engagement, lapses and slips of attention, complacency, mental fatigue, and stress). Emerging technologies for advanced data analytics grounded in neuroscience provide new capability that can enhance warfighter development and mission success by embedding neurofeedback into live and synthetic Naval training environments, providing novel analytical features and data to adapt training in near-real time and accelerate learning at the point of need.

The U.S. Navy seeks to identify a major step forward in neuro-enhanced AI systems to reduce time-to-proficiency and predict Sailor readiness within the unique maritime military environment. This envisioned capability will leverage and further develop Commercial Off-the-Shelf (COTS) neurotechnologies along with complimentary biosensors (e.g., electrocardiography [ECG], electromyography [EMG], eye tracking) and behavioral monitoring tools for Navy-specific use cases to interface with personnel, enabling adaptive and responsive system interaction based on near real-time human state data.

This SBIR topic will prioritize two key demonstrated factors in support of its objective: (1) the ability to collect neural, physiological, and behavioral data in parallel with operators using a desktop or higher fidelity simulator; and (2) the ability to analyze and interact with that data, both in near real-time and post-hoc, using an advanced language-understanding system coupled with an extensive foundational model of the human psychophysiology and/or behavior to provide feedback. This effort will complement existing Navy initiatives, such as those led by NAVAIR, NAWCAD, and NAWCTSD, enhancing existing learning environments through the addition of a brain-based performance layer.

The platform will deliver an autonomous solution for near real-time feedback, improved after-action reporting, and guided adaptation of training scenarios via open data standards that can be used to improve understanding of Sailor state (static and dynamic), which will be imperative for improving warfighter performance and training towards an ever-evolving mission in the future fight.

PHASE I: Design and validate a strategy for integrating the neuro-enhanced AI system with existing Navy training architectures (e.g., NAWCTSD training systems or LVC frameworks). Define and characterize mission-relevant cognitive states predictive of optimal warfighter performance. Develop a system architecture that fuses neurophysiological, behavioral, and mission/environmental data for predictive insight.
Deliver system architecture documentation, a feasibility analysis of neuro-enhanced AI system data integration with Naval training systems, a preliminary data model for cognitive and physiological performance state prediction, and a prototype development roadmap for Phase II.

PHASE II: Build and demonstrate a working prototype of the system integrated within a Navy-relevant training environment. Instrument a Naval operational team (e.g., aircrew, ship bridge) for real-time neurophysiological data collection and adaptive training response. Implement a neuro-enhanced advanced language understanding system for AI-driven coaching, guiding warfighter and instructors in near real-time.
Deliver an Institutional Review Board (IRB) application/approval; a cybersecurity and RMF compliance report; data strategy documentation and integration with existing Navy training platforms; a live data collection event demonstrating improvements in performance and mission readiness; updated data exchange framework using API or Navy-compliant standards.
DON will provide Phase II awardees with the appropriate guidance and assistance for human research protocols though the performer will be responsible for obtaining any required Institutional Review Board (IRB) determinations. IRB determination as well as processing, submission, and review of all paperwork required for human subject use can be a lengthy process. As such, no human research will be allowed without IRB review and work will not be authorized until approval has been obtained, typically as an Option to be exercised during Phase II.

PHASE III DUAL USE APPLICATIONS: Validate system effectiveness for improving warfighter performance and readiness; demonstrate adaptive capabilities with AI-based recommendations; achieve authority to operate (ATO) with Navy training platform(s). Support transition of the SBIR-developed neuro-enhanced AI system.
Validated capabilities will be relevant for:
• Naval Aviation: Enhancing pilot and flight officer readiness and cognitive workload management.
• Surface Fleet Training: Real-time feedback during complex shipboard simulations.
• Submarine Operations: Monitoring mental states during long-duration missions.
• Medical Teams Afloat: Enhancing high-stakes clinical team performance.
• Special Operations Forces: Resilience training and peak performance optimization
Commercial applications include aviation, e-sports, medical simulation, and elite training environments where human performance optimization is critical.

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KEYWORDS: Neuroanalytics; Human Performance; Predictive Analytics; Brain-Computer Interface (BCI); AI-Enabled Coaching; Real-Time Training Adaptation; Cognitive Load Monitoring; Aircrew Readiness; Adaptive Learning; Large Language Model; NAVAIR; NAWCTSD

DON26BZ01-NV034 TITLE: Effects of Additive Loading on Electromagnetic Properties in 3D Printing

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Hypersonics (SHY)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Assess the effects of additives into 3D-printed input materials that are structurally and thermally viable for weapon system components, to determine the changes to electromagnetic (EM) properties that can be achieved based on how the additives change the material properties of 3D printed materials, and changes required to the 3D-printing process to ensure sufficient additive concentration to achieve relevant EM property changes. The end goal of this research is to establish what EM behavior effects are possible with relevant material properties for weapon systems and what additive composition are needed to obtain them. An initial use case of an antenna radome for a weapon system navigation receiver will be explored.

DESCRIPTION: Many different 3D printing techniques are currently employed today and the use of this technology has progressed from niche, one-off manufacturing to producing large components, printing directly onto complex-shaped objects, and even mass manufacture. The majority of the printing that is performed, however, focuses on pure polymer materials. There is a need to develop technologies to attenuate electromagnetic (EM) radiation for relevant purposes specific to many military applications. Pure polymer materials traditionally used for 3D printing do not attenuate Radio Frequency (RF) and are often transparent to key frequencies. The incorporation of additives into the polymer input materials can change the EM properties of the bulk material as evidenced by initial research by the Naval Surface Warfare Center Dahlgren Division. The full benefit applied to more relevant applications needs to be addressed. The work in this SBIR topic is meant to determine what EM attenuation behaviors are possible with the incorporation of additives, for materials intended for use in relevant environments. This includes analyzing changes to the physical properties of the produced materials to determine how the thermal and mechanical properties as well as the printability of the materials are affected, to include changes needed to the printing process to create more relevant effects.

PHASE I: Produce additive incorporated 3D-material substrates and conducting characterization of the electromagnetic changes. (Note: The form of the materials will depend on the printing techniques employed, but could include filaments, powders, or resin materials, selected based on applicability to the expected operating environment for weapon system antenna radome.)

PHASE II: Print antenna radome representative samples with different additives and additive concentrations to assess the EM property control potential along with structural and thermal performance. Impacts to the printing process will also be assessed to determine if modifications to 3D printer

software/hardware are required to reach full benefit. These assessments will inform the selection of final material and additives for Phase III.

PHASE III DUAL USE APPLICATIONS: Print a full-scale antenna radome prototype, with additive selection and concentration, to meet specified performance parameters for frequency transmission and rejection. Antenna radome prototypes will be characterized for EM, structural, and thermal performance prior to testing an actual weapon system. Rapid printing of prototypes using validated material specifications and printing methodologies will also be conducted to demonstrate the feasibility of in-theater replacement part manufacture with modified EM response characteristics.
Given the ever increasing spectrum usage & crowding a dual use application would be antenna radome designs that provide a high rejection, tight bandpass to mitigate non-desired frequency interference. Additional dual use application would be tuning for thermal performance for commercial antenna applications in high solar load environment where the additives would be tailored to improve heat dissipation and reduce impact of ultra-violet radiation degradation to structural material properties of the radome.

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KEYWORDS: Additive; Manufacturing; Electromagnetic Properties; EM; 3D Print; Nanomaterials; Transparency; Reflection; Emission

DON26BZ01-NV035 TITLE: Integrated Multidisciplinary Design, Analysis, and Optimization Framework for Hypersonic Boost-Glide Weapons

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Hypersonics (SHY)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Hypersonics;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and demonstrate an integrated multidisciplinary design, analysis, and optimization (MDAO) framework for hypersonic boost-glide weapons that enables concurrent optimization of vehicle geometry, mission trajectory, and control strategy by leveraging existing modeling tools, incorporating reduced-order models, applying artificial intelligence and machine learning (AI/ML) to accelerate design and reduce computational cost, and providing early insights into system cost estimation, manufacturability, and technology development roadmaps.

DESCRIPTION: The Department of the Navy (DON) requires advanced simulation and optimization capabilities to accelerate the conceptual design and mission planning of hypersonic boost-glide weapons. These systems must deliver long-range strike capabilities, survive extreme thermal and structural environments, and maintain maneuverability for terminal effectiveness against defended targets. Designing such vehicles is highly complex due to the strong coupling between aerodynamic heating, structural loading, control authority, system mass, and mission trajectory.

Conventional design approaches treat these disciplines in isolation and in sequence, often resulting in suboptimal performance, prolonged development timelines, and increased costs. MDAO methods offer a more integrated approach, enabling concurrent consideration of key factors and improved trade space exploration. However, coupling high-fidelity models across multiple domains creates significant computational challenges. Practical MDAO frameworks must incorporate reduced-order models, surrogate approximations, and robust optimization techniques that balance computational efficiency and modeling accuracy [Refs 1, 3, 4].

This SBIR topic seeks innovative tools and methods that support an integrated MDAO framework for the design and optimization of hypersonic boost-glide weapons. Solutions should enable concurrent optimization of vehicle geometry, mission trajectory, and control strategies while accounting for launch platform constraints such as volume, mass, interface requirements, and environmental loads. The framework should also address internal system considerations such as payload integration, guidance and control subsystems, and thermal protection. The capability should support conceptual-level design and deliver outputs that inform system cost estimation, manufacturability, technology development roadmaps, and risk reduction strategies.

Proposals should demonstrate capabilities in the following areas:

• Aerodynamic and trimmed flight analysis to predict forces and moments over a broad range of Mach numbers, including control surface deflection effects and geometric deformation. Integration with existing computational fluid dynamics tools is encouraged.

• Aerothermal modeling to estimate heating loads and surface temperatures, including convective and radiative heat transfer and thermal protection system behavior.
• Structural analysis to evaluate stresses, strains, and deformation under combined aerodynamic and thermal loads, with support for high-temperature materials and composite structures.
• Mass properties and internal system layout to optimize placement of payloads, sensors, power systems, and thermal subsystems while maintaining center-of-gravity control and packaging feasibility.
• Trajectory and control optimization to evaluate and enhance flight performance while meeting constraints on range, maneuverability, survivability, and terminal accuracy.
• System-level integration into an existing or proposed MDAO architecture such as ADAPT [Ref 2] or OpenMDAO, with support for geometry parameterization, solver coupling, multi-objective optimization, and design variable management.
• Uncertainty quantification and robust optimization to evaluate sensitivity to variations in input parameters, models, or environmental conditions, and to ensure resilient design outcomes.
• AI/ML methods to accelerate convergence, construct reduced-order models, support adaptive sampling, and enable data-driven design exploration. Solutions may also include optimizing the objective function itself by learning weighting factors for multi-objective problems, generating surrogate models for expensive simulations, discovering improved formulations via symbolic regression, or adaptively refining the objective function as new information becomes available.
Proposed solutions should leverage existing tools, frameworks, and prior government investments wherever feasible. The resulting toolset should support traceability between design inputs and mission-level measures of effectiveness, helping guide early-phase trade studies and enabling faster transition to detailed design and development.

PHASE I: Develop a prototype MDAO framework for hypersonic boost-glide weapons. Integrate key modules and demonstrate coupling with existing architectures such as ADAPT or OpenMDAO. Apply the framework to optimize a representative boost-glide vehicle, capturing control surface deflection effects and geometric deformations over a notional trajectory. Evaluate computational efficiency, model fidelity, and extensibility. Prepare a Phase II plan.

PHASE II: Develop a fully integrated MDAO framework that enables co-design of vehicle geometry, trajectory, and control strategies for hypersonic boost-glide weapons. Incorporate launch platform constraints and model in-flight geometric deformations, control surface deflections, and effects such as ablation. Demonstrate manufacturability and cost-informed design on a non-canonical configuration. Leverage AI/ML to accelerate optimization, support surrogate modeling, and enable adaptive, data-driven design exploration. Validate the framework on realistic scenarios and implement workflow automation to support repeatable, efficient design cycles.

PHASE III DUAL USE APPLICATIONS: Transition the MDAO framework and supporting modules to practical applications within the Department of War and commercial aerospace sectors. Conduct extensive validation and optimization across a broad range of hypersonic vehicle configurations and flight conditions. Support integration into existing design and analysis workflows to enable use within operational design environments. Collaborate with industry and DOW stakeholders to ensure compliance with deployment standards. Develop comprehensive training materials, user documentation, and technical support resources to enable adoption by both expert and non-expert users.

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KEYWORDS: Hypersonics; Multidisciplinary Design, Analysis, and Optimization; MDAO; Computational Fluid Dynamics; Artificial Intelligence / Machine Learning; AI/ML; Software Tools; Aerodynamics

DON26BZ01-NV036 TITLE: Solid-Fuel Rotating Detonation Ramjet (SF-RDR) for High-Speed Propulsion

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Hypersonics (SHY)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Hypersonics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

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OBJECTIVE: Develop a throttleable solid-fuel Rotating Detonation Ramjet Engine (SFRDE) system by integrating a controllable gas generator to precisely regulate fuel supply, enabling stable and efficient Rotating Detonation Engine (RDE) operation.

DESCRIPTION: The Department of Navy (DON) seeks innovative solid-fuel detonation-based propulsion solutions that can deliver superior performance and operational flexibility. The RDE is a promising candidate to replace current constant-pressure combustion systems, due to its high-thermal efficiency, wide-operating Mach range, short combustion time, and small volume. However, to fully realize the benefits of an RDE for naval applications, particularly in the context of ramjet operation, the ability to operate an RDE on solid fuels and precisely control thrust output is crucial. This SBIR topic focuses on developing a throttleable solid-fuel rotating detonation ramjet (SFRDE) system, enabling dynamic adjustments to a coupled gas generator to enable optimal performance across a wide range of mission profiles.

To date, RDEs have been demonstrated to operate at ramjet relevant conditions; however, the applicability of RDEs to ramjet cycles has largely focused on the use of gaseous or liquid fuels [Refs 1, 2]. The use of solid fuels in RDEs presents additional complexities. Fuel formulations must be carefully tailored to provide detonable fuel at ramjet relevant temperatures. The use of a gas generator to provide the combustible mixture could potentially lead to solid particles clogging the fuel injectors. The design of the gas generator is also crucial to provide a mixture adequate for sustained detonability and coupling with the RDE inlet. Recent studies have demonstrated the viability integration of solid propellants and rotating detonation engines through the use of gas generators [Ref 3]. The proposed research should address the following two key areas to achieve a throttleable SFRDE:

Throttleable Gas Generator Development: Design and develop a compact, lightweight, and throttleable gas generator capable of precisely controlling the flow rate and composition of the fuel and/or oxidizer supplied to the RDE. Additional considerations should include the selection of appropriate gas generator propellants based on performance, stability, and safety considerations, as well as consideration of ignition methods suitable for the gas generator.

Combustion Chamber Design: Optimize the rotating detonation engine combustion chamber design for stable rotating detonation wave propagation and efficient mixing of the gas generator's output with the primary oxidizer stream. Design considerations should include injector geometry and placement to promote rapid mixing and flame stabilization; chamber geometry to facilitate detonation wave initiation and propagation; and thermal management strategies for both the gas generator and combustion chamber.

PHASE I: Design, develop, and demonstrate: (1) a throttleable gas generator subsystem and (2) that the gas generator provides a combustible mixture detonable at ramjet relevant temperatures. Solutions are preferred that are capable of demonstrating a subscale SFRDE system with the throttleable gas generator. Demonstrations should achieve sustained detonation operation for nominal durations of 0.5 to 3.0 seconds after reaching steady state. The subscale should trace directly to the conceptual design and planning for tactical-scale prototype to be developed under Phase II.

PHASE II: Develop and fabricate a tactical-scale prototype of a throttleable SFRDE. Performance and operation will be demonstrated under ramjet-relevant conditions during ground-test. The tactical prototype shall reflect the geometry and functionality of an operational system but is not required to be flight-weight. The ground-test campaign nominally will characterize performance across throttle settings, including thrust, specific impulse, combustion efficiency, and response time. If a limited ground test campaign is conducted, data collected should provide sufficient information to validate system-level performance models to assess the SFRDE's impact on overall ramjet cycle performance. Awardee(s) should provide a manufacturing readiness assessments and initial cost estimates of Low-Rate Initial Production (LRIP).

PHASE III DUAL USE APPLICATIONS: Further improve the SFRDE system and expand on the experimental demonstration for Navy-relevant conditions and vehicle geometries. These demonstrations shall be conducted via ground test facilities at simulated ramjet operating conditions. If supported by transition partners, Phase III may include development of flight-representative hardware or subcomponents and associated validation for integration into a future flight demonstration. Awardee(s) should improve manufacturing readiness assessments and cost estimates provided during Phase II. The commercial potential of this device lies in the component fabrication and potential secondary applications. The development of rotating detonation combustor hardware, and lessons learned in combustion diagnostic system integration will be directly applicable to future advanced solid-fuel propulsion system efforts. The sub-systems and technologies developed could be used across a broad range of power-generation, thermal management, and aerospace applications. The system has applicability to energy production research and development efforts ongoing in RDEs by industry and government agencies, including NASA and the Department of Energy.

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KEYWORDS: Solid Fuel; Rotating Detonation Engine; Gas Generator; Detonation; High-Speed Propulsion; Ramjet

DON26BZ01-NV037 TITLE: Synthetic Alkali Atom Vapor Density for Atom-Based Sensors

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber;Microelectronics;Quantum Science

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

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OBJECTIVE: Simplify the thermal management of practical atom-based quantum sensors based on alkali atoms by creating a passive atom source operated at thermal equilibrium based on a synthetic alkali vapor density for rubidium or cesium atoms.

DESCRIPTION: Quantum sensors based on atoms offer the opportunity to produce measurements with excellent sensitivity or long-term stability, making them attractive use in atomic clocks, magnetometers, or inertial sensors. In these sensors, the atomic vapor represents the sensing media where variations in signal magnitude from fluctuations in atom number can lead to instability or loss of sensitivity. Maintaining consistent signal throughout environmental conditions represents one of several key design criteria for atom-based sensors for use outside the laboratory.
Many atom-based sensors rely on heavy alkali atoms, specifically rubidium and cesium. This is because of the simplified, hydrogen-like energy level structure, the availability of narrow-linewidth semiconductor diode lasers on the relevant D1 (795/895 nm) and D2 (780/852 nm) transitions, the accessibility of commercial microwave electronics at the 3-10 GHz hyperfine splittings, and the ease of production of vapor phase atoms at modest temperatures. The temperature dependence of the alkalis [Ref 1] leads to thermal stabilization at 80-130°C (ideal for vapor cells at 10e12-10e14/cc) or closer to room temperature (ideal for atom trapping at 10e8-10e10/cc). These temperatures rarely align with thermal profiles of other aspects of the system, requiring additional design at the expense of size, weight, and power (SWaP). Active approaches to alkali regulation have been demonstrated to manipulate the vapor to a non-equilibrium state. These approaches involve forced chemical reactions, intercalated graphite, alkali impregnated materials glasses [Refs 2,3]. In each case, a feedback loop must respond to measurements of the vapor density, leading to extra sensor complexity.
An equilibrium vapor density represents the simplest atom source which can be synthetically adjusted to an elevated temperature through a mixture [Ref 4]. Here, a primary species mixed with a secondary species reduces the equilibrium vapor density of both species by the mixing ratio following Raoult's Law [Ref 5]. Selecting a lower vapor density secondary species limits the negative impact of additional atom-atom collisions. Such an approach can be applied to laser-cooled systems in addition to vapor cells to enable equilibrium operation at elevated system temperature, providing tight thermal regulation at low power.

PHASE I: Develop and demonstrate a method to produce a predetermined mixture of primary and secondary alkali species allowing for reduction of equilibrium vapor density of the primary species.

Mixtures consistent with supporting laser cooling at elevated temperatures from 30-85°C should be demonstrated corresponding to ~10-10,000× reductions in the primary species. Spectroscopic (or equivalent) determinations of the primary species density in the mixture should be evaluated against unmixed samples of the primary alkali species. A detailed synthesis approach for each mixture along with vapor density evaluations will be submitted.
In the Phase I Option, if exercised, stability of the mixtures against thermal cycling should be demonstrated for the mixtures produced in the Phase I Base.

PHASE II: Produce mixtures capable of supporting laser cooling and trapping at elevated temperatures over the 30-85°C range. Mixtures will be produced in or transferred into chambers that support optical access, magnetic fields, and ultra-high vacuum conditions compatible with atom trapping for evaluation. Spectroscopic (or equivalent) evaluation of the primary alkali reduction will be determined as in Phase I. In addition, atom trapping performance will be evaluated to determine number of atoms and loading time constant at a range of temperatures around the target temperature. The proposer may partner with other organizations if atom-trapping evaluation capability is not available in house. A plan to produce delivery mechanisms for mixtures in a variety of warm atom and cold atom scenarios will be submitted.
Deliver at a minimum three (3) samples (containing > 1 mg each) of the atom-trapping material in a proposed delivery mechanism to the Navy at the conclusion of Phase II.

PHASE III DUAL USE APPLICATIONS: Based on the demonstrations and continual advancement of laser cooling technologies, the atom source should lead to dramatic improvements in the SWaP of cold atom systems. Support the Navy in transitioning the technology to Navy use.
The end product technology could be leveraged to adapt atom-based sensors to a variety of thermal environments to support biomedical, communications, and navigation applications.

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KEYWORDS: Quantum sensing, magneto-optical trap, atom source, atomic clock, atom interferometer, atomic magnetometer

DON26BZ01-NV038 TITLE: High-power, Long Coherence Length Blue Laser

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber;Microelectronics;Quantum Science

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a blue wavelength, high-power laser with a long coherence length capable of high pulse repetition frequencies.

DESCRIPTION: In recent years, blue laser diode technology has enabled improved data storage, enhanced fluorescence imaging, metal processing, and other applications [Ref 1]. Lasers in this wavelength band also fall within the 'optical window' of water and will experience less attenuation than other wavelength bands [Ref 2]. The wavelength band will also experience less diffraction compared to other communication wavelengths [Ref 3]. This SBIR topic seeks to develop a blue laser capable of high pulse repetition rates and long coherence length light while maintaining a high optical power.
Target specifications for the desired product include:
• High optical power output: 10 W continuous wave
• Optical wavelength: 425 nm to 475 nm
• Long coherence length: > 10 m
• High pulse repetition frequency: > 100 MHz
• Laser will need to operate continuously and reliably for lifetime of 2000 days

PHASE I: Perform a design and materials study to assess the feasibility of the proposed technology or process to meet the target specifications listed above. A final report must include an assessment of:
• Preliminary design and simulation of laser technology
• The size, weight, and power (SWaP) implications of the proposed technology
• Pathway to meet the lifetime target specification including accelerated life testing
• The scalability of the approach for low quantity prototypes, low-rate production, and full rate production.
The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build laser prototype in Phase II.

PHASE II: Build and demonstrate the laser technology and characterize its performance against the target goals of optical power, wavelength, coherence length, pulse repetition frequency, beam quality, and continuously and reliable laser lifetime.
Deliver five (5) representative lasers to the Navy at the conclusion of Phase II that can be further tested.

PHASE III DUAL USE APPLICATIONS: Based on the prototypes developed in Phase II, continue development to assist the Government in integrating the technology with relevant technologies. Beyond Navy applications, the proposed laser technology will be relevant for a range of commercial and scientific applications including holography, spectroscopy, and medical sciences.

REFERENCES:

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KEYWORDS: Diode laser; blue laser; fiber amplifier; pulsed laser; high power laser; coherence length

DON26BZ01-NV039 TITLE: Reentry Test Body Telemetry Antenna

COMPONENT TECHNOLOGY PRIORITY AREA(S): Hypersonics;Nuclear;Space Technology

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design, develop, and test a reentry body antenna or antenna system capable of transmitting high speed, real time, inflight, encrypted data. The data transmission should be in bands alternate to S band such as the K & Ka bands and communicate with geostationary satellites used as a pass-through mechanism to relay the encrypted data to ground.

DESCRIPTION: The development of a next-generation telemetry communications antenna for Navy Submarine Launched Ballistic Missile (SLBM) reentering test bodies is critical in advancing developmental technologies being evaluated on flight tests. While common ground tests such as wind tunnels, arc jets, and vibration provide insights into predictable reentry environments, flight testing remains the gold standard in evaluating reentry bodies (RBs) and their onboard technologies. Flight tests evaluate a reentry body's ability to withstand the harsh and sometimes unpredictable environments of flight to include launch, separation, and reentry.

The current technology to monitor SLBM payloads during flight include a transmitter/receiver system between the reentry body and ground stations. Data is captured during flight and transmitted to the ground in the S band (2-4 GHz), making data transfer slower than higher frequency bands [Ref 3]. Due to the S band being a highly populated frequency band and the power on the RB required to telemeter data in the S band back down to the ground receiver, midflight data transmission is both slow and costly. Additionally, since the transmitter/receiver system today is only between the RB and ground station, real time data transmission is lost during a portion of the flight when the RB is the furthest away from the ground, otherwise commonly known as "over the top" of the flight trajectory as well as during reentry when the body enters plasma blackout. To solve this problem, the technology proposed should use alternate frequency bands, such as K and Ka bands (18-40 GHz) and make use of geostationary satellites as a pass-through mechanism to capture real time data from the RB and telemeter the encrypted data back down to the ground at high speeds in order to minimize data transmission latency and loss. The use of alternate frequency bands allows for high data rate information exchange [Ref 1]. This new technology would solve the issue of losing real-time data transmission midflight.

By having real-time, high-speed data throughout the duration of flight on a flight test, the Navy can better understand technology performance throughout the various environments and environment transitions and can more effectively diagnose issues or failures resulting in faster technology maturation.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain at least a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order

to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Prepare a detailed plan to accomplish the objective to include 1) A clear and concise definition of the problem; 2) Definition of the System Requirements; 3) Proposed technology solution 4) Draft technology solution specification; 5) Identification of trade-off studies to be studied and how the trade studies will influence product design; 6) Optimized structural and thermal designs; 7) detailed plan to achieve prototype development and testing; 8) detailed plan to achieve system integration and qualification. Attempt to meet all requirements while achieving optimal encrypted data transmission performance in reliability, form factor minimization, and connectivity while meeting all integration requirements. AES256 is a standard.
In summary, the Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Implement the program plan described in the Phase I deliverable. Three prototypes should be delivered. The first prototype should be able to be incorporated onto a test flight such as a sounding rocket or full scale test on a Multi-Service Advanced Capability Hypersonics Test Bed (MACH-TB) platform to evaluate its performance [Ref 4]. The second prototype will be used for destructive ground test activities required for qualification and the third prototype will be used for non-destructive ground test and qualification activities.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: The final product will be an antenna or antenna system that is matured to its final form factor, qualified to Trident II D5LE environments. The product shall meet all size, weight, and power constraints for implementation onto a reentry test body.
A possible non-DoW use for this effort can be improved communications for commercial space payloads in multiple bands.

REFERENCES:

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KEYWORDS: Telemetry; flight test; reentry body; encrypted data transmission; high frequency data transmission; geostationary satellite; high speed data

DEPARTMENT OF THE NAVY (DON)
Small Business Innovation Research (SBIR)
DoW 2026 SBIR BAA Release 1
Direct to Phase II (DP2) Proposal Submission Instructions

IMPORTANT

- **The following instructions apply to Direct to Phase II (DP2) SBIR topic only:**
  - **DON26BZ01-DV001 through DON26BZ01-DV005**

- Submitting small business concerns (SBCs) are encouraged to thoroughly review the DoW SBIR/STTR Program Broad Agency Announcement (BAA) and register for the DSIP Listserv to remain apprised of important programmatic changes.
  - The DoW Program BAA is located at: https://www.defensesbirsttr.mil/SBIR-STTR/Opportunities/#announcements. Select the tab for the appropriate BAA cycle.
  - Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

- The information provided in the DON Proposal Submission Instructions takes precedence over the DoW Instructions posted for this BAA.

- A submitting SBC **MUST** use the DP2 Phase I Feasibility proposal template for Volume 2. This template is specific to DON DP2 topics and meets DP2 submission requirements. The DP2 Phase I Feasibility proposal template can be found at https://navysbir.com/links_forms.htm.

- Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF) or any combination of these are eligible to submit proposals in response to DON topics advertised in this BAA. Information on Majority Ownership in Part and certification requirements at time of submission for these proposing SBCs are detailed in the section titled ADDITIONAL SUBMISSION CONSIDERATIONS.

- The DON may consider the following FAR and Non-FAR contracting strategies when issuing Phase II awards: Cost Plus Fixed Fee (CPFF), Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT).

- This BAA is issued under regulations set forth in Federal Acquisition Regulation (FAR) 35.016 and awards will be made under "other competitive procedures". The policies and procedures of FAR Subpart 15.3 shall not apply to this BAA, except as specifically referenced in it. All procedures are at the sole discretion of the Government as set forth in this BAA. Submission of a proposal in response to this BAA constitutes the express acknowledgement to that effect by the proposing SBC.

INTRODUCTION

The DON SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DON's Fleet through research and development (R&D) topics that have dual-use potential, but primarily address the needs of the DON. More information on the programs can be found on the DON

SBIR/STTR website at www.navysbir.com. Additional information on DON's mission can be found on the DON website at www.navy.mil.

The Department of War (DoW), including the Department of the Navy (DON), may issue an SBIR award to an SBC under Phase II , without regard to whether the SBC received a Phase I award for such project. Prior to such an award, the head of the agency, or their designee, must issue a written determination that the SBC has demonstrated the scientific and technical merit and feasibility of the technology solution that appears to have commercial potential (for use by the government or in the public sector). The determination must be submitted to the Small Business Administration (SBA) prior to issuing the Phase II award. As such, DON issues this portion of the BAA in accordance with the requirements of the Direct to Phase II (DP2) authority. Only those proposing SBCs that are capable of meeting the DP2 proposal requirements may participate in this DP2 BAA. No Phase I awards will be issued to the designated DP2 topic.

For questions regarding this BAA, use the information in Table 1 to determine who to contact for what types of questions.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS BAA

Type of Question	When	Contact Information
Program and administrative	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	BAA Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoW SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoW SBIR/STTR Program BAA for details.
	BAA Open	DoW SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoW SBIR/STTR Program BAA for details.
Electronic submission to the DoW SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific BAA instructions and forms	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DON SYSTEMS COMMAND (SYSCOM) SBIR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
DON26BZ01-DV001 to DON26BZ01-DV004	Ms. Kristi DePriest	Naval Air Systems Command (NAVAIR)	navair-sbir@us.navy.mil

DON26BZ01-DV005	Mr. Jason Schroepfer	Naval Sea Systems Command (NAVSEA)	NSSC_SBIR.fct@navy.mil

Each DON SBIR DP2 topic requires documentation to determine that Phase I feasibility, described in the Phase I section of the topic, has been met.

The DON SBIR DP2 is a two-step process:

> STEP ONE: Prepare and Submit a Phase I Feasibility Proposal (instructions and link to template provided below). The purpose of the Phase I Feasibility Proposal is for the proposing SBC to provide documentation to substantiate that both Phase I feasibility and the scientific and technical merit described in the topic have been met. The Phase I Feasibility Proposal must: demonstrate that the proposing SBC performed Phase I-type research and development (R&D) and provide a concise summary of Phase II objectives, work plan, related research, key personnel, transition/commercialization plan, and estimated costs. Feasibility documentation MUST NOT be solely based on work performed under prior or ongoing federally funded SBIR/STTR work. The government will evaluate Phase I Feasibility Proposals and select SBCs to submit a Full DP2 Proposal. Demonstrating proof of feasibility is a requirement for a DP2 award. The SBC must submit a Phase I Feasibility Proposal to be considered for selection to submit a Full DP2 Proposal.
>
> STEP TWO: If selected, the cognizant SYSCOM Program Office will contact the SBC directly to provide instructions on how to submit a Full DP2 Proposal.

DON SBIR reserves the right to make no awards under this DP2 BAA. All awards are subject to availability of funds and successful negotiations. Proposing SBCs must read the topic requirements carefully. The Government is not responsible for expenditures by the proposing SBC prior to award of a contract. For 2026.BZ Release 1 topics designated as DP2, DON will accept only Phase I Feasibility Proposals (described below).

DP2 PROPOSAL SUBMISSION REQUIREMENTS

The following section details requirements for submitting a compliant DON SBIR DP2 Proposal to the DoW SBIR/STTR Programs.

(NOTE: Proposing SBCs are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.)

DoW SBIR/STTR Innovation Portal (DSIP). Proposing SBCs are required to submit proposals via the DoW SBIR/STTR Innovation Portal (DSIP); and follow proposal submission instructions in the DoW SBIR/STTR Program BAA on the DSIP at https://www.dodsbirsttr.mil/submissions. Proposals submitted by any other means will be disregarded. Proposing SBCs submitting through DSIP for the first time will be asked to register. It is recommended that proposing SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposals that are not successfully certified electronically in DSIP by the Corporate Official prior to BAA Close will NOT be considered submitted and will not be evaluated by DON. Proposals that are encrypted, password protected, or otherwise locked in any portion of the submission will be REJECTED unless specifically directed within the text of the topic to which you are submitting. Please refer to the DoW SBIR/STTR Program BAA for further information.

Eligibility. Each proposing SBC must:

- Have demonstrated feasibility of Phase I-type R&D work
- Have submitted a Phase I Feasibility Proposal for evaluation
- Meet Offeror Eligibility and Performance Requirements as defined in the Proposal Fundamentals section of the DoW SBIR/STTR Program BAA
- Comply with primary employment requirements of the principal investigator (PI) during the Phase II award including, employment with the SBC at the time of award and during the conduct of the proposed project. Primary employment means that more than one-half of the PI's time is spent in the employ of the SBC
- Register in the System for Award Management (SAM) as defined in the Certifications and Registrations section of the DoW SBIR/STTR Program BAA. To register, visit https://sam.gov/

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoW SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).**
  - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
    - A submitting SBC MUST use the DP2 Phase I Feasibility proposal template for Volume 2. The DP2 Phase I Feasibility proposal template can be found at https://navysbir.com/links_forms.htm.

      This template is specific to DON DP2 topics and meets DP2 submission requirements:
      - Not to exceed 30 pages, regardless of page content; Phase I Proof of Feasibility portion not to exceed 20 pages, Snapshot of Proposed Phase II Effort portion not to exceed 10 pages
      - Single column format, single-spaced typed lines
      - Standard 8 ½" x 11" paper
      - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
      - No font size smaller than 10-point

  - Additional information:
    - A font size smaller than 10-point is allowable for headers, footers, imbedded tables, figures, images, or graphics that include text. However, proposing SBCs are cautioned that if the text is too small to be legible it will not be evaluated.

- **Cost Volume (Volume 3).** The text fields related to costs for the proposed effort must be answered in the Cost Volume of the DoW Submission system (at https://www.dodsbirsttr.mil/submissions/), however, proposing SBCs DO NOT need to download and complete the separate cost volume template when submitting the DON SBIR Phase I Feasibility Proposal. Proposing SBCs are to include a cost estimate in the Order of Magnitude Cost Estimate Table (example below) within the Snapshot of Proposed Phase II Effort portion of the Technical Volume (Volume 2). Please refer to Table 3 below for guidance on cost and period of performance. Costs for the Base and Option are to be separate and identified on the Proposal Cover Sheet and in the Order of Magnitude Cost Estimate Table in the Technical Volume (Volume 2).

Order of Magnitude Cost Estimate Table			
Line Item – Details	**Estimated Base Amount**	**Estimated Option Amount**	**Total Estimated Amount Base + Option**

Direct Labor (fully burdened) – Prime			
Subcontractors/Consultants			
Material			
Travel & ODC			
G&A			
FCCM			
Fee/Profit			
TABA (NTE $25K, included in total amount)			
Total Estimated Costs			

TABLE 3: COST & PERIOD OF PERFORMANCE

Topic Number	Base		Option		Total (NTE)
	Cost (NTE)	POP (NTE)	Cost (NTE)	POP (NTE)	
DON26BZ01-DV001 to DON26BZ01-DV004	$1,000,000	30 mos.	$400,000	12 mos.	$1,400,000
DON26BZ01-DV005	$700,000	12 mos.	$1,300,000*	24 mos.*	$2,000,000*

* Step Two: for the Full Phase II submission, if selected, DON26BZ01-DV005 will require the Phase II Option 1 and Phase II Option 2 to be detailed separately:

- Phase II Option 1: Cost $700,000, Period of Performance 12 months
- Phase II Option 2: Cost $600,000, Period of Performance 12 months

- Additional information:
  - For Phase II a minimum of 50% of the work is performed by the proposing SBC. The percentage of work requirement must be met in the Base costs as well as in the Option costs. The percentage of work is measured by both direct and indirect costs. To calculate the minimum percentage of work for the proposing SBC the sum of all direct and indirect costs attributable to the proposing SBC represent the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) is the denominator. The subcontractor percentage is calculated by taking the sum of all costs attributable to the subcontractor as the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) as the denominator. **NOTE:** G&A, if proposed, will only be attributed to the proposing SBC.
  - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
  - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
  - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3.

- **Company Commercialization Report (Volume 4)**. DoW collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and

Requirements section of the DoW SBIR/STTR Program BAA for details to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DON may or will require to process a proposal, if selected, for contract award.

  All proposing SBCs must review and submit the following items, as applicable:
  - — **Majority Ownership in Part.** Proposing SBCs which are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, are eligible to submit proposals in response to DON topics advertised within this BAA. Complete the certification as detailed under ADDITIONAL SUBMISSION CONSIDERATIONS.

  - Additional information:
    - — Proposing SBCs may include the following administrative materials in Supporting Documents (Volume 5); a template is available at https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing SBC may want to include in Volume 5:
      - Additional Cost Information to support the Cost Volume (Volume 3)
      - SBIR/STTR Funding Agreement Certification
      - Data Rights Assertion
      - Disclosure of Information (DFARS 252.204-7000)
      - Prior, Current, or Pending Support of Similar Proposals or Awards
      - Foreign Citizens
    - — Details of Request for Discretionary Technical and Business Assistance (TABA), if proposed, is to be included under the Additional Cost Information section if using the DON Supporting Documents template.
    - — Do not include documents or information to substantiate the Technical Volume (Volume 2) (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.
    - — A font size smaller than 10-point is allowable for documents in Volume 5; however, proposing SBCs are cautioned that the text may be unreadable.

- **Fraud, Waste and Abuse Training Certification (Volume 6)**. DoW requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this BAA to assess security risks presented by SBCs seeking a Federally funded award. SBCs must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details.

DP2 EVALUATION AND SELECTION

The following section details how the DON SBIR/STTR Programs will evaluate Phase I Feasibility proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DON SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoW and DON SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing SBC has met eligibility requirements and followed the instructions for Proposal Cover Sheet as specified in the DoW SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).** The DON will evaluate and select Phase I Feasibility proposals using the evaluation criteria specified in the Method of Selectin and Evaluation Criteria section of the DoW SBIR/STTR Program BAA, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation and proposals will not be compared to one another. Cost is not an evaluation criterion and will not be considered during the evaluation process; the DON will only do a compliance review of Volume 3. Due to limited funding, the DON reserves the right to limit the number of awards under any topic.

  The Technical Volume (Volume 2) will undergo a compliance review (prior to evaluation) to verify the proposing SBC has met the following requirements or the proposal will be REJECTED:
  - A submitting SBC MUST use the DP2 Phase I Feasibility proposal template for Volume 2. The DP2 Phase I Feasibility proposal template can be found at https://navysbir.com/links_forms.htm.
    This template is specific to DON DP2 topics and meets DP2 submission requirements:
    - Not to exceed 30 pages, regardless of page content; Phase I Proof of Feasibility portion not to exceed 20 pages, Snapshot of Proposed Phase II Effort portion not to exceed 10 pages
    - Single column format, single-spaced typed lines
    - Standard 8 ½" x 11" paper
    - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
    - No font size smaller than 10-point, except as permitted in the instructions above.

- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will undergo a compliance review to verify the proposing SBC has met the following requirements or the proposal will be REJECTED:
  - Must not exceed values for the Base and Option (refer to Table 3).
  - Must meet minimum percentage of work; a minimum of 50% of the work is performed by the proposing SBC. The percentage of work requirement must be met in the Base costs as well as in the Option costs.

- **Company Commercialization Report (Volume 4).** The CCR (Volume 4) will not be evaluated by the DON nor will it be considered in the award decision. However, all proposing SBCs must refer to the DoW SBIR/STTR Program BAA to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing SBC has included items in accordance with the DP2 SUBMISSION INSTRUCTIONS section above.

- **Fraud, Waste, and Abuse Training Certificate (Volume 6).** Not evaluated.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Due Diligence Program to Assess Security Risks. Refer to the DoW SBIR/STTR Program BAA to ensure compliance with Volume 7 requirements.

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing SBCs to consider during proposal preparation and submission process.

Due Diligence Program to Assess Security Risks. The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of War, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by SBCs seeking a Federally funded award. Please review the Certifications and Registrations section of the DoW SBIR/STTR Program BAA for details on how DoW will assess security risks presented by SBCs. The Due Diligence Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The SBIR and STTR Policy Directive section 9(b) allows the DON to provide TABA (formerly referred to as DTA) to its awardees. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; and commercializing the SBIR/STTR product or process, including intellectual property protections. Proposing SBCs may request, in their Cost Volume (Volume 3), to contract these services themselves through one or more TABA providers in an amount not to exceed the values specified below. The Phase II TABA amount is up to $25,000 per award, is to be included as part of the award amount and is limited by the established award values for Phase II by the SYSCOM (i.e., within the $2,000,000 or lower limit specified by the SYSCOM). The amount proposed for TABA cannot include any profit/fee by the proposing SBC and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the SBIR proposing SBC. A Phase II project may receive up to an additional $25,000 for TABA as part of one additional (sequential) Phase II award under the project for a total TABA award of up to $50,000 per project. A TABA Report, detailing the results and benefits of the service received, will be required annually by October 30.

Request for TABA funding will be reviewed by the DON SBIR/STTR Program Management Office.

If the TABA request does not include the following items the TABA request will be denied.

- TABA provider(s) (firm name)
- TABA provider(s) point of contact, email address, and phone number
- An explanation of why the TABA provider(s) is uniquely qualified to provide the service
- Tasks the TABA provider(s) will perform (to include the purpose and objective of the assistance)
- Total TABA provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA must <u>NOT</u>:

- be subject to any indirect costs, profit, or fee by the SBIR proposing SBC
- propose a TABA provider that is the SBIR proposing SBC
- propose a TABA provider that is an affiliate of the SBIR proposing SBC
- propose a TABA provider that is an investor of the SBIR proposing SBC
- propose a TABA provider that is a subcontractor or consultant of the requesting SBC otherwise required as part of the paid portion of the research effort (e.g., research partner, consultant, tester, or administrative service provider)

TABA requests must be included in the proposal as follows:

- Phase II:
  - — DON Phase II Cost Volume (provided by the DON SYSCOM) - the value of the TABA request.
  - — Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.

Proposed values for TABA must NOT exceed:

- Phase II: A total of $25,000 per award, not to exceed $50,000 per Phase II project

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing SBC shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design, production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). An SBC whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DON Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DON Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Majority Ownership in Part. Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, **are eligible** to submit proposals in response to DON topics advertised within this BAA.

For proposing SBCs that are a member of this ownership class the following must be satisfied for proposals to be accepted and evaluated:

a. Prior to submitting a proposal, proposing SBCs must register with the SBA Company Registry Database.
b. The proposing SBC within its submission must submit the Majority-Owned VCOC, HF, and PEF Certification. A copy of the SBIR VC Certification can be found on https://navysbir.com/links_forms.htm. Include the SBIR VC Certification in the Supporting Documents (Volume 5).
c. Should a proposing SBC become a member of this ownership class after submitting its proposal and prior to any receipt of a funding agreement, the proposing SBC must immediately notify the Contracting Officer, register in the appropriate SBA database, and submit the required certification, which can be found on https://navysbir.com/links_forms.htm.

System for Award Management (SAM). It is strongly encouraged that proposing SBCs register in SAM, https:// sam.gov, by the Close date of this BAA, or verify their registrations are still active and will not expire within 60 days of BAA Close. Additionally, proposing SBCs should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. An SBC

selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Cybersecurity Maturity Model Certification (CMMC) Program. DoW has established the CMMC Program to verify that awardees have implemented required security measures necessary to safeguard Federal Contract Information (FCI) and Controlled Unclassified Information (CUI). CMMC Level requirements are identified within each topic. Proposing SBCs should carefully review and consider the CMMC requirements as compliance may impact proposed costs and technical approach. Please review the DoW SBIR/STTR Program BAA for additional information on the CMMC Program.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.240-7997, in order to be considered for award, an SBC is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit https://www.sprs.csd.disa.mil/nistsp.htm. For in-depth tutorials on these items please visit https://www.sprs.csd.disa.mil/webtrain.htm.

Human Subjects, Animal Testing, and Recombinant DNA. If the use of human, animal, and recombinant DNA is included under a DP2 proposal, please carefully review the requirements at: https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I BAA Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Debriefs. Requests for a debrief must be made within 15 calendar days of select/non-select notification via email as specified in the select/non-select notification. Please note debriefs are typically provided in writing via email to the Corporate Official identified in the proposal of the proposing SBCs within 60 days of receipt of the request. Requests for oral debriefs may not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the debrief request.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Pre-award agency protests related to the terms of the BAA must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DON Topics may be obtained from the DON SYSCOM Program Managers listed in Table 2 above. For protests filed with the GAO, a copy of the protest must be submitted to the appropriate DON SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DON reserves the right to limit the number of awards under any topic. Any notification received from the DON that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct cost analysis, confirm eligibility of the proposing SBC, and to take other relevant steps necessary prior to making an award.

Contract Types. The DON will consider the following for award: Cost Plus Fixed Fee (CPFF), Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT).

Contract Deliverables. Contract deliverables are typically progress reports and final reports. Required contract deliverables must be uploaded to https://www.navysbirprogram.com/navydeliverables/.

Transfer Between SBIR and STTR Programs. Section 4(b)(1)(i) of the SBIR and STTR Policy Directive provides that, at the agency's discretion, projects awarded a Phase I under a BAA for SBIR may transition in Phase II to STTR and vice versa.

Navy SBIR Transition Program (STP). All Phase II awardees have the opportunity to participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: https://navystp.com. Phase II awardees will be contacted separately regarding this program.

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm that developed the technology as a result of a Phase I or Phase II award. The DON will give Phase III status to any award that falls within the above-mentioned description. Consequently, DON will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DON protect the rights of the SBIR/STTR firm.

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Direct to Phase II
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DON26BZ01-DV001 TITLE: DIRECT TO PHASE II: Extended Range 10-inch Air Launched Rocket

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Implement Highly Loaded Grain (HLG) propulsion technology into an existing 10-inch diameter rocket motor to create a tactically relevant, extended range rocket motor.

DESCRIPTION: The U.S. Navy is pursuing enhancements to the performance, range, and tactical flexibility of existing 10-inch rocket motor systems. A key enabler of this objective is the maturation and application of HLG propulsion technology. HLG designs maximize total impulse within volume-constrained tactical solid propellant systems while enabling adaptable thrust-time profiles, including boost-sustain variants.
This Direct to Phase II SBIR topic seeks integration of HLG technology into an existing 10-inch diameter rocket motor, thereby increasing performance and advancing the Technology Readiness Level (TRL) and Manufacturing Readiness Level (MRL) of the HLG propulsion approach.
Key Technical Guidelines:

1. Rocket Motor Case: 10-inch diameter tactical casing with boat-tail geometry based on the High-speed Anti-Radiation Missile (HARM) aft-end structure
2. Grain Design: HLG-formulated geometry tailored for constrained volume and thrust shaping
3. Ballistics Software: CLWire ballistic simulation software provided by Naval Air Warfare Center Weapons Division (NAWCWD)
4. Risk Posture: Low to moderate for non-HLG-specific subsystems; medium risk for nozzle/igniter design
5. Performance Objective: Total impulse increase of approximately 30% over legacy baseline
6. Thrust Profile: Support both all-boost and boost/sustain regimes; comply with NAWCWD performance parameters including Maximum Expected Operating Pressure (MEOP) and thrust onset rates
7. Propellant Formulation: Aluminized solid propellant: Ammonium Perchlorate (AP) / Aluminum (Al) / Hydroxyl-Terminated Polybutadiene (HTPB) binder
8. Materials Compatibility: Maximize re-use of existing materials for insulation, liners, oxidizers, and binders
9. Environmental Qualification: Thermal: –65 °F to +160 °F (–53.9 °C to +71.1 °C); Structural: withstand shock and vibration in accordance with military deployment profiles
10. Nozzle & Igniter Development: Moderate risk with identified maturation path toward tactically viable configurations

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able

to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: For a Direct to Phase II topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort and developed a concept for a workable prototype or design to address, at a minimum, the basic requirements of the stated objective above. The below actions would be required to satisfy the requirements of Phase I:

1. Ballistic Design: Evidence of preliminary or detailed grain geometry development, performance modeling (e.g., with CLWire or similar), and total impulse optimization.
2. Motor Fabrication: Documentation of hardware build efforts, including grain casting, case integration, and materials characterization relevant to the HLG configuration.
3. Static Test Results: Data from one or more static firings that validate thrust-time profiles, ignition performance, MEOP survivability, and total impulse enhancement attributed to the HLG propulsion technology.
4. Importantly, feasibility documentation must not rely solely on work conducted under prior or ongoing federally funded SBIR/STTR awards. Applicants are required to demonstrate that the proposed concept has been advanced through non-SBIR/STTR-funded efforts, indicating technical maturity sufficient for immediate Phase II execution.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit and Phase I feasibility described in Phase I above has been met (i.e., the small business must have performed Phase I-type research and development related to the topic NOT solely based on work performed under prior or ongoing federally funded SBIR/STTR work) and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above.

PHASE II: Focus on developing, documenting, fabricating, and validating a tactical solid rocket motor that integrates HLG propulsion technology, in accordance with Government technical guidelines and performance objectives.

1. Initial Concept Design and Detailed Design Review (DDR)
   - Develop an initial system design incorporating: (1) Ballistic modeling using CLWire (Government-furnished); (2) Thermal and structural insulation design; (3) Tactically relevant nozzle architecture; and (4) Igniter configuration suited for aluminized propellant initiation.
   - Document the full design concept for review in a DDR to be assessed against Government-agreed technical requirements and performance metrics.
   - Government acceptance of DDR exit criteria is required prior to initiating fabrication.
2. Fabrication and Assembly (following DDR approval)
   - Finalize design details, generate component drawings, and fabricate tooling for both component and propellant casting.
   - Perform propellant mixing and casting per specified aluminized solid formulation (AP/Al/HTPB), ensuring compatibility with insulation and liner materials.
   - Utilize a Government-supplied flight weight motor case and HLG-specific materials, as requested and made available by the program office.

3. As-Manufactured Validation and Testing

- Conduct an Item Under Test (IUT) review to compare the as-built motor configuration with the as-designed concept.
- Present findings to the Government for validation and alignment with performance expectations.
- Upon concurrence, proceed to static fire testing of the motor in a controlled test environment.

Work in Phase II may become classified. Please see note in the Description.

PHASE III DUAL USE APPLICATIONS: Mature the Phase II rocket motor concept for higher fidelity static fire demonstrations. The developed rocket motors will incorporate flight representative subcomponents (e.g., nozzle, insulation, ignition system, etc.) while still optimizing the propulsion design to maximize system range. Demonstrate multiple static firings to assess the environmental robustness of the rocket motors and performance relative to the technical guidelines provided by the Government. A final report will be provided that documents the design and testing results, provides a Technology Readiness Level (TRL) assessment, and outlines a path to further mature the technology.
The developed propulsion technology with have application to space launch and space-based systems.

REFERENCES:

1. "Naval Air Systems Command/Naval Air Warfare Center Weapons Division (NAWCWD): 2023 Highlights." https://www.navair.navy.mil/nawcwd/sites/g/files/jejdrs601/files/document/%5Bfilename%5D/NAWCWD%20Fact%20Sheet%202023.pdf
2. PMA-242 Mission Statement and Description for AARGM. NAVAIR. https://www.navair.navy.mil/product/AARGM
3. "National Industrial Security Program Executive Agent and Operating Manual (NISP), 32 U.S.C. § 2004.20 et seq. 1993." https://www.ecfr.gov/current/title-32/subtitle-B/chapter-XX/part-2004

KEYWORDS: Solid Propulsion; Energy Management; Rocket Motor; Extended Range; Propellant; Hypersonics

DON26BZ01-DV002 TITLE: DIRECT TO PHASE II: Domestic Production of Zirconium and Hafnium Metal Organic Precursors

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Hypersonics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and demonstrate a pilot-scale manufacturing process for producing high purity tetrakis(dimethylamido)zirconium(IV) (TDMAZ), tetrakis(dimethylamido)hafnium(IV) (TDMAH) and related metal dimethylamide compounds, with a targeted annual production capacity exceeding 6,000 kg of TDMAZ.

DESCRIPTION: The Department of the Navy is seeking a domestic source of critical chemical feedstocks including TDMAZ, TDMAH, and other metal dimethylamide compounds. These chemical feedstocks can be used as metal organic precursors for atomic layer deposition (ALD), chemical vapor deposition (CVD), and chemical vapor infiltration (CVI) of metal oxides, nitrides, and carbonitrides used in microelectronics and ceramic manufacturing [Refs 1-3]. While TDMAZ is a vital ceramic precursor for the electronics and semiconducting industry, this effort will also support the use of TDMAZ for the preparation of metal nitrides and carbonitrides for ceramics and ceramic matrix composites.
This SBIR topic seeks to establish a domestic manufacturing capability for the production of > 6,000 kg/year of TDMAZ. Synthesis of TDMAZ and other metal dimethylamides often involves pyrophoric and air/water sensitive reagents, and the proper storage and handling of these reagents is crucial for the development of a cost-effective and large-scale manufacturing process. Along with the production volumes mentioned above, the metal precursors must have a purity > 99% and a target retail price of < $4,000/kg of TDMAZ, preferably < $2,500/kg. The proposed manufacturing facility must be located in the United States or US territories, and the company owning and operating this manufacturing facility must be wholly US owned and based.

PHASE I: For a Direct to Phase II topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort and developed a concept for a workable prototype or design to address, at a minimum, the basic requirements of the stated objective above. The below actions would be required to satisfy the requirements of Phase I:
Proposing SBCs will have demonstrated expertise in the synthesis of air and water-sensitive compounds and access to facilities that will enable them to produce zirconium and hafnium organic precursors at the kilogram scale.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit and Phase I feasibility described in Phase I above has been met (i.e., the small business must have performed Phase I-type research and development related to the topic NOT solely based on work performed under prior or ongoing federally funded SBIR/STTR work) and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above.

PHASE II: Identify a chemical process to manufacture TDMAZ and perform a safety analysis on the handling, storage, and waste management of air and water sensitive reagents and intermediates used in production scale synthesis. Facility upgrades to allow for safety compliance during manufacture of metal dimethylamides is allowed. The ability to expand and manufacture other metal dimethylamide compounds, such as TDMAH is encouraged.

Demonstrate a scalable approach for the synthesis of TDMAZ and provide 1 kg of material for analysis and internal verification by Navy researchers. Demonstrate pilot scale production of at least 50 kg TDMAZ/month. It is expected that the optimized approach will be consistent with a target production of > 6,000 kg TDMAZ/year. Conduct a technoeconomic analysis based on production at the > 6,000 kg/year scale. Identification of manufacturing or production issues and business model modifications required to further improve the process (e.g., reduced cost, increased availability, safety) will be documented. At the end of the Phase II Base period, 100 kg of TDMAZ produced using the new pilot scale manufacturing process will be delivered to the Government.

PHASE III DUAL USE APPLICATIONS: Establish a facility to produce > 6,000 kg of TDMAZ/year. Conduct several production runs to demonstrate the ability to supply zirconium and hafnium carbide precursors at < $2,500/kg to the DoW through the Defense Logistics Agency.
Zirconium and hafnium carbide are used in commercial aerospace structures, microelectronics, cutting tool bits, and coatings in nuclear reactors.

REFERENCES:

1. Hausmann, D.M. and Gordon, R.G. "Surface Morphology and Crystallinity Control in the Atomic Layer Deposition (ALD) of Hafnium and Zirconium Oxide Thin Films." J. Cryst. Growth, 241, pp. 251-261, 2003. https://www.sciencedirect.com/science/article/abs/pii/S0022024802021334
2. Fix, R.; Gordon, R.G. and Hoffman, D.M. "Chemical Vapor Deposition of Titanium, Zirconium, and Hafnium Nitride Thin Films." Chem. Mater., 3, pp. 1138-1148, 1991. https://pubs.acs.org/doi/10.1021/cm00018a034
3. Lamm, B.W. and Mitchell, D.J. "Chemical Vapor Deposition of Zirconium Compounds: A Review." Coatings, 13, pp. 266, 2023. https://www.mdpi.com/2079-6412/13/2/266

KEYWORDS: Zirconium Carbide; Hafnium Carbide; Ceramic Precursors; Pilot Manufacturing; Technoeconomic Analysis; High-Temperature Coatings

DON26BZ01-DV003 TITLE: DIRECT TO PHASE II: AI/ML Assisted Field Troubleshooting in Avionics Optical Network

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Network Systems-of-Systems;Sustainment;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design, develop, and integrate a portable artificial intelligence/ machine learning (AI/ML)-enabled diagnostic module compatible with existing Optical Backscattering Reflectometer (OBR) and Optical Time Domain Reflectometer (OTDR) mainframes. The module will be engineered to support in-field optical network troubleshooting and management for high-speed communication systems.

DESCRIPTION: Current airborne military (mil-aero) core avionics, electro-optical (EO), communications, and electronic warfare systems are experiencing continuous growth in bandwidth demand, coupled with stringent requirements to reduce Size, Weight, and Power (SWaP). Earlier-generation multimode optical fibers have replaced traditional shielded twisted-pair wire and coaxial cable, offering increased electromagnetic interference (EMI) immunity, higher bandwidth and throughput, and notable reductions in aircraft size and weight.

However, maintenance and troubleshooting of these advanced optical networks remain highly dependent on traditional telecommunication test equipment. Identifying and resolving faults—such as fiber breaks, fractures, and high-loss terminations—requires locating and distinguishing anomalies within meter-level precision, whereas modern avionic information-processing networks demand centimeter-level spatial resolution from source to detector.

Fault detection must extend beyond typical Weapons Replaceable Assembly (WRA) interfaces to identify:

- Backplane/module degradation
- Line replaceable module-to-optical transceiver faults
- Polymer waveguide failures
- Inline sensor (fiber grating) issues
- Optical link loss across concatenated waveguide segments

Frequent airframe panel removal during fault isolation disrupts aircraft availability and mission readiness—especially for stealth platforms—highlighting the need for faster, more accurate, and less intrusive diagnostics.

To overcome these limitations, a portable AI/ML-enabled troubleshooting device is proposed to support field diagnostics across military airborne fiber-optic systems. The device will leverage next-generation reflectometry technologies and machine intelligence to enhance fault resolution precision and technician efficiency.

Key Capabilities:

- AI-Augmented Fault Detection
  - Real-time identification of defects (breaks, voids, misalignments, link degradation)
  - Pattern recognition and anomaly classification using historical signature databases

• AI-Driven Virtual Assistants
    o On-device or network-connected chatbots providing guided maintenance workflows
    o Embedded AR interface for overlaying diagnostics on test hardware in real time
• Advanced Troubleshooting Metrics
    o Spatial resolution to centimeter scale across multiple fiber types
    o Predictive maintenance algorithms to reduce unplanned network downtime
• Plug-and-Play Integration
    o Fully compatible with existing portable OTDR/OBR mainframes
    o Support for both multimode (50/125, 62.5/125, 100/140 µm) and single mode (9/125 µm) fiber types
    o GUI developed for intuitive field use across all operational conditions
• Wavelength and Environmental Resilience
    o Operational wavelength support: SWDM and CWDM
    o Designed for MIL-PRF-28800 Class 2 with select Class 1 enhancements
    o Operational temperature range: –40°C to +95°C
    o Resistant to mechanical shock, altitude variation, vibration, humidity, and thermal cycling

The device will build upon a fusion of legacy and emerging fiber-optic diagnostic technologies, including:

• Optical Time Domain Reflectometry (OTDR)
• Optical Backscatter Reflectometry (OBR)
• Photon-Counting OTDR (PC-OTDR)
• Low Correlation OTDR (LC-OTDR)
• Pseudo Random Sequence (PRS) Correlation OTDR (C-OTDR)
• Optical Frequency Domain Reflectometry (OFDR)

PHASE I: For a Direct to Phase II topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort and developed a concept for a workable prototype or design to address, at a minimum, the basic requirements of the stated objective above. The below actions would be required to satisfy the requirements of Phase I:
Concept Development: Developed a concept for a viable prototype or design solution that addresses, at a minimum, the core technical and performance objectives outlined in the stated topic.
Feasibility Demonstration: Designed, developed, and demonstrated the technical feasibility of a low-cost, AI/ML-based plug-in module compatible with portable OBR and OTDR mainframes. The solution must meet applicable aviation support equipment requirements, including ruggedization, thermal compatibility, and interface standards.
Performance Modeling and Simulation: Modeled and simulated the plug-in module's performance under high-speed application conditions, validating its functionality across relevant operational scenarios and wavelengths.
Design Packaging: Delivered a conceptual packaged design of the plug-in module, incorporating mechanical footprint, connector interface, and Graphical User Interface (GUI) considerations to support seamless integration into current field-deployable test equipment.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit and Phase I feasibility described in Phase I above has been met (i.e., the small business must have performed Phase I-type research and development related to the topic, but from non-SBIR funding sources) and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above. Documentation should include all relevant information including, but not limited to technical reports, test data, prototype designs/models, and performance goals/results. Work submitted within the feasibility documentation must have been substantially performed by the offeror and/or the principal

investigator (PI). Read and follow all of the DON SBIR FY26 Release 3 Direct to Phase II Broad Agency Announcement (BAA) Instructions. Phase I proposals will NOT be accepted for this topic.

PHASE II: Design, construct, and validate a functional AI/ML-enabled plug-in module prototype. Focus on transitioning the concept design into an operational system capable of meeting the rigorous demands of military optical diagnostics.
Include in the Prototype Design and Fabrication the following:
• Engineering of a robust plug-in module design based on Phase I feasibility studies and modeling outcomes.
• Integrating AI/ML processing hardware, signal acquisition architecture, and interfaces into a fully packaged prototype.
• Ensuring form-factor compliance with portable OTDR and OBR mainframes, including connector integrity, mechanical footprint, and GUI usability.
• Compiling system-level test data and validating against entry criteria for Technology Readiness Level (TRL) 6.

PHASE III DUAL USE APPLICATIONS: Collaborate with defense avionics industries as well as support equipment companies to accelerate transition to production.
Commercial telecommunication systems, fiber-optic networks, and data centers will benefit from the development of the AI/MIL based OBR and OTDR. These applications will be able to easily test/diagnose optical networks.

REFERENCES:

1. "MIL-PRF-28800F; Test Equipment for use with Electrical and Electronic Equipment , General Specification for." http://everyspec.com/MIL-PRF/MIL-PRF-010000-29999/MIL-PRF-28800F_18207/
2. Villalba, Sergi; Casas, Joan R. "Application of optical fiber distributed sensing to health monitoring of concrete structures." Mechanical Systems and Signal Processing, Volume 39, Issues 1-2, August-September 2013, pp. 441-451. https://www.sciencedirect.com/science/article/abs/pii/S0888327012000283
3. Tosi, D.; Molardi, C.; Blanc, W.; Paixão, T.; Antunes, P. and Marques, C. "Performance Analysis of Scattering-Level Multiplexing (SLMux) in Distributed Fiber-Optic Backscatter Reflectometry Physical Sensors." Sensors, 20(9), 2595. https://doi.org/10.3390/s20092595
4. Liu, X.; Lun, H.; Fu, M.; Fan, Y.; Yi, L.; Hu, W. and Zhuge, Q. "AI-Based Modeling and Monitoring Techniques for Future Intelligent Elastic Optical Networks." Applied Sciences, 10(1), 363. https://doi.org/10.3390/app10010363
5. Cho, J. Y. et al. "DeepALM: Holistic Optical Network Monitoring based on Machine Learning," 2022 Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2022, pp. 1-3. https://ieeexplore.ieee.org/document/9748360
6. Chan, Eric Y.; Beranek, Mark W. and Harres, Daniel N. "A Novel Gb/s Transceiver with OTDR Built-in-test (BIT) for Health Monitoring of Local Area Networks." Optical Fiber Communication Conference and Exposition and The National Fiber Optic Engineers Conference, OSA Technical Digest Series (CD) (Optica Publishing Group, 2007), paper OWU2. https://ieeexplore.ieee.org/document/4348948
7. Straub, M. et al. "AI-based OTDR event detection, classification and assignment to ODN branches in passive optical networks." 49th European Conference on Optical Communications (ECOC 2023), Hybrid Conference, Glasgow, UK, 2023, pp. 1146-1149. doi: 10.1049/icp.2023.2469. https://ieeexplore.ieee.org/document/10484603

KEYWORDS: Optical Backscattering Reflectometer; OBR; Optical Time Domain Reflectometer; OTDR; Augmented Reality; AR; Low correlation OTDR; LC-OTDR; Pseudo Random Signal; PRS; Correlation Optical Time Domain Reflectometer; C-OTDR

DON26BZ01-DV004 TITLE: DIRECT TO PHASE II: Highly Loaded Grain in 2.75" Form Factor

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design, develop, and demonstrate a Highly Loaded Grain (HLG) technology in a 2.75 inch rocket motor to extend range in a tactically relevant form factor.

DESCRIPTION: The objective of this SBIR Direct to Phase II topic is to utilize HLG to increase the range available in a 2.75” rocket motor and advance the Technology Readiness Level (TRL) and Manufacturing Readiness Level (MRL) of HLG technology. HLG is a propellant technology that improves total impulse in a given volume, as well as provides capability for mission flexibility. The Mk66 is a low cost 2.75” rocket motor utilizing minimum smoke propellent and is in use with unguided rockets and the Advanced Precision Kill Weapon System (APKWS II) All Up Round (AUR). Increasing the range available at an affordable cost in a Mk66 motor case is needed to pace emerging threats.
Key Technology Guidelines:

1. Rocket motor case: 2.75" Mk 66 case
2. Grain design: HLG propulsion technology
3. Ballistics software: CLWire provided by the Naval Air Warfare Center Weapons Division (NAWCWD)
4. Risk posture: Low/moderate risk for non-HLG specific components
5. Total Impulse: Increase by 30%
6. Thrust Profile: Implement all-boost and boost/sustain thrust profile with performance guidelines provided by NAWCWD (Maximum Expected Operating Pressure (MEOP) and initial thrust dictated by legacy Mk 66 system)
7. Propellant: Objective: Min-Smoke, Threshold: Reduced Smoke
8. Materials: Maximize compatibility/usage of existing rocket motor materials (propellant oxidizers and binders, insulation, liners, etc.)
9. Environments: thermal (-65 °F to 160 °F) (-53.9 °C to 71.1 °C) and mechanical environments (shock/vibe) required to enter military usage.
10. Nozzle and igniter: medium risk with path towards tactical design

PHASE I: For a Direct to Phase II SBIR topic, the Government expects that the small business has accomplished Phase I-type feasibility work and can document within the proposal submission to indicate previous research and development work has been conducted to design, implement, and test HLG propulsion technology in a "Phase I-type" effort. This work would include ballistic design, motor fabrication, and static test results. The feasibility documentation MUST NOT be solely based on work performed under prior or ongoing SBIR/STTR effort.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific

and technical merit and Phase I feasibility described in Phase I section above has been met (i.e., the small business must have performed Phase I-type research and development related to the topic NOT solely based on work performed under prior or ongoing federally funded SBIR/STTR work) and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above.

PHASE II: Develop an initial concept design incorporating the following elements: ballistics, insulation, nozzle, and igniter. This design will be formally documented and presented in a Detailed Design Review (DDR) to evaluate compliance with the technical requirements established in coordination with the Government. Approval of the DDR and its associated exit criteria is a prerequisite for advancing motor fabrication. Upon acceptance, the awardee will initiate fabrication activities, which include detailed design finalization, component and cast tooling production, and propellant mix and casting.
Following fabrication, the as-manufactured motor will undergo an Item Under Test (IUT) evaluation, during which its performance and specifications will be assessed relative to the original design concept. This review will be submitted to the Government for validation and approval. Subsequently, the motor will be static fired, contingent upon mutual agreement between the Government and the awardee.
Upon completion, the awardee will submit a final report to Naval Aviation Warfare Center Weapons Division (NAWCWD). This report will document the prototype's design, fabrication process, and test results. It will also identify any low-maturity technology areas and introduce a plan to further develop these technologies during Phase III.

The Government will furnish the motor case, HLG materials, and technical support as requested by the awardee throughout the award process.

PHASE III DUAL USE APPLICATIONS: Utilizing Phase II results, refine and execute risk reduction and technology maturation efforts to develop the design to an overall high TRL, and integrate into a full system, progressing towards potential integration into an existing program of record.
Dual uses include development of high-rate minimum-smoke propellant and HLG development toward rotary-wing applications.
Potential commercial industry utilization include pace-based applications, such as satellite thrusters/solid propulsion systems.

REFERENCES:

1. BAE Systems. "APKWS® laser-guidance kits successfully tested by U.S. counter-drone office." March 28, 2023. https://www.baesystems.com/en/article/apkws--laser-guidance-kits-successfully-tested-by-u-s--counter-drone-office
2. Harper, Jon. "Ukraine to get more laser-guided munitions to kill drones." Defensescoop, November 3, 2023. https://defensescoop.com/2023/11/03/ukraine-to-get-more-laser-guided-munitions-to-kill-drones/
3. NAVAIR. "Advanced Precision Kill Weapon System (APKWS)." https://www.navair.navy.mil/product/APKWS
4. NAWCWD. "NAWCWD Fact Sheet 2023". https://www.navair.navy.mil/nawcwd/sites/g/files/jejdrs601/files/document/%5Bfilename%5D/NAWCWD%20Fact%20Sheet%202023.pdf

KEYWORDS: Highly Loaded Grain; Small Diameter System; Solid Rocket Motor; Affordable Mass; Range Increase; Propellant Development

DON26BZ01-DV005 TITLE: DIRECT TO PHASE II: Advanced Damage Criticality Analysis

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a passive Structural Health Monitoring (SHM) system to identify, locate, and characterize the severity of defects and cracks due to fatigue loading or impacts based on novel or advanced technologies with a basis in physics and avoiding qualitative assumptions.

DESCRIPTION: The Navy seeks an effective passive Structural Health Monitoring (SHM) system for Navy ship hulls and other structures that can monitor defects, such as crack growth from fatigue or impacts, and provide actionable information about the severity of the defect in an automated manner, i.e., in real time. Such fatigue cracks develop and grow in Navy ship hull welds and plating from cyclical life-cycle stresses and event-driven forces from severe sea states, collisions, and groundings.
The U.S. Navy and other navies around the world have installed SHM systems to monitor hull structural health but almost all are based on using strain gauges to monitor stresses on the hull and inferring crack growth based on fatigue life calculations. For example, the Military Sealift Command (MSC) has worked with the American Bureau of Shipping (ABS) and installed SHM systems consisting of strain gauges and accelerometers on several ships in the T-EPF class, which monitor hull deflection and dynamic movement due to the ship's loading and the sea states encountered. The data from these sensors is being fed into a digital twin model developed to calculate structural stresses for managing vessel survivability and to minimize operating risk.
There have been some attempts to develop fiber optics sensors to measure strain or Acoustic Emission (AE) sensors to monitor fatigue cracks directly. These approaches have seen varying levels of success, yet, better systems are needed. There may even be some applications for LiDAR use to improve success probability. The Navy is particularly interested in locating and characterizing the severity or criticality of a defect if one is detected. Currently there is not a system available on the commercial market.
The Navy's need for such hull monitoring capability has become more important with the introduction of high-speed and catamaran vessels, which are more prone to hull cracking due to the designs of the ships, materials of the hull, and stresses experienced in high seas. An ideal system would be capable of monitoring large areas of the ship's hull with sensing devices that provide cost effective coverage with the following capabilities:

- Detect and identify the location of crack growth signals in the hull if they exist in the presence of ship's background noise without producing false positives or negatives.
- Produce results in an automated manner, i.e., real time, so they are immediately available to the operating crew.
- Provide insight as to the severity of the crack growth considering the complex geometries found in hull structures with varying thicknesses and stiffeners.

The Navy would benefit from understanding structural risks in real time with the goal of minimizing the possibility of incurring structural damage at sea. The SHM system the Navy needs should provide meaningful information on ship structural health and reduce inspection and maintenance costs during repair availabilities by identifying areas of concern or damage in advance.

PHASE I: For a Direct to Phase II topic, the Government expects that the small business has accomplished the following in a Phase I-type feasibility effort and developed a concept for a workable prototype or design to address, at a minimum, the basic requirements for identifying, locating, and characterizing crack growth in ship hull structures in an automated, real-time manner.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit and describes the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above.

PHASE II: Develop and deliver a passive SHM prototype solution (hardware/software/firmware) using novel or advanced technologies for use by the Navy in demonstrating the ability to monitor large area hull structures and identify, locate, and characterize crack growth in ship hull structures in an automated, real-time manner.

PHASE III DUAL USE APPLICATIONS: Assist the Navy in transitioning the technology for Navy use. Provide and field a passive SHM capability based on advanced technologies that will be used for identifying, locating, and characterizing crack growth in ship hull structures in an automated, real-time manner. Provide Navy personnel with training on how to utilize the system for the collection of data. Work with Navy personnel on how to install and operate the system until such time as they intend to assume that role.
In a manner like shipboard hull structural monitoring, the advanced passive SHM system could be employed in other useful applications, such as the following:
• Monitor ship hulls covered with acoustic tiles or other coatings/coverings for loose or missing tiles due to failure of the tile adhesive or other material defects.
• Monitor ship hulls with known areas of cracking or corrosion to determine when repairs are dictated and when other maintenance should occur.
• Monitor large Aboveground Storage Tanks (AST), common to both military and civilian petrochemical storage, to identify and locate AST bottom plate leaks.
This technology would also apply to commercial ship hull monitoring and SHM of offshore platform structures, such as oil drilling and production rigs.

REFERENCES:

1. "GUIDANCE NOTES ON STRUCTURAL MONITORING USING ACOUSTIC EMISSIONS OCTOBER 2016." American Bureau of Shipping (ABS). https://ww2.eagle.org/content/dam/eagle/rules-and-guides/current/other/239_gn_structural_monitoring_using_acoustic_emissions_2016/AET_GN_e.pdf
2. "ABS Guide for Hull Condition Monitoring System, July 2020." https://ww2.eagle.org/content/dam/eagle/rules-and-guides/current/conventional_ocean_service/73_Hull_Condition_Monitoring_2016/hull-condition-monitoring-guide-july20.pdf
3. "ABS Guide for Smart Functions for Marine Vessels and Offshore Units, June 2022." https://ww2.eagle.org/content/dam/eagle/rules-and-guides/current/other/307_smart_functions_marine_offshore_2022/smart-guide-jun22.pdf
4. "Structural health monitoring by use of sensor data." Petroleumstilsynet, February 21, 2024. https://www.havtil.no/contentassets/93a0efce534a46ec8d8d08344cbbfa53/structural-health-monitoring-by-use-of-sensor-data.pdf

KEYWORDS: Structural Health Monitoring (SHM); Guided Wave Analysis; Guided Wave Ultrasound; Ultrasonic Sensors Passive Monitoring; Acoustic Emission

DEPARTMENT OF THE AIR FORCE
SMALL BUSINESS INNOVATION RESEARCH (SBIR)
DoW 2026 SBIR BAA Release 1
PHASE I PROPOSAL SUBMISSION INSTRUCTIONS

The Air Force intends these Phase I proposal submission instructions to clarify the Department of War (DoW) Broad Agency Announcement (BAA) as it applies to the topics solicited herein. **Offerors must ensure proposals meet all requirements of the SBIR Program BAA posted on the Defense SBIR/STTR Innovation Portal (DSIP) at the proposal submission deadline date/time.**

Applicants are encouraged to thoroughly review the DoW Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- Full component-specific instructions and topic descriptions are available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate BAA cycle.

Please ensure all e-mail addresses listed in the proposal are current and accurate. The DAF is not responsible for ensuring notifications are received by firms changing mailing address/e-mail address/company points of contact after proposal submission without proper notification to the DAF. **If changes occur to the company mail or email addresses or points of contact after proposal submission, the information must be provided to the AF SBIR/STTR One Help Desk.** The message shall include the subject line, "FY26 SBIR BAA Address Change".

Points of Contact:
General information related to the AF SBIR/STTR program and proposal preparation instructions, contact the AF SBIR/STTR One Help Desk at usaf.team@afsbirsttr.us. All applicants have ample opportunity to request clarifying information. **The DAF encourages applicants to request clarifying information as early as possible, as delays in such requests constrain the DAF's ability to provide satisfactory resolution to applicant concerns.**

- Questions regarding the DSIP electronic submission system, contact the DoW SBIR/STTR Help Desk at dodsbirsupport@reisystems.com.
- For technical questions about the topics during the pre-announcement and open period, please reference the DoW FY26 SBIR BAA.
- Air Force SBIR/STTR Contracting Officer (CO):
  - Mr. Daniel J. Brewer, Daniel.Brewer.13@us.af.mil

General information related to the AF Small Business Program can be found at the AF Small Business website, http://www.airforcesmallbiz.af.mil/. The site contains information related to contracting opportunities within the AF, as well as business information and upcoming outreach events. Other informative sites include those for the Small Business Administration (SBA), www.sba.gov, and the Procurement Technical Assistance Centers (PTACs), http://www.aptacus.us.org. These centers provide Government contracting assistance and guidance to small businesses, generally at no cost.

PHASE I PROPOSAL SUBMISSION

The DoW FY26 SBIR BAA, https://www.dodsbirsttr.mil/submissions/login, includes all program requirements. Phase I efforts should address the feasibility of a solution to the selected topic's requirements.

PHASE I PROPOSAL FORMAT

Complete proposals must include all of the following:
Volume 1: Cover Sheet

Note: If selected for funding, the proposal's technical abstract and discussion of anticipated benefits will be publicly released. Therefore, do not include proprietary information in this section.
Volume 2: Technical Volume
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents
Volume 6: Fraud, Waste, and Abuse Training
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

COVER SHEET (VOLUME 1)
Complete the proposal Cover Sheet in accordance with the instructions provided via DSIP. The technical abstract should include a brief description of the program objective(s), a description of the effort, anticipated benefits and commercial applications of the proposed research, and a list of key words/terms. The technical abstract of each successful proposal will be submitted to the Office of the Secretary of War(OSW) for publication and, therefore, must not contain proprietary or classified information.

TECHNICAL VOLUME (VOLUME 2):
The Technical Volume should include all graphics and attachments but should not include the Cover Sheet, which is completed separately as Volume 1. The Phase I technical volume (uploaded in Volume 2) shall contain the required elements found below. Ensure that all graphics are distinguishable in black and white.

The Phase I Technical Volume page/slide limits identified for the topics do not include the Cover Sheet, Cost Volume, Cost Volume Itemized Listing (a-h). The Technical Volume must be no smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. Only the Technical Volume and any enclosures or attachments count toward the page limit. In the interest of equity, pages/slides in excess of the stated limits will not be reviewed. The documents required for upload into Volume 5, "Other", do not count toward the specified limits.

These instructions supplement the FY26 SBIR BAA. In addition to the requirements found in the FY26 SBIR BAA, applicants are required to provide the following information in Volume 2:

Key Personnel: Identify in the Technical Volume all key personnel who will be involved in this project; include information on directly related education, experience, and citizenship.

- A technical resume of the principal investigator, including a list of publications, if any, must be included. Only one principal investigator/project manager can be designated to a proposal at any given time.
- Concise technical resumes for subcontractors and consultants, if any, are also useful.
- Identify all U.S. permanent residents to be involved in the project as direct employees, subcontractors, or consultants.
- Identify all non-U.S. citizens expected to be involved in the project as direct employees, subcontractors, or consultants. For all non-U.S. citizens, in addition to technical resumes, please provide countries of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project, as appropriate. Additional information may be requested during negotiations in order to verify the foreign citizen's eligibility to participate on a contract issued as a result of this announcement. **Note:** Do not upload information such as Permanent Resident Cards (Green Cards), birth certificates, Social Security Numbers, or other PII to the DSIP system.

Phase I Statement of Work Outline
NOTE: The DAF uses the work plan outline as the initial draft of the Phase I Statement of Work (SOW). Therefore, **do not include proprietary information in the work plan outline**. To do so will necessitate a request for revision, if selected, and may delay award.

Include a work plan outline in the following format:
Scope: List the effort's major requirements and specifications.
Task Outline: Provide a brief outline of the work to be accomplished during the Phase I effort.
Milestone Schedule
Deliverables
Progress reports
Final report with SF 298

COST VOLUME (VOLUME 3)

Cost information should be provided by completing the Cost Volume in DSIP and including the Cost Volume Itemized Listing specified below. The Cost Volume detail must be adequate to enable Air Force personnel to determine the purpose, necessity and reasonability of each cost element. Provide sufficient information (a.-g. below) regarding funds use. The DSIP Cost Volume and Itemized Cost Volume Information will not count against the specified page limit. The itemized listing also may be submitted in Volume 5 under the "Other" dropdown option.

a. **Direct Cost Materials**: Justify costs for materials, parts, and supplies with an itemized list containing types, quantities, prices and where appropriate, purpose. Material costs may include the costs of such items as raw materials, parts, subassemblies, components, and manufacturing supplies.

b. **Other Direct Costs**: This category includes, but is not limited to, specialized services such as machining, milling, special testing or analysis, and costs incurred in temporarily using specialized equipment. Proposals including leased hardware must include an adequate lease v. purchase justification.

c. **Direct Labor**: Identify key personnel by name, if possible, or by labor category, if not. Direct labor hours, labor overhead and/or fringe benefits, and actual hourly rates for each individual are also necessary for the CO to determine whether these hours, fringe rates, and hourly rates are fair and reasonable.

d. **Travel**: Travel costs must relate to project needs. Break out travel costs by trip, number of travelers, airfare, per diem, lodging, etc. The number of trips required, as well as the destination and purpose of each, should be reflected. Recommend budgeting at least one trip to the Air Force location managing the contract.

e. **Subcontracts**: Involvement of university or other consultants in the project's planning and/or research stages may be appropriate. If so, describe in detail and include information in the Cost Volume. The proposed total of consultant fees, facility lease/usage fees, and other subcontract or purchase agreements may not exceed **one-third of the total contract price** or cost (do not include profit in the calculation), unless otherwise approved in writing by the CO. The SBIR funded work percentage calculation considers both direct and indirect costs after removal of the SBC's proposed profit. Support subcontract costs with copies of executed agreements. The documents must adequately describe the work to be performed. At a minimum, include a Statement of Work (SOW) with a corresponding detailed Cost Volume for each planned subcontract.

f. **Special Tooling, Special Test Equipment, and Material**: The inclusion of equipment and materials will be carefully reviewed relative to need and appropriateness to the work proposed. Special tooling and

special test equipment purchases must, in the CO's opinion, be advantageous to the Government and relate directly to the effort. These toolings or equipment should not be of a type that an offeror would otherwise possess in the normal course of business. These may include items such as innovative instrumentation and/or automatic test equipment.

g. **Consultants**: Provide a separate agreement letter for each consultant. The letter should briefly state what service or assistance will be provided, the number of hours required, and the hourly rate.

NOTE: If no exceptions are taken to an offeror's proposal, the Government may award a contract without negotiations. Therefore, the offeror's initial proposal should contain the offeror's best terms from a cost or price and technical standpoint. If there are questions regarding the award document, contact the Phase I CO identified on the cover page. The Government reserves the right to reopen negotiations later if the CO determines doing so to be necessary.

COMPANY COMMERCIALIZATION REPORT (VOLUME 4)

Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the FY26 SBIR BAA for full details on this requirement. Information contained in the CCR will not be considered by the Air Force during proposal evaluations.

SUPPORTING DOCUMENTS VOLUME (VOLUME 5)

The following documents may be required if applicable to your proposal:

1. DD Form 2345: For proposals submitted under export-controlled topics, either International Traffic in Arms or Export Administration Regulations (ITAR/EAR), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website, http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP/DD2345Ins tructions.aspx. DD Form 2345 approval will be required if proposal is selected for award.
2. Verification of Eligibility of Small Business Joint Ventures (Attachment 3 to the FY26 SBIR BAA)
3. Technical Data Rights Assertions (if asserting data rights restrictions)

FRAUD, WASTE, AND ABUSE TRAINING (VOLUME 6)

Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

DISCLOSURES OF FOREIGN AFFILIATIONS OR RELATIONSHIPS TO FOREIGN COUNTRIES (VOLUME 7)

Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoW SBIR Program BAA.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)

The Air Force does not participate in the Discretionary Technical and Business Assistance (TABA) Program. Proposals submitted in response to DAF topics shall not include TABA.

AIR FORCE PROPOSAL EVALUATIONS

Proposals will be evaluated for overall merit in accordance with the criteria discussed in the FY26 SBIR BAA. DAF is seeking varying technical/scientific approaches and/or varying and new technologies that would be responsive to the problem statement(s) and area(s) of interest in the topic. Multiple procurements are planned and anticipated to be awarded as a result of the topic, each proposal is considered a separate procurement and will be evaluated on its own merit, and that the Government may award all, some, or none of the proposals. Any per-award or per-topic funding caps are budgetary estimates only, and more or less funding may become available. Funding decisions are made with complete disregard to the other awards under the same topic.

In accordance with 15 USC 638(vv) (Section 4 of the SBIR and STTR Extension Act of 2022), and the Deputy Secretary of Defense Memorandum; Subject: Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program dated May 13, 2024, the DAF will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. The DAF will use information provided by the small business concern in response to the Disclosure of Foreign Affiliations or Relationships to Foreign Countries and the proposal to conduct a risk-based due diligence review on the cybersecurity practices, patent analysis, employee analysis, and foreign ownership of a small business concern, including the small business concern and employees of the small business concern to a foreign country, foreign person, foreign affiliation, or foreign entity. The DAF will also assess proposals utilizing open-source analysis and analytical tools, for the nondisclosures of the information set forth in 15 U.S.C. 638(g)(13). If DAF assesses that a small business concern has security risk(s), DAF will review the proposal, the evaluation, and the security risks and may decide not to select the proposal for award based upon a totality of the review.

MAJORITY OWNERSHIP IN PART BY MULTIPLE VENTURE CAPITAL, HEDGE FUND, AND PRIVATE EQUITY FIRMS

Small business concerns that are owned in majority part by multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds are not eligible to submit applications or receive awards for DAF Topics.

PERFORMANCE OF WORK REQUIREMENTS AND LOCATION OF WORK

For Phase I, a minimum of two-thirds of the research or analytical effort must be performed by the Awardee. The DAF measures percentage of work by both direct and indirect costs, not including profit. Occasionally, the DAF will consider deviations from this performance of work requirement. **Requests for Performance of Work deviations must be made twice: prior to submission during the topic open period and as part of the initial proposal submission.** For requests prior to the initial proposal submission, the DAF will consider the request and approve or disapprove requesting applicants to proceed with DSIP submission. Upon proposal receipt, the DAF will again consider such requests for approval for the resultant award.

All R/R&D work must be performed in the United States. Based on a rare and unique circumstance, the DAF may approve a particular portion of the R/R&D work to be performed or obtained in a country outside of the United States. The awarding Funding Agreement officer must approve each specific condition in writing. Applicants seeking this approval must make such a request with their initial proposal submission. The DAF will not consider these requests prior to proposal submission.

DAF USE OF SUPPORT CONTRACTORS

Restrictive notices notwithstanding, proposals may be handled for administrative purposes only, by support contractors. These support contractors may include, but are not limited to TEC Solutions, Inc., APEX, Oasis Systems, Riverside Research, Peerless Technologies, HPC-COM, Mile Two, Montech,

Wright Brothers Institute, and MacB (an Alion Company). In addition, only Government employees and technical personnel from Federally Funded Research and Development Centers (FFRDCs) MITRE and Aerospace Corporations working under contract to provide technical support to AF Life Cycle Management Center and Space and Missiles Centers may evaluate proposals. All support contractors are bound by appropriate non-disclosure agreements. Contact the AF SBIR/STTR CO Daniel J. Brewer (Daniel.Brewer.13@us.af.mil) with concerns regarding the use of support contractors.

PROPOSAL STATUS AND FEEDBACK

The Principal Investigator (PI) and Corporate Official (CO) indicated on the Proposal Cover Sheet will be notified by e-mail regarding proposal selection or non-selection. Small Businesses will receive a notification for each proposal submitted. Please read each notification carefully and note the Proposal Number and Topic Number referenced.

Automated feedback will be provided for Phase I proposals designated Not Selected. Additional feedback may be provided at the sole discretion of the DAF.

IMPORTANT: Proposals submitted to the DAF are received and evaluated by different organizations, handled by topic. Each organization operates within its own schedule for proposal evaluation and selection. Updates and notification timeframes will vary. If contacted regarding a proposal submission, it is not necessary to request information regarding additional submissions. Separate notifications are provided for each proposal.

The Air Force anticipates that all proposals will be evaluated and selections finalized within approximately 90 calendar days of solicitation close. Refrain from contacting the BAA CO for proposal status before that time.

Refer to the DoW SBIR Program BAA for procedures to protest the Announcement.
As further prescribed in FAR 33.106(a), FAR 52.233-2, Protests should be submitted to: Air Force SBIR/STTR Contracting Officer Daniel J. Brewer, Daniel.Brewer.13@us.af.mil. The DAF uses its best efforts to resolve concerns raised by interested parties at the contracting officer level through open and frank discussions. For agency protests, the DAF provides for inexpensive, informal, procedurally simple, and expeditious resolution of protests.

DAF SUBMISSION OF FINAL REPORTS

All Final Reports will be submitted to the awarding DAF organization in accordance with Contract instructions. Companies will not submit Final Reports directly to the Defense Technical Information Center (DTIC).

PHASE II PROPOSAL SUBMISSIONS

DAF organizations may request Phase II proposals while Phase I technical performance is ongoing or at any time after the conclusion of the period of performance. This decision will be based on the awardee's technical progress, as determined by an DAF Technical Point of Contact review using the Phase II review criteria outlined above.

Phase II is the demonstration of the technology found feasible in Phase I. Only Phase I awardees are eligible to submit a Phase II proposal. All Phase I awardees will be sent a notification with the Phase II proposal submittal date and detailed Phase II proposal preparation instructions. If the physical or email addresses or firm points of contact have changed since submission of the Phase I proposal, correct information shall be sent to the DAF SBIR/STTR One Help Desk. Phase II dollar values, performance periods, and proposal content will be specified in the Phase II request for proposal.

NOTE: The DAF primarily makes SBIR Phase I and II awards as Firm-Fixed-Price contracts. However, awardees are strongly urged to work toward a Defense Contract Audit Agency (DCAA)-approved accounting system. If the company intends to continue work with the DoD, an approved accounting system will allow for competition in a broader array of acquisition opportunities, including award of Cost-Reimbursement types of contracts. Please address questions to the Phase II CO, if selected for award.

All proposals must be submitted electronically via DSIP by the date indicated in the Phase II proposal instructions. Note: Only ONE Phase II proposal may be submitted for each Phase I award.

AIR FORCE SBIR/STTR PROGRAM MANAGEMENT IMPROVEMENTS

The DAF reserves the right to modify the Phase II submission requirements. Should the requirements change, all Phase I awardees will be notified. The DAF also reserves the right to change any administrative procedures that will improve management of the DAF SBIR/STTR Program at any time.

DAF
DoW 2026 SBIR BAA
Phase I Topic Index
Release 1

Topic Number	Topic Name	Award Maximum Value*	Award Maximum Duration**	Technical Volume (Volume 2) Page/Slide Limit***
DAF26BZ01-NV003	Low-Cost Modular Payload Vehicle for Agile Electronic Warfare Swarms with Ground Launch Capability	$275,000.00	6	20
DAF26BZ01-NV006	Intelligent Threat Aware Autonomy	$300,000.00	6	20
DAF26BZ01-NV008	Runtime Assured Autonomy	$300,000.00	6	20
DAF26BZ01-NV500	Autonomous Space Cargo Network (ASCN): AI-Driven Logistics Automation and Digital Twin Integration for Space Mobility	$75,000.00	3	10
DAF26BZ01-NV501	Commercial-Derives Insights for Novel Tactical Surveillance, Reconnaissance, and Tracking (TacSRT) Capabilities	$175,000.00	3	5

*Proposals in excess of this amount will not be considered for evaluation or award.
**Proposals in excess of this duration will not be considered for evaluation or award.
***Pages/slides in excess of this number will not be considered during evaluations.

DAF26BZ01-NV003 TITLE: Low-Cost Modular Payload Vehicle for Agile Electronic Warfare Swarms with Ground Launch Capability

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a low-cost, versatile sUAS platform (Group 3 and below) specifically designed to accommodate modular payloads and capable of ground launch. This platform should enable agile electronic warfare applications in swarms. This topic is intended to develop a standalone solution that can be integrated with a variety of payloads, either by modifying an existing sUAS platform or by developing a new platform from the ground up.

DESCRIPTION: The effective deployment of electronic warfare (EW) capabilities relies on agile and adaptable platforms that can rapidly integrate and deploy a variety of payloads. Current sUAS platforms often lack the modularity and flexibility required to support the rapid evolution of EW technology. This topic addresses the need for a low-cost, versatile sUAS platform specifically designed to accommodate modular payloads and designed for ground launch, enabling rapid deployment in diverse operational environments. Proposals may consider either modifying an existing, commercially available sUAS platform to meet the requirements of this topic, or developing a new platform optimized for modularity and ground launch.
The key innovation is the development of a sUAS platform (either new or modified) with a standardized payload interface that allows for rapid integration and swapping of different payloads. This modular design, combined with ground launch capability, will enable:

- Rapid Payload Integration: Simplified and standardized interfaces for connecting power, data, and control signals to the payload.
- Payload Agnosticism: The ability to accommodate a wide range of payload sizes, weights, and power requirements.
- Enhanced Mission Flexibility: The ability to quickly reconfigure the sUAS for different missions by swapping payloads.
- Simplified Logistics: Reduced maintenance and support costs through standardized components and interfaces.
- Ground Launch Compatibility: Robust design specifically for compatibility with ground launch systems, enabling rapid deployment from ground-based platforms, even in challenging terrain.

The sUAS platform should be optimized for operation in low to medium sized swarms (3-10 units), allowing for coordinated EW effects. The design should also prioritize low cost, ease of use, reliability, and the following Key Performance Parameters (KPPs):

- Payload Capacity: Minimum of 5 lbs
- Endurance: Minimum flight time of 45 minutes with a 5 lb payload.
- Range: Minimum operational range of 100 kilometers.
- Ground Launch System Compatibility: Compatible with a readily available ground launch system (e.g., pneumatic launcher, rail system).

- Deployment Time: Capable of being launched and operational within 5 minutes of arrival at the launch site.

PHASE I: Develop a concept for a modular payload vehicle, either by modifying an existing platform or designing a new one, including detailed specifications for the payload interface, power system, communication system, and control system. The proposal should demonstrate the feasibility of achieving rapid payload integration, payload agnosticism, the stated KPPs, and successful ground launch. The proposal should detail the integration challenges and requirements, whether modifying an existing platform or developing a new one. Documentation should include preliminary designs, performance simulations, component selection rationale, and a ground launch integration plan.

PHASE II: Develop, integrate, and demonstrate a prototype modular payload vehicle. Specific goals include:
- Demonstrate rapid payload integration and swapping capabilities.
- Achieve a target cost of [Insert Specific Cost Target based on your research] per unit.
- Demonstrate achievement of the stated KPPs.
- Develop a user-friendly interface for controlling and managing the sUAS and its payload.
- Utilize open-source software and off-the-shelf hardware components where possible.
- Demonstrate successful integration and launch from a representative ground launch system.
- Evaluate the impact of ground launch conditions on the performance and reliability of the sUAS platform and its payload.

PHASE III DUAL USE APPLICATIONS: Successful development of a modular payload vehicle will have significant applications in both military and commercial sectors. Potential applications include:
- Rapidly deployable sensor platforms for environmental monitoring.
- Versatile delivery platforms for logistics and supply chain management.
- Adaptable inspection platforms for infrastructure maintenance.
- Enhanced situational awareness for first responders, rapidly deployable from ground-based platforms.
- Rapidly deployable communication and EW capabilities for disaster relief efforts, even in areas with limited infrastructure.

The core technologies developed in this SBIR have broad applicability across both defense and commercial sectors, fostering economic growth and societal benefit.

REFERENCES:

1. Department of Defense, Unmanned Systems Integrated Roadmap, FY2013-2038; https://apps.dtic.mil/sti/tr/pdf/ADA592015.pdf
2. 2021 U.S. Department of Defense Counter-Small Unmanned Aircraft Systems Strategy; https://media.defense.gov/2021/Jan/07/2002561080/-1/-1/1/DEPARTMENT-OF-DEFENSE-COUNTER-SMALL-UNMANNED-AIRCRAFT-SYSTEMS-STRATEGY.PDF
3. Department of the Air Force Operational Imperatives; https://www.af.mil/Portals/1/documents/2023SAF/OPERATIONAL_IMPARITIVES_INFOGRAPHIC.pdf

KEYWORDS: sUAS; Modular Payload; Ground Launch; Electronic Warfare; Rapid Integration; Standardized Interface; Swarm Operations

DAF26BZ01-NV006 TITLE: Intelligent Threat Aware Autonomy

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE:-The objectives are to do: 1. Weapon Engagement Zone (WEZ) Modeling: Develop models to represent the area where a weapon can effectively engage targets. This involves considering factors like weapon range, vehicle movement, and threat trajectories, to provide risk measures for path planning and weapons employment.2. WEZ Avoidance: Develop path planning algorithms for ACPs to navigate safely through dynamic WEZs, minimizing risk while reaching objectives efficiently. This requires real-time solutions that can handle multiple static and moving threats.3. Advanced Weaponeering: Optimize weapon usage for ACPs to maximize target capture and neutralization. This includes assigning appropriate weapons to targets, considering target movement and the overall mission context.4. Mutual Support: Investigate how multiple ACPs can cooperate effectively in adversarial situations. This includes coordinated movement to avoid threats and collaborative weapon engagement for increased effectiveness.

DESCRIPTION: To address future Air Force strategic needs, an increasing number of advanced systems with intelligent autonomy are being envisioned. Intelligent autonomy is central to systems involving advanced automation, artificial intelligence, machine learning, adaptive control architectures, and heightened performance compared to the state of the art. A critical need for enabling these future autonomous systems are behaviors that can be leveraged by higher level cognition or mission managers to achieve collaborative mission execution for ACPs. The question that needs to be asked is, “Provided that systems have all the data available to them from sensors and mission objectives, what is it that the systems actually have to do to be successful in their mission?” It is clear that the sensing and available of data is a critical requirement for making informed decisions, this may entail a deep investigation on coupling behaviors with sensing capability; but, the focus of this effort is more toward the thinking and action than the sensing of the sense-think-act process flow. Near term objectives of this work are to invest in basic and applied research to building on the accomplished R&D, address specific identified technical challenges and tools for solving Intelligent Threat Aware Autonomy (ITA2) objectives. Far term objectives involve advanced technology development to constrict ITA2 avionics packages, perform real-time hardware and flight testing of ITA2 products, manufacture vehicles capable of performing ITA2 or hardware that interfaces with current ACPs, and flight test on Air Force / DoD commercial platforms. Intelligent Threat Aware Autonomy (ITA2) is aimed at finding ways to take measured risks and enable autonomous systems to achieve air superiority in threat laden environments. Multiple facets of this project are to be investigated including: ways of measuring risk from ensuing threats, leveraging own-ship weapon models for capturing targets of interest, avoiding adversarial threats, addressing limited communication range and navigational error, quantifying mutual support and types of mutual support, and measures of force through collaboration and teaming. Lastly, the addressing of uncertainty of own-ship(s) states, target vehicle(s) states, operations boundaries, target vehicle capability, and other forms of uncertainties such as communication delay and environmental disturbances (wind) are important for obtaining reliable and robust behaviors.

Vehicle control is performed by providing the vehicles desired aim-points or waypoint plans in three-dimensional space. The inner loop control systems of aircraft is out of scope of this work; rather, interfacing with current / existing vehicle control technologies is expected though the use of aim-points. This reduces the burden of developing the necessary vehicle control commands such as normal acceleration, roll-rate, and throttle. Furthermore, it leverages the most state of the art methods for performing vehicle control and AI enabling technologies.

PHASE I: In Phase 1, focus should be on initial developments of proposed solutions to one or more of the design challenges. Alternate solutions should be considered and the most promising approaches identified. Development and analysis in desktop simulation environments with representative aircraft platforms (kinematic or dynamic in nature) can be performed to assist transition to Phase II. Investigations in methodology and computational performance should motivate the use of flight-worthy methods in later phases. Feasibility studies should be conducted regarding proposed solution approaches. Initial design and analysis studies in desktop simulation environments should be performed. Based on initial analyses and experimental results, recommendations for further R&D and a Phase II technology development plan should be completed. Surrogate models representing Air Force platforms of interest can be used in Phase I. No government furnished data or equipment should be required. Air Force customers/stakeholders and specific Air Force technology applications of interest should be identified. These should be technologies in which advancements in ITA2 will provide significant benefit.

PHASE II: In Phase II, design details and experimental test plans should be significantly expanded. Development and analysis in higher fidelity desktop simulation environments with representative platform applications should be performed. Develop realistic use cases that exercise ITA2 functionality and demonstrate benefits or capabilities. The ITA2 system should be able to avoid adversarial threats, engage on targets of interest, and be able to quantify relevant mission parameters for a mission manager to select from available behaviors. Success will be defined by demonstrating the benefits of the advanced ITA2 technology as compared to current state of the art methods such as circumnavigation around threat regions or pure-pursuit to intended targets. Evaluation of ITA2 Technology will entail review of real-time functionality and test/demonstrate the technologies in a software/hardware integration laboratory environment. It may be valuable to repeat some or all of the capstone experiments that were performed in desktop simulations. Cost and schedule permitting, port developed real time code to flight processors and perform initial flight demonstrations with surrogate sUAS platform(s), again testing capstone experiments.

Depending on contractual arrangements, government furnished data or equipment could be provided in the form of simulation models or equipment supporting laboratory or flight testing. At this stage, systems used to demonstrate the developed ITA2 technologies should closely align with Air Force programs of interest that employ advanced, adaptive and intelligent autonomy. Technology transfer plans should be constructed showing how the developed Phase II products can directly support such programs in preparations for Phase III efforts.

PHASE III DUAL USE APPLICATIONS: In Phase III, teaming arrangements should be made with airframe/avionics manufacturers to develop/finalize ITA2 system design(s) in a pre-production phase. Required safety analysis and testing for eventual certification should be performed. Phase III activities should directly support Air Force programs of interest with flight testing and demonstrations on full scale vehicles. This program is integrating ACP technology with Autonomy Government Reference Architecture (A-GRA) missions systems to enable ACP transitions. A successful program will deliver a prototype suite of technologies to enable autonomous ACPs with enhanced capabilities for Air Force missions. However, trust in the autonomy will be paramount for close-in manned-unmanned operations

and ITA2 will be a key enabling technology to provide the required level of trust in the unmanned systems.

Follow-on Phase III activities should expand applications to other branches of the military and DoD customers. Clear applications include counter uncrewed aerial system defense, cruise missile defense, high-value airborne asset defense, high-speed interception of incoming threats, suppression of enemy air defenses, destruction of enemy air defenses, combat air patrol, etc. Methods and approaches should be suitable for pre-mission planning, real-time mission execution, and wargaming to inform commander's intent. ITA2 applications should be extended to various groups of air platforms; but, group 5 systems are of particular interest to the DoD.

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1. Buzikov, Maksim E., and Andrey A. Galyaev. "Time-optimal interception of a moving target by a Dubins car." Automation and Remote Control 82 (2021): 745-758
2. Cockayne, E. J., and G. W. C. Hall. "Plane motion of a particle subject to curvature constraints." SIAM Journal on Control 13.1 (1975): 197-220
3. Dillon, Patrick M., et al. "Optimal Trajectories for Aircraft Avoidance of Multiple Weapon Engagement Zones." Journal of Aerospace Information Systems 20.8 (2023): 520-525
4. Manyam, Satyanarayana G., et al. "Coordinating defender path planning for optimal target-attacker-defender game." AIAA Scitech 2019 Forum. 2019
5. Von Moll, Alexander, and Isaac Weintraub. "Basic engagement zones." Journal of Aerospace Information Systems 21.10 (2024): 885-891
6. Weintraub, Isaac E. "Optimal Defense of High Value Airborne Assets." Phd Dissertation, Air Force Institute of Technology, Wright-Patterson AFB, OH, (2021)
7. Weintraub, Isaac E., et al. "An Optimal Engagement Zone Avoidance Scenario in 2- D." AIAA SciTech 2022 Forum. 2022
8. Wolek, Artur, et al. "Sampling-Based Risk-Aware Path Planning Around Dynamic Engagement Zones." arXiv preprint arXiv:2403.05480 (2024)

KEYWORDS: Autonomous Collaborative Platforms; Collaborative Combat Aircraft; Risk Aware Autonomy; Threat Aware Autonomy; Artificial Intellgence; Weapon Engagement Zone; WEZ; WEZ Aware Operations; Offensive Coutner Air; Defensive Counter Air.

DAF26BZ01-NV008 TITLE: Runtime Assured Autonomy

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: The Need for Advanced Autonomy: The Air Force has gained wide interest in fully autonomous, unmanned air platforms operating in teams making collaborative decisions to successfully complete missions. Highest level, real-time decision making will be the responsibility of advanced autonomy. This autonomy will include both flight-level autonomy and mission-level autonomy. Flight-level autonomy functions will generate local commands that keep the vehicle operating safely. Mission-level autonomy functions will continuously deliver courses of action (COAs) to each platform in the fleet, commanding mission progress in real time. Although all vehicles in the fleet will have instantiations of the mission-level autonomy functions, COAs will typically be generated by a chosen fleet leader.

The Need for Runtime Assured Autonomy: Autonomy approaches under current development can be highly complex and nondeterministic in their behaviors. AFRL is currently developing approaches for autonomously executed missions using complex event processing techniques. This class of autonomy will be difficult, if not impossible, to fully certify from an airworthiness perspective, and therefore cannot be trusted to correctly operate under all mission conditions. Further, the capabilities of artificial intelligence and autonomy are rapidly increasing with continually updated versions and design iterations expected to occur throughout the operational lifecycles of unmanned systems. Such protocols are clearly not amenable to the time consuming and expensive airworthiness certification process.

To address this hurdle, Runtime assured autonomy (RTAA) functions will be needed to perform runtime monitoring of the autonomy and enact procedures to mitigate any adverse effects due to errors in the autonomy design. The safety and performance protections provided by RTAA will lessen the certification burden, allowing rapid fielding of autonomy functions.

Topic Objective: The objective of this topic is to develop innovative approaches to RTAA systems that protect the individual platform and the fleet against undiscovered design errors in the autonomy functions. The focus should be on use cases in which the RTAA determines whether the autonomy is generating infeasible, incorrect, and/or non-optimal solutions (e.g., commanded paths or task allocation) that may affect mission progress and effectiveness.

DESCRIPTION: Several of the Air Force's Operational Imperatives call for unmanned platforms to support manned platforms. The Advanced Battle Management System, Moving Target Engagement, Tactical Air Dominance and Global Strike imperatives all call for less expensive, attritable uncrewed platforms to aid in executing complex battle missions. These uncrewed systems cannot always be guaranteed to be controlled by remote human operators due to loss of radio communications or saturated operator workload. Full autonomy will need to fill the gap when human command/control cannot. To address future Air Force tactical and strategic needs, an increasing number of advanced systems with intelligent autonomy are being envisioned. Intelligent autonomy is central to systems involving a wide range of advanced adaptation, reconfiguration, autonomous decision making and contingency management.

Assured autonomy is the requirement that the autonomy operates safely and correctly under all circumstances and mission scenarios. RTAA fulfills this Air Force technology need, providing continuous monitoring/mitigation of autonomy functions to deliver required assurances of safe flight and correct mission execution. There are considerable challenges to developing a working RTAA system. The two key functions of the RTAA are:

1. Fault detection & isolation: The RTAA system must be able to determine if the autonomy is correctly producing COAs and other commands, which is especially difficult if agnostic of the autonomy function details. Developing strategies that can indirectly detect and isolate autonomy design faults in dynamic environments will be key to developing the RTAA system. Faults within the autonomy will need to be determined through the effects those faults have on the platform's safety, performance, and/or mission effectiveness. RTAA fault determination may come from comparing the current actions of the autonomy with nominal functional or performance requirements (e.g., what defines correct behavior), sanity checks, rubrics, rule sets, etc.

2. Mitigation response: If the RTAA determines that errors in the design of the autonomy functions are adversely affecting flight and mission decisions, it must then activate proper recovery or reversionary protocols. This may include first commanding the vehicle to a failsafe loiter point, then clearing functional states and restarting the autonomy functions. As a last resort, the RTAA may activate return-to base or ditch procedures. If available, the RTAA may switch to simpler, reversionary autonomy functions that can continue the mission either temporarily until the advanced autonomy is back online, or to mission completion, if capable.

The two main functional levels of an RTAA system are:

1. Platform/fleet safety: Here, the RTAA typically treats the autonomy functions as a black box and simply monitors the platform and fleet for safety violations. The RTAA will monitor, for example, 1) flight envelope parameters such as angle of attack, angular rates, g-loading, etc., determining if their values remain within prescribed limits, 2) flight corridor values, determining if the vehicles are within their prescribed airspace and location for path deconfliction, and 3) path commands generated by the autonomy functions to determine if the vehicle's maneuvering capabilities can fly the commanded path. If it is determined that safety violations are ensuing, (and assuming no hardware faults or other contingencies are causing unsafe conditions), then the RTAA will deactivate the autonomy functions and

activate simpler reversionary controllers or procedures designed to bring the vehicle/fleet back to a safe state.

2. Autonomy function performance: Here, the RTAA is monitoring for correct and/or optimal performance of the autonomy itself. The RTAA must determine if the autonomy functions are, for example, 1) generating correct COAs, including safe, optimal and deconflicted paths, 2) commanding proper asset allocation and reassignment of platform roles, if necessary (e.g., send the vehicle with the most fuel to the furthest mission point, or use the fastest vehicle for the most time-critical objective, etc.), 3) replanning mission objectives accordingly due to unforeseen changes in the environment (inclement weather, observed adversarial threats, etc.), changes in the commander's intent (uploaded changes to mission objectives, etc.) or other unforeseen contingencies, and 4) addressing other relevant mission aspects to maximize mission effectiveness.

PHASE I: This SBIR topic directly aligns with several current Air Force Operational Imperatives involved in autonomy and uncrewed air platforms. Phases I and II directly support R&D efforts in AFRL and larger Air Force efforts involving autonomous collaborative platforms (ACPs). Successful outcomes of Phases I and II will directly support future 6.3 efforts in autonomous combat operations, tactical teaming, advanced contingency management, and runtime assurance.

Phase I should be a feasibility study of proposed solutions, focusing on initial design ideas of architectures and approaches for the RTAA system. The effort should identify technical challenges, risks, and design requirements. The architecture should explicate functional design elements of the integrated components, interface requirements, required communication pathways, and required sensor suites. Functional designs of the two main elements of the RTAA system should be addressed:

RTAA fault determination function: The RTAA system should continually perform information acquisition, gathering relevant information from onboard sensors, information transmitted from local fleetmates, and broadcasted information from tactical and strategic sources (e.g., ground command base, satellites, AWACs, etc.). RTAA subsystems will then perform knowledge extraction, fusing and filtering the gathered information and delivering this knowledge to fault determination functions within the RTAA system.

Although monitoring for platform safety violations will be a critical part of the RTAA system, for this effort, focus should be primarily on determining the autonomy's performance in delivering correct or optimal COAs. Use cases should be developed that cover a range of faults in the autonomy design causing corrupt COA generation and subsequent erroneous actions. Use cases involving contingencies may provide rich scenarios for experimental studies (platform hardware faults, pop-up threats, unforeseen mission changes, etc.). Here, strategies should be constructed to determine if the autonomy is incorrectly responding to contingencies. Strategies may involve fault detection and isolation techniques, heuristic mechanisms, or other formal methods. Accurate determination is critical and the RTAA function should not be producing false alarms, shutting down correctly operating autonomy functions that are responding to off-nominal contingency conditions.

Mission scenarios for use case development may include ISR, patrol, supply delivery, high-value escort, weapon engagement/deployment, enemy suppression, etc. Contested areas of operation may include high-risk zones, no-fly zones, natural and urban terrain, with red team air and ground assets, etc.

RTAA mitigation function: If the RTAA system determines the autonomy is not operating correctly, it will activate recovery procedures to re-acquire safe and correct operations. This may involve simple reversionary functions that bring the platform or mission to a safe state. Here too, Phase I should be a feasibility study of proposed solutions, focusing on initial design ideas. Recovery procedures under various scenarios/context should be outlined and risks assessed. Under what conditions should the mission be abandoned and the fleet commanded to return to base? Or, should a failsafe loiter point be targeted as the autonomy functions are cleared and restarted? Can vehicles continue with the mission under a set of pre-defined COAs commanded in succession?

Phase I objectives: Initial architecture and functional design approaches may be delivered as documented descriptions. Initial, low-order design and analysis studies in desktop simulation environments can be performed to support the proposed set of use cases. A Phase II technology development plan should be completed based on Phase I results.

Respondents can propose use of their own models and simulation environments in Phase I. No government furnished data is required. However, awardees should expect significant interaction and involvement with the Air Force during initial planning for Phase II and beyond to align with specific platforms, architectures and missions of interest to the Air Force.

PHASE II: In Phase II, design architectures should be significantly matured and align with the Air Force's Autonomy-Government Reference Architecture (A-GRA). Technology maturation should be a significant part of the Phase II, with design iteration and testing in higher fidelity desktop simulation environments with representative platform applications and mission scenarios of interest to the Air Force. Develop realistic use cases that exercise the functionality of the RTAA fault detection and subsequent mitigation measures. Benefits of the recovery processes and operation over a wide range of scenarios should be demonstrated. Capstone demonstrations should be constructed showcasing the utility of the technology advancements.

The technology readiness level of the developed products should then be matured further, constructing real-time functionality and testing the developed technologies in a real-time software integration laboratory environment. Repeat capstone experiments that were performed in desktop simulations.

Depending on contractual arrangements, government furnished data or equipment could be provided in the form of simulation models or equipment supporting laboratory bench testing. At this stage, systems used to demonstrate the developed technologies should closely align with Air Force programs of interest that employ advanced intelligent autonomy. Technology transfer plans should be constructed showing how the developed Phase II products can directly support such programs in preparations for Phase III efforts.

PHASE III DUAL USE APPLICATIONS: Phase III efforts should be pursued in Air Force programs, other DoD branches and commercial endeavors. The AFWERX's STRATFI/TACFI programs could serve as potential starting points, supporting identified Air Force customers in, for example, 1) AFRL directorates), 2) AFWERX's Autonomy Prime with application to low-cost unmanned aircraft system test beds, 3) Agility Prime with application to air mobility systems with autonomy, or 4) current and future air wings that focus on platforms and systems driven by autonomy. For example, the 412th Test Wing at Edwards Air Force Base has conducted multiple autonomous flight tests and research projects to advance the capabilities of unmanned aircraft systems.

Proposed activities should directly support these Air Force customers and their programs of interest with advanced technology development and flight testing. Teaming arrangements should be made with airframe/avionics manufacturers to develop/finalize the system designs in a pre-production phase. The Phase II-developed real time code should be ported to flight processors and initial flight demonstrations with surrogate sUAS platforms should be performed, again testing capstone experiments.

Follow-on Phase III programs should focus on final design developments, completely expanding the RTAA products for full envelop operation and integrated with fully matured contingency management and flight and mission autonomy systems. Required V&V, safety analysis and testing for eventual certification should be performed at this stage.

Commercialization efforts should be pursued in parallel, teaming with industry to license the developed code and/or manufacture relevant avionics subsystems. The developed products could support DoD platforms and mission systems of interest, or advanced civilian applications, such as urban/advanced air mobility (UAM/AAM). These vehicles are incorporating non-traditional electric or hybrid propulsion vertical takeoff and landing capabilities (eVTOL/hVTOL). These aircraft are being developed for both manned and unmanned operations, typically utilizing a single onboard pilot, remote pilot, or fully autonomous control. Mission applications include personnel recovery/ delivery, medical evacuation, resupply/distribution, patrol, search and rescue, etc., with applications in law enforcement, civil air patrol, firefighting, disaster/humanitarian relief, border patrol, bridge/building/utility inspections, environmental services, agriculture, etc.

With these applications, trust in the onboard autonomy will be critical. Often the onboard pilot will have limited flight training (e.g., an EMT or first responder) and he/she will not have sufficient experience to correctly respond to complex contingencies that may arise. Fully automatic contingency management integrated with the onboard autonomy will be required. Further, operations over densely populated urban areas will require significant evidence that the autonomy will be bounded to safe/correct actions. RTAA systems will be a key enabling technology to provide this evidence.

RTAA applications should be extended to ground vehicles, self-driving cars, and other autonomous modes of transportation. Other applications may include industrial systems, medical devices, robotic applications and any functions requiring assured intelligent autonomy.

Wherever autonomy is needed, systems that assure that autonomy will always do the right thing will also be needed. RTAA is not only a required technology, it is an enabling technology for future systems driven

by autonomy. Sending unchecked autonomy out into the world will simply not be allowed. Societal regulations will not allow untrusted autonomy to control machinery that can harm people or cause physical damage. RTAA will be required without doubt.

REFERENCES:

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3. Bak, S., et. al. "Using Run-Time Checking to Provide Safety and Progress for Distributed Cyber-Physical Systems," Proc. IEEE International Conference on Embedded and Real-Time Computing Systems and Applications, 2013.
4. Clark, M., Koutsoukos, X., Kumar, R., Lee, I., Pappas, G., Pike, L., Porter, J., Sokolsky, O., "A Study on Run Time Assurance for Complex Cyber Physical Systems," AFRL Final Report, April, 2013.
5. Schierman, J., Ward, D., Dutoi, B., et al., "Run-Time Verification and Validation for Safety-Critical Flight Control Systems," AIAA Paper 2008- 6338, Proceedings of the AIAA Guidance, Navigation, and Control Conference, Honolulu, Hawaii, Aug., 2008.
6. L. R. Hook, W. Ryan, M. A. Skoog and J. Fuller, "Exploring the Potential of Automatic Safety Systems in General Aviation," 2023 IEEE Aerospace Conference, Big Sky, MT, USA, 2023, pp.1-10, doi: 10.1109/AERO55745.2023.10115864.
7. Skoog, M., Hook, L., Ryan, W. (2020). Leveraging ASTM Industry Standard F3269-17 for Providing Safe Operations of a Highly Autonomous Aircraft. 1-7.10.1109/AERO47225.2020.9172434.
8. Hook, L., Skoog, M., Garland, M., Ryan, W., Sizoo, D., VanHoudt, J. (2018). Initial considerations of a multi-layered run time assurance approach to enable unpiloted aircraft. 1-11.10.1109/ AERO. 2018.8396622.
9. Hobbs, K., Mote, M., Abate, M., Coogan, S., Feron, E., "Runtime Assurance for Safety-Critical Systems: An Introduction to Safety Filtering Approaches for Complex Control Systems," IEEE Control Systems Magazine 43 (2), 28-65, 2023.

KEYWORDS: Runtime assured autonomy; mission autonomy; mission contingencies; autonomous collaborative platforms; combat collaborative aircraft; attritable uncrewed aircraft

DAF26BZ01-NV500 TITLE: Autonomous Space Cargo Network (ASCN): AI-Driven Logistics Automation and Digital Twin Integration for Space Mobility

COMPONENT TECHNOLOGY PRIORITY AREA(S): Sustainment & Logistics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with section 3.5 of the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: The objective of this effort is to develop and assess the feasibility of an Autonomous Space Cargo Network (ASCN) centered on Autonomous Mobile Robots (AMRs) to modernize cargo handling and logistics operations at the Space Joint Movement Complex (SJMC) and across the Department of War's (DoW) space mobility enterprise. The ASCN will prioritize robotic mobility, autonomous cargo transport, and modular robotic integration, enabling real-time cargo movement, autonomous load execution, and resilient logistics in contested and commercial environments. Supporting technologies, such as Artificial Intelligence (AI) for task orchestration and digital twins for system modeling, will be used to evaluate AMR coordination, predictive maintenance, and system optimization. This effort will establish the foundational architecture and performance requirements for a scalable, cybersecure, and interoperable logistics framework for future space sustainment missions.

DESCRIPTION: The SJMC is envisioned as the central logistics hub for DoW space operations, supporting rapid deployment, sustainment, and agile mobility. Current cargo handling and logistics processes are heavily manual, lack real-time adaptability, and are not optimized for space-based supply chains or contested logistics environments. To address these capability gaps, the ASCN will deliver a hardware-centric logistics automation platform built around Autonomous Mobile Robots (AMRs). The ASCN will combine autonomous robotics, intelligent decision-support, and digital twin technology to enable full-spectrum cargo management from warehouse to orbital interface while increasing speed, precision, and resilience. This effort will lay the foundation for a modular, scalable space logistics infrastructure aligned with U.S. Space Force (USSF) sustainment strategy. Key capabilities include:

1. Autonomous Cargo Handling & Transport Optimization
- Robotic forklifts, pallet movers, and modular AMRs for autonomous loading/unloading
- Sensor-rich navigation systems for dynamic obstacle avoidance and precision docking
- Fleet coordination for multi-robot cargo movement across terrestrial and orbital logistics nodes
2. AI-Driven Logistics Command & Control
- AI used for task orchestration, load prioritization, and mission responsiveness
- Integration with the Spaceport of the Future's Common Operating Picture (SPOF COP)
3. Machine Learning for Mission Adaptability
- Predictive analytics for resource positioning and contingency planning

- Visibility and orchestration across all classes of supply
4. Commercial & Military Logistics Interoperability
- Compatibility with U.S. Transportation Command (USTRANSCOM), Space Systems Command (SSC), Defense Logistics Agency (DLA), and commercial launch providers
- Joint protocol development for space cargo integration
5. Cybersecure & Resilient Robotics Architecture
- Blockchain-secured logistics tracking and tamper prevention
- Quantum-resistant algorithms and Zero Trust cybersecurity framework

This effort will lay the foundation for a future-ready, modular, and scalable space logistics infrastructure, aligned with U.S. Space Force (USSF) sustainment strategy and capable of supporting both terrestrial and orbital cargo networks.

PHASE I: The objective of Phase I is to develop a conceptual design and functional prototype for an AMR-based ASCN system that integrates robotic cargo handling and logistics optimization for space-focused missions and/or environments. Phase I will focus on system architecture, workflow modeling, early prototyping, and integration requirements—not full system development. Technical approach considerations include:

- Simulate AMR-enhanced cargo workflows within the SJMC, focusing on load planning and routing
- Design and model robotic automation frameworks for autonomous cargo handling and dynamic load stabilization
- Assess integration with DoW logistics platforms and commercial space operations
- Conduct stakeholder engagements to refine system requirements
- Execute preliminary load balance and maneuverability testing using 463L pallets
- Perform energy efficiency and power requirement analyses

Phase I deliverables include:
- Conceptual design document detailing AMR system architecture, and integration pathways.
- Initial prototype demonstration showcasing cargo management logic, load balancing, and sensing capabilities.
- Feasibility study covering integration potential with DoW systems, environmental resilience, scalability, and power/energy assessments.
- Phase II roadmap outlining test campaigns, system modifications for larger payloads, and milestones for operational prototype development.

PHASE II: The Phase II objective is to design, develop, and demonstrate a fully operational prototype of the ASCN capable of autonomous cargo handling, intelligent logistics coordination, and mission adaptability in contested, austere logistics, or space mission focused environments. This phase will validate the system's ability to improve cargo throughput, reduce human intervention, and integrate with both DoW and commercial logistics systems. Technical focus areas include:

- Build and integrate a full-scale ASCN prototype with autonomous robotic handlers, real-time cargo identification, and secure communications.
- Implement advanced AI models for adaptive cargo prioritization, dynamic routing, and autonomous load planning.

- Integrate with existing DoW logistics platforms such as USTRANSCOM, the Spaceport of the Future's Common Operating Procedure (COP) Logistics Module, and commercial systems where applicable.
- Conduct operational testing in representative logistics environments (e.g., Space Launch facilities, Distribution Hubs).
- Collect performance data for load accuracy, handling speed, mission responsiveness, energy usage, and system reliability.
- Demonstrate predictive maintenance and mission adaptability functions under simulated disruption scenarios.

Phase II deliverables include:
- Fully functional AMR-based ASCN prototype
- Operational test campaign report with performance metrics and logistics improvements
- AI performance evaluation for AMR coordination and decision support
- Integration documentation with SSC, DLA, USTRANSCOM, and commercial platforms
- Phase III transition plan, outlining commercialization strategy, scaling pathways, and targeted end-user adoption timelines.

PHASE III DUAL USE APPLICATIONS: For Phase III, military applications include:

- Deploy AMRs across USSF logistics operations for autonomous cargo handling
- Integrate with DoW logistics ecosystems for multi-domain cargo movement
- Provide real-time logistics decision support for Joint Logistics planners
- Enhance sustainment for Agile Combat Employment (ACE) and distributed operations
- Support mission rehearsal and planning via integration with Spaceport of the Future Common Operating Picture (COP) Logistics Module

Commercial applications include:
- Offer AMR-based cargo handling and warehouse robotics to aerospace and space launch sectors
- Enable robotic logistics optimization for commercial supply chains and intermodal hubs
- Deliver cybersecure, AI-managed inventory and transport systems to spaceports and research facilities
- Position AMRs for future lunar and orbital supply chain networks

Transition plan considerations include:
- Military Transition through operational implementation at SSC logistics nodes and USTRANSCOM-managed facilities, with support from DLA for broad sustainment integration.
- Commercial Licensing to logistics automation firms, aerospace manufacturers, and spaceport operators, supported by targeted pilot deployments.
- Technology Integration with enterprise AI/machine learning (ML) platforms used in military logistics and commercial warehouse management systems.

The anticipated Technology Readiness Level (TRL) for each phase is the following:
- Phase I: TRL 3 to 4 – Analytical and laboratory-based proof of concept
- Phase II: TRL 5 to 6 – System/subsystem prototype demonstrated in relevant environment
- Phase III: TRL 7 to 9 – System demonstration in operational environment and full deployment.

REFERENCES:

1. U.S. Space Force. (2023, March). Mission sustainment strategy. Office of the Deputy Chief of Space Operations for Operations, Cyber, and Nuclear (SF/S4O). https://www.dau.edu/sites/default/files/webform/documents/26816/2023_%20USSF%20Mission%20Sustainment%20Strategy%20efile_signatures.pdf.

2. United States Space Force. (2022, December). Space Doctrine Publication 4-0: Sustainment. Space Training and Readiness Command (STARCOM). https://www.starcom.spaceforce.mil/Portals/2/SDP%204-0%20Sustainment%20(Signed).pdf?ver=jFc_4BiAkDjJdc49LmESgg%3D%3D.

KEYWORDS: Autonomous Cargo Logistics for Space; AI-Driven Cargo Management System; Autonomous Space Mobility & Sustainment; Digital Twin for Space Logistics Optimization; Cybersecure AI Logistics Framework

DAF26BZ01-NV501 TITLE: Commercial-Derived Insights for Novel Tactical Surveillance, Reconnaissance, and Tracking (TacSRT) Capabilities

COMPONENT TECHNOLOGY PRIORITY AREA(S): Space Technology

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with section 3.5 of the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: The objective of this Phase I effort is to identify, assess, and demonstrate the feasibility of novel, non–missile-warning space and/or ground enabled sensing and analytic capabilities that can deliver rapid, commercially derived insights with meaningful operational utility. The effort seeks concepts that enhance geospatial tactical awareness, reduce operational risk, and provide operators with timely, relevant, and resilient information in contested environments. Phase I will evaluate scientific and technical feasibility, characterize expected performance, and define the minimum viable capability that can be matured into a rapidly fieldable prototype in Phase II.

DESCRIPTION: This topic seeks to rapidly field non–missile-warning, space and/or ground enabled sensing and analytic capabilities that enhance warfighter decision speed. Space Force Components and Combatant Commands increasingly depend on commercially derived, space and/or ground enabled insights, but existing systems lack the responsiveness, automation, and sensing diversity needed for real-time tactical awareness. Adversary advancements and dynamic operational environments have outpaced traditional acquisition approaches, creating critical gaps that Tactical Surveillance, Reconnaissance, and Tracking (TacSRT) is working to solve.

Aligned with the Space Force Commercial Space Strategy, this topic solicits innovative, unclassified concepts across the sensing-to-analysis continuum—including data collection, phenomenology exploitation, analytic fusion, and information delivery—that can deliver meaningful operational utility within one year. Proposed solutions may introduce new sensing or analytic methods or significantly advance existing commercial approaches. An initial operational capability (IOC) is defined as a functional prototype that provides testable outputs directly to operators.

Solutions may include hardware, software, analytic tools, sensing concepts, data-processing architectures, or integrated workflows. Stand-alone capabilities and service-based models are acceptable, and performers may leverage commercial space-as-a-service or existing commercial space infrastructure. Approaches must deliver timely, operationally relevant insights without requiring government development of new space hardware.

Representative in-scope areas include novel phenomenology sensing, automated exploitation pipelines, multi-sensor fusion, change detection, activity characterization, material or environmental signature analysis, deep maritime or littoral monitoring, rapid-revisit analytics, unconventional sensing approaches, space-to-air or space-to-ground tipping and cueing, high-cadence environmental insight, incorporation of AI/ML, and fusion of structured or unstructured data. Out-of-scope areas include missile warning/tracking, kinetic interceptors, satellite buses, and launch vehicles. The overarching intent is to operationalize commercial capabilities rapidly and ensure warfighters receive meaningful, unique insights at the speed of need. The intent of this effort is not focused on Operational Planning Product (OPP) generation through the Global Data Marketplace but targets a new innovative solution (view Reference 1 for additional context).

PHASE I: Phase I will determine the technical merit, scientific feasibility, and operational applicability of proposed non–missile-warning space-enabled sensing or analytic capabilities. Over a three-month Period of Performance (PoP), performers will identify the core technical approach, characterize expected performance, and validate feasibility through targeted analysis, modeling, simulation, or initial prototype demonstrations. Activities may include:

- Characterizing sensing or analytic methods and defining the minimum viable capability.
- Conducting trade studies, modeling, data analysis, or small-scale experiments.
- Assessing performance in relevant operational scenarios aligned to SRT-supported Components.
- Evaluating pathways for integration into existing commercial-derived workflows or architectures.
- Identifying technical risks, operational constraints, data dependencies, and mitigation strategies.
- Outlining expected capability maturity achievable within a 12-month timeframe, from Phase I contract award through Phase II

Phase I must culminate in:
1. A clear feasibility assessment supported by technical evidence.
2. A Phase II plan aligned to the schedule, including delivery of an initial operational capability and full prototype within the Phase II PoP.
3. Defined operating parameters and anticipated performance metrics.
4. An integration and transition concept demonstrating how the capability can support SRT mission needs.

Performers must demonstrate that Phase II development can begin immediately to ensure continuity and minimizing administrative latency.

PHASE II: Phase II will mature the feasible concepts identified in Phase I into a deployable, testable prototype. As the principal R&D effort, Phase II will include design refinement, system development, integration activities, and operationally relevant testing focused on delivering a measurable improvement in tactical insight or decision advantage. Activities may include:

- Developing and demonstrating a prototype that produces actionable outputs.
- Conducting system integration, to include data pipelines, processing architectures, or delivery mechanisms.
- Evaluating performance across key operating parameters such as latency, persistence, coverage, accuracy, resilience, and usability.

- Testing in relevant or operationally representative environments, including SRT-supported workflows.
- Iterating capability in collaboration with Space Force Components and end users.
- Documenting performance, reliability, and scalability for transition planning.

Success criteria for Phase II include:
- Demonstrated prototype capability that materially improves the sensing or analytic options available to the warfighter.
- Ability to integrate outputs into operational workflows with minimal burden on users.
- Evidence that commercial markets can support long-term sustainment or scaling.
- Achievable path to Phase III transition, including identification of customers, funding mechanisms, and required approvals.

For Phase II proposals, the target should be to develop a working prototype for immediate operational demonstration and transition as soon as possible and adhere to the expected capability maturity timeframe developed in Phase I.

PHASE III DUAL USE APPLICATIONS: Phase III will pursue full transition of the capability to operational use through non-SBIR/STTR funding streams. This phase may include final integration, expanded testing, scaling to larger user bases, or adaptation for additional mission partners. Capabilities entering Phase III should target a Technology Readiness Level (TRL) of 6-7, demonstrating functionality in relevant environments and readiness for operational deployment. Activities may include:

- Full integration into SRT operational workflows or related architectures.
- Transition planning with acquiring organizations, including SYD 810/ETT, and their stakeholders.
- Compliance with security, accreditation, data, and interoperability requirements.
- Engagement with Combatant Commands and Partner Nations for expanded adoption.
- Identifying dual-use opportunities across defense, civil, and commercial markets to strengthen long-term sustainability.

Phase III success is achieved when the capability is fielded at scale, provides recurring operational value, and reduces warfighter risk by delivering timely, commercial-derived insights that expand sensing diversity and strengthen decision advantage.

REFERENCES:

1. https://www.ssc.spaceforce.mil/Newsroom/Article/3791948/space-force-leverages-commercial-data-analytics-to-aid-combatant-commands-in-ne.

KEYWORDS: TacSRT; space sensing; analytics; surveillance; reconnaissance; tracking; terrestrial; space-to-ground; fusion; change detection; downlink; automated exploitation; maritime awareness; environmental monitoring; decision making; situational awareness; AI-ML

DEPARTMENT OF AIR FORCE (DAF)
SMALL BUSINESS INNOVATION RESEARCH (SBIR)
DoW 2026 SBIR BAA Release 1
DIRECT-TO-PHASE-II (D2P2)
PROPOSAL SUBMISSION INSTRUCTIONS

The DAF intends these proposal submission instructions to clarify the Department of War (DoW) Broad Agency Announcement (BAA) as it applies to the topics solicited herein. **Firms must ensure proposals meet all requirements of the SBIR Program BAA posted on the Defense SBIR/STTR Innovation Portal (DSIP) at the proposal submission deadline date/time.**

Applicants are encouraged to thoroughly review the DoW Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- Full component-specific instructions and topic descriptions are available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Please ensure all e-mail addresses listed in the proposal are current and accurate. The DAF is not responsible for ensuring notifications are received by firms changing mailing address/e-mail address/company points of contact after proposal submission without proper notification to the DAF. **If changes occur to the company mail or email addresses or points of contact after proposal submission, the information must be provided to the DAF SBIR/STTR One Help Desk.** The message shall include the subject line, "FY26 SBIR BAA Address Change".

Points of Contact:

- For general information related to the DAF SBIR/STTR program and **proposal preparation instructions**, contact the DAF SBIR/STTR One Help Desk at usaf.team@afsbirsttr.us. All applicants have ample opportunity to request clarifying information. **The DAF encourages applicants to request clarifying information as early as possible, as delays in such requests constrain the DAF's ability to provide satisfactory resolution to applicant concerns.**
- For questions regarding the **DSIP electronic submission system**, contact the DoW SBIR/STTR Help Desk at dodsbirsupport@reisystems.com.
- **For technical questions about the topics** during the pre-announcement and open period, please reference the FY26 SBIR BAA.
- Air Force SBIR/STTR Contracting Officer (CO):
  Mr. Daniel J. Brewer, Daniel.Brewer.13@us.af.mil

General information related to the DAF Small Business Program can be found at the DAF Small Business website, http://www.airforcesmallbiz.af.mil/. The site contains information related to contracting opportunities within the DAF, as well as business information and upcoming outreach events. Other informative sites include those for the Small Business Administration (SBA), www.sba.gov, and the Procurement Technical Assistance Centers (PTACs), http://www.aptacus.us.org. These centers provide Government contracting assistance and guidance to small businesses, generally at no cost.

DIRECT TO PHASE II

15 U.S.C. §638 (cc), as amended by the SBIR AND STTR EXTENSION ACT OF 2022, allows DoD to make a SBIR Phase II award to a small business concern with respect to a project, without regard to whether the small business concern was provided an award under Phase I of an SBIR

program with respect to such project. DAF is conducting a "Direct to Phase II" implementation of this authority for these FY26 SBIR topics and does not guarantee D2P2 opportunities will be offered in future solicitations. Each eligible topic requires documentation to determine whether the feasibility requirement described in the Phase I section of the topic has been met.

DIRECT TO PHASE II PROPOSAL SUBMISSION

The DoW FY26 SBIR Broad Agency Announcement, https://www.dodsbirsttr.mil/submissions/login, includes all program requirements. D2P2 efforts should address the feasibility of a solution to the selected topic's requirements.

The complete proposal must be submitted electronically through DSIP. Ensure the complete technical volume and additional cost volume information is included in this sole submission. The preferred submission format is Portable Document Format (.pdf). Graphics must be distinguishable in black and white. **VIRUS-CHECK ALL SUBMISSIONS.**

INTRODUCTION: D2P2 proposals must follow the steps outlined below:

1. Applicants must create a Cover Sheet in DSIP; follow the Cover Sheet instructions provided in the DoW FY26 SBIR BAA.Applicants must provide documentation satisfying the Phase I feasibility requirement* to be included in the Phase II proposal. Applicants must demonstrate completion of research and development through means other than the SBIR/STTR Programs to establish the feasibility of the proposed Phase II effort based on the criteria outlined in the topic description.
2. Applicants must submit D2P2 proposals using the instructions below.

*NOTE: DAF will not consider the applicant's D2P2 proposal if the applicant fails to demonstrate technical merit and feasibility have been established. It will also not be considered if it fails to demonstrate the feasibility effort was substantially performed by the applicant and/or the principal investigator (PI). Refer to the topics' Phase I descriptions for minimum requirements needed to demonstrate feasibility. Feasibility documentation cannot be based upon or logically extend from any prior or ongoing federally funded SBIR or STTR work. Note, feasibility documentation can be included as part of Volume or Volume 5.

DIRECT TO PHASE II PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS

A. Proposal Requirements. A D2P2 proposal shall provide sufficient information to persuade the DAF the proposed technology advancement represents an innovative solution to the scientific or engineering problem worthy of support under the stated criteria.

B. Proprietary Information. Information constituting a trade secret, commercial/financial information, confidential personal information, or data affecting National Security must be clearly marked. It shall be treated in confidence to the extent permitted by law. Be advised, in the event of proposal selection, the Work Plan will be incorporated into the resulting contract by reference. Therefore, DO NOT INCLUDE PROPRIETARY INFORMATION in the work plan. See the DoW BAA regarding proprietary information marking.

C. General Content. Proposals should be direct, concise, and informative. Type shall be no smaller than 11-point on standard 8 ½ X 11 paper, with one-inch margins and pages consecutivelynumbered. Applicants are discouraged from including promotional and non-programmatic items. If included, such material will count toward the page limit.

DIRECT TO PHASE II PROPOSAL FORMAT

Complete proposals must include all of the following:

Volume 1: Cover Sheet
Note: If selected for funding, the proposal's technical abstract and discussion of anticipated benefits will be publicly released. Therefore, do not include proprietary information in this section.
Volume 2: Technical Volume
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents, e.g. DoD Form 2345 (if applicable), Militarily Critical Data Agreement (if applicable); etc.
Volume 6: Fraud, Waste, and Abuse Training Completion
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

Phase II proposals require a comprehensive, detailed description of the proposed effort. DAF D2P2 efforts are to be proposed in accordance with the information in these instructions. Commercial and military potential of the technology under development is extremely important. Proposals emphasizing dual-use applications and commercial exploitation of resulting technologies are sought.

All D2P2 research or research and development (R/R&D) must be performed by the small business and its team members in the United States, as defined in the DoW SBIR BAA. The Principal Investigator's (PI's) primary employment must be with the small business concern at the time of awardand during the entire period of performance. Primary employment means more than one-half the PI's time is spent in the small business' employ. This precludes full-time employment with another entity. Only one principal investigator/project manager can be designated to a proposal at any given time.

Knowingly and willfully making false, fictitious, or fraudulent statements or representations may be a felony under18 U.S.C. Section 1001, punishable by a fine up to $250,000, up to five years in prison, or both.

Please note the FWA Training must be completed prior to proposal submission. When training is complete and certified, DSIP will indicate completion of the Volume 6 requirement. The proposal cannot be submitted until the training is complete. The DAF recommends completing submission early, as site traffic is heavy prior to solicitation close, causing system lag. **Do not wait until the last minute.** The DAF will not be responsible for proposals not completely submitted prior to the deadline due to system inaccessibility unless advised by DoD. The DAF will not accept alternative means of submission outside of DSIP.

COVER SHEET (VOLUME 1)

Complete the proposal Cover Sheet in accordance with the instructions provided via DSIP. The technical abstract should include a brief description of the program objective(s), a description of the effort, anticipated benefits and commercial applications of the proposed research, and a list of keywords/terms. The technical abstract of each successful proposal will be submitted to the Office of the Secretary of War (OSW) for publication and, therefore, must not contain proprietary or classified information.

TECHNICAL VOLUME (VOLUME 2)

The technical proposal includes all items listed below in the order provided. Refer to topic index for page limitations. Pages in excess of this count will not be considered by the Government in evaluations.

(1) **Table of Contents:** A table of contents should be located immediately after the Cover Sheet.

(2) **Glossary:** Include a glossary of acronyms and abbreviations used in the proposal.

(3) **Milestone Identification:** Include a program schedule with all key milestones identified.

(4) **Identification and Significance of the Problem or Opportunity:** Briefly reference thespecific technical problem/opportunity to be pursued under this effort.

(5) **Phase II Technical Objectives:** Detail the specific objectives of the Phase II work and describe the technical approach and methods to be used in meeting these objects. The proposal should also include an assessment of the potential commercial application for eachobjective.

(6) **Work Plan:** The work plan shall be a separate and distinct part of the proposal package, using a page break to divide it from the technical proposal. It must contain a summary description of the technical methodology and task description in broad enough detail to provide contractual flexibility. The following is the recommended format for the work plan; begin this section on a new page. **DO NOT include proprietary information.**

a) 1.0 – Objective: This section is intended to provide a brief overview of thespecialty area. It should explain the purpose and expected outcome.
b) 2.0 – Scope: This section should provide a concise description of the work to beaccomplished, including the technology area to be investigated, goals, and majormilestones. The key elements of this section are task development and deliverables, i.e., the anticipated end result and/or the effort's product. This section must also be consistent with the information in Section 4.0 below.
c) 3.0 – Background: The applicant shall identify appropriate specifications, standards, andother documents applicable to the effort. This section includes information or explanation for, and/or constraints to, understanding requirements. It may include relationships to previous, current, and/or future operations. It may also include techniques previously determined ineffective.
d) 4.0 – Task/Technical Requirements: The detailed individual task descriptions must be developed in an orderly progression with sufficient detail to establish overall program requirements and goals. The work effort must be segregated into major tasks and identified in separately numbered paragraphs.

Each numbered major task should delineate the work to be performed by subtask. The work plan MUST contain every task to be accomplished in definite, realistic, and clearlystated terms. Use "shall" whenever the work plan expresses a binding provision. Use "should" or "may" to express a declaration or purpose. Use "will" when no contractor requirement is involved, i.e., "... power will be supplied by the Government."

(7) **Deliverables:** Include a section clearly describing the specific sample/prototype hardware/ software to be delivered, as well as data deliverables, schedules, and quantities. Be aware of the possible requirement for unique item identification IAW DFARS 252.211-7003, ItemIdentification and Valuation, for hardware. If hardware/ software will be developed but not delivered, provide an explanation. At a minimum, the following reports will be required under ALL Phase II contracts.

a) Scientific and Technical Reports: Rights in technical data, including software, developed under the terms of any contract resulting from a SBIR Announcement generally remain with the contractor. The Government obtains SBIR/STTR data rights in all data developed or generated under the SBIR/STTR contract for a period of 20 years, commencing at contract award. Upon expiration of the 20-year SBIR/STTR license, the Government has Government purpose rights to the SBIR data.

i. Final Report: The first page of the final report will be a single-page project summary, identifying the work's purpose, providing a brief description of the effort accomplished, and listing potential result applications. The summary may be published by DoD. Therefore, it must not contain any proprietary or classified information. The

remainder of the report should contain details of project objectives met, work completed, results obtained, and technical feasibility estimates.

ii. Status Reports: Status reports are due quarterly at a minimum.

b) Additional Reporting: AF may require additional reporting documentation including:

i. Software documentation and users' manuals;
ii. Engineering drawings;
iii. Operation and maintenance documentation
iv. Safety hazard analysis when the project will result in partial ortotal development and delivery of hardware; and
v. Updates to the commercialization results.

(8) **Related Work:** Describe significant activities directly related to the proposed effort, including any previous programs conducted by the Principal Investigator, proposing firm, consultants, or others, and their application to the proposed project. Describe how these activities interface with the proposed project and discuss any planned coordination with outside sources. Also list any applicant-identified subject matter experts, regardless of affiliation, providing comments regarding the applicant's knowledge of the state-of-the-art in the specific approach proposed. Describe previous work not directly related to the proposed effort but similar. Provide the following:

a. Short Description
b. Client for which work was performed (including individual to be contacted and phone number)
c. Date of completion

(9) <u>Commercialization Potential</u>:

a) The DoD requires a commercialization plan be submitted with the Phase II proposal,specifically addressing the following questions:

i. What is the first planned product to incorporate the proposed technology?
ii. Who are the probable customers, and what is the estimated market size?
iii. How much money is needed to bring this technology to market and how will it be raised?
iv. Does your firm have the necessary marketing expertise and, if not, how will your firm compensate?
v. Who are the probable competitors, and what price/quality advantage is anticipated by your firm.

b) The commercialization strategy plan should briefly describe the commercialization potential for the proposed project's anticipated results, as well as plans to exploit it. Commercial potential is evidenced by:

i. The existence of private sector or non-SBIR/STTR Governmentalfunding sources demonstrating commitment to Phase II efforts/results.
ii. The existence of Phase III follow-on commitments for the research subject.
iii. The presence of other indicators of commercial technology potential,including the firm's commercialization strategy.

c) If awarded a D2P2, the awardee will be required to update periodically the commercialization results of the project via SBA. These updates will be required at completion of the effort, and subsequently when the contractor submits a new SBIR/STTR proposal to DoW. Firms not submitting a new proposal to DoW will be requested to provide updates annually after the D2P2 completion.

d) Note, the "Commercialization Plan" and the "Company Commercialization Report" are distinct documents. The Company Commercialization Report (CCR) comprises Volume 4 as separately indicated in these instructions.

(10)**<u>Relationship with Future R/R&D Efforts</u>**:

a) State the anticipated results of the proposed approach, specifically addressing plans for Phase III, if any.
b) Discuss the significance of the D2P2 effort in providing a basis for the Phase III R/R&D effort, if planned.

D. **<u>Key Personnel</u>:** In the technical volume, identify all key personnel involved in the project. Include information directly related to education, experience, and citizenship. Atechnical resume for the Principal Investigator, including publications, if any, must also be included. Concise technical resumes for subcontractors and consultants, if any, are also useful. Identify all non-U.S. citizens expected to be involved in the project as direct employees, subcontractors, or consultants. For these individuals, in addition to technical resumes, please provide countries of origin, type of visas or work permits held, and identify the tasks they are anticipated to perform.

Foreign Nationals (also known as Foreign Persons) means any person who is NOT:

a. a citizen or national of the United States; or

b. a lawful permanent resident; or
c. a protected individual as defined by 8 U.S.C. § 1324b

ALL applicants proposing to use foreign nationals MUST follow theFY26 SBIR BAA and disclose this information regardless of whether the topic is subject to ITAR restrictions.

When the topic area is subject to export control, these individuals, if permitted to participate, are limited to work in the public domain. Further, tasks assigned must not becapable of assimilation into an understanding of the project's overall objectives. This prevents foreign citizens from acting in key positions, such as Principal Investigator, Senior Engineer, etc. Additional information may be requested during negotiations in order to verify foreign citizens' eligibility to perform on a contract awarded under this BAA.

The following will apply to all projects with military or dual-use applications developing beyond fundamental research (basic and applied research ordinarily published and sharedbroadly within the scientific community):

(1) The Contractor shall comply with all U. S. export control laws and regulations, including the International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, andthe Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, in the performance of this contract. In the absence of available license exemptions/exceptions, the Contractor shall be responsible for obtaining the appropriate licenses or other approvals, if required, for exports of (including deemed exports) hardware, technical data,and software, or for the provision of technical assistance.
(2) The Contractor shall be responsible for obtaining export licenses, if required, before utilizing foreign persons in the performance of this contract, including instances where thework is to be performed on-site at any Government installation (whether in or outside the
United States), where the foreign person will have access to export-controlled technologies,including technical data or software.
(3) The Contractor shall be responsible for all regulatory record keeping requirements associated with the use of licenses and license exemptions/exceptions.
(4) The Contractor shall be responsible for ensuring that these provisions apply to its subcontractors.

E. **Facilities/Equipment:** Describe instrumentation and physical facilities necessary and available to carry out the D2P2 effort. Justify equipment to be purchased (detail in cost proposal). State whether proposed performance locations meet environmental laws and regulations of Federal, state, and local Governments for, but not limited to, airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulkwaste disposal practices, and handling and storage of toxic and hazardous materials.

F. **Consultants/Subcontractors**: Private companies, consultants, or universities may be involved in the project. All should be described in detail and included in the cost proposal. In accordance with the Small Business Administration (SBA) SBIR Policy Directive, a minimum of 50% of the R/R&D must be performed by the proposing firm, unless otherwise approved in writing by the Contracting Officer. These

requests can only be made upon proposal submission. Signed copies of all consultant or subcontractor letters of intent must be attached to the proposal. These letters should briefly state the contribution or expertise being provided. Include statements of work and detailed cost proposals. Include information regarding consultant or subcontractor unique qualifications. Subcontract copies and supporting documents do not count against the Phase II page limit. Identify any subcontract/consultant foreign citizens per E above.

G. **Prior, Current, or Pending Support of Similar Proposals or Awards:**

WARNING: While it is permissible, with proper notification, to submit identical proposals or proposals containing a significant amount of essentially equivalent work forconsideration under numerous Federal program solicitations, it is unlawful to enter into contracts or grants requiring essentially equivalent effort. Any potential for this situation must be disclosed to the solicitation agency(ies) before award. If a proposal submitted in response to this BAA is substantially the same as another proposal previously, currently, or in the process of being funded by another Federal agency/DoD Component or the DAF, the applicant must so indicate on the Cover Sheet and provide the following:

a) The name and address of the Federal agency(ies) or DoD Component(s) towhich proposals were or will be submitted, or from which an awarded is expected or has been received;
b) The proposal submission or award dates;
c) The proposal title;
d) The PI's name and title for each proposal submitted or award received; and
e) Solicitation(s) title, number, and date under which the proposal was or will besubmitted, or under which an award is expected or has been received.
f) If award was received, provide the contract number.
g) Specify the applicable topics for each SBIR proposal submitted or award received.

NOTE: If this section does not apply, state in the proposal, "No prior, current, or pending support for proposed work."

COST VOLUME (VOLUME 3)

A detailed cost proposal must be submitted. Cost proposal information will be treated as proprietary. Proposed costs must be provided by both individual cost element and contractor fiscal year (FY) in sufficient detail to determine the basis for estimates, as well as the purpose, necessity, and reasonableness of each. This information will expedite award if the proposal is selected. Generally, Firm-Fixed-Price contracts are appropriate for Phase II awards. In accordance with the SBA SBIR/STTR Policy Directive, Phase II contracts must include profit or fee.

Cost proposal attachments do not count toward proposal page limitations. The cost proposalincludes:

a) **Direct Labor:** Identify key personnel by name, if possible, and labor category, if not. Direct labor hours, labor overhead, and/or fringe benefits, and actual hourly rates for each individual are also necessary for the CO to determine whether these hours, fringe rates, and hourly rates are fair and reasonable.

b) **Direct Cost Materials:** Costs for materials, parts, and supplies must be justified and supported. Provide an itemized list of types, quantities, prices, and, where appropriate, purpose. If computer or software purchases are planned, detailed information such as manufacturer, price quotes, proposed use, and support for theneed will be required.

c) **Other Direct Costs:** This includes specialized services such as machining or milling, special test/analysis, and costs for temporary use/lease of specialized facilities/ equipment. Provide usage (hours) expected, rates, and sources, as well as brief discussion concerning the purpose and justification. Proposals including leased hardware must include an adequate lease versus purchase rationale.

d) **Special Tooling, Special Test Equipment, and Material:** The inclusion of equipment and materials will be carefully reviewed relative to need and appropriateness to the work proposed. Special tooling and special test equipment purchases must, in the CO's opinion, be advantageous to the Government and relate directly to the effort. These toolings or equipment should not be of a type that an applicant would otherwise possess in the normal course of business. These may include such items as innovative instrumentation and/or automatic test equipment.

e) **Subcontracts:** Subcontract costs must be supported with copies of subcontract agreements. Agreement documents must adequately describe the work to be performed and cost bases. The agreement document should include a SOW, assignedpersonnel, hours and rates, materials (if any), and proposed travel (if any). A letter from the subcontractor agreeing to perform a task or tasks at a fixed price is not considered sufficient. The proposed total of all consultant fees, facility leases or usage fees, and other subcontract or purchase agreements may not exceed one-half of the total contract price, unless otherwise approved in writing by the Contracting Officer.

The prime contractor must accomplish price analysis, including reasonableness, of the proposed subcontractor costs. If based on comparison with prior efforts, identify the basis upon which the prior prices were determined reasonable. If price analysis techniques are inadequate or the FAR requires subcontractor cost or pricing data submission, provide a cost analysis. Cost analysis includes but is not limited to, consideration of materials, labor, travel, other direct costs, and proposed profit rates.

f) **Consultants:** For each consultant, provide a separate agreement letter briefly stating the service to be provided, hours required, and hourly rate, as well as ashort, concise resume.

g) **Travel:** Each effort should include, at a minimum, a kickoff or interim meeting. Travel costs must be justified as required for the effort. Include destinations, numberof trips, number of travelers per trip, airfare, per diem, lodging, ground transportation, etc. Per Diem and lodging rates may be found in the Joint Travel Regulation (JTR),Volume 2, www.defensetravel.dod.mil.

h) **Indirect Costs:** Indicate proposed rates' bases, e.g., budgeted/actual rates per FY, etc. The proposal should identify the specific rates used and allocation bases to which they are applied. Do not propose composite rates; proposed rates and applications per FY throughout the anticipated performance period are required.

i) **Non-SBIR Governmental/Private Investment:** Non-SBIR Governmental and/or private investment is allowed. However, it is not required, nor will it be a proposal evaluation factor.

NOTE: If no exceptions are taken to an applicant's proposal, the Government may award a contract without exchanges. Therefore, the applicant's initial proposal should contain the applicant's best terms from a cost or price and technical standpoint. If there are questions regarding the award document, contact the Phase I CO identified on the cover page. The Government reserves the right to reopen exchanges later if the CO determines doing so to be necessary.

COMPANY COMMERCIALIZATION REPORT (VOLUME 4)

Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW FY26 SBIR BAA for full details on this requirement. Information contained in the CCR will not be considered by the DAF during proposal evaluations.

SUPPORTING DOCUMENTS VOLUME (VOLUME 5)

The following document is required for all applicants under this release:

1. The Regulatory Compliance Initial Safety Assessment must be completed and submitted as part of offeror's Volume 5 proposal submission. The required document and version can be located here. Do not submit previous versions of this document.

The following documents may be required if applicable to your proposal:

1. DD Form 2345: For proposals submitted under export-controlled topics either International Traffic in Arms or Export Administration Regulations (ITAR/EAR), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website, http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP/DD2345Ins tructions.aspx. DD Form 2345 approval will be required if proposal if selected for award.
2. Verification of Eligibility of Small Business Joint Ventures (Attachment 3 to the DOW SBIR FY26 BAA)
3. Technical Data Rights Assertions (if asserting data rights restrictions)

Feasibility Documentation (required for all proposal submissions, contained within Volume 5, not subject to page limitations)

1. D2P2 proposals require a comprehensive, detailed effort description. Proposals should demonstrate sufficient technical progress or problem-solving results to warrant more extensive RDT&E. Developing technologies with commercial and military potential is extremely important. Particularly, DAF is seeking proposals emphasizing technologies' dual-use applications and commercialization.
2. * NOTE: The applicant shall provide information to enable the agency to make the 15 U.S.C. 638(cc) determination of scientific and technical feasibility and merit. Applicants are required to provide information demonstrating scientific and technical merit and feasibility has been established. The DAF will not review the Phase II proposals if it is determined the applicant 1) fails to demonstrate technical merit and feasibility are established or 2) the feasibility documentation does not support substantial performance by the applicant and/or the PI. Refer to the Phase I description within the topic to review the minimum requirements needed to

demonstrate scientific and technical feasibility. **Feasibility documentation cannot be based upon or logically extend from any prior or ongoing federally funded SBIR or STTR work.**

3. If appropriate, include a reference or works cited list as the last page.
4. Feasibility efforts detailed must have been substantially performed by the applicant and/or the PI. If technology in the feasibility documentation is subject to intellectual property (IP) rights, the applicant must provide IP rights assertions. Additionally, applicants shall provide a short summary for each item asserted with less than unlimited rights describing restriction's nature and intellectual property intended for use in the proposed research. Please see FY26 SBIR BAA for technical data rights information.
5. DO NOT INCLUDE marketing material. Marketing material will NOT be evaluated.

FRAUD, WASTE AND ABUSE TRAINING (VOLUME 6)

Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

DISCLOSURES OF FOREIGN AFFILIATIONS OR RELATIONSHIPS TO FOREIGN COUNTRIES (VOLUME 7)

Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoW Program BAA.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)

The DAF does not participate in the Discretionary Technical and Business Assistance (TABA) Program. Proposals submitted in response to DAF topics should not include TABA.

METHOD OF SELECTION AND EVALUATION CRITERIA

D2P2 proposals are evaluated on a competitive basis by subject matter expert scientists, engineers, or other technical personnel. Throughout evaluation, selection, and award, confidential proposal and evaluation information will be protected to the greatest extent possible. D2P2 proposals will be disqualified and not evaluated if the Phase I equivalency documentation does not establish the proposed technical approach's feasibility and technical merit.

Proposals will be evaluated for overall merit in accordance with the criteria outlined in the FY26 SBIR BAA Section 6.0. DAF is seeking varying technical/scientific approaches and/or varying and new technologies that would be responsive to the problem statement(s) and area(s) of interest in the topic. Multiple procurements are planned and anticipated to be awarded as a result of the topic, each proposal is considered a separate procurement and will be evaluated on its own merit, and that the Government may award all, some, or none of the proposals. Any per-award or per-topic funding caps are budgetary estimates only, and more or less funding may become available. Funding decisions are made with complete disregard to the other awards under the same topic.

In accordance with 15 USC 638(vv) (Section 4 of the SBIR and STTR Extension Act of 2022), and the Deputy Secretary of Defense Memorandum; Subject: Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program dated May 13, 2024, the DAF will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. The DAF will use information provided by the small business concern in response to the Disclosure of Foreign Affiliations or Relationships to Foreign Countries and the proposal to conduct a risk-based due diligence review on the cybersecurity practices, patent analysis,

employee analysis, and foreign ownership of a small business concern, including the small business concern and employees of the small business concern to a foreign country, foreign person, foreign affiliation, or foreign entity. The DAF will also assess proposals utilizing open-source analysis and analytical tools, for the nondisclosures of the information set forth in 15 U.S.C. 638(g)(13). If DAF assesses that a small business concern has security risk(s), DAF will review the proposal, the evaluation, and the security risks and may decide not to select the proposal for award based upon a totality of the review.

MAJORITY OWNERSHIP IN PART BY MULTIPLE VENTURE CAPITAL, HEDGE FUND, AND PRIVATE EQUITY FIRMS

Small business concerns that are owned in majority part by multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds are not eligible to submit applications or receive awards for DAF Topics.

PERFORMANCE OF WORK REQUIREMENTS AND LOCATION OF WORK

For Phase I, a minimum of two-thirds of the research or analytical effort must be performed by the Awardee. The DAF measures percentage of work by both direct and indirect costs, not including profit. Occasionally, the DAF will consider deviations from this performance of work requirement. **Requests for Performance of Work deviations must be made twice: prior to submission during the topic open period and as part of the initial proposal submission.** For requests prior to the initial proposal submission, the DAF will consider the request and approve or disapprove requesting applicants to proceed with DSIP submission. Upon proposal receipt, the DAF will again consider such requests for approval for the resultant award.

All R/R&D work must be performed in the United States. Based on a rare and unique circumstance, the DAF may approve a particular portion of the R/R&D work to be performed or obtained in a country outside of the United States. The awarding Funding Agreement officer must approve each specific condition in writing. Applicants seeking this approval must make such a request with their initial proposal submission. The DAF will not consider these requests prior to proposal submission.

DAF USE OF SUPPORT CONTRACTORS

Restrictive notices notwithstanding, proposals may be handled for administrative purposes only, by support contractors. These support contractors may include, but are not limited to APEX, Peerless Technologies, Engineering Services Network, HPC- COM, Mile Two, REI Systems, MacB (an Alion company), Montech, Oasis, Astrion/Oasis, and Infinite Management Solutions. In addition, only Government employees and technical personnel from Federally Funded Research and Development Centers (FFRDCs) MITRE and Aerospace Corporations working under contract to provide technical support to AF Life Cycle Management Center and Space Force may evaluate proposals. All support contractors are bound by appropriate non-disclosure agreements. **Contact the DAF SBIR/STTR Contracting Officer (Daniel.Brewer.13@us.af.mil) with concerns about any of these contractors**.

PROPOSAL STATUS AND FEEDBACK

The Principal Investigator (PI) and Corporate Official (CO) indicated on the Proposal Cover Sheet will be notified by e-mail regarding proposal selection or non-selection. Small Businesses will receive a notification for each proposal submitted. Please read each notification carefully and note the Proposal Number and Topic Number referenced.

Automated feedback will be provided for proposals designated Not Selected. Additional feedback may be provided at the sole discretion of the DAF.

IMPORTANT: Proposals submitted to the DAF are received and evaluated by different organizations, handled by topic. Each organization operates within its own schedule for proposal evaluation and selection. Updates and notification timeframes will vary. If contacted regarding a proposal submission, it is not necessary to request information regarding additional submissions. Separate notifications are provided for each proposal.

DAF anticipates that all proposals will be evaluated and selections finalized within approximately 90 calendar days of solicitation close. Please refrain from contacting the DAF for proposal status before that time.

Refer to the FY26 SBIR BAA for procedures to protest the Announcement.
As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: Air Force SBIR/STTR Contracting Officer Daniel J. Brewer, Daniel.Brewer.13@us.af.mil and to the individual PCO listed on your firm's selection notification.

DAF SUBMISSION OF FINAL REPORTS

All Final Reports will be submitted to the awarding DAF organization in accordance with Contract instructions. Companies will not submit Final Reports directly to the Defense Technical Information Center (DTIC).

DAF
DoW 2026 SBIR BAA
Direct to Phase II Topic Index
Release 1

Topic Number	Topic Name	Award Maximum Value*	Maximum Duration (in months)**	Maximum Technical Volume (Volume 2) Page/Slide Limit***
DAF26BZ01-DV001	Homeland Convoy Counter Drone Operations	$ 2,000,000.00	15	35
DAF26BZ01-DV002	Autonomous Leader-Follower UAS Formation for Enhanced Mission Resilience and Reduced Operator Workload	$ 2,000,000.00	15	35
DAF26BZ01-DV005	AI Framework for Multimodal Scene Construction and Data Generation	$ 2,000,000.00	24	35
DAF26BZ01-DV007	CHORD - Collaborative Human Autonomy Operational Review	$ 1,700,000.00	21	35

*Proposals in excess of this amount will not be considered for award.
**Proposals in excess of this duration will not be considered for award. Note that this duration includes both “technical” and “reporting” periods.
***Pages/slides in excess of this number will not be considered during proposal evaluations.

DAF26BZ01-DV001 TITLE: Homeland Convoy Counter Drone Operations

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a system that minimizes the threat a UAS poses to military convoys in non-permissive environments that have the most restrictive rules of engagement.

DESCRIPTION: The United States Military and its US government interagency partners have requirements to move personnel and equipment on public roads that are at risk of surveillance and improvised attacks by drones. The open nature of our free society provides anonymity to our fellow citizens and potential bad actors that makes UAS defense along public roads very challenging. The legal and beneficial use of UAS complicates the problem that all UAS in vicinity of US military operations in the homeland cannot be determined to be a threat by their presence alone.
Convoys require a system that can identify UAS operations that either follow, or are on a course for physical intercept, of the vehicles in the convoy and have the ability to disrupt the flight of that UAS to ensure safe passage. While kinetic (Mass striking Mass) solutions are legal they must be a final measure and must be employed in a way that minimizes risk to US citizens in vicinity of the convoy. This is especially difficult when a bad actor uses the cover of an urban or suburban area to employ their system to maximize clutter to sensors and complicate the engagement process (Rules of engagement and establishment of intent) of kinetic solutions.
The system is expected to detect UAS around the convoy and establish their trajectories with respect to the convoy. Human teaming with the detection system should be intuitive and cognitively low effort, similar to an alarm system in a security operations center. This system should have the ability to interfere with the UAS operations using Electronic Warfare or other non-kinetic means and have the ability to employ kinetic effectors that produce effects that have a low probability of collateral damage.
The system can require its own vehicle for proof of concept but must have accommodations to be an add-on to an existing vehicle in the convoy. Proposals can be partial solutions to this problem but must have detailed and reasonable explanations on what would be expected of the portions of the systems that are not within the scope of their proposal. The system must be Software Open Systems Architecture (SOSA) compliant so the requirement owners can add or complement successful offeror's solutions with GOTS software and hardware. The Medusa GOTS MOSA/SOSA government reference architecture (GRA) is the planned target and will be available to any performer, post selection, if requested. The GRA uses the Open Mission Networking Interface (OMNI) and the New Architecture Technology Server (NATS) publish and subscribe message system, and could be integrated with a core part of the offeror's development. Making some assumptions or reserving some details for how the GRA will be part of the effort is acceptable as few additional details will be provided prior to contract award. Proposals will not be selected if they are closed and fully proprietary, or are architectures that do not allow other technologies development by the US government and their contractors to be added to the system.

PHASE I: This is a Direct to Phase 2 (D2P2) topic. Phase 1 like proposals will not be evaluated and will be rejected as nonresponsive. For this D2P2 topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort via some other means (e.g. IRAD, or other funded work). It must have developed a concept for a workable prototype or design to address at a minimum the basic capabilities of the stated objective above. Proposal must show, as appropriate to the proposed effort, a demonstrated technical feasibility or nascent capability to meet the capabilities of the stated objective. Proposal may provide example cases of this new capability on a specific application. The documentation provided must substantiate that the proposer has developed a preliminary understanding of the technology to be applied in their Phase II proposal to meet the objectives of this topic. Documentation should include all relevant information including, but not limited to, technical reports, test data, prototype designs/models, and performance goals/results.

PHASE II: Develop a system that minimizes the threat a UAS poses to military convoys in non-permissive environments that have the most restrictive rules of engagement.

i. Develop or use Medusa GRA software architecture that connects sensors, effectors and microservices
ii. Develop or select sensors that provide awareness of UAS around convoys that may travel as fast as 70mph
iii. Design a system that takes sensor data and uses it to determine the presences and trajectory with respect to the convoy
iv. Design a defeat system that uses EW, Kinetic (low collateral options) and other options that have the ability to change the UAS's flight path
v. Prototype a complete system and its vehicle as required to do a demonstration of the system
vi. Design and develop (or adopt) a SOSA with the details needed to prove its compliance with DoD goals
vii. Produce documentation of the system and its internal and external interfaces
viii. Demonstrate the system at a location of your choice or in collaboration (if available) with any US military or AFRL activity that can be negotiated after award

Complete the design of the system, demonstrate performance of a prototype system through field testing, and deliver the prototype for subsequent evaluation by the government.

PHASE III DUAL USE APPLICATIONS: The Government has an interest in transition of the demonstrated concept to airfield operations and cargo delivery, but offer options for weapons loading and other aerial port operations in both austere and well-supported locations. Solutions may have application to commercial air cargo operations, warehouse material handling operations, and construction.

REFERENCES:

1. Department of the Air Force Operational Imperatives, https://www.af.mil/Portals/1/documents/2023SAF/OPERATIONAL_IMPARITIVES_INFOGRAPHIC.pdf
2. Schroeder, C. Martinez, G. Galloway; USAF Pallet and Dunnage System Evaluation, 1997 https://apps.dtic.mil/sti/pdfs/ADA356785.pdf
3. Management and control of Intermodal Containers and System 463L Equipment, Defense Transportation Regulation – Part VI https://www.ustranscom.mil/dtr/part-vi/dtr_part_vi_608.pdf

KEYWORDS: Radar; Sensing on the move; Signal analysis; day/night cameras; Machine Learning; Artificial Intelligence; Lidar; Low Collateral Effects Interceptors; Racing Quadrotors; Common Launch Tube; April Tags; Video Stabilization; GPS Jamming;

DAF26BZ01-DV002 TITLE: Autonomous Leader-Follower UAS Formation for Enhanced Mission Resilience and Reduced Operator Workload

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: The objective of this topic is to develop and demonstrate an affordable robust and reliable autonomous leader-follower UAS formation capability, enabling a single First-Person View (FPV) pilot to command and control multiple UAS effectively. This system should incorporate: seamless pilot reassignment in case of lead UAS failure, synchronized terminal guidance capabilities, and an innovative stasis mode for follower units to conserve energy and maintain position. Furthermore, the system must include advanced target designation features, allowing the lead UAS to mark targets for autonomous execution by follower units

DESCRIPTION: The increasing complexity and scale of modern military operations demand unmanned aerial systems (UAS) capable of operating autonomously and collaboratively. Current UAS deployments often require dedicated operators for each platform, resulting in high personnel costs and increased cognitive burden on the warfighter. There is a critical need for UAS technologies that can significantly reduce operator workload while enhancing mission effectiveness and resilience, particularly in contested environments where communication and control links may be degraded or disrupted. The ability for a single pilot to effectively manage multiple autonomous UAS in a coordinated formation, with built-in redundancy and adaptive control, represents a significant advancement in UAS capabilities.
Offerors are encouraged to explore innovative approaches to autonomous UAS formation control, incorporating advanced AI algorithms, resilient communication networks (e.g., Neuron Mesh Networks), and robust sensor fusion techniques. The proposed solution should address challenges related to maintaining formation integrity, adapting to dynamic environments, and operating in GPS-denied or contested environments. Innovative approaches to stasis mode are encouraged, optimizing power consumption while maintaining situational awareness. Development should include:

- AI-based Autonomous Control Algorithms: For leader-follower formation flight, obstacle avoidance, and target engagement.
- Resilient Communication Network: A robust and secure communication network enabling seamless data sharing and control within the UAS formation (potentially leveraging Neuron Mesh Network technologies).
- Synchronized Terminal Guidance: Algorithms for coordinated approach and landing of multiple UAS at designated targets.
- Stasis Mode: An energy-efficient mode allowing follower UAS to maintain position and situational awareness while minimizing power consumption.
- Target Designation System: A user-friendly interface for the pilot to designate targets for autonomous execution by follower units.
- Pilot Reassignment Capability: A mechanism for automatic and seamless transfer of lead UAS control to a follower unit in case of failure or loss of communication

PHASE I: This is a Direct to Phase 2 (D2P2) topic. Phase 1 like proposals will not be evaluated and will be rejected as nonresponsive. For this D2P2 topic, the Government expects that the small business will have accomplished the following in a Phase I-type effort via some other means (e.g. IRAD, or other funded work). It must have developed a concept for a workable prototype or design to address at a minimum the basic capabilities of the stated objective above. The proposal must show, as appropriate to the proposed effort, a demonstrated technical feasibility or nascent capability to meet the capabilities of the stated objective. The proposal may provide example cases of this new capability on a specific application. The documentation provided must substantiate that the proposer has developed a preliminary understanding of the technology to be applied in their Phase II proposal to meet the objectives of this topic. Documentation should include all relevant information including, but not limited to technical reports, test data, prototype designs/models, and performance goals/results

PHASE II: This topic seeks to build upon prior research and development demonstrating the feasibility of autonomous UAS formation control and AI-based pilot assistance. Specifically, prior work has demonstrated the ability to control multiple simulated UAS with a single operator using AI-powered flight management systems. This Phase II effort will focus on transitioning these algorithms to real-world UAS platforms, integrating them with a resilient communication network, and validating their performance in realistic operational environments. The mid-term field test with a minimum of 3 UAS is a critical milestone demonstrating the advancement of the technology towards TRL 6.
Phase II Deliverables:

- A fully functional and integrated prototype of the autonomous leader-follower UAS formation system.
- A comprehensive technical report documenting the system's architecture, algorithms, performance, and limitations.
- A detailed user manual for the system, including instructions for setup, operation, and maintenance.
- A detailed transition strategy outlining the steps required to transition the technology to operational use.
- Demonstration of the system's capabilities in relevant operational environments.

Desired Key Performance Parameters (KPPs):

- Number of UAS controlled by a single pilot: Minimum 3 UAS
- Formation maintenance accuracy: < 1 meter deviation from designated position.
- Pilot reassignment time: < 5 seconds.
- Terminal guidance accuracy: < 1 meter CEP (Circular Error Probable).
- Stasis mode power consumption: < 10% of normal operating power.
- Target designation accuracy: < 0.5 meter deviation from designated target coordinates.
- Communication range: > 5 kilometers.
- Autonomous operation time (without pilot intervention): > 30 minutes.

Special Considerations: Consideration should be given to cybersecurity vulnerabilities and mitigation strategies.

PHASE III DUAL USE APPLICATIONS: Beyond military applications, this technology offers significant dual-use potential. Civilian applications include:

- Search and Rescue: Coordinated UAS teams can rapidly survey disaster areas, locate survivors, and deliver aid.
- Precision Agriculture: Multiple UAS can monitor crop health, optimize irrigation, and apply targeted treatments, improving yields and reducing resource consumption.
- Infrastructure Inspection: Autonomous formations can inspect bridges, pipelines, and power lines more efficiently and safely than traditional methods.

- Security and Surveillance: Coordinated UAS teams can provide enhanced security for critical infrastructure, large events, and border patrol operations.

The core technologies developed in this SBIR have broad applicability across both defense and commercial sectors, fostering economic growth and societal benefit.

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2. 2021 U.S. Department of Defense Counter-Small Unmanned Aircraft Systems Strategy https://media.defense.gov/2021/Jan/07/2002561080/-1/-1/1/DEPARTMENT-OF-DEFENSE-COUNTER-SMALL-UNMANNED-AIRCRAFT-SYSTEMS-STRATEGY.PDF
3. Department of the Air Force Operational Imperatives, https://www.af.mil/Portals/1/documents/2023SAF/OPERATIONAL_IMPARITIVES_INFOGRAPHIC.pdf

KEYWORDS: Autonomous UAS; Leader-Follower Formation; AI-Powered Control; Resilient Communication; Terminal Guidance; Contested Environments

DAF26BZ01-DV005 TITLE: AI Framework for Multimodal Scene Construction and Data Generation

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: The objective is to develop a capability for generating geo-specific, sensor-independent scenes for multimodal (RF and EO/IR) synthetic data generation by leveraging geo-spatial information, time-of-day, seasonal data, and measured databases, overcoming limitations in existing models and radiometric data.

DESCRIPTION: The DoD requires large-scale, high-fidelity background scenes to advance autonomous systems and Artificial Intelligence and Machine Learning (AI/ML) capabilities. These scenes are critical for providing realistic, context-rich environments that enable AI/ML and/or autonomous systems to learn, adapt, and perform effectively in real-world, dynamic conditions. A critical component of this effort is the ability to generate dynamic, high-fidelity background scenes that realistically model operational environments. Unlike traditional synthetic data generation, which often focuses on isolated sensor outputs, scene generation must create a coherent, interactive world where autonomous agents can navigate, perceive, and process imagery based on their movement and decision-making.
This presents several challenges. First, scene generation requires accurate modeling of complex environmental factors such as terrain variation, urban structures, vegetation, weather conditions, and electromagnetic propagation—all of which impact sensor performance. Additionally, ensuring spatial and temporal consistency across multimodal data (e.g., RF and EO/IR) is far more demanding than simply generating independent synthetic datasets. Autonomous systems rely on their ability to interpret changes in the environment dynamically, requiring realistic physics-based interactions between sensors and the scene. Further, aligning RF and EO/IR perspectives within the same scenario for sensor fusion introduces an added layer of complexity, demanding precise calibration of sensor viewpoints, occlusions, and atmospheric effects.

To accurately model such complex environments, scene generation tools must not only produce synthetic RF and EO/IR data but also ensure that these representations align with real-world sensor measurements. When the underlying environment is well-characterized, scene generation tools can generate multimodal imagery alongside ground truth labels, providing ready-made datasets for AI/ML models and autonomous agents. However, their effectiveness is often constrained by the availability of accurate models and measured databases that capture the necessary radiometric and electromagnetic characteristics of the environment. Addressing these limitations requires the development of software that integrates geospatial data, time-of-day, seasonal variations, measured databases, and land cover data to generate detailed representations of the environment. Furthermore, this software must support standardized scene formats compatible with existing simulation tools such as FLITES (EO/IR) and Xpatch (RF), allowing for flexible resolution and fidelity adjustments based on scenario requirements. Finally, a structured approach should be proposed to refine synthetic scene renderings as real-world measurements become available, improving realism and scene fidelity over time.

PHASE I: Since this is a Direct-to-Phase-II (D2P2) topic, the Government expects that the applicant has demonstrated the ability to automatically build background scenes for either RF or EO/IR simulation tools, such as Xpatch (AFRL/RY) and FLITES (AFRL/RW), using commonly available resources and knowledge of a given geographic region; demonstrated the potential to use background scenes to generate geo-specific, synthetic EO/IR or RF training data for AI/ML models or train/evaluate autonomous systems.

PHASE II: Develop prototype software that ingests geographic coordinates and/or other identifying information on a region-of-interest on Earth, sources representative data on that region, and builds geo-specific background scene data this is compatible with both RF and EO/IR simulation tools. Develop a standardized interface / data format that is compatible with existing RF and EO/IR simulation tools, such as Xpatch and FLITES. Establish and document relevant use-cases. Demonstrate the software's ability to improve AI/ML model performance on geolocation, ATR, scene understanding, and other key tasks in a relevant environment. Plan and coordinate one or more demonstrations to provide proof of concept determination. Perform experiments and analyze results to establish the adequacy of the solution approach and minimize transition risk. Contact potential customers and transition partners to support Phase III activities. Deliver prototype software to AFRL.

PHASE III DUAL USE APPLICATIONS: Add additional classified data sources and work with multiple end-users to provide additional specific capabilities required. Further development will refine the scene generation capability to increase accuracy and fidelity for specific end-users. Commercialize this capability within the DoD and private sectors. Programs of record that would benefit from this effort include but are not limited to FLITES (Fast Line-of-sight Imagery for Targets and Exhaust-plume Signatures), Xpatch (RF signature prediction), JSTAR (Joint Surveillance Target Attack Radar System), Synthetic Scene Generation Model (SSGM), and AFSIM (Advanced Framework for Simulation, Integration, and Modeling). End-users include but not limited to all DoD components.

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3. Hazlett, M., Andersh, D. J., Lee, S. W., Ling, H., & Yu, C. L. (1995, June). XPATCH: a high-frequency electromagnetic scattering prediction code using shooting and bouncing rays. In Targets and Backgrounds: Characterization and Representation (Vol. 2469, pp. 266-275). SPIE

KEYWORDS: Scene generation; artificial intelligence; radio frequency; electro optical; infrared; multimodal; synthetic data; geo-specific

DAF26BZ01-DV007 TITLE: CHORD - Collaborative Human Autonomy Operational Review

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Future Autonomous Collaborative Platforms (ACPs) will introduce AI-enabled uncrewed aircraft into the fleet. These platforms will assume significant tactical decision-making responsibilities and operate alongside traditional crewed aircraft. This paradigm shift complicates knowledge elicitation for post-mission debriefing, as it necessitates understanding both human and autonomous aircraft decision-making processes. This introduces a new research challenge: effectively logging the necessary information from human and ACP decision-actions for debriefing and presenting it to warfighters through innovative human-machine interfaces (HMIs). The primary objective of this topic is to prototype and develop debriefing approaches that effectively fuse the decision-making chains of both human operators and multiple autonomous ACPs, presenting that information clearly and concisely.

DESCRIPTION: Mission debriefing for manned and remotely piloted aircraft (RPA) and crewed aircraft in military operations is currently conducted manually by warfighters. This typically involves verbal communication and classroom-style discussions, with little to no AI or software assistance for reflecting on missions, identifying lessons learned, or pinpointing areas for improvement. As pilots are the primary tactical decision-makers, verbal communication sessions are essential for eliciting and understanding their decision-making processes. As teams of ACPs begin making tactical decisions with a high level of autonomy, it is unknown what information needs to be logged during mission and how that information should be displayed so that the warfighter can audit and understand after mission debriefing, what decisions, tactics, techniques, and procedures (TTPs) the autonomous systems acted on. This topic looks to advance existing debriefing tools for replaying mission execution and enhance them with additional functionality targeting debriefing of autonomous ACPs.

A secondary focus of this topic is to identify data input requirements from autonomy that would be necessary for support debriefing of autonomy. Modern methods for autonomous decision-making tend to employ black-box deep learning algorithms with limited transparency, leading to lack of trust and assurance that autonomous agent decisions comply with the Law of Armed Conflict (LOAC). XAI (Explainable Artificial Intelligence) is actively researching techniques to make black box models more understandable while other areas are using more transparent symbolic methods that are rooted in explicit rules to perform reasoning and problem-solving. An ACP will likely include a combination of inherently explainable and low transparency algorithms for different decision-making processes. Information needs for debriefing that will be identified in this topic should guide autonomy development with regards to autonomy logging/reporting for debriefing and algorithm practicality.

A consideration for DP2 participation is the demonstration of an existing debriefing tool that the proposer has developed that is used in military operations or that it is being developed under a recognized US DoD program. This will allow for a solid foundation for which CHORD can build upon that focuses

specifically on debriefing of human machine teaming for ACPs. An expectation of common debriefing functionality such as data playback, a digital map display, timeline, event logs, and data visualization of vehicle fuel, health, and status will be necessary for DP2 consideration. It is not essential that the existing debriefing tool has been applied to unmanned systems, and debriefing tools in non-air or crewed vehicle domains will be considered.

While logging and video playback of ACP mission execution are critical components of debriefing functionality, they will likely be inadequate for truly understanding ACP decision-making. Software analytics, AI tools, and novel HMI designs will be necessary to answer key questions, such as: What tactical decisions did the ACP make? When were these decisions made? What was the rationale or considerations behind the ACP's decisions? As ACPs assume greater responsibility in tactical decision-making, it is crucial to conduct research and develop software tools that enable warfighters to understand and trust these autonomous systems.

Core Research Questions:

1. What types of information must be logged and exchanged between ACPs and the warfighter during post-mission debriefing to support transparency and trust in autonomous operations?
2. Do current government reference architectures and standards adequately support the information exchange requirements for debriefing ACP teams?
3. How should information from ACPs be structured and visualized within the HMI to align with warfighter cognitive models and situational awareness needs?
4. What HMI features for debriefing best support comprehension of ACP autonomy decision chains, contextual reasoning, and deviations from expected behavior?
5. What types of software or AI-enabled analytics tools would be most useful to summarize, explain, and visualize autonomous decision-making by ACPs?

PHASE I: This topic is for Direct to Phase II (DP2) submission. The Department of Air Force will accept Direct to Phase II proposals for the cost of up to $1,700,000 for a 18-month period of performance. To submit a DP2 proposal, proposers must justify the scientific and technical merit, feasibility, and potential military and/or commercial applications of their proposed work. This justification requires detailed documentation, such as technical reports, test data, prototype designs/models, and achieved performance goals/results. The following are prerequisites for DP2 consideration:

- Proposer must demonstrate an existing debriefing tool or after-action review (AAR) tool that they have developed, either already deployed in military operations or developed under a recognized U.S. DoD program.
- Though most of the work will be executed at UNCLASSIFIED or CUI, there is the potential for work to be executed at a SECRET level. Proposers must be able to acquire and maintain a secret level facility and Personnel Security Clearances.
- Proposer must demonstrate knowledge of and access to subject matter expertise regarding Air Force doctrine, tactics, techniques, and procedures.
- Proposer must exhibit strong technical merit and possess a team with expertise in software development, architecture, deployment, front-end web development, autonomous systems, government reference architectures, and artificial intelligence.
- Proposer must demonstrate a clear understanding of the complexities involved in debriefing missions where both humans and machines are making decisions.
- Proposer must present a strong commercialization strategy, including a well-defined plan for bringing a debriefing software solution to market.

PHASE II: Provide a design, prototype, and integrated demonstration of a debriefing tool built on a government reference architecture. Potential features to explore for debriefing UI includes: components that give insight into ACP team decision making, visualize ACP team and individual state and

performance, provide situation awareness into the environment the ACPs are interacting with, allows conversing with the debriefing tool using conversation AI (e.g., LLMs), visualize uncertainty and mission anomalies, and allows users to explore "what-if" scenarios by modifying key parameters of the mission (e.g., number of ACPs, ACP payloads, weather conditions) and re-run the mission in a simulated environment to see how the ACP would have responded are all of interest in this phase 2 topic. Performers will have opportunities to develop other transformational debriefing concepts for debriefing teams of ACPs.

Deliverables include:

1. A government reference architecture compliant debriefing tool developed in a modern web-based UI development framework.
2. An integrated demonstration of the debriefing tool with multi-vehicle simulation.
3. Technical Design Document (TDD) of debriefing software including:
   - Technical whitepaper detailing key findings.
   - UML software documentation
   - UX/UI wireframes and mockups (e.g., Figma, Balsamiq, Adobe XD).
4. Government reference architecture change proposals addressing identified gaps regarding the interfaces between autonomy and mission debriefing.

PHASE III DUAL USE APPLICATIONS: Developing mission and autonomy debriefing software through this SBIR will directly benefit future Autonomy-Capable Platforms (ACPs) reliant on autonomous decision-making. The resulting robust and versatile solution will enable CHORD to dramatically improve the efficiency and trustworthiness of post-mission analysis for both military and commercial autonomous systems.

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KEYWORDS: Mission Debriefing; After-Action Review; Autonomy; UX/UI; Human-Machine Teaming; Trust

Defense Advanced Research Projects Agency (DARPA)
DoW 2026 Small Business Innovation Research (SBIR)
Annual Broad Agency Announcement (BAA)
Proposal Submission Instructions

INTRODUCTION

DARPA's mission is to create and prevent technological surprise for our national security. As part of this mission, DARPA makes high-risk, high-reward investments in science and technology that have the potential to disrupt current understandings and/or approaches. The topics that follow focus on technical domains important to DARPA's mission, pursuing innovative research concepts that fall within one of its technical offices. More information about DARPA's technical domains and research topics of interest can be found at https://www.darpa.mil/about/offices. DARPA offers free resources through DARPAConnect to help potential performers navigate DARPA, including "Tips for DARPA Proposal Success." Join DARPAConnect at www.DARPAConnect.us to leverage on-demand learning and networking resources.

The pace of discovery in both science and technology is accelerating worldwide, resulting in new fields of study and the identification of scientific areas ripe for small business utilization through the SBIR and Small Business Technology Transfer (STTR) programs. Small businesses are critical for developing technology to support national security. Proposers are encouraged to consider whether the Research/Research and Development (R/R&D) being proposed to Department of War (DoW) Components also has private sector potential, either for the proposed application or as a base for other applications.

Proposers responding to a topic in this BAA must follow all general instructions provided in the DoW SBIR Program BAA. DARPA requirements, in addition to or deviating from the DoW SBIR Program BAA, are provided in the instructions below. All DARPA SBIR proposals must be submitted electronically through the Defense SBIR/STTR Innovation Portal (DSIP) as described in the Proposal Preparation and Submission sections of these instructions. It is recommended that organizations register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposers are encouraged to submit proposals as early as possible to avoid unexpected delays due to a high volume of traffic during the final hours before a BAA closes. *DARPA will not accept any late proposals.*

Proposers are encouraged to thoroughly review the DoW SBIR Program BAA and register for the Defense SBIR/STTR Innovation Portal (DSIP) Listserv to remain apprised of important programmatic and contractual changes.

- The DoW SBIR Program BAA is located at: https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Please select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at https://secure.campaigner.com/CSB/Public/Form.aspx?fid=667492&ac=g9gk.

Specific questions on the administration of the DARPA Program and these proposal preparation instructions should be directed to DARPA Small Business Programs Office at SBIR_BAA@darpa.mil. DSIP Topic Q&A will NOT be available for these DARPA topics. Technical questions related to improving the understanding of a topic's requirements must be submitted by All white papers are due at 12:00 pm ET on the closing date for this release.

This release contains an open topic. As outlined in section 7 of the SBIR and STTR Extension Act of 2022, innovation open topic activities—

(A) Increase the transition of commercial technology to the DoW;
(B) Expand the small business nontraditional industrial base;
(C) Increase commercialization derived from investments of the DoW; and
(D) Expand the ability for qualifying small business concerns to propose technology solutions to meet the needs of the DoW.

Unlike conventional topics, which specify the desired technical objective and output, open topics can use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

A small business concern may only submit one (1) proposal to each open topic. If more than one proposal from a small business concern is received for a single open topic, only the most recent proposal to be certified and submitted prior to the submission deadline will receive an evaluation. All prior proposals submitted by the small business concern for the same open topic will be marked as nonresponsive and will not receive an evaluation.

Proposals submitted in response to this topic will follow a two-step submission process.

STEP ONE- White paper submission: Proposing small business concerns must certify and submit, by the deadline listed above, the following proposal volumes in DSIP:

1. All Firm-level Forms. On the Defense SBIR/STTR Innovation Portal (DSIP) at https://www.dodsbirsttr.mil/submissions/, prepare the Firm-level Forms – Firm Certifications, Audit Information, and Company Commercialization Report (CCR).
2. Supporting Documents (Volume 5). ***A white paper must be uploaded to Volume 5 outlining the proposed effort.*** The white paper must not exceed three (3) pages in length. The header on each page of the white paper should contain your company name, topic number, and proposal number assigned by DSIP when the proposal was created. The header may be included in the one-inch margin. Content requirements of the white paper are provided in the Description section of the topic.

NOTE: At step one of this process, proposers will NOT complete Volume 1 (Proposal Coversheet), Volume 2 (Technical Volume), Volume 3 (Cost Volume), or Volume 6 (Fraud, Waste and Abuse training). The Company Commercialization Report will be required in the Firm-level Forms but will not be provided in Volume 4.

White papers will be screened by DARPA to determine suitability for full proposal submission, based on responsiveness to the requirements outlined in the Description section of the topic, and an understanding of the capability gap. **Note:** reviewers' assessment comments will be made available for all firms that submit a white paper. DARPA does not, however, offer any further feedback to firms for white paper submissions.

STEP TWO: Firms that have submitted white papers will be notified of recommendation for a full Phase I proposal submission within 30 days of the deadline for white papers. Full Phase I proposals will follow the guidelines provided in the DoW Program BAA, with additional details and deviations below.

PHASE I PROPOSAL GUIDELINES
Phase I proposals for topic* DPA26BZ01-NP001 *will be accepted ONLY from firms that have submitted a white paper that DARPA has determined meets the requirements of the Description section of the topic. Proposals from firms that have not been recommended for Phase I submission will be marked non-responsive and will not be evaluated.

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in Appendix A.

Current Release Award Structure by Topic

Topic Number	Phase I			
	Technical Volume			
	White Paper	Slide Deck	Award Amount	Period of Performance (PoP)
DPA26BZ01-NP001	10 pages	5 pages	$150,000	4 months
DPA26BZ01-NV002	10 pages	5 pages	$250,000	8 months
DPA26BZ01-DV003	10 pages	5 pages	$300,000	6 months

Technical Volume (Volume 2) – White Paper & Slide Deck Format
The white paper shall not exceed 10 pages, and the slide deck shall not exceed five (5) pages. For information on the content of each of these elements of the Technical Volume and the commercialization strategy, please see Appendix A.

Cost Volume (Volume 3)
Please see the chart above for award amounts listed by topic. Proposers are required to use the applicable cost proposal template as instructed in Appendix A.

Content of the Cost Volume
Proposers should refer to the DARPA Phase I Proposal Instructions, provided in Appendix A.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW SBIR Program BAA for full details on this requirement. Information contained in the CCR will not be considered by DARPA during proposal evaluations.

Supporting Documents (Volume 5)
In addition to the documents required by DoW, small businesses may also submit additional documentation to support the Technical Volume (Volume 2) and the Cost Volume (Volume 3) in Volume 5. See Appendix A for required certifications that must be included in Volume 5. For additional information, see the SBIR Annual Program BAA at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission.

DIRECT TO PHASE II PROPOSAL GUIDELINES
Proposers should refer to the DARPA Direct to Phase II Proposal Instructions provided in Appendix B.

Current Release Award Structure by Topic

Standard Proposal Format

Topic Number	Direct to Phase II				
	Technical Volume	Award Amount	Period of Performance (PoP)	Option Amount	Option PoP
DPA26BZ01-DV003	35 pages	$1,500,000	18 months	$500,000	6 months

Technical Volume (Volume 2) –Standard Format (35 pages)
If a proposer can provide adequate documentation to substantiate that the scientific and technical merit and feasibility described in the Phase I section of the topic has been met and describes the potential commercial applications, the Direct to Phase II (DP2) authority allows the Department of Defense (DoW) to make an award to a small business concern under Phase II of the SBIR program without regard to whether the small business concern was provided an award under Phase I of an SBIR program. This topic is accepting DP2 proposal submissions.

DP2 Feasibility Documentation shall not exceed 10 pages. DP2 Technical Proposal shall not exceed 20 pages. Phase II commercialization strategy shall not exceed five (5) pages. This should be the last section of the Technical Volume.

White Paper & Slide Deck Proposal

Topic Number	Direct to Phase II					
	Technical Volume		Award Amount	Period of Performance (PoP)	Option Amount	Option PoP
	White Paper	Slide Deck				
DPA26BZ01-DV004	20 pages	15 slides	$1,500,000	18 months	$500,000	6 months

Technical Volume (Volume 2) – White Paper & Slide Deck Format (35 pages)
If a proposer can provide adequate documentation to substantiate that the scientific and technical merit and feasibility described in the Phase I section of the topic has been met and describes the potential commercial applications, the Direct to Phase II (DP2) authority allows DoW to make an award to a small business concern under Phase II of the SBIR program without regard to whether the small business concern was provided an award under Phase I of an SBIR program. This topic is accepting DP2 proposal submissions.

The white paper shall not exceed 20 pages, and the slide deck shall not exceed 15 pages. For information on the content of each of these elements of the Technical Volume and the commercialization strategy, please see Appendix B: DARPA DIRECT TO PHASE II (DP2) INSTRUCTIONS.

Content of the Technical Volume
Proposers should refer to the DARPA DP2 Proposal Instructions provided in Appendix B.

Cost Volume (Volume 3)
Please see the chart above for award amounts listed by topic. Proposers are required to use the applicable cost proposal template as instructed in Appendix B

Content of the Cost Volume
Proposers should refer to the DARPA DP2 Proposal Instructions provided in Appendix B.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW SBIR Program BAA for full details on this requirement. DARPA will not consider information contained in the CCR during proposal evaluations.

Supporting Documents (Volume 5)
In addition to the documents required by DoW, small businesses may submit additional documentation to support the Technical Volume (Volume 2) and the Cost Volume (Volume 3) in Volume 5. See Appendix B for required certifications that must be included in Volume 5. For additional details, please refer to the DoW SBIR Program BAA
.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission.

PHASE II PROPOSAL GUIDELINES
Phase II proposals may only be submitted by Phase I awardees. Should DARPA have funding available and decide to proceed with a Phase II, proposers awarded a Phase I contract will be eligible to submit a proposal for Phase II and will be contacted by the DARPA Small Business Programs Office at the appropriate time during their Phase I period of performance. Phase II proposals will be evaluated in accordance with the evaluation criteria listed in the applicable DoW SBIR Program BAA. Phase II selection(s) are at the sole discretion of the Government and are subject to funding availability and Phase I performance. Phase II Instructions will be provided by DARPA upon notification for Phase II submission.

Current Release Award Structure by Topic

Topic Number	Phase II				
	Technical Volume	Award Amount	Period of Performance (PoP)	Option Amount	Option PoP
DPA26BZ01-NP001	25 pages	$1,300,000	12 months	$500,000	6 months
DPA26BZ01-NV002	25 pages	$1,800,000	18 months	n/a	n/a
DPA26BZ01-DV003	25 pages	$1,800,000	18 months	$500,000	6 months

Technical Volume (Volume 2)
The Technical Volume shall not exceed 25 pages. The Phase II commercialization strategy shall not exceed 5 pages. This should be the last section of the Technical Volume and is included in the 25-page total. Any pages in the Technical Volume over 25 pages will not be considered in proposal evaluations.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
DARPA does not offer TABA funding.

MAJORITY OWNERSHIP IN PART BY MULTIPLE VENTURE CAPITAL, HEDGE FUND, AND PRIVATE EQUITY FIRMS
Proposers that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, are eligible to submit proposals in response to DARPA topics advertised within this BAA.

For proposers that are a member of this ownership class, the following must be satisfied for proposals to be accepted and evaluated:

a. Prior to submitting a proposal, firms must register with the SBA Company Registry Database.
b. The proposer, within its submission, must submit the Majority-Owned VCOC, HF, and PEF Certification. A copy of the SBIR VC Certification can be found on https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics. Include the SBIR VC Certification in the Supporting Documents (Volume 5).
c. Should a proposer become a member of this ownership class after submitting its proposal and prior to any receipt of a funding agreement, the proposer must immediately notify the Contracting Officer, register in the appropriate SBA database, and submit the required certification which can be found at the link referenced in sec b. above.

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the DoW SBIR Program BAA. DARPA will conduct an evaluation of each conforming proposal. Proposals that do not comply with the requirements detailed in this BAA and the research objective(s) of the corresponding topic are considered non-conforming and, therefore, are not evaluated nor considered for award.

Using the evaluation criteria, the Government will evaluate each proposal in its entirety, documenting the strengths and weaknesses relative to each evaluation criteria. Based on these identified strengths and weaknesses, the Government will determine the proposal's overall selectability for funding. Proposals will not be evaluated against each other during the evaluation process but rather evaluated on their own individual merit to determine how well the proposal meets the criteria stated in this BAA and the corresponding DARPA topic.

Awards will be made to proposers whose proposals are determined to be the most advantageous to the Government, consistent with instructions and evaluation criteria specified in the DoW SBIR Program BAA and availability of funding.

For the purposes of this proposal evaluation process, a selectable proposal is defined as follows:

Selectable: A selectable proposal is a proposal that the Government has evaluated against the evaluation criteria listed in the DoW SBIR Program BAA and DARPA topic, and the strengths of the overall proposal outweigh its weaknesses. Additionally, there are no accumulated weaknesses that would require extensive negotiations and/or a resubmitted proposal.

For the purposes of this proposal evaluation process, a non-selectable proposal is defined as follows:

Non-Selectable: A proposal is considered non-selectable when the Government has evaluated the proposal against the evaluation criteria listed in the DoW SBIR Program BAA and DARPA topic, and the strengths of the overall proposal do not outweigh its weaknesses.

Proposing firms will be notified of selection or non-selection status for a Phase I or Direct to Phase II award within 90 calendar days of the closing date of the BAA. The Corporate Official (CO) indicated on the Proposal Cover Sheet will be notified by e-mail regarding proposal selection or non-selection. In accordance with the Small Business Administration (SBA) SBIR/STTR Policy Directive, Appendix I, paragraph 4, Method of Selection and Evaluation Criteria, subparagraph (d) Release of Proposal Review Information, DARPA will provide a technical evaluation narrative to the proposer for each proposal submitted in response to a topic. An informal feedback session may be requested by the proposing firm via e-mail at sbir@darpa.mil. The informal feedback is provided at the sole discretion of DARPA.

It is the policy of DARPA to treat all proposals as source selection information and to disclose their contents only for the purpose of evaluation. Restrictive notices notwithstanding, during the evaluation process, submissions may be handled by support contractors for administrative purposes and/or to assist with technical evaluation. All DARPA support contractors are expressly prohibited from performing DARPA-sponsored technical research and are bound by appropriate nondisclosure agreements. Input on technical aspects of the proposals may be solicited by DARPA from other Government and/or non-Government consultants/experts who are strictly bound by the appropriate nondisclosure requirements. No submissions in response to the BAA will be returned. Upon completion of the evaluation and selection process, an electronic copy of each proposal received will be retained at DARPA.

Proposal titles, abstracts, anticipated benefits, and keywords of proposals that are selected for contract award will undergo a DARPA Policy and Security Review. Proposal titles, abstracts, anticipated benefits, and keywords are subject to revision and/or redaction by DARPA. Final approved versions of proposal titles, abstracts, anticipated benefits, and keywords may appear on the DoW SBIR/STTR awards website and/or the SBA's SBIR/STTR award website (https://www.sbir.gov/awards).

Refer to the DoW SBIR Program BAA for procedures to protest this BAA. As further prescribed in the Federal Acquisition Regulation (FAR 33.106(b), FAR 52.233-3), protests regarding the selection decision should be submitted to:

DARPA
Contracts Management Office (CMO)
675 N. Randolph Street
Arlington, VA 22203
E-mail: CMO_SBIRProtests@darpa.mil and sbir@darpa.mil

AWARD AND CONTRACT INFORMATION

1. General Award Information

For all Phase I awards, multiple awards are anticipated. DARPA will only award purchase orders under FAR Part 13.

For Phase II awards, multiple awards are anticipated. DARPA may award Federal Acquisition Regulation (FAR)-based Government contracts (Firm-Fixed Price or Cost-Plus Reimbursement) or Other

Transactions (OT) for Prototype agreement (under the authority of 10 U.S.C. § 4021) subject to approval of the Contracting Officer or Agreements Officer, respectively.

- Proposers must state their requested contract type in their proposal
    - If requesting a FAR based Cost-Plus Reimbursement type contract, proposers shall include their Defense Contract Management Agency (DCMA) Final Determination Letter showing approval of the company's Accounting System.
    - If requesting a OT for Prototype, proposers shall include a completed OT in their proposal (Please see Model OT for Prototype provided on the DARPA Small Business site (https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics). NOTE: Proposers at a minimum need to complete areas as color coded in the template. If proposers wish to negotiate any articles or attachments of the OT, please provide redlines and comments as to why each redline is proposed.
    - If requesting a OT for Prototype, proposers shall include completed OT Certifications, provided on the DARPA Small Business site at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.
    - No additional action is required to be taken by proposers requesting a FAR based Firm-Fixed Price type contract.

The resources made available for each topic issued under this BAA will depend on the quality of the proposals received and the availability of funds.

The Government reserves the right to select for negotiation all, some, one, or none of the proposals received in response to this BAA and to make awards with or without communications with proposers. Additionally, the Government reserves the right to award all, some, one, or none of the options on the contract(s)/agreement(s) based on available funding and the performer's technical performance. If warranted, portions of resulting awards may be segregated into pre-priced options. Additionally, DARPA reserves the right to accept proposals in their entirety or to select only portions of proposals for award. In the event that DARPA desires to award only portions of a proposal, negotiations may be opened with that proposer. The Government reserves the right to fund proposals in phases with options for continued work, as applicable.

The Government reserves the right to request any additional, necessary documentation once it makes the award instrument determination. The Government reserves the right to remove a proposal from award consideration should the parties fail to reach an agreement on award terms, conditions, and price within a reasonable time, and/or the proposer fails to provide requested additional information within three (3) business days.

It is DARPA's goal to award SBIRs and STTRs as expeditiously as possible. In an effort to do so, please review and ensure the DARPA SBIR/STTR Pre-Award Checklist located on the DARPA Small Business website at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-overview/ is completed prior to being selected for award.

In all cases, the Government Contracting/Agreements Officer reserves the right to select award instrument type, regardless of instrument type proposed, and to negotiate all instrument terms and conditions with selectees. DARPA will apply publication or other restrictions, as necessary, if it determines that the research resulting from the proposed effort will present a high likelihood of disclosing performance characteristics of military systems or manufacturing technologies that are unique and critical to defense. Any award resulting from such a determination will include a requirement for DARPA permission before publishing any information or results on the program. For more information on publication restrictions, see the DoW SBIR Program BAA.

All proposers should carefully read the following information about OT awards:

The flexibility of the OT award instrument is beneficial to the program because the performer will be able to apply its commercial best practices as required to carry out the research project that may be outside of the Federal Acquisition Regulation (FAR) process-driven requirements. Streamlined practices will be used, such as milestone-driven performance, intended to reduce time and effort on award administration tasks and permit performers to focus on the research effort and rapid prototyping. Because of this ability, OTs provide the Agreements Officer the flexibility to create an award instrument that contains terms and conditions that promote commercial transition, reduce some administratively burdensome acquisition regulations, and meet SBIR/STTR program goals.

Please see https://acquisitioninnovation.darpa.mil/what-are-ots for more information on OTs.

1. **Cybersecurity Maturity Model Certification (CMMC)**

Firms engaging in Controlled Unclassified Information (CUI), Export Controlled (EXPT), or International Traffic in Arms Regulations (ITAR) work for DARPA must have CMMC Level 2 self-assessment certification. Please see https://www.sprs.csd.disa.mil/nistsp.htm for more information. Many resources exist to assist companies with meeting this criterion, one of which is Project Spectrum (http://www.projectspectrum.io). Users can create accounts and gain access to useful tools, including educational videos and material, as well as a cyber self-assessment.

2. **Transition and Commercialization Support Program (TCSP)**

DARPA will provide services to Phase II or DP2 awardees upon contract execution through TCSP at no cost to awardees. The TCSP goal is to maximize the potential for SBIR/STTR companies to move their technology beyond Phase II and into other research and development programs for further maturity or into solutions or products for DoW acquisition programs, other Federal programs, and/or the commercial market. Please visit https://www.darpa.mil/work-with-us/communities/small-business/transition-commercialization for more information on DARPA TCSP.

3. **Embedded Entrepreneurship Initiative**

Awardees of SBIR funding pursuant to this BAA may be eligible to participate in the DARPA Embedded Entrepreneurship Initiative (EEI). An invitation to participate in EEI is at the sole discretion of the Government based on evaluation of technical and commercial factors and is subject to program balance and the availability of funding. EEI is a limited scope program offered by DARPA, at the Agency's discretion, to a small subset of awardees. The goal of DARPA's EEI is to increase the likelihood that DARPA-funded technologies take root in the U.S. and provide new capabilities for national defense. EEI supports DARPA's mission by accelerating the transition of innovations out of the lab and into new capabilities for the DoW. EEI investment supports development of a robust and deliberate Go-to-Market strategy for DARPA-funded advanced technology, into high-value products and capabilities for the government and commercial markets, and positions DARPA awardees to attract U.S. private investment. The following is for informational and planning purposes only and does not constitute solicitation of proposals to EEI.

There are three elements to DARPA's EEI: (1) A Senior Commercialization Advisor (SCA) from DARPA who works with the Program Manager (PM) to examine the business case for the awardee's technology and uses commercial methodologies to identify steps toward achieving a successful transition of technology to the government and commercial markets; (2) Connections to potential U.S. industry and private investor partners via EEI's Investor Working Groups; and (3) Additional funding to hire an embedded entrepreneur to achieve specific milestones in a Go-to-Market strategy for transitioning the technology into products that serve both defense and commercial markets. This embedded entrepreneur's

qualifications should include business experience within the target industries of interest, experience in commercializing early-stage technology, and the ability to communicate and interact with technical and non-technical stakeholders, and customers. Funding for EEI is typically no more than $310,000 per awardee over the duration of the award. An awardee will attend one commercialization workshop, and also may apportion EEI funding to hire more than one embedded entrepreneur, if achieving the milestones requires a unique expertise that can be obtained without exceeding the awardee's total EEI funding.

EEI Application Process:
After receiving an SBIR/STTR Phase II award, awardees interested in being considered for EEI should notify their DARPA PM during the period of performance. If the DARPA PM determines that EEI could be of benefit to transition the technology to product(s) the Government needs, the PM will refer the performer to the DARPA Commercial Strategy Team. The SCA will then contact the performer, assess fitness for EEI and determine, in consultation with the PM, and Commercial Strategy Team, whether or not to invite the performer to participate in the EEI. Factors that are considered in determining fitness for EEI include DoW/Government need for the technology; competitive approaches to enable a similar capability or product; risks and impact of the Government's being unable to access the technology from a sustainable source; Government and commercial markets for the technology; cost and affordability; manufacturability and scalability; supply chain requirements and barriers; regulatory requirements and timelines; intellectual property and Government use rights, and available funding.

After SCA review, the Commercial Strategy Team may request the SBIR/STTR awardee to submit additional tasks for review.

EEI awards are at the sole discretion of DARPA and are subject to program balance and the availability of funding. For more information, please refer to the EEI website https://www.darpa.mil/about/offices/commercial-strategy.

ADDITIONAL INFORMATION
DARPA intends to use electronic mail for all correspondence regarding these topics. Questions related to the technical aspect of the research objectives and awards specifically related to a topic should be e-mailed to SBIR_BAA@darpa.mil. Please reference the topic number in the subject line. All questions must be in English and must include the name, e-mail address, and the telephone number of a point of contact.

DARPA will attempt to answer questions in a timely manner; however, questions submitted within seven (7) calendar days of the proposal due date listed herein may not be answered. DARPA will post a consolidated Frequently Asked Questions (FAQ) document. To access the posting please visit: https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics. Under the topic number summary, there will be a link to the FAQ. The FAQ will be updated on an ongoing basis until one week prior to the proposal due date.

Technical support for the Defense SBIR/STTR Innovation Portal (DSIP) is available Monday through Friday, 9:00 a.m. – 5:00 p.m. ET. Requests for technical support must be emailed to with a copy to SBIR_BAA@darpa.mil.

DARPA
DoW 2026 SBIR BAA
Release 1
Topic Index

DPA26BZ01-NP001	Broadening Availability of Regimens for K-9s (BARK) - Open Topic
DPA26BZ01-NV002	Photonic-electronic Panels Integration (PEPI)
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DPA26BZ01-DV004	Extremity Platform for On-demand Surgical Implantation and Tissue Integration with Osteochondral Neogenesis

DPA26BZ01-NP001 TITLE: Broadening Availability of Regimens for K-9s (BARK) - Open Topic

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Biomanufacturing, Contested Logistics Technologies

COMPONENT TECHNOLOGY PRIORITY AREA(S): Biotechnology

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop medical products that are interoperable and compatible across humans and dogs to meet the health needs of both human warfighters and military working dogs.

DESCRIPTION: Military working dogs (MWDs) are critical to national security, serving as force multipliers by enhancing the lethality and survivability of their teams through their unique senses, agility, and autonomy. Providing effective medical care for MWDs is critical but imposes significant challenges. Not all human medical products work for canines (K-9s), who sometimes require separate, specialized veterinary products. MWD handlers must therefore carry extra medical equipment to provide emergency care for their K-9 partner, exacerbating equipment loads and impeding operational capacity. Specialized equipment also expands medical materiel requirements and dependence on the less-resourced veterinary market, encumbering acquisition and sustainment activities.

Medical technologies that are interoperable and compatible across humans and dogs can address unmet needs of valuable MWDs while lessening the burden on medics, logisticians, and other contributors to force health protection. Species-interoperable medical technologies, particularly those supporting acute and tactical care, will improve lifesaving medical care for these MWDs while mitigating logistical and operational burdens of treating both human and K-9 warfighters.

The Defense Advanced Research Projects Agency (DARPA) is soliciting medical technologies that are interoperable and compatible across humans and dogs. Technologies of greatest interest allow for the replacement of existing products in medical sets with interoperable products, reducing the total amount of medical supplies—expanding capability without expanding the kits. Examples of specific technologies of interest include, but are not limited to:

- Filters for donor plasma capable of removing cross-reactive antigens that impede compatibility across species;
- Universal synthetic plasma designed with all necessary functional components for transfusion;
- Sensors and form-factors that enable physiological monitoring (e.g., core body temperature and blood pressure) or triage in both humans and dogs;
- Medical devices (e.g., splints, backboards, tourniquets, mechanical ventilators) designed to be rapidly adjustable at point of care for flexible use across anatomies;
- Interoperable medical countermeasures (MCMs) and form-agnostic personal protective equipment to mitigate or protect against chemical, biological, radiological, and nuclear threats; and
- Delivery mechanisms that can modulate dosing, including through excipients or combination devices (e.g., autoinjectors), to enable universal use of pharmaceuticals and MCMs.

The aim of the solicitation is to create a demonstrative prototype that can quickly progress to pre-clinical or clinical testing during a contracted SBIR Phase II period of performance. Research that merely tests existing products, including collection of data to titrate dosages or support label expansion of a marketed product to an additional species, is unlikely to achieve the degree of technical innovation a successful proposal should demonstrate. New drug discovery is discouraged but may be considered in particularly compelling cases. Products enabled by proposed research should feasibly be safe and effective in both humans and canines—to be verified in future trials—for use cases where no single existing product serves

both patient sets. Technologies of interest should be capable of achieving substantially equivalent or superior performance in humans compared to currently approved options.
An initial white paper describing the technical approach is required and will be evaluated. If DARPA selects a white paper for further evaluation, the Government will issue an invitation to submit a full proposal. The technical white paper should include an overview of the proposed concept with details to support feasibility. The overview should address the bullets below, which are listed in order of importance:

- Proposed concept: Describe the proposed research and medical product. Outline the design and operation of the main components that are being proposed for development and mode(s) of action. Use clear calculations, preliminary data, or mechanistic justifications to support feasibility of the proposed concept.
- Concept of employment: Identify how the proposed product could be employed. Provide details on the intended use, indication, and effect, and the prevalence and impact of the addressed medical condition. What benefits, including new capabilities or improved metrics, does the proposed solution provide compared to current commercial off-the-shelf (COTS) options (e.g., commercially available human- or canine-specific products)?
- Path to market: Identify relevant predicate or otherwise established products that have been approved, licensed, authorized, or cleared by the U.S. Food and Drug Administration (FDA) or other relevant regulatory authority. Provide a general plan or strategy for securing market access and ensuring regulatory compliance for both humans and canines if the proposed technology is successful and shown to be safe and effective. For any special FDA programs noted (e.g., Breakthrough Device designation, 513(g)), ensure that the white paper describes how the medical product meets the program requirements.
- Scalability: Provide a brief analysis of the feasibility of scaling the technology across both the DoW and industry. Are the achievable production costs low enough to merit widespread adoption, especially if alternative human- or canine-specific products are available? Is the system sufficiently familiar or intuitive to medics, handlers, and clinicians that large-scale deployment and administration wouldn't require significant training or modification of protocols? What are the projected maintenance and storage requirements, operational availabilities, and service lifetimes, and, if applicable, how do they compare with COTS alternatives?

PHASE I: During Phase I, the company should research and develop innovative technologies to make medical products that are interoperable and compatible to meet the health needs of both humans and canines. The company should develop, or apply in a novel way, technology that directly enables one or more medical products that address use cases where no single existing product serves both human and canine patient sets. The company should determine technical feasibility of the conceptualized medical product and design a preliminary prototype. Feasibility demonstration may comprise modeling and simulation, where appropriate. All foundational scientific testing necessary to proceed to future animal testing should be performed during Phase I, and success criteria (e.g., biomarkers, clinical or surrogate endpoints, etc.) for such future testing should be established. The company should also perform an analysis of predicted performance achievable through the technical approach and identify whether this performance could be substantially equivalent or superior to currently approved options for human use, if any exist.
Testing during Phase I should be conducted in vitro or in silico, and proposals should justify the approach and applicability to humans and canines. Human subject research (HSR) and use of animal subjects in the proposed research is not allowed in Phase I. The DoW defines an animal subject as a living or dead vertebrate, a larval fish or amphibian, or an egg-laying vertebrate after hatching. For details, refer to DoW Instruction 3216.01: Use of Animals in DoD Conducted and Supported Research and Training at: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/321601p.pdf.
Proposers should recommend technical milestones that will be used to demonstrate their progress to DARPA throughout Phase I. Phase I fixed payable milestones for this program should include, in the final

month of effort, a Final Phase I Report. This written report should, at a minimum, detail the approach, provide a comprehensive summary of all key testing and analysis of results conducted during the effort, compare the approach with alternative COTS and state-of-the-art technologies, and include a feasibility analysis addressing the technical, financial/commercial, clinical, and regulatory considerations impacting potential implementation at scale. The report should also set the stage for Phase II by substantiating, if justified, that continued design and development could feasibly result in a prototype demonstration and feasible work plan to achieve a fieldable prototype during Phase II. Monthly financial and technical summary reports will also be required throughout the period of performance (PoP).

PHASE II: Building upon results from Phase I, the company should further develop, fabricate, demonstrate, test, and validate a prototype. The company should experimentally establish the prototype's performance parameters in pre-clinical settings, in accordance with the success criteria developed in Phase I. User testing should be conducted to establish, or inform future work towards, operational viability in fieldable environments and use cases. Prototype(s) may be provided to appropriate DoW entities for further testing and evaluation by personnel. In addition, companies should provide a detailed regulatory and commercialization strategy for DoW and non-DoW applications, which will likely require engagement with the FDA during (or prior to) the Phase II PoP.

Proposed topics may be identified by the program manager as research involving Human and/or Animal Use. In accordance with DoW policy, human and/or animal subjects in research conducted or supported by DARPA shall be protected. Although these protocols should not be needed to carry out Phase I, significant lead time is required to prepare the documentation and obtain approval to avoid delay of a Phase II award. Please visit https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-overview/ and click on the Human Research Guidelines link or the Animal Research Guidelines link to understand what is required to comply with human protocols and animal protocols. Proposers are encouraged to separate research tasks from tasks involving human and/or animal use in the technical proposal and cost proposal to avoid potential delay of contract award.

Proposers should recommend technical milestones that will be used to demonstrate their progress to DARPA throughout Phase II. Phase II fixed milestones for this program should include, in the final month of effort, a Final Phase II Report. This written report should, at a minimum, document all final prototype designs and production methods/protocols, methods and results of all testing—including those related to or indicative of tolerability, safety, and efficacy—and any scientific advances that have been achieved under the program. The Report should include quantitative metrics on decision-making benefits, costs, risks, and schedule for implementation of a full prototype capability based on the experimental demonstration, as well as a go-to-market strategy informed by these parameters. A quantitative comparison to alternative COTS and state-of-the-art technologies should be included in the decision-making metrics. Monthly financial and technical summary reports will also be required throughout the PoP.

PHASE III DUAL USE APPLICATIONS: Potential dual-use applications encompass those of corresponding medical products currently used to treat human and canine military and civilian patients. Technologies should expand the current applications of established medical products. Medical technologies, particularly those supporting acute and tactical care, may be used in emergency medicine spanning hospital, ambulatory, and first-aid settings. In DoW/military contexts, such equipment could replace current products found in the medical sets and kits of medics and dog handlers, as well as products used globally in Military Medical Treatment Facilities. Applications are also abundant in veterinary medicine, providing medical care for companion animals, service animals, and civilian working dogs (including operational canines, like those serving in civilian law enforcement, search and rescue, and emergency response) in addition to MWDs.

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KEYWORDS: Medical Equipment and Supplies; Medicine and Health Sciences; Military Dogs; Combat Casualty Care; Military Medicine; Battlefield Medicine; Veterinary Medicine; Biomedical Technology

DPA26BZ01-NV002 TITLE: Photonic-electronic Panels Integration (PEPI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software, Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and demonstrate a prototype for photonic-electronic panel technology for the integration of photonic chips (PICs) that contain high-density three-dimensional chip-to-chip and intra-chip optical routing technology developed by the DARPA Heterogeneous Adaptively Produced Photonic Interfaces (HAPPI) program. Moving from planar to 3D optical routing is an enabling technology for PICs and when applied at the panel level enables (1) dense connectivity of multiple photonic and electrical integrated circuits and (2) a platform for seamless co-integration of optical and electrical interconnections and external interfaces.

DESCRIPTION: The HAPPI program is developing three-dimensional chip-to-chip and intra-chip optical routing that are compatible with standard microelectronics manufacturing processes. The program scope includes multiple routing planes within a photonic integrated circuit (PIC) or photonic interposer, connections between routing layers capable of traversing substrate thickness, and surface methods for coupling light from one photonic chip to another. This technology will enable packages that include multiple PICs with high density optical interconnects.Notably outside of the HAPPI program scope is waveguide-to-fiber coupling and larger scale chip-to-chip connectivity at the system board level. A commercial opportunity exists to create a platform similar to an electrical printed circuit board (PCB) platform that incorporates 3-dimensional optical routing planes, co-integration of optical and electrical routing, interconnections to photonic and electronic integrated circuits and an interface with fiber arrays for long-reach signaling.This SBIR will develop photonic-electronic panels for multi-chip integration capability as a manufacturable, commercially available, low-cost technology. Solutions developed under this topic shall meet a within panel optical interconnect pitch of 100-300 um that show non-blocking routing as well as co-integration with electronic wiring. A >16 channel detachable fiber attach unit (FAU) shall be developed as a method to introduce photonic signals to the panel. A minimum of two integrated circuits (ICs) must be mounted to the panels that integrate with the embedded signaling pathways. This mounting scheme could employ flip chip bonding IC, optical bumps, or other equivalent surface mount like bonding technologies 2. IC components shall be chosen such that they can demonstrate the ability to route both optical and electronic signals and are available commercially or from a foundry. The finalized integration of FAU, panel, and ICs is expected to meet the point-to-point optical link budget as specified in Table 1 below. In addition to meeting the link requirements for insertion loss (terminating at the PIC) the designed routing must demonstrate built-in optical loopback, e.g. a continuous optical loop from insertion point at the FAU through the system, in and out of a PIC, and returning to back to the same FAU. A notional schematic is included in Figure 1 to provide a visualization of the panel concept, the required loopback path, and highlight the two optical interfaces (OI-1 and OI-2) and waveguide (WG) components of the link budget. The panel prototype units delivered at the end of Phase II must be self-contained, environmentally robust, and meet minimum performance metrics.

PHASE I: The HAPPI program is developing three-dimensional chip-to-chip and intra-chip optical routing that are compatible with standard microelectronics manufacturing processes. The program scope includes multiple routing planes within a photonic integrated circuit (PIC) or photonic interposer, connections between routing layers capable of traversing substrate thickness, and surface methods for coupling light from one photonic chip to another. This technology will enable packages that include multiple PICs with high density optical interconnects.Notably outside of the HAPPI program scope is waveguide-to-fiber coupling and larger scale chip-to-chip connectivity at the system board level. A commercial opportunity exists to create a platform similar to an electrical printed circuit board (PCB)

platform that incorporates 3-dimensional optical routing planes, co-integration of optical and electrical routing, interconnections to photonic and electronic integrated circuits and an interface with fiber arrays for long-reach signaling.This SBIR will develop photonic-electronic panels for multi-chip integration capability as a manufacturable, commercially available, low-cost technology. Solutions developed under this topic should meet a within panel optical interconnect pitch of 100-300 um that show non-blocking routing as well as co-integration with electronic wiring. A >16 channel detachable fiber attach unit (FAU) should be developed as a method to introduce photonic signals to the panel. A minimum of two integrated circuits (ICs) must be mounted to the panels that integrate with the embedded signaling pathways. This mounting scheme could employ flip chip bonding IC, optical bumps, or other equivalent surface mount like bonding technologies. A minimum of two IC components should be chosen such that they can demonstrate the ability to route both optical and electronic signals and are available commercially or from a foundry. The finalized integration of FAU, panel, and ICs is expected to meet the point-to-point optical link budget as specified in Table 1 below. In addition to meeting the link requirements for insertion loss (terminating at the PIC) the designed routing must demonstrate built-in optical loopback, e.g. a continuous optical loop from insertion point at the FAU through the system, in and out of a PIC, and returning to back to the same FAU. A notional schematic is included in Figure 1 to provide a visualization of the panel concept, the required loopback path, and highlight the two optical interfaces (OI-1 and OI-2) and waveguide (WG) components of the link budget. The panel prototype units delivered at the end of Phase II must be self-contained, environmentally robust, and meet minimum performance metrics. [insert Table 1 and Fig 1 below this text]

PHASE II: Phase II will cover the fabrication of the Phase I designs that demonstrate a detachable fiber array unit connection to a panel board that fans out into multilayer routing and couples to on board IC components. Realized designs should incorporate two ICs that demonstrate the ability to route both photonic and electronic signals on the same panel platform. In addition to meeting the link loss requirements specified, the panel prototype should include at least one photonic link with continuous loopback. A packaging compatibility report should be included to outline the methods and constraints for bonding/coupling the chosen ICs to their panel systems. As Phase II is focused on fabrication and assembly, milestones include interim reports on individual components and links as the project proceeds toward final system assembly. By the end of Phase II, the Performers must deliver ten panel prototype units meeting the program metrics and a Final Program Report detailing the fabrication process and characterization data for delivered prototypes*.* All prototypes should be provided with adequate instructions to support government testing and evaluation using standard laboratory equipment.

Schedule/Milestones/Deliverables

- Kickoff: System Requirements Review (SRR), to include risk analysis
- Month 3: Documented completion of finalized panel design and component layouts; IC selection and sourcing. Report on fabrication and assembly schedule.
- Month 6: Market Study Report on application requirements, potential customers, and commercialization strategy.
- Month 9: Interface Compatibility Report describing the IC to panel interface design and assembly including compatibility requirements and design constraints
- Month 12: Report describing photonic-electronic panel post fabrication results, and assembly plan for IC attach and FAU I/O.
- Month 18: Demonstration of photonic-electronic panels meeting program metrics; Final Report for Phase II Period to include characterization data for the link loss.

PHASE III DUAL USE APPLICATIONS: Phase III work is typically oriented towards commercialization of SBIR/STTR research or technology with funding obtained from either the private sector, a non-SBIR/STTR Government source, or both, to develop the technology into a viable product for sale in military or private sector markets. It is envisioned that the technology developed under the

SBIR program will have dual-use commercial and Department of War (DOW) applications. In the commercial space, the vision is for these panel sized interposers to enable interconnectivity of PICs with broader external systems. Optical interfaces between components are expected to be key enablers in this area as the ability to move signal in and out of the microsystem efficiently is critical. In the DOW domain, the optical panel platforms are anticipated to be a key building block for high-performance co-packaged optical systems for photonic computing, signal processing, sensor arrays, and high bandwidth communication modules.

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1. Heterogenous Adaptively Produced Photonic Interfaces (HAPPI) BAA:https://sam.gov/opp/a926fcb22cda4dec8ae2b9fa1ba1b26a/view2. Photonic Bump Bonds:https://navysbir.us/n25_A/N25A-T022.htm

KEYWORDS: Photonic, Advanced Microsystem Packaging, Optical Packaging, Optical Fiber Attach, Photonic Integrated Circuits, Optoelectronics

DPA26BZ01-DV003 TITLE: SWiFT: Smart Whole Blood Field Transfusion system

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Biomanufacturing

COMPONENT TECHNOLOGY PRIORITY AREA(S): Biotechnology

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and demonstrate a self-contained, autonomous device capable of blood collection, temporary storage (<4hr), diagnostic testing, and transfusion, with continuous donor/recipient monitoring and alerts to issues, revolutionize safe, efficient and effective blood collection and transfusion in both military and civilian settings.

DESCRIPTION: The current processes for blood collection and transfusion are complex and labor-intensive with real-time decision-making presently executed by numerous skilled medical professionals and varied systems and technologies, and prone to human error. Data from Combat Training Centers (CTC) demonstrate that less than a 60% success rate at WB transfusion and that the average time from initiating a walking blood bank to completing the transfusion of one unit of blood is over 40 minutes. This data is also obtained without significant duress for the trainees – no patient is dying and there is not actual risk to the medical providers (i.e., no actual indirect or direct fire). Multiple factors contribute to this including the identification of suitable veins for venipuncture, to ensuring accurate typing and crossmatching of blood, to the physical requirements of priming IV tubing, agitating the blood collection bag, maintaining gravity for drainage… each step presents opportunities for mistakes that at best risk lost units of blood (and lost donors) and at worst have life-threatening consequences. These outcomes and considerations highlight the intrinsically complex difficulties of the process and greater impact to expected outcomes in Department of War (DoW) workflows with and for its warfighters.This SBIR topic seeks innovative proposals for a single, integrated device that addresses these challenges through complete automation and continuous monitoring of both the donor and recipient. The device should be easily applicable to a wide range of patient sizes and ages and must autonomously collect blood (assess adequacy of vascular access and provide pump action, real-time mixing of citrate anticoagulant to ensure each drop of blood is usable in the event of a “short” donation, continuous unit agitation), type and screen the donated blood, type and crossmatch the recipient to ensure viability for donation, and transfuse blood with heated circuit. Additional options include: pressure transfusion to rapidly transfuse blood, direct donor-to-recipient transfusion, up-to-double unit collection from a single donor, autonomous venipuncture for other procedures requiring vascular access, blood unit extraction for other uses. All equipment to perform all functions are housed within the device and there is absolutely no need to access or manipulate anything. In configuration one, attach the device IV tubing to the donor’s IV (and/or the recipients IV) and push “go.” In configuration 2, place the device on the respective donors or recipients for venous access and push “go.” The device should also incorporate cost permissive non-invasive sensors to continuously monitor donor and recipient vital signs (e.g., HR, SpO2, respirations, blood pressure, temperature). The device must be programmed to recognize abnormal vital sign patterns or other concerning indications of a blood transfusion reaction and automatically alert medical personnel. The goal is to create a self-contained unit that can be applied to a blood donor or recipient and, with minimal human intervention, perform all necessary steps for safe and effective blood collection and transfusion. The continuous monitoring and alert system will further enhance safety by providing an early warning of potential adverse events. This will reduce variability and errors, increase precision and reproducibility, enhance efficiency and efficacy, eliminate the need to manage individual equipment elements, and free up medical personnel for other critical activities. The proposed device should integrate and miniaturize existing technologies, incorporating robust sensors, AI-driven decision logic, and secure data connectivity. The device must meet stringent safety and reliability standards, with built-in fail-safes and comprehensive monitoring capabilities. The device should also be ruggedized for use in austere

environments. Successful proposals will demonstrate a clear path toward a deployable prototype that can significantly improve blood transfusion services in both military and civilian settings. Target cost for such a device should be less than $300 at market price, weigh less than 5 lbs when empty, be rugged enough to drop from a standing height, and be designed to fit in a “golden hour cooler” if the blood will not be used immediately and to be discarded after transfusion is complete.The devices must include end-to-end automation of the process and anticipate a single product submission for FDA approval.

The essential steps to automate (and integrate) include:

1. Vascular access (placement of IV catheter in patient).
2. Miniaturization of automated vascular access.
3. Human hooks donor/recipient up IV tubing to IV/IO.
4. Assessment of the suitability for drainage of a placed vascular access (IV or IO).
5. Assessment of donor blood type.
6. Infectious disease screen.
7. Infusion of appropriate amount of anticoagulant (citrate) in real-time (avoids problem of underfilling).
8. Agitation.
9. Stoppage when bag is “full” (avoids problem of overfilling and over collecting in the event that donation and transfusion are synchronous).
10. *Individual must remove and reset/connect the device to recipient
11. Assessment of recipient blood type (by finger/skin prick, no IV/IO required) - synchronous or asynchronous execution.
12. Assessment of recipient vascular access - synchronous or asynchronous execution.
13. Determine Whether to Initiate Transfusion ALLOWS TRANFUSION ONLY AFTER CONDITIONS MET (matching blood type, access sufficient) – avoids anaphylaxis, lost time, lost blood.
14. Transfuses blood on pump (faster than by gravity).
15. Transfuses calcium after blood transfusion complete to reverse anticoagulant effect of citrate (optional).
16. Report successful completion of procedure.
17. Secure or Discontinue IV catheter.

PHASE I: Feasibility and Conceptual DesignObjective: To establish the technical and practical feasibility of the SWiFT system by conducting a comprehensive systems integration study, developing a detailed conceptual design, and creating a risk-mitigated roadmap for Phase II prototype development.

Key ActivitiesTechnology & Component Down-Selection:

- Conduct a trade study of existing COTS (Commercial Off-The-Shelf) and emerging technologies for each critical subsystem: robotic venipuncture, microfluidic blood typing/crossmatching, compact refrigeration, automated fluid handling, and non-invasive vital sign monitoring.
- Evaluate and select candidate components based on performance, size, weight, power (SWaP) constraints, ruggedness, and biocompatibility.
- Perform a systems integration analysis to identify primary interface challenges, data communication protocols, and power management strategies for a unified device.

Conceptual Design & System Architecture:

- Develop a detailed 3D CAD model and conceptual design of the integrated SWiFT system, outlining the placement of all major subsystems.
- Define the end-to-end autonomous workflow, from donor vein acquisition to recipient transfusion, including all safety interlocks and error-handling procedures.

- Architect the intelligent alert system, defining specific physiological thresholds (e.g., for hypotension, tachycardia, signs of transfusion reaction) and the logic for escalating alerts to medical personnel.

Risk Reduction & Phase II Planning:
- Identify the highest-risk technical challenges (e.g., miniaturizing the refrigeration unit, ensuring reliable venipuncture across diverse patient populations) and propose specific risk-mitigation strategies.
- Develop a detailed project plan for Phase II, including a work breakdown structure, schedule, and resource allocation for prototype fabrication and testing.
- Outline the preliminary regulatory strategy, identifying the likely FDA classification and key requirements for future clinical validation.Phase I Milestones:
- Month 2: Technology Assessment Report: A comprehensive report detailing the results of the trade study, the down-selected components for each subsystem, and the systems integration assessment.
- Month 4: Conceptual Design Review (CDR): Presentation of the complete conceptual design package, including CAD models, system architecture diagrams, and the defined autonomous workflow and alert logic.
- Month 6: Final Feasibility & Phase II Plan: Delivery of the final report summarizing the overall feasibility, risk-mitigation strategies, preliminary regulatory pathway, and the detailed, actionable plan for Phase II development.

PHASE II: Prototype Development and ValidationObjective: To develop, integrate, and validate a fully functional, ruggedized alpha prototype (TRL 6) of the SWiFT system, demonstrating its capability to perform the end-to-end transfusion process autonomously and safely in a laboratory setting.

Key Activities:Alpha Prototype Fabrication & Integration:
- Procure all selected components and fabricate custom parts required for the SWiFT prototype chassis and internal structure.
- Integrate the mechanical, electrical, and fluidic subsystems into a single, cohesive benchtop unit.
- Develop the core control software for process automation, data logging, and system diagnostics.
- Integrate the non-invasive vital sign sensors and develop the intelligent alert system software, testing its ability to trigger alerts based on simulated physiological data.

System Refinement & Ruggedization:
- Conduct iterative testing of the integrated benchtop prototype to identify and resolve software bugs, hardware conflicts, and workflow inefficiencies.
- Based on benchtop testing, design and fabricate a second-generation "alpha-ruggedized" prototype, focusing on miniaturization, durability, and a user-friendly interface for field operation.
- Perform environmental testing on the ruggedized prototype to assess its resilience to shock, vibration, and temperature extremes relevant to military operational environments.

Verification & Validation (V&V) Testing:
- Develop a comprehensive V&V test plan with clear pass/fail criteria for every system function.
- Conduct extensive laboratory testing using blood analogues and whole blood samples (per appropriate protocols) to validate the system's performance in blood collection, typing, crossmatching, and transfusion accuracy.
- Validate the vital sign monitoring and alert system against calibrated medical simulators to confirm its accuracy and reliability in detecting adverse events.

Phase II Milestones:

- Month 9: Benchtop Prototype Integration Complete: A functional, integrated benchtop prototype capable of executing all core software commands and mechanical actions is complete.
- Month 18: Alpha-Ruggedized Prototype & CDR: Completion of the second-generation ruggedized prototype. A Critical Design Review (CDR) is held to approve the design for final validation testing.

Phase II Milestones (Option):

- Month 21: Full System V&V Report: Submission of a comprehensive report detailing the results of all laboratory and environmental testing, with performance data validating the entire end-to-end system and the intelligent alert functionality.
- Month 24: Final Demonstration & Transition Plan: A live, end-to-end demonstration of the final ruggedized prototype for project stakeholders. Delivery of a detailed plan for Phase III, outlining the path to FDA clearance, manufacturing scale-up, and commercial transition.

PHASE III DUAL USE APPLICATIONS: Commercial: Emergency Medical Services (EMS), remote clinics, hospitals, blood banks. Improved efficiency and safety in blood transfusions, reduced labor costs, expanded access to blood products in underserved areas, and enhanced patient safety through continuous monitoring and alerts. Disasters and Mass Casualty Incidents.DoW/Military: Tactical combat casualty care, forward operating bases, mass casualty events. Rapid and reliable blood transfusions in austere environments, reduced reliance on external blood supplies, improved survival rates for wounded warfighters, and enhanced real-time patient monitoring capabilities.

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KEYWORDS: Automation, Blood Transfusion, Medical Device, Robotics, Sensors, Diagnostics, Autotransfusion, Blood banking, Vital Sign Monitoring, Patient Safety, Wireless Alerts

DPA26BZ01-DV004 TITLE: Extremity Platform for On-demand Surgical Implantation and Tissue Integration with Osteochondral Neogenesis

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Biomanufacturing

COMPONENT TECHNOLOGY PRIORITY AREA(S): Biotechnology

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop an on-demand regenerative medicine platform for complete finger restoration following trauma, enabling full functional recovery and eliminating the need for traditional finger and joint repair surgeries.

DESCRIPTION: Joint injuries represent a critical threat to military readiness and long-term service member health. Post-traumatic osteoarthritis (PTOA) affects service members at a 12-fold higher rate than civilians, with joint injuries accounting for over 25% of all military medical evacuations from theater. Current treatment paradigms rely on joint replacement with metal and plastic implants that are fundamentally incompatible with military service requirements, preventing return to full duty and participation in high-impact activities essential for combat readiness. The economic burden exceeds $25 billion annually when accounting for direct medical costs, lost productivity, and disability compensation. Service members with PTOA face years of progressive disability, revision surgeries, and often permanent functional limitations. Hand and finger trauma from IED blasts affect 68% of service members' ability to return to full duty, while foot and ankle injuries severely limit mobility and load-bearing capacity crucial for deployment readiness. Current technology has demonstrated complete regeneration of 5cm segmental bone defects and full-thickness cartilage restoration in large animal models. However, effective reconstruction of dense connective tissues like tendon and ligament critical for the restoration of function has not been demonstrated. Unlike mechanical implants, regenerated tissue grows with the patient, self-repairs minor damage, and maintains the complex biomechanical properties optimized by evolution. This enables service members to return to unrestricted duty after experiencing a traumatic joint injury. This SBIR seeks a revolutionary approach: achieving complete biological finger regeneration by converging technologies that restore native tissue architecture and function, moving beyond current methods of mechanical stabilization and repair. Proposals should detail methods to produce scaffolds that provide an anatomically precise framework for regeneration and to deliver the bioactive proteins and therapeutic factors required to promote the growth of bone, cartilage, ligaments, tendons, and the blood vessels necessary to sustain the new tissue. The platform will be able to address finger and hand injuries critical to military function, restoring fine motor control and full range of motion to hands. Focusing on the finger joints will provide a proof-of-concept for regenerating multiple integrated tissues through tissue-engineered approaches. This smaller-scale model is advantageous because it has lower load-bearing requirements than large joint reconstructions (such as the knee, shoulder, or hip), making it an ideal initial objective. Success metrics include complete regeneration of the finger digit verified by imaging, restoration of full range of motion and fine motor capacity appropriate. The technology will be immediately translatable to civilian populations, addressing the 1 million Americans annually who suffer major extremity injuries and establishing a new standard of care that makes prosthetics and metal implants obsolete.

PHASE I: This topic is soliciting Direct to Phase II (DP2) proposals only. This project is suitable for Direct to Phase II consideration if strong preliminary data from the proposing firm demonstrates feasibility of the protein engineering platform, tendon/ligament scaffold integration approach, and 3D printing approach. DP2 justification would be based on existing proof-of-concept data showing bone and dense connective tissue regeneration, established manufacturing processes, and clear regulatory pathway

through FDA Breakthrough Designation precedent. Direct to phase II performers must demonstrate that they have previously completed the following:

- Design and fabrication of "patient-specific" scaffolds
- Conduct in vitro studies assessing cell viability, proliferation, and differentiation on scaffolds
- Perform pilot animal study to evaluate biocompatibility and early tissue regeneration
- Develop surgical protocols for scaffold implantation and tissue integration

PHASE II: Objective: Complete animal studies, establish GMP manufacturing, and prepare for clinical trials.
Activities:

- Develop optimized growth factor delivery profiles for bone, ligament, tendon, cartilage, synovium, and vascular regeneration
- Conduct pivotal IDE-enabling studies in animal joint regeneration models
- Complete biocompatibility testing for all scaffold components
- Establish GMP-compliant manufacturing processes for patient-specific implants
- Develop Chemistry, Manufacturing, and Controls (CMC) documentation
- (Option) Prepare and submit Investigational Device Exemption (IDE) application to FDA

Phase II Milestones:

- Month 6: Demonstrate optimized growth factor delivery profiles for bone, ligament, tendon, cartilage, synovium, and vascular regeneration
- Month 12: Complete pivotal animal studies demonstrating safety, efficacy, and total joint regeneration
- Month 18: Establish GMP manufacturing and complete biocompatibility testing
- Month 24 (Option): Submit IDE application to FDA and receive approval

PHASE III DUAL USE APPLICATIONS: : Commercial: Treatment of traumatic joint injuries, and congenital joint disorders in civilian populations. Market applications include orthopedic surgery, sports medicine, and trauma centers. Potential to replace current $50+ billion joint replacement industry with regenerative solutions offering superior long-term outcomes. Department of War (DoW)/Military: Restoration of full combat readiness for service members with hand trauma, elimination of medical separations due to debilitating hand injuries, reduced long-term healthcare costs, and improved veteran quality of life. Applications span combat casualty care, training injuries, and occupational trauma across all service branches.

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KEYWORDS: Regenerative medicine, joint regeneration, tissue engineering, bioactive scaffolds, military medicine, combat injuries, ligament and tendon regeneration, bone regeneration

Appendix A: DARPA PHASE I PROPOSAL INSTRUCTIONS

I. Introduction

NOTE: do NOT submit a Phase I proposal against topic DPA26BZ01-NP001 unless you have submitted a white paper and have been notified to submit a Phase I proposal.

A complete proposal submission consists of:

Volume 1: Proposal Cover Sheet
Volume 2: Technical Volume
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents
- a. Verification of Eligibility of Small Business Joint Ventures (Attachment 3), if applicable
- b. Data Rights Assertions (if applicable)
- c. Other supporting documentation

Volume 6: Fraud, Waste and Abuse Training
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

The Defense SBIR/STTR Innovation Portal (DSIP) provides a structure for building the proposal volumes and submitting a consolidated proposal package. If this is your first time submitting an SBIR proposal using DSIP, please review detailed training guides at https://www.dodsbirsttr.mil/submissions/learning-support/training-materials. It is the responsibility of the proposing firm to ensure that a complete proposal package is certified and submitted by the close date listed in the topic to which they are responding. ***DARPA will not accept late proposals*.**

To assist in proposal development, templates for Volume 2: Technical Volume and Volume 3: Cost Volume have been provided as attachments on the DARPA Small Business website at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics. Use of the DARPA Cost Proposal template is mandatory.

Proposers should ensure that they have an accurate and active entity registration on SAM.gov, and those engaging in International Traffic in Arms Regulations (ITAR) or Controlled Unclassified Information (CUI) work for DARPA must have Level 2 certification. Please see https://www.sprs.csd.disa.mil/nistsp.htm for more information. Many resources exist to assist companies with meeting this criterion, one of which is Project Spectrum (http://www.projectspectrum.io). Users can create accounts and gain access to useful tools, including educational videos and material, as well as a cyber self-assessment.

II. Proprietary Information

Proposers that include in their proposals data that they do not want disclosed to the public for any purpose, or used by the Government except for evaluation purposes, shall follow instructions in the DoW SBIR BAA regarding marking propriety proposal information.

III. Phase I Proposal Instructions

a. Classification and Marking of Proposal Submission

1. **All proposals are required to be UNCLASSIFIED or CUI. Do not include any**

classified information in your proposal submission.

2. Do not include any proprietary information (PROPIN) on the Proposal Coversheet (Volume 1), as this may be released publicly, if selected for award.

3. PROPIN or any other CUI information may be included in the Technical Volume (Volume 2), as needed. Ensure that the document is marked appropriately as CUI with the appropriate CUI Control Block on the first page of Volume 2.

4. The Cost Volume (Volume 3) should be marked CUI for PROPIN as it includes the proposer's budget/cost information.

5. Volumes 4-7 should be marked as appropriate based on content included in each volume. For additional CUI marking information refer to https://www.archives.gov/cui/registry/category-list to properly identify the categories of CUI.

b. Proposal Cover Sheet (Volume 1)

The Cover Sheet must include a brief technical abstract of no more than 3000 characters that describes the proposed research and development project with a discussion of anticipated benefits and potential commercial applications. If your proposal is selected for award, the technical abstract and discussion of anticipated benefits may be publicly released.

c. Format of the Technical Volume (Volume 2) – White Paper & Slide Deck

1. The Technical Volume must include two parts, PART ONE: white paper, and PART TWO: slide deck, combined as a single Portable Document Format (PDF) for upload to DSIP.

2. Type of File: The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the Technical Volume file. If a virus is detected, it may cause rejection of the proposal. Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

3. Length: The length of the white paper shall not exceed ten (10) pages, and the slide deck shall not exceed five (5) pages/slides. The Government will not consider pages in excess of the page count limitations.

1. Layout: Number all pages of your proposal consecutively. Font size should not be smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. The header on each page of the Technical Volume should contain your company name, topic number, and proposal number assigned by DSIP when the Cover Sheet was created. The header may be included in the one-inch margin.

d. Content of the Technical Volume (Volume 2) – White Paper & Slide Deck (15 pages)

See **Section II** of the Phase I Template – Volume 2: Technical Volume at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics for the slide deck and white paper template.

White Paper (not to exceed 10 pages). Provide the following information:

1. Goals and Impact: Clearly describe what is being proposed and what difference it will make (qualitatively and quantitatively), including a brief discussion on how this directly relates to the topic.

2. Technical Plan: Provide an explicit, detailed description of the Phase I approach. The Statement of Work should indicate what tasks are planned, how and where the work will be conducted, a schedule of major events, and the final product(s) to be delivered. The Phase I effort should attempt to determine the technical feasibility of the proposed concept. The methods planned to achieve each objective or task should be discussed explicitly and in detail.

3. Management and Capabilities: Designate key personnel who will be involved in the Phase I effort. Provide a summary of expertise of the team, including subcontractors and key personnel. Describe the organizational experience in this technology area, previous work not directly related to the proposed effort but similar, existing intellectual property required to complete the project, and any specialized facilities to be used as part of the project. List Government-furnished materials or data assumed to be available. Describe any specialized facilities to be used as part of the project, the extent of access to these facilities, and any biological containment, biosafety, and certification requirements.

4. Transition and Commercialization Plan (not to exceed five (5) pages):

    a) Describe the commercial product or DoW system to be developed.
    b) Discuss the potential end users – DoW, Federal, and/or private sector customers. Discuss your business model for this technology (i.e., how do you anticipate generating revenue with this technology?). Who are you selling to, directly or indirectly, a supplier, an integrator, or an end user?
    c) Describe your company's funding history. Discuss how much additional funding above this proposed effort (include additional required technology development, staffing requirements, infrastructure requirements, intellectual property (IP) strategy costs, etc.) will be required to bring this technology to market and how you anticipate going about getting that funding (e.g., Government S&T contracts, investment).
    d) Describe the timeline to maturity for sales or transition to an end user. Describe your IP strategy.
    e) Describe the technology, market, team and business risks associated with this proposed effort and your plan to mitigate these risks.

Slide Deck (not to exceed five (5) slides). Provide the following information (convert the completed deck to a PDF and attach it to the white paper):

1. What are you trying to do and how does this directly relate to the topic?

2. Technology and commercial product: Specifically, what are you proposing to produce – software, system, application? Be specific on what your proposed technology development is targeting as an end state.

3. How is the technology approached today? Who is doing the research, development and delivering products/services? What are the current limitations in the technology and commercial marketplaces?

4. Management: Overview of team, facilities and qualifications.

5. Technical summary quad chart: Use template provided at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

NOTE: All letters of recommendation, CVs, and Data Rights Assertions can be loaded in Volume 5: Supporting Documents.

Advocacy Letters (OPTIONAL)* Feedback received from potential Commercial and/or DoW customers and other end-users regarding their interest in the technology to support their capability gaps. Advocacy letters that are faxed or e-mailed separately will NOT be accepted.

Letters of Intent/Commitment (OPTIONAL)* Relationships established, feedback received, support and commitment for the technology with one or more of the following: Commercial customer, DoW Program Management (PM)/Program Executive Office (PEO), a Defense Prime, or vendor/supplier to the Primes and/or other vendors/suppliers identified as having a potential role in the integration of the technology into fielded systems/products or those under development. Letters of Intent/Commitment that are faxed or e-mailed separately will NOT be accepted.

*Advocacy Letters and Letters of Intent/Commitment are optional, and should ONLY be submitted to substantiate any transition or commercialization claims made in the commercialization strategy. Please DO NOT submit these letters just for the sake of including them in your proposal. These letters DO NOT count against any page limit.

In accordance with section 3-102d of DoW Directive 5500.07, Joint Ethics Regulation, May 15, 2024, letters from Government personnel will NOT be considered during the evaluation process.

e. Format of Cost Volume (Volume 3)

Proposers are required to use the Phase I – Volume 3: Cost Proposal Template (Excel Spreadsheet) provided at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

f. Content of the Cost Volume (Volume 3)

Some items in the Cost Breakdown Guidance below may not apply to the proposed project. If such is the case, there is no need to provide information on every item.

For Phase I proposals, proposers should NOT provide documentation to substantiate how all proposed costs were derived. However, proposers should be prepared to provide such documentation should the Contracting Officer request this documentation. If any substantiating documentation is requested by the Contracting Officer, it is important to respond as quickly as possible to the request to avoid delays in contract negotiation.

Examples of substantiating documentation are as follows: If you proposed travel cost to attend a project-related meeting or conference, and used a travel website to compare flight costs, include a screen shot of the comparison. Similarly, if you proposed to purchase materials or equipment, and used the internet to search for the best source, include your market research for

those items. You do not necessarily have to propose the cheapest item or supplier, but you should be able to explain your decision to choose one item or supplier over another. It's important to provide enough information to allow contracting personnel to understand how the proposer plans to use the requested funds.

Cost Breakdown Guidance:

- List all key personnel by name as well as by number of hours dedicated to the project as direct labor.
- Special tooling and test equipment and material costs may be included. The inclusion of equipment and material will be carefully reviewed relative to need and appropriateness for the work proposed. The purchase of special tooling and test equipment must, in the opinion of the Contracting Officer, be advantageous to the Government and should be related directly to the specific topic. These may include such items as innovative instrumentation and/or automatic test equipment. Title to property furnished by the Government or acquired with Government funds will be vested with DARPA; unless it is determined that transfer of title to the contractor would be more cost effective than recovery of the equipment by the DARPA.
- Cost sharing is permitted for proposals under this BAA; however, cost sharing is not required, nor will it be an evaluation factor in the consideration of a proposal.
- If Subcontractors will be performing Fundamental Research under the effort please incorporate the following into proposal: 1) a separate statement of work (SOW) outlining the specific work that the proposer finds to qualify as Fundamental Research; OR 2) Within Prime contractor SOW identify which tasks are to be performed that are fundamental research.
- Proposers should complete both tabs within the Cost Proposal Spreadsheet (Cost Model & Milestone Chart)

For more information about cost proposals and accounting standards associated with contract awards, see the Defense Contract Audit Agency (DCAA) publication titled "Audit Process Overview – Information for Contractors" at http://www.dcaa.mil.

Please note, a separate, more detailed cost proposal spreadsheet will be provided for any Phase II Proposals.

g. Company Commercialization Report (Volume 4)

The Company Commercialization Report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. The Company Commercialization Report (CCR) is required for Phase I and Direct to Phase II proposals. Please refer to the DoW SBIR Program BAA for full details on this requirement. Information contained in the CCR will not be considered by DARPA during proposal evaluations.

h. Supporting Documents (Volume 5)

In addition to required DoW documentation and certifications, small businesses may also submit additional documentation to support the Technical Volume (Volume 2) and the Cost Volume (Volume 3) in Volume 5. See Appendix A Introduction for required certifications that must be included in Volume 5. For additional information, see the SBIR Annual Program Broad Agency Announcement (BAA) at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations

i. Fraud, Waste and Abuse (Volume 6)

The Fraud, Waste and Abuse (FWA) training is required for Phase I and Direct to Phase II proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA in your firm. This training material must be thoroughly reviewed once per year. Plan ahead and leave ample time to complete this training based on the proposal submission deadline. Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. §100), punishable by a fine of up to $10,000, up to five years in prison, or both. Understanding the indicators and types of fraud, waste and abuse that can occur is critical for the SBIR/STTR awardees' role in preventing the loss of research dollars.

j. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)

In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission (NOTE: PDF uploads will no longer be accepted). Full proposal submissions cannot be certified and submitted by the Corporate Official until Volume 7 is fully completed and the webform is submitted.

For additional details, please refer to the DoW SBIR Program BAA.

APPENDIX B: DARPA DIRECT TO PHASE II (DP2) PROPOSAL INSTRUCTIONS

I. Introduction

A complete proposal submission consists of:

Volume 1: Proposal Cover Sheet
Volume 2: Technical Volume (feasibility documentation and technical proposal)
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents
- a. Subcontract Pricing Considerations (if applicable)
- b. Data Rights Assertions (if applicable)
- c. Verification of Eligibility of Small Business Joint Ventures (Attachment 3), if applicable
- d. Other supporting documentation

A completed proposal submission in DSIP does NOT indicate that the mandatory supporting documents have been uploaded. It is the responsibility of the proposing small business concern to ensure that the mandatory documents listed above have been uploaded and included with the proposal submission.

Volume 6: Fraud, Waste and Abuse Training
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

The Defense SBIR/STTR Innovation Portal (DSIP) provides a structure for building the proposal volumes and submitting a consolidated proposal package. If this is your first time submitting an SBIR or STTR proposal using DSIP, please review detailed training guides at https://www.dodsbirsttr.mil/submissions/learning-support/training-materials. It is the responsibility of the proposing firm to ensure that a complete proposal package is certified and submitted by the close date listed in the topic to which they are responding. ***DARPA cannot accept late proposals.***

To assist in proposal development, templates for Volume 2: Technical Volume and Volume 3: Cost Volume have been provided as attachments to the announcement posted at https://www.dodsbirsttr.mil/submissions/login. Use of these templates is mandatory.

Proposers should ensure that they have an accurate and active entity registration on SAM.gov, and those engaging in International Traffic in Arms Regulations (ITAR) or Controlled Unclassified Information (CUI) work for DARPA must have Level 2 CMMC certification. Please see https://www.sprs.csd.disa.mil/nistsp.htm for more information. Many resources exist to assist companies with meeting this criterion, one of which is Project Spectrum (http://www.projectspectrum.io). Users can create accounts and gain access to useful tools, including educational videos and material, as well as a cyber self-assessment.

II. Proprietary Information

Proposers that include in their proposals data that they do not want disclosed to the public for any purpose, or used by the Government except for evaluation purposes, shall follow instructions in the DoW BAA regarding marking propriety proposal information.

III. DP2 Proposal Instructions

a. Classification and Marking of Proposal Submission

6. **All proposals are required to be UNCLASSIFIED or CUI. Do not include any classified information in your proposal submission.**

7. Do not include any proprietary information (PROPIN) on the Proposal Coversheet (Volume 1), as this may be released publicly, if selected for award.

8. PROPIN or any other CUI information may be included in the Technical Volume (Volume 2), as needed. Ensure that the document is marked appropriately as CUI with the appropriate CUI Control Block on the first page of Volume 2.

9. The Cost Volume (Volume 3) should be marked CUI for PROPIN as it includes the proposer's budget/cost information.

10. Volumes 4-7 should be marked as appropriate based on content included in each volume. For additional CUI marking information refer to https://www.archives.gov/cui/registry/category-list to properly identify the categories of CUI.

b. Proposal Cover Sheet (Volume 1)

The Cover Sheet must include a brief technical abstract of no more than 3000 characters that describes the proposed R&D project with a discussion of anticipated benefits and potential commercial applications. Do not include proprietary or classified information in the Proposal Cover Sheet. If your proposal is selected for award, the technical abstract and discussion of anticipated benefits may be publicly released.

c. Format of Technical Volume (Volume 2) – Standard Proposal Format

1. The Technical Volume must include two parts, PART ONE: Feasibility Documentation and PART TWO: Technical Proposal. See **Sec I** of the Direct to Phase II Template – Volume 2: Feasibility Documentation and Technical Proposal at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

2. Type of file: The Technical Volume must be a single Portable Document Format (PDF) file, including graphics. Perform a virus check before uploading the Technical Volume file. If a virus is detected, it may cause rejection of the proposal. **Do <u>not</u> lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document**.

3. Length: The length of each part of the technical volume (Feasibility Documentation and Technical Proposal) will be specified by the corresponding topic. The Government will not consider pages in excess of the page count limitations.

4. Layout: Number all pages of your proposal consecutively. Font size should not be smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. The header on each page of the Technical Volume should contain your company name, topic number, and proposal number assigned by DSIP when the Cover Sheet was created. The header may be included in the one-inch margin.

d. Content of the Technical Volume (Volume 2) – Standard Proposal Format

PART ONE: Feasibility Documentation

1. Provide documentation to substantiate that the scientific and technical merit and feasibility described in the Phase I section of the topic has been met and describe the potential commercial applications. Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results.
2. Maximum page length for feasibility documentation will be specified by the topic. If you have references, include a reference list or works cited list on the last page of the feasibility documentation. This will count towards the page limit.
3. Work submitted within the feasibility documentation must have been substantially performed by the proposer and/or the Principal Investigator.
4. If technology in the feasibility documentation is subject to Intellectual Property (IP), the proposer must either own the IP or must have obtained license rights to such technology prior to proposal submission, to enable it and its subcontractors to legally carry out the proposed work. Documentation of IP ownership or license rights shall be included in the Technical Volume of the proposal.
5. Include a one-page summary on Commercialization Potential addressing the following:

a) Does the company have marketing expertise and, if not, how will that expertise be brought into the company?

b) Describe the potential for commercial (Government or private sector) application and the benefits expected to accrue from this commercialization.

DO NOT INCLUDE marketing material. Marketing material will NOT be evaluated.

PART TWO: Standard Technical Proposal

Significance of the Problem. Define the specific technical problem or opportunity addressed and its importance.

1. **Phase II Technical Objectives.** Enumerate the specific objectives of the Phase II work and describe the technical approach and methods to be used in meeting these objectives.

2. **Phase II Statement of Work.** The statement of work should provide an explicit, detailed description of the Phase II approach, indicate what is planned, how and where the work will be carried out, a schedule of major events, and the final product to be delivered. The methods planned to achieve each objective or task should be discussed explicitly and in detail. This section should be a substantial portion of the total proposal.
   a. Human/Animal Use: Proposers proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential delay of contract award.
   b. Phase II Option Statement of Work (if applicable, specified in the corresponding topic). The statement of work should provide an explicit, detailed description of the activities planned during the Phase II Option, if exercised. Include how and where the work will be carried out, a schedule of major events, and the final product to be delivered. The methods planned to achieve each objective or task should be discussed explicitly and in detail.

3. **Related Work.** Describe significant activities directly related to the proposed effort, including any conducted by the PI, the proposer, consultants or others. Describe how these activities interface with the proposed project and discuss any planned coordination with outside sources. The proposal must persuade reviewers of the proposer's awareness of the state of the art in the

specific topic. Describe previous work not directly related to the proposed effort but similar. Provide the following: (1) short description, (2) client for which work was performed (including individual to be contacted and phone number), and (3) date of completion.

5. **Relationship with Future Research or Research and Development.**
    a. State the anticipated results of the proposed approach if the project is successful.
    b. Discuss the significance of the Phase II effort in providing a foundation for Phase III research and development or commercialization effort.
    c. Identify the applicable clearances, certifications and approvals required to conduct Phase II testing and outline the plan for ensuring timely completion of said authorizations in support of Phase II research or research and development effort.

6. **Key Personnel.** Identify key personnel who will be involved in the Phase II effort including information on directly related education and experience. A concise resume of the PI, including a list of relevant publications (if any), must be included. All resumes count toward the page limitation. Identify any foreign nationals you expect to be involved on this project.

7. **Foreign Citizens.** Identify any foreign citizens or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. Refer to DoW SBIR BAA for more information.
   Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. § 552a), if applicable, and the Freedom of Information Act (5 U.S.C. § 552(b)(6)).

8. **Facilities/Equipment.** Describe available instrumentation and physical facilities necessary to carry out the Phase II effort. Items of equipment to be purchased (as detailed in the cost proposal) shall be justified under this section. Also state whether or not the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name) and local Governments for, but not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices and handling and storage of toxic and hazardous materials.

9. **Subcontractors/Consultants.** Subcontractor means any supplier, distributor, vendor, firm, academic institution, research center, or other person or entity that furnishes supplies or services pursuant to a subcontract, at any tier. Involvement of a university or other subcontractors or consultants in the project may be appropriate. If such involvement is intended, it should be identified and described according to the Cost Breakdown Structure at https://www.dodsbirsttr.mil/submissions/learning-support/firm-templates. Please refer to DoW SBIR BAA for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.

10. **Prior, Current or Pending Support of Similar Proposals or Awards.** If a proposal submitted in response to this topic is substantially the same as another proposal that was funded, is now being funded, or is pending with another Federal Agency, or another or the same DoW Component, you must reveal this on the Proposal Cover Sheet and provide the following information:
    a. Name and address of the Federal Agency(s) or DoW Component to which a proposal was submitted, will be submitted, or from which an award is expected or has been received.

b. Date of proposal submission or date of award.
c. Title of proposal.
d. Name and title of the PI for each proposal submitted or award received.
e. Title, number, and date of BAA(s) or solicitation(s) under which the proposal was submitted, will be submitted, or under which award is expected or has been received.
f. If award was received, state contract number.
g. Specify the applicable topics for each proposal submitted or award received.

Note: If this does not apply, state in the proposal "No prior, current, or pending support for proposed work."

11. **Transition and Commercialization Strategy.** DARPA is equally interested in dual use commercialization of SBIR/STTR projects that result in products sold to the U.S. military, the private sector market, or both. DARPA expects explicit discussion of key activities to achieve this result in the transition and commercialization strategy part of the proposal. The Technical Volume of each Direct to Phase II proposal must include a transition and commercialization strategy section. The Phase II transition and commercialization strategy shall not exceed five (5) pages and will NOT count against the proposal page limit.

    Information contained in the commercialization strategy section will be used to determine suitability for participation in EEI. Selection for participation in EEI will be made independently following selection for SBIR/STTR award. Please refer to item 3 of the Award and Contract Information section of these Instructions for more information on the DARPA EEI and additional proposal requirements.

    The transition and commercialization strategy should include the following elements:

    a) **A summary of transition and commercialization activities conducted during Phase I, and the Technology Readiness Level (TRL) achieved.** Discuss the market, competitive landscape, potential stakeholders and end-users, and how the preliminary transition and commercialization path or paths may evolve during the Phase II project. Describe key proposed technical milestones during Phase II that will advance the technology towards product such as prototype development, laboratory and systems testing, integration, testing in operational environment, and demonstrations.
    b) **Problem or Need Statement.** Briefly describe what you know of the problem, need, or requirement, and its significance relevant to a Department of Defense application and/or a private sector application that the SBIR/STTR project results would address. Is there a broader societal need you are trying to address? Please describe.
    c) **Description of Product(s) and/or System Application(s).** Identify the commercial product(s) and/or DoW system(s), or system(s) under development, or potential new system(s). Identify the potential DoW end-users, Federal customers, and/or private sector customers who would likely use the technology.
    d) **Business Model(s)/Procurement Mechanism(s).** Discuss your current business model hypothesis for bringing the technology to market. Describe plans to license, partner, or self-produce your product. How do you plan to generate revenue? Describe the resources you expect will be needed to implement your business models. Discuss your plan and expected timeline to secure these resources. Understanding DARPA's goal of creating and sustaining a U.S. military advantage, describe how you intend to develop your product and supply chains to enable this differentiation.
    e) **Target Market.** Describe the market and addressable market for the innovation. Describe the customer sets you propose to target, their size, their growth rate, and the key

reasons they would consider procuring the technology. Discuss the business economics and market drivers in the target industry. Describe competing technologies existent today on the market as well as those being developed in the lab. How has the market opportunity been validated? Describe the competition. How do you expect the competitive landscape may change by the time your product/service enters the market?

f) **Funding Requirements.** Describe your company's funding history. How much external financing have you raised? Describe your plans for future funding sources (internal, loan, angel, venture capital, etc.).

g) **Transition and Commercialization Risks.** Describe the major technology, market and team risks associated with achieving successful transition of the DARPA funded technology. DARPA is not afraid to take risks but we want to ensure that our awardees clearly understand the risks in front of them. What are the key risks in bringing your innovation to market? What actions do you plan to undertake to mitigate these risks?

h) **Expertise/Qualifications of Team/Company Readiness.** Describe the expertise and qualifications of your management, marketing/business development and technical team that will support the transition of the technology from the prototype to the commercial market and into Government operational environments. Has this team previously taken similar products/services to market? If the present team does not have this needed expertise, how do you intend to obtain it? What is the financial history and health of your company (e.g., availability of cash, profitability, revenue growth, etc.)?

i) **Anticipated Transition and Commercialization Results.** Include a schedule showing the anticipated quantitative transition and commercialization results from the Phase II project at one year after the start of Phase II, at the completion of Phase II, and after the completion of Phase II (i.e., amount of additional investment, sales revenue, etc.). After Phase II award, the company is required to report actual sales and investment data in its Company Commercialization Report at least annually.

12. **Data Rights Assertions**

See Direct to Phase II Template – Volume 2: Feasibility Documentation and Technical Proposal at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics, for details on data rights assertions.

Advocacy Letters (OPTIONAL)* Feedback received from potential Commercial and/or DoW customers and other end-users regarding their interest in the technology to support their capability gaps. Advocacy letters that are faxed or e-mailed separately will NOT be accepted.

Letters of Intent/Commitment (OPTIONAL)* Relationships established, feedback received, support and commitment for the technology with one or more of the following: Commercial customer, DoW PM/PEO, a Defense Prime, or vendor/supplier to the Primes and/or other vendors/suppliers identified as having a potential role in the integration of the technology into fielded systems/products or those under development. Letters of Intent/Commitment that are faxed or e-mailed separately will NOT be accepted.

*Advocacy Letters and Letters of Intent/Commitment are optional and should ONLY be submitted to substantiate any transition or commercialization claims made in the commercialization strategy. Please DO NOT submit these letters just for the sake of including them in your proposal. These letters DO NOT count against any page limit.

In accordance with section 3-209 of DoD 5500.7-R, Joint Ethics Regulation, letters from Government personnel will NOT be considered during the evaluation process.

e. Format of the Technical Volume (Volume 2) – White Paper & Slide Deck

1. The Technical Volume must include two parts, PART ONE: white paper and PART TWO: slide deck, combined as a single Portable Document Format (PDF) for upload to DSIP. See **Sec II** of the Direct to Phase II Template – Volume 2: Feasibility Documentation and Technical Proposal at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

2. Type of file: The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the Technical Volume file. If a virus is detected, it may cause rejection of the proposal. Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

4. Length: The length of each part of the technical volume (white paper and slide deck) will be specified by the corresponding topic. The Government will not consider pages in excess of the page count limitations.

5. Layout: Number all pages of your proposal consecutively. Font size should not be smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. The header on each page of the Technical Volume should contain your company name, topic number, and proposal number assigned by DSIP when the Cover Sheet was created. The header may be included in the one-inch margin.

f. Content of the Technical Volume (Volume 2) – White Paper & Slide Deck

White Paper (NTE 20 pages). Provide the following information:
Goals and Impact: Clearly describe what is being proposed and what difference it will make (qualitatively and quantitatively), including a brief discussion on how this directly relates to the topic.

1. Phase I Feasibility: This topic is accepting Direct to Phase II proposals ONLY. To be eligible, proposers must demonstrate that the documented feasibility work as required in the topic has been achieved outside of the SBIR program.

2. Technical Plan: Outline and address all technical areas and challenges inherent in the approach and possible solutions for overcoming potential problems. Provide specific objectives, metrics, and milestones at intermediate stages to demonstrate a plan for accomplishment of the project objectives. Propose additional appropriate qualitative and quantitative metrics specific to the approach, as needed. Intermediary milestones should occur at no greater than 1-month increments.

3. Management and Capabilities: Designate key personnel who will be involved in the Phase II effort. Provide a summary of expertise of the team, including subcontractors and key personnel. Describe the organizational experience in this technology area, previous work not directly related to the proposed effort, but similar, existing intellectual property required to complete the project, and any specialized facilities to be used as part of the project. List Government-furnished materials or data assumed to be available. Describe any specialized facilities to be used as part of the project, the extent of access to these facilities, and any biological containment, biosafety, and certification requirements.

4. Transition and Commercialization Plan:
   a) Describe the commercial product or DoW system to be developed.

b) Discuss the potential end users – DoW, Federal, and/or private sector customers. Discuss your business model for this technology (i.e., how you anticipate generating revenue with this technology?). Who are you selling to directly or indirectly, a supplier, an integrator, or an end user?
c) Describe your company's funding history. Discuss how much additional funding above this proposed effort (include additional required technology development, staffing requirements, infrastructure requirements, IP strategy costs, etc.) that will be required to bring this technology to market and how you anticipate going about getting that funding (e.g., Govt S&T contracts, investment).
d) Describe the timeline to maturity for sales or transition to an end user. Describe your IP strategy.
e) Describe the technology, market, team and business risks associated with this proposed effort and your plan to mitigate these risks.

Slide Deck (not to exceed 15 slides). Provide the following information (convert the completed deck to a PDF and attach it to the white paper):

1. What are you trying to do and how does this directly relate to the topic?

2. Technology and commercial product: Specifically, what are you proposing to produce – software, system, application? Be specific on what your proposed technology development is targeting as an end state.

3. How is the technology approached today? Who is doing the research, development and delivering products/services? What are the current limitations in the technology and commercial marketplaces?

4. Technical and commercial value proposition: How have you substantiated the feasibility of your approach? What is innovative in your approach and how does it compare to the state-of-the-art? Why do you think it will be successful both from a technical and commercial perspective? If you are successful, what difference will it make? Discuss your proposed business model – how do you expect to generate revenue from your technology?

5. Technical and commercial risks: What are the key technical and commercial challenges and how do you plan to address/overcome these?

6. Technical and commercial market analysis: Who will care and what will the impact be if you are successful? What/who are the markets/industries/integrators/stakeholders that would/should care?

7. Cost, schedule and milestones: Provide a summary of your cost volume. Provide a summary of your schedule and milestones. How much will your proposed effort cost in total? How long will it take? What are your technical milestones for achieving the proposed efforts? What are your transition and commercialization plan milestones? Discuss how much funding will be required to bring your proposed technology to market and execute your proposed transition and commercialization plan. Include any funding raised to date and expected plans for raising any additional required funding (Government contracting revenue, product sales, internal R&D investment, loan, angel or Venture Capital investment, etc.). Describe timeline to maturity for operational use or commercial sales.

8. Management: Overview of team, facilities and qualifications.

9. Technical summary quad chart: Use template provided at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

10. Commercialization summary quad chart: Use the DARPA Transition and Commercialization Strategy Plan (TCSP) Quad template, located at https://www.darpa.mil/work-with-us/communities/small-business/transition-commercialization/.

NOTE: All letters of recommendation, CVs, and Data Rights Assertions (see Direct to Phase II Template – Volume 2: Technical Volume at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics for details on data rights assertions) can be loaded in Volume 5: Supporting Documents.

g. Format of Cost Volume (Volume 3)

Proposers are required to use the Direct to Phase II – Volume 3: Cost Proposal Template (Excel Spreadsheet) provided at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics. The Cost Volume (and supporting documentation) DOES NOT count toward the page limit of the Technical Volume.

Subcontractors should use this document for unsanitized cost proposals and send them to SBIR_BAA@darpa.mil.

h. Content of the Cost Volume (Volume 3)

Some items in the Cost Breakdown Guidance below may not apply to the proposed project. If such is the case, there is no need to provide information on every item.

ALL proposed costs should be accompanied by documentation to substantiate how the cost was derived. For example, if contractors:

- Propose direct labor costs, contractors could provide current paystubs for proposed employees, or a rate agreement with DCMA, etc.
- Propose consultant costs, contractors could provide historical invoices, current contract with consultant, etc.
- Propose to purchase materials or equipment; you could provide historical invoices, current quotes, market research for those items, etc.

The above list is purely for informational purposes and does not limit the proposer from proposing other costs. Again, however, all costs must be accompanied by substantiating documentation.

Proposers do not necessarily have to propose the cheapest item or supplier, but you should explain your decision to choose one item or supplier over another. It is important to provide enough information to allow contracting personnel to understand how the proposer plans to use the requested funds. If selected for award, failure to include the documentation with your proposal will delay contract negotiation, and the proposer will be asked to submit the necessary documentation to the Contracting Officer to substantiate costs (e.g., cost estimates for equipment, materials, and consultants or subcontractors). It is important to respond as quickly as possible to the Contracting Officer's request for documentation.

<u>Cost Breakdown Guidance</u>:

- List all key personnel by name as well as by number of hours dedicated to the project as direct labor. Special tooling and test equipment and material cost may be included. The inclusion of equipment and material will be carefully reviewed relative to need and appropriateness for the

work proposed. The purchase of special tooling and test equipment must, in the opinion of the Contracting Officer, be advantageous to the Government and should be related directly to the specific topic. These may include such items as innovative instrumentation and/or automatic test equipment. Title to property furnished by the Government or acquired with Government funds will be vested with DARPA; unless it is determined that transfer of title to the contractor would be more cost effective than recovery of the equipment by the DARPA.

- Cost sharing is permitted for proposals under this announcement; however, cost sharing is not required nor will it be an evaluation factor in the consideration of a proposal.
- All subcontractor costs and consultant costs must be detailed at the same level as prime contractor costs in regard to labor, travel, equipment, etc. All subcontractor costs must be substantiated with Subcontractor Pricing Considerations. Enter this information in the Explanatory Material section of the online cost proposal form. **NOTE: If proposing subcontractors, contractors must satisfy the requirement of FAR Part 15.404-3(b), Subcontract Pricing Considerations, stated below:**

  (b) The prime contractor or subcontractor shall-
  (1) Conduct appropriate cost or price analyses to establish the reasonableness of proposed subcontract prices;
  (2) Include the results of these analyses in the price proposal

  The Supporting Documents Volume (Volume 5) may be used for this documentation.
- If Subcontractors will be performing Fundamental Research under the effort please incorporate the following into proposal: 1) a separate SOW outlining the specific work that the proposer finds to qualify as Fundamental Research; OR 2) Within Prime contractor SOW identify which tasks are to be performed that are fundamental research.

For more information about cost proposals and accounting standards, see the DCAA publication titled "Audit Process Overview – Information for Contractors" available at http://www.dcaa.mil.

i. Company Commercialization Report (Volume 4)

The Company Commercialization Report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. The Company Commercialization Report (CCR) is required for Phase I and Direct to Phase II proposals. Please refer to the DoW SBIR Program BAA for full details on this requirement. Information contained in the CCR will not be considered by DARPA during proposal evaluations.

j. Supporting Documents (Volume 5)

In addition to required DoW documentation and certifications, small businesses may also submit additional documentation to support the Technical Volume (Volume 2) and the Cost Volume (Volume 3) in Volume 5. See Appendix B Introduction and the SBIR Annual Program Broad Agency Announcement (BAA) at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations for **required** certifications that must be included in Volume 5.

k. Fraud, Waste and Abuse (Volume 6)

The Fraud, Waste and Abuse (FWA) training is required for Phase I and Direct to Phase II proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA in your firm. This training material must be thoroughly reviewed once per year. Plan ahead and leave ample time to complete this training based on the proposal submission

deadline. Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. § 1001), punishable by a fine of up to $10,000, up to five years in prison, or both. Understanding the indicators and types of fraud, waste and abuse that can occur is critical for the SBIR/STTR awardees' role in preventing the loss of research dollars.

l. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission (NOTE: PDF uploads will no longer be accepted). Full proposal submissions cannot be certified and submitted by the Corporate Official until Volume 7 is fully completed and the webform is submitted.

Defense Health Agency (DHA)
Small Business Innovation Research (SBIR)
DoW 2026 SBIR BAA Release
Proposal Submission Instructions

INTRODUCTION

The Defense Health Agency (DHA) SBIR Program seeks small businesses with strong research and development capabilities to pursue and commercialize medical technologies.

Proposers responding to a topic in this solicitation must follow all general instructions provided in the Department of War (DoW) SBIR Program solicitation. DHA requirements in addition to or deviating from the DoW Program solicitation are provided in the instructions below.

Only Government personnel will evaluate proposals submitted under this solicitation cycle with the exception of the technical personnel from Knowesis Inc. who will provide technical analysis in the evaluation of proposals submitted against DHA topics:

- Military Working Dog Whole Blood Product or Substitute
- Detection and Treatment of Traumatic Brain Injury in Military Working Dogs
- Environmental Exposure Detection technologies and/or Decontamination Treatments for Military Working Dogs

Proposers are encouraged to thoroughly review the DoW solicitation and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The full DoW solicitation is available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate solicitation cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Specific questions pertaining to the administration of the DHA SBIR Program and these proposal preparation instructions should be directed to: DHA SBIR Program Management Office (PMO) Email: usarmy.detrick.medcom-usamrmc.mbx.dhpsbir@health.mil

For technical questions about a topic during the pre-release period, contact the Topic Author(s) listed for each topic in the BAA. To obtain answers to technical questions during the formal BAA period, visit the Topic Q&A: https://www.dodsbirsttr.mil/submissions/login.

PHASE I PROPOSAL GUIDELINES

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in the DoW solicitation. The following proposal instructions supplement those found within the DoW SBIR FY26 R1 Program BAA.

Proposal Coversheet (Volume 1)

The proposal coversheet is prepared on DSIP and in accordance with the DoW solicitation.

Technical Volume (Volume 2)
The technical volume is not to exceed 20 pages and must follow the formatting requirements provided in the DoW SBIR solicitation. Do not duplicate the electronically-generated Cover Sheet or insert information associated with the Technical Volume into other sections of the proposal, as these will count toward the 20-page limit.

Only the electronically-generated Cover Sheet and Cost Volume are excluded from the 20- page limit. Technical Volumes that exceed the 20-page limit will be deemed non-compliant and will not be evaluated.

Cost Volume (Volume 3)
The Phase I Base amount must not exceed $250,000 over a 6-month period of performance. Costs for the Base and Option must be separated and clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.

Please review the updated Percentage of Work (POW) calculation details included in the DoW solicitation. DHA will occasionally accept deviations from the POW requirements with written approval from the Funding Agreement officer.

Travel must be justified and should relate to direct Research Development Test & Evaluation (RDT&E) Technology Readiness Level (TRL)-increasing task costs. Travel costs must include the purpose of the trip(s), number of trips, origin and destination, length of trip(s), and number of personnel.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW solicitation for full details on this requirement. Information contained in the CCR will be considered by DHA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoW solicitation for more information.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. The Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoW Program BAA.

PHASE II PROPOSAL GUIDELINES

Phase II proposals may only be submitted by Phase I awardees. Phase II is the demonstration of the technology found feasible in Phase I. Specific due date, content, and submission requirements of the

Phase II proposal will be provided by the DHA SBIR PMO; typically during month five of the Phase I contract.

Due to limited funding, the DHA SBIR Program reserves the right to limit awards under any topic. Only proposals considered to be of superior quality will be funded. Small businesses submitting a proposal are required to develop and submit a Commercialization Strategy describing feasible approaches for transitioning and/or commercializing the developed technology in their Phase II proposal, including plans to pursue appropriate seed-stage investment. This plan shall be included in the Technical Volume.

The Cost Volume must contain a budget for the entire 24-month Phase II period in an amount not to exceed the maximum dollar amount of $1,300,000. Budget costs must be submitted using the Cost Volume format (accessible electronically on the DoW submission site) and shall be presented side-by-side on a single Cost Volume Sheet.

DHA SBIR Phase II proposals have seven volumes: Proposal Cover Sheets; Technical Volume; Cost Volume; Company Commercialization Report; Supporting Documents; Fraud, Waste, and Abuse; Disclosures of Foreign Affiliations or Relationships to Foreign Countries.

The Technical Volume has a 40-page limit including: table of contents, pages intentionally left blank, references, letters of support, appendices, technical portions of subcontract documents (e.g., statements of work and resumes) and any attachments. Technical Volumes that exceed the 40-page limit will be deemed non-compliant and will not be evaluated.

DIRECT TO PHASE II (DP2) PROPOSAL GUIDELINES

Offerors may submit Direct to Phase II proposals only if allowed pursuant to the topic posting. DP2 proposals are different than traditional DHA SBIR Phase I proposals. The chart below explains some of these differences under this solicitation.

	STANDARD DHA SBIR PROCESS	DHA SBIR DP2 PROCESS
PHASE 1 FUNDING LEVEL	$250,000	None
PHASE 1 TECHNICAL *POP DURATION	6 months	None
PHASE 2 FUNDING LEVEL	$1,300,000	$1,300,000
PHASE 2 TECHNICAL *POP DURATION	24 months	24 months

*POP= Period of Performance

DIRECT TO PHASE II (DP2) PROPOSAL SUBMISSION GUIDELINES

Direct to Phase II topics require documentation to determine that Phase I feasibility described in the Phase I section of the topic has been met.

Use data to substantiate your claims. Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results.

Feasibility documentation cannot be based upon any prior or ongoing federally funded SBIR or STTR work and DP2 proposals MUST NOT logically extend from any prior or ongoing federally funded SBIR or STTR work.

DP2 proposals must include all volumes, not to exceed maximum page limit, and must follow the formatting requirements provided in the DoW SBIR Program BAA. Technical Volumes that exceed the page limit will be deemed non-compliant and will not be evaluated. The following proposal instructions supplement those found within the DoW SBIR FY26 R1 Program BAA.

a. DoW Proposal Cover Sheet (Volume 1)

b. Technical Volume (Volume 2):

Part 1: Phase I Justification (20 Pages Maximum)

Part 2: Phase II Technical Proposal (40 Pages Maximum)

c. Cost Volume (Volume 3)

d. Company Commercialization Report (Volume 4)

e. Supporting Documents (Volume 5)

f. Fraud, Waste, Abuse (Volume 6)

g. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)

Technical Volume (Volume 2):

Phase I Justification: Offerors are **required** to provide evidence that the scientific and technical merit and feasibility have been established as described in the topic's description and Phase I.

Technical Proposal:

1. Results of current work – Discuss the objectives of your effort, the research conducted, findings or results, and estimates of technical feasibility. Provide evidence that the scientific and technical merit and feasibility have been established as described in the topic's description and Phase I.
2. Technical objectives and approach – List specific technical objectives of the Direct to Phase II research and describe the technical approach in detail to be used to meet these objectives.
3. Work plan – The plan should indicate what is planned, how and where, a schedule of major events, and the final product to be developed.
4. Related work – Describe significant activities directly related to the proposed effort, including those conducted by the Principal Investigator, the proposing firm, consultants, or others. Report how the activities interface with the proposed project and discuss any planned coordination with outside sources. The proposers' awareness of the state-of-the art in the technology and associated science must be demonstrated.
5. Relationship with future research or Research and Development – State the anticipated results of the proposed approach if the project is successful. Discuss the significance of the effort in providing a foundation for Phase III research or research and development effort.
6. Technology transition and commercialization strategy – Describe your company's strategy for converting the proposed SBIR into acquisition in DoW, other Federal Agencies, and/or private sector markets with widespread commercial use, as well as the product or service expected to result from this effort. Note: The commercialization strategy is separate from the Commercialization Report. The strategy addresses how you propose to commercialize this research, while the Company Commercialization Report covers what you have done to commercialize the results of past awards.

7. Key personnel – Identify key personnel, including the Principal Investigator, who will be involved and have impact in the effort. List directly related education and experience and relevant publications (if any) of key personnel. A concise resume of the Principal Investigator(s) must be included.
8. Foreign Citizens – Identify any foreign citizens or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. Proposing small business concerns frequently assume that individuals with dual citizenship or a work permit will be permitted to work on an SBIR project and do not report them. A proposal may be deemed non-responsive if the requested information is not provided. Therefore, proposing small business concerns should report all individuals expected to be involved on this project that are considered a foreign national as defined in the DoW Program BAA. You may be asked to provide additional information during negotiations to verify the foreign citizen's eligibility to participate in a SBIR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)).
9. Facilities/Equipment – Justify items of equipment to be purchased (as detailed in the cost proposal), including Government Furnished Equipment (GFE). All requirements for government furnished equipment or other assets, as well as associated costs, must be determined and agreed to during contract negotiations. State whether the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name) and local governments for, but not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.
10. Consultants – Involvement of university, academic institution, or other consultants in the project may be appropriate. If such involvement is intended, it should be described in detail and identified in the Cost Volume.

Cost Volume (Volume 3):
The Cost Volume must contain a budget for the entire 24-month Direct to Phase II period and up to the maximum dollar amount of $1,300,000. Costs must be separated and clearly identified on the Proposal Cover Sheet (Volume 1) and in the Cost Volume (Volume 3).

Please review the updated Percentage of Work (POW) calculation details included in the DoW Program BAA. DHA will occasionally accept deviations from the POW requirements with written approval from the Funding Agreement Officer.

Travel must be justified and relate to the project needs for direct Research Development Test & Evaluation (RDT&E) Technology Readiness Level (TRL) and increasing costs. Travel costs must include the purpose of the trip(s), number of trips, origin and destination, length of trip(s), and number of personnel.

Company Commercialization Report (Volume 4):
Completion of the CCR of the proposal submission in DSIP is required. Information contained in the CCR will be considered by DHA during proposal evaluations. Please refer to the DoW SBIR Program BAA for full details on this requirement.

Supporting Documents (Volume 5):
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoW Program BAA for more information.

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DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
The DHA SBIR Program does not participate in Technical and Business Assistance. Contractors shall not submit proposals that include Technical and Business Assistance.

The DHA SBIR Program has a Commercialization Readiness Program that provides technical and commercialization assistance to small businesses that have Phase I and Phase II projects.

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the DoW solicitation.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 days of the closing date of the topic. Non-selected companies may request feedback within 15 calendar days of the non-select notification. The Corporate Official identified in the firm's proposal shall submit the feedback request to the SBIR Office at usarmy.detrick.medcom-usamrmc.mbx.dhpsbir@health.mil. Feedback is provided in an official PDF via email to the Corporate Official identified in the firm proposal within 60 days of receipt of the request. Requests for oral feedback will not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the feedback request.

NOTE: Feedback is not the same as a FAR Part 15 debriefing. Acquisitions under this solicitation are awarded via "other competitive procedures". Therefore, offerors are neither entitled to nor will they be provided FAR Part 15 debriefs.

Refer to the DoW solicitation for procedures to protest the Announcement.

As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to:
Ms. Samantha L. Connors SBIR/STTR Chief, Contracting Officer
DHA Contracting Activity
Email: Samantha.l.connors.civ@health.mil

AWARD AND CONTRACT INFORMATION
Phase I awards will total up to $250,000 for a 6-month effort and will typically be Firm-Fixed-Price Purchase Orders.

Direct to Phase II awards will total up to $1,300,000 for a 24-month effort and will typically be Firm-Fixed-Price contracts. If a different contracting type is preferred, such as cost-plus, the rational as to why must be included in the proposal. Awardees will be informed of Technical Point of Contact/Contract Officer Representative upon award.

Data Deliverables (Contract Data Requirements Lists – CDRLS):
Data can be in the form of test data, computer software, algorithms, design details, progress reports, technical data, financial or management reports, or any information required by contract to be delivered. Multiple DD Form 1423, Contract Data Requirements Lists (CDRLS), that are included in the contract award identify potential data requirements in a contract that require the contractor to formally deliver the data (contractual obligation) to the Government. Each CDRL will reference a Data Item Description (DID) that describes data content, format, media, and intended use of a single data product. Each DID is uniquely numbered to identify the data deliverables.

ADDITIONAL INFORMATION

RESEARCH INVOLVING HUMAN SUBJECTS, HUMAN SPECIMENS/DATA, OR ANIMAL RESEARCH
Prior to contract award when an IRB is indicated, proposers must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human subjects, human specimens, or research with animals. If necessary, approvals are not obtained within two months of notification of selection, the decision to award may be terminated.

Offerors are expressly forbidden to use, or subcontract for the use of, laboratory animals in any manner without the express written approval of the DHA Research and Development (DHA R&D) Animal Care and Use Review Office (ACURO). Written authorization to begin research under the applicable protocol(s) proposed for this award will be issued in the form of an approval letter from the DHA R&D ACURO to the recipient. Modifications to previously approved protocols require re-approval by ACURO prior to implementation.

Research under this award involving the use of human subjects, to include the use of human anatomical substances or human data, shall not begin until the DHA R&D Office of Human Research Oversight (OHRO) provides formal authorization. Written approval to begin a research protocol will be issued from the DHA R&D OHRO, under separate notification to the recipient. Written approval from the DHA R&D OHRO is required for any sub-recipient using funds from this award to conduct research involving human subjects. If the Offeror intends to submit research funded by this award to the U.S. Food and Drug Administration, Offerors shall propose a regulatory strategy for review.

*NOTE: Exempt animal or human research use shall also reflect ‘yes’ on the proposal coversheet for DHA R&D ACURO and OHRO records.

Non-compliance with any provision may result in withholding funds and or termination of the award.

FEDERAL FACILITY USE
The DHA SBIR Program highly discourages small business concerns (SBCs) from subcontracting to a federal facility and/or utilizing for testing due to the significant lead time required to secure approval, which could substantially delay the performance of the award. Use of federal facilities is prohibited without an approved waiver from the DHA SBIR Office.

An SBC whose proposed work includes federal facility use is required to provide a written justification, uploaded to the Supporting Documents (Volume 5), that includes the following information:

1. An explanation of why the SBIR/STTR research project requires the use of the federal facility, including data that verifies the absence of non-federal U.S. facilities, in support of the overall mission and research area.
2. Evidence that there is no applicable U.S. facility that has the ability or expertise to perform the specified work.
3. Why the Federal Agency will not and cannot fund the use of the Federal facility or personnel for the SBIR/STTR project with non-SBIR/STTR money.

The DHA SBIR Program has the right of refusal. Companies that fail to meet requirements specified above will be at risk of delays to award or funding.

If the proposal is selected, the DHA Contracting Activity (DHACA) will assist in establishing the waiver for DHA SBIR/STTR Office approval. If approved, the proposer will subcontract directly with the federal facility and not a third-party representative.

Transfer of funds between a company and a Military Lab must meet additional requirements:

(1) The DoW Intramural Researcher must obtain a letter from his/her commanding officer or Military Facility director authorizing his/her participation in the Extramural Research project. This letter must be included in the.

(2) The DoW Intramural Researcher must also coordinate with his/her local RM office (or equivalent) to prepare a sound budget and justification for the estimated costs. Where there are no DoW-established reimbursement rates [e.g., institution review board (IRB) fees, indirect cost rates, etc.], the Military Facility's RM office (or equivalent) must provide details of how the proposed rates were determined. The DoW-generated budget and justification must be included in the proposal.

(3) The proposal must include a proposed financial plan for how the Military Facility's Intramural Research costs will be supported [i.e., directly funded by DoW, resources (other than award funds) provided by the Awardee to the Military Facility, or award funds provided by the Awardee to the Military.

INTERNATIONAL TRAFFIC IN ARMS REGULATION (ITAR)
For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and nonprofit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases, such as Phase III, if the university/research institution is the source of critical knowledge, effort, or infrastructure.

SYSTEM FOR AWARD MANAGEMENT (SAM)
As stated in FAR Provision 52.204-7 System for Award Management, proposing Small Business Concerns must be registered in SAM, https://sam.gov, at the time of submission and at the time of award, during performance, and through final payment of any contract resulting from this solicitation.
CYBERSECURITY MATURITY MODEL CERTIFICATION (CMMC)

On September 10, 2025, the DoW published the final Defense Federal Acquisition Regulation Supplement (DFARS) rule that formally integrated CMMC 2.0 into defense contracts. The new DFARS 252.204-7021 clause inserts CMMC requirements directly into contracts, making cybersecurity an essential part of doing business with the Department.

Phase I Requirements
The CMMC level required for Phase I awards is **CMMC Level 1 (Self)**. This CMMC level, or higher (see 32 CFR part 170), is required prior to award for each contractor information system that will process, store, or transmit Federal contract information (FCI) or controlled unclassified information (CUI) during performance of the contract.

This requires the completion of an annual self-assessment against the 17 security controls specified in FAR 52.204-21. Proposers must have a current (less than one year old) self-assessment and a senior official's affirmation of compliance posted in the Supplier Performance Risk System (SPRS) prior to award.

Phase II Requirements
The CMMC level required for Phase II and Direct to Phase II awards is **CMMC Level 2 (Self)**. This CMMC level, or higher (see 32 CFR part 170), is required prior to award for each contractor information system that will process, store, or transmit Federal contract information (FCI) or controlled unclassified information (CUI) during performance of the contract.

This requires the implementation of all 110 security controls outlined in NIST SP 800-171 Rev. 2. Proposers must have a current (less than three years old) CMMC Level 2 self-assessment, along with a corresponding System Security Plan (SSP), posted in the Supplier Performance Risk System (SPRS) prior to award. An annual affirmation of compliance by a senior official is also required.

If a proposer has not yet met all 110 controls, they may still be eligible for award with a Plan of Action & Milestones (POA&M). The POA&M must be posted to SPRS alongside the self-assessment and must detail a plan to close all remaining gaps within 180 days of the assessment date. Please note, certain high-priority controls cannot be on a POA&M.

DHA SBIR DoW 2026 BAA
Topic Index
Release 1

DHA26BZ01-NV001	Military Working Dog Whole Blood Product or Substitute
DHA26BZ01-NV002	Detection and Treatment of Traumatic Brain Injury in Military Working Dogs
DHA26BZ01-NV003	Environmental Exposure Detection Technologies and/or Decontamination Treatments for Military Working Dogs
DHA26BZ01-NV004	Development of Wearable Device for Detecting Biomarkers and Treating Viral and Bacterial Infections
DHA26BZ01-DV005	Development of Small Molecules with Antifungal Properties for Safe Use in Human Patients (Direct to Phase II)

DHA26BZ01-NV001 TITLE: Military Working Dog Whole Blood Product or Substitute

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Biomanufacturing (BIO)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Operational Medicine

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Develop a whole blood product or substitute to aid in hemorrhage control for Military Working Dogs (MWD) after battlefield injury that can be used near the point of injury (POI) and throughout the continuum of care to reduce morbidity and mortality.

DESCRIPTION: This topic is in support of the DoD Working Dog Strategic Research Plan concerning solutions for bleeding control and coagulopathy support.[1] The Military Working Dog (MWD) provides a unique and important service to the warfighter. MWDs serve as sentries, perform tracking and patrol, and are used for the detection of explosives. These activities come with a high risk of injury. Uncontrolled hemorrhage following traumatic injury accounts for over 45% of all MWD battlefield deaths[2]. The current standard of care for hemorrhage in the MWD is to provide immediate fluid therapy through the delivery of crystalloid fluids as the first-line treatment, which is then followed by a synthetic colloid or hypertonic saline. These treatments also require the administration of supplemental oxygen to maintain appropriate oxygen levels and for the survival of the MWD[3]. To improve their survival rates, the development of a shelf stable canine whole blood product or substitute is a critical priority.
The goal of this topic is to develop a stable canine whole blood product and/or substitute (i.e. hemoglobin or polymer oxygen carriers), intended for canine use at both POI and throughout the continuum of care. The product should have a shelf-life of greater than 3 years and be thermal stable (-9°C to 60°C) to ensure accessibility in operational environments. The product must primarily replicate the oxygen carrier characteristics of whole blood and demonstrate the ability to be used safely and effectively to treat blood loss following traumatic injury. This research topic does not support the use of canines for testing purposes. Any animal testing would require use of suitable animal models that would approximate the response of a canine.
Blood products derived from canine donors must be negative for canine red blood cell antigens DEA 1.1 and DEA 1.2. Donor animals must also be tested for blood borne diseases including canine brucellosis, hemobartonellosis, Borrelia burgdorferi (Lyme disease), Dirofilaria immitis (heartworm disease), Ehrlichia canis, Rocky Mountain spotted fever, Coccidioides immitis, Babesia canis, Babesia gibsoni, Mycoplasma haemocanis and plasma levels of von Willebrand factor. All donor animals must be current on immunizations for canine distemper, hepatitis, parainfluenza, leptospirosis, parvovirus, Bordatella, coronavirus and rabies virus as applicable.

PHASE I: Identify a blood substitute solution that can meet the performance and chemical specifications for a canine[4] whole blood product or substitute. The primary specifications must include the ability to perform oxygen exchange, possess an oxygen capacity of approximately 24 ml/dL, and be compatible with the blood chemistry of a canine4.The proposed solution must have a shelf life of greater than 3 years and be thermally stable (-9°C to 60°C). This phase should address the feasibility of the proposed solution and end with the development of a plan for the testing of the proposed solution. The plan will include appropriate studies using in vitro and in vivo models to determine whether the solution can address the various functional aspects of whole blood that are required for being a replacement. The plan must include the descriptions and methods of determining the study endpoints to be measured for each of the specifications. The plan must address the risks and potential payoffs of the technologies that are being investigated and recommend the best option to achieve the objective. Any in vivo studies are to be planned using a suitable animal model that approximates the canine. No canine research will be performed for this topic area.

PHASE II: Develop the solution as described in Phase I to generate a shelf-stable (> 3 years) and thermal stable (-9°C to 60°C) whole blood product or substitute that possess the primary performance (ability to perform oxygen exchange, possess an oxygen capacity of approximately 24 ml/dL) and chemical characteristics4 of canine blood. Methods for development must be amenable to large scale production and be able to produce a sustainable commercial market. Conduct the testing plan as outlined in Phase I. Testing will include an in vivo evaluation of the technology compared to crystalloid fluids (current standard of care), plasma, and whole blood in a model of severe hemorrhage/hemorrhagic shock. The size of the study must be appropriately powered to ensure that the results are statistically significant. The study must include the evaluation of the solution for the acute phase of care through 72 hours for hemorrhage/hemorrhagic shock. Any in vivo studies must be performed on an animal model that approximates the canine. No canine research will be performed for this topic area. Outcome measures of the in vivo evaluation must include survival, evaluation of tissue oxygenation, correction of coagulopathies, prevention of endothelial injury and vascular reactivity, end organ function, inflammatory markers or toxicity measures, evaluation of coagulation parameters, and evaluation of metabolic parameters. Successful prototype solutions must develop final product specification documents that list all product components and their concentrations, instructions for storage, and instructions for preparation (if required) and use of the whole blood product or substitute.

PHASE III DUAL USE APPLICATIONS: The end goal is to achieve FDA Center for Veterinary Medicine approval for this product. Transition the prototype from an advanced development stage to a fully vetted product ready for real-world use. Refine the development of a commercialization plan that may include development of different pathways, including both military and private sectors. This product should be applicable to a broad spectrum of civilian use markets, including remote and emergency veterinary facilities. In addition, the work may result in technology transition to Acquisition Program managers within DoW These efforts will be crucial in turning the prototype into a commercially viable product that can be rapidly deployed in military operations.

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3. Department of the Army, Headquarters. Veterinary Service: Tactics, techniques, and Procedures. Field manual No. 4-02.18. Pages 5-7 to 5-23. December 2004. 4. Clinical Practice Guidelines for Military Working Dogs. 12 December 2018, CPG ID_16

KEYWORDS: Hemorrhage control; whole blood products; whole blood substitutes; oxygen carriers; canine; military working dog

DHA26BZ01-NV002 TITLE: Detection and Treatment of Traumatic Brain Injury in Military Working Dogs

COMPONENT TECHNOLOGY PRIORITY AREA(S): Operational Medicine

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Evaluate previously developed traumatic brain injury (TBI) detection and treatments methods that can be repurposed for use in military working dogs (MWDs) after suffering from battlefield injuries.

DESCRIPTION: This topic is in support of the DoD Working Dog Strategic Research Plan concerning mitigation, strategies, and treatments for the detection and treatment of TBI.[1] Due to the high-risk nature of MWD operations, TBI is a common injury. TBI in the MWD carries an extremely high mortality rate with a prehospital mortality of over 40% for severe TBI cases. It is estimated that 25-40% of all MWD trauma cases are accompanied by TBI, but there is limited data concerning the short- and long-term effects of TBI on the performance and health of the MWD. Current clinical detection methods for TBI in the MWD are by the observation of altered mentation (coma, stupor, depression, lethargy, inappropriate behavior or responses) of the MWD and by use of the modified veterinary Glasgow coma scale or with physical evidence of head trauma (e.g., lacerations, abrasions, bruising, swelling, pain, bleeding from the nose or ears). Current treatment guidelines for TBI in MWDs are largely based on treatment recommendations for humans and are primarily supportive measures to maintain blood pressure, oxygen levels, proper ventilation, and body temperature to mitigate secondary injuries[2,3,4]. There have been many TBI detection methods and treatment strategies developed for humans that have shown promising results in rodent and large animal models[5]. The objective of this SBIR is to review research that was performed in rodents, canines, or other large animal models that could be repurposed for the detection and treatment of TBI specifically in MWDs. This research topic does not support the use of canines for testing purposes. Any animal testing would require use of suitable animal model that would approximate the response of a canine.

PHASE I: Identify TBI detection and/or TBI treatment methodologies that could be used for canine physiology based on previous TBI research. Develop a solution that builds on one or more of these methodologies. This solution must address the level(s) of TBI severity (mild, moderate, and severe) that the detection or treatment solution will address. Determine the technical feasibility of performing the proposed TBI detection and/or treatment concept in a Role 1 or Role 2 setting within a 72-hour window post injury. Define key components and milestones needed to develop the proposed solution. Develop a research plan that can be used to conduct in vitro and/or in vivo feasibility studies. This plan must include study descriptions and methods for determining study endpoints which will be used to measure the effectiveness of the detection or treatment methodology. The developed plan must also address any potential risks that may occur as part of any in vivo studies. The size of any studies performed must be appropriately powered to ensure that the results are statistically significant. Any in vivo studies are to be planned using a suitable animal model that approximates the canine. No canine research will be performed for this topic area. The plan must include the descriptions and methods of determining the study endpoints to be measured for successful prototypes. The expectation is that the outcome of this phase will be a developed plan for prototype development and for proof-of-concept testing.

PHASE II: Develop a working TBI detection prototype and/or TBI treatment based on the solution outlined in Phase I that can be used for laboratory feasibility testing. Conduct the feasibility studies as described in the outcome documentation of the Phase I project to demonstrate the feasibility of the prototype and document the results which can be used to determine success in an appropriate in vivo model. No canine research will be performed for this topic area. The testing plan as described in Phase I

for any in vivo study must compare the proposed solution to current TBI detection methods (altered mentation criteria and/or veterinary Glasgow coma scale) or TBI treatment methods (as defined by the Management Algorithm for TBI for MWDs3) as a base measure of effectiveness. Additional outcome measures based on the method to test can include detection time, detection rate, rate of survival, permanent behavioral changes, ability to return to duty, and/or days until return to duty. Successful prototype solutions must develop final product specification documents that list all product components and their concentrations, instructions for storage, and instructions for preparation (if required) and use. Develop an initial plan for product approval through the appropriate regulatory pathway of the FDA Center for Veterinary Medicine (CVM).

PHASE III DUAL USE APPLICATIONS: The end goal is to achieve FDA CVM approval for the prototype developed in Phase II. Optimize the TBI detection prototype and/or the TBI treatment and demonstrate its usefulness with end users. Use feedback from end-users to further optimize the prototypes. Transition the prototype from an advanced development stage to a fully vetted product ready for real-world use. Refine the development of a commercialization plan that may include development of different pathways, including both military and private sectors. This product should be applicable to a broad spectrum of civilian use markets, including emergency veterinary facilities, other government agencies (DHS, FEMA, ATF, etc.), law enforcement and search and rescue canines. In addition, the work may result in technology transition to Acquisition Program managers within DoW. These efforts will be crucial in turning the prototype into a commercially viable product that can be rapidly deployed in military operations.

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KEYWORDS: Traumatic brain injury (TBI), military working dog (MWD), canine, treatment, detection, assay

DHA26BZ01-NV003 TITLE: Environmental Exposure Detection Technologies and/or Decontamination Treatments for Military Working Dogs

COMPONENT TECHNOLOGY PRIORITY AREA(S): Operational Medicine

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Develop decontamination treatments for military working dogs that have been exposed to toxic industrial chemicals and materials through the performance of their duties.

DESCRIPTION: This topic is in support of the DoD Working Dog Strategic Research Plan concerning mitigation, countermeasures and treatments for toxin/toxic exposures[1]. In modern military operations, military working dogs (MWDs) are at risk of exposure by many different types of hazardous materials. These include toxic industrial chemicals (TICs) and materials (TIMs) such as hydrocarbons, polychlorinated biphenyls, glycols, hazardous metals, gases (hydrogen cyanide, hydrogen sulfide, freon, carbon monoxide, etc.), acids and alkali substances. Techniques for the decontamination of hazardous material exposures to the surface of the MWD are well defined[2,3,4]. Although there are useful treatment options for external decontamination, there are few treatment options for toxic exposures that have been absorbed into the body of the MWD.
The objective of this topic is to develop new treatments for MWDs against hazardous materials that have been absorbed into the body either through the skin or mucous membranes, by inhalation, or ingestion. Current systemic treatments employed to care for MWDs include supportive antibiotic therapy for sulfur mustard, atropine injections for nerve agents, and Narcan for narcotics, but there are limited treatment options available for TIC/TIM exposures[4,5,6]. Systemic treatments for the MWD should be able to be performed by veterinarians and their support personnel (trained animal care specialists (68T) in Role 1 and/or veterinary medical and surgical teams (VMST) in Role 2). Potential MWD systemic treatments could include but are not limited to kits containing indicators or detectors of TIC/TIM exposure with easily identifiable injectable treatments for the identified contaminant (indicator/detector) and/or hemoperfusion systems and filters that can be used to remove contaminants from the blood (systemic). This research topic does not support the use of canines for testing purposes. Any animal testing would require use of a suitable animal model that would approximate the response of a canine.

PHASE I: This phase is to propose either a TIC/TIM indicator/detector that can be used in conjunction with current treatments and/or propose a systemic treatment system for MWD toxic exposures. For indicator/detector technologies, identify potential technologies that are lightweight for use in the field and can ascertain exposures to specific TICs/TIMs and determine what decontamination treatment options to use for MWD survival. For systemic treatments, determine the potential of materials and methods which could be developed into a system for the systemic treatment of TIC/TIM exposures in canines. All proposed treatments must have the potential to reduce the specific TIC/TIM to levels that are safely below the LD50[5,6,7] of the TIC/TIM based on average weight of a canine (80-100 lbs). A listing of potential toxins and LD50 levels can be found in the MERCK Veterinary Manual Toxicology[7]. Down select the best candidate materials and methods for inclusion in an exposure indicator or detection system and/or systemic treatments for development and testing in Phase II based on operational relevance. Develop a plan for the testing of candidate technologies and treatments using both in vitro and in vivo studies (as necessary). Any in vivo studies are to be planned using a suitable animal model that approximates the canine. No canine research will be performed for this topic area. The plan must include the descriptions and methods of determining the study endpoints to be measured for successful prototypes. Estimate the potential effects of the proposed systems and processes on MWDs and their handlers.

PHASE II: For TIC/TIM indicator/detector technologies associated with the treatment approach, develop and optimize a workable prototype that can be used in both laboratory and field testing. Perform the

testing plan for the evaluation of the prototype as described in Phase I. Demonstrate the effectiveness, sensitivity, and specificity of the technologies for the detection of the specified contaminants. Demonstrate the ability for the technology to be used in the appropriate field setting by the working dog handler or Role 1 or Role 2 veterinary staff once trained.
For systemic treatment products, develop and optimize a workable prototype for laboratory testing. Validate the proof of concept by demonstrating the effectiveness of the developed prototype against TICs/TIMs on appropriate surrogate materials and evaluate its effectiveness by performing the testing plan developed as part of Phase I on appropriate efficacy models. Verify that contamination levels are reduced to or below performance objective levels. Demonstrate the practicality of use at the specified Role of Care in which the treatment is intended to occur. If any in vivo studies were planned, they must be performed on an animal model that approximates the canine. No canine research will be performed for this topic area.
Estimate transportability and storage stability of the developed detection and/or treatment prototype. Develop final product specification documents that include a list of all system components and their requirements and instructions for deployment and stowage.

PHASE III DUAL USE APPLICATIONS: The end goal is to achieve FDA Center for Veterinary Medicine approval for the prototype developed in Phase II. Optimize the TIC/TIM indicator or detector prototypes and/or systemic treatments for use with additional hazardous materials and demonstrate its effectiveness with end users. Use feedback from end-users to further optimize the prototypes. Transition the prototype from an advanced development stage to a fully vetted product ready for real-world use. Refine the development of a commercialization plan that may include development of different pathways, including both military and private sectors. This product should be applicable to a broad spectrum of civilian use markets, including emergency veterinary facilities, other government agencies (DHS, FEMA, ATF, etc.), law enforcement and search and rescue canines. In addition, the work may result in technology transition to Acquisition Program Managers within DoW. These efforts will be crucial in turning the prototype into a commercially viable product that can be rapidly deployed in military operations.

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KEYWORDS: Canine, military working dog (MWD), decontamination, hazardous materials, chemical warfare agents (CWA), toxic industrial chemicals (TICs), toxic industrial materials (TIMs)

DHA26BZ01-NV004 TITLE: Development of Wearable Device for Detecting Biomarkers and Treating Viral and Bacterial Infections

COMPONENT TECHNOLOGY PRIORITY AREA(S): Operational Medicine

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Develop a non-invasive wearable device that can discretely detect biomarkers for and provide initial broad-spectrum treatment for pan-viral and pan-bacterial infections. If fielded for military use, it may require additional security measures.

DESCRIPTION: The DHA Strategic Research Plan (SRP): Environmental Exposures (June 2024) lists two capability requirements under the "Assess" and "Treat" capability areas that align with this proposal: Environmental Detection and Health Risk Assessments under Assess and Environmental Exposures Treatment under Treat. In addition, the DHA SRP: Military Infectious Diseases (May 2024) lists three capability requirements under the "Prevent", "Treat", and "Enable" capability areas that align with this proposal: Prevention of Military Relevant Endemic and Emerging Infectious Diseases under Prevent, Treatment of Military Relevant Endemic and Emerging Infectious Diseases under Treat, and Core Competencies under Enable.
The Department of the Air Force (DAF) is looking for an advanced, non-invasive (does not break the skin or physically enter the body) wearable device (i.e., flash/continuous glucose style monitoring) capable of qualitatively detecting all-viral and all-bacterial infections using discrete biomarkers for such infections: TRAIL, MxA, CD46, IP-10, PTX3, or other non-blood based biomarkers (saliva, sweat, etc.) for viral infections and CRP, PCT, IL-6, IL-8, CD35, CD55, CD64, pro-ADM, or other non-blood based biomarkers (saliva, sweat, etc.) for bacterial infections. The end goal is a wearable device that discretely detects viral and bacterial infections and renders initial, broad-spectrum anti-viral or anti-bacterial treatment(s) at austere operational environments where no immediate medical countermeasures and no other detection capabilities are available until casualties are evacuated to locations with more robust medical resources for additional and specific differentiation and treatment. At a higher echelon of care, medical personnel must be able to receive data from the device to find out what category of threats (viral or bacterial) has triggered a biomarker detection and what corresponding treatments have been rendered to the affected force before providing more advanced care.
By continuously monitoring validated biomarkers, this device will empower warfighters to detect and respond to biological threats early, enhancing their survivability and operational effectiveness in high-threat theaters and mitigating risks to mission and force. This Air Force Medical Command initiative improves force health protection and ensures mission success. Dual-use functionality of this technology will focus on civilian healthcare systems.

PHASE I: Phase I will provide proof of feasibility on biomarker detection capabilities and initial treatment integration under various operational parameters. Efficacy must be demonstrated in progress documents via reproducible testing results against, at minimum, the bacterial and viral biomarkers listed in the description. Specifically, provide a detailed analysis of selected biomarkers (and their thresholds to indicate infections) and biosensor technologies, demonstrating their critical role in detecting and differentiating viral and bacterial infections as well as initial treatment considerations for such infections. This investigation will systematically address key attributes such as device detection capabilities, continuous monitoring in austere operational environments, and operator interface design. Additionally, results will outline specific power supply requirements for the system and propose resilient solutions that guarantee the continuity of health status data, whether the device is actively deployed or in storage. This Phase will demonstrate the feasibility of a wearable device for DAF warfighters and explore significant opportunities for standardization across the Department of War (DoW).

PHASE II: Phase II will focus on development and testing of one to five wearable prototypes: Biomarker Detection: Qualitatively detect pan-viral and pan-bacterial infection biomarkers to achieve 95% sensitivity and 95% specificity or higher. Durability and Design: Ruggedized and capable of operating in a wide range of environments and all-weather conditions (i.e., 0°F to 120°F operating temperatures and up to 100% humidity). The device must be able to withstand physical impacts (i.e., repeated contacts with body armor or repeated drop from 6 ft on hard surface) as well as prevent chafing and potential detachment (i.e., no thicker than 5 mm). Materials must be fully compatible with military uniforms and equipment. Monitoring Frequence and Power: Continuous monitoring with a maximum interval of 5 minutes, and the power must continuously operate for a minimum of 30 days without a need to recharge. Solar power is acceptable. User Interface: Infection alert status must be a simple, intuitive format, using color codes, audible alerts and/or vibration, even when the device is worn under protective gear, allowing warfighters to rapidly manipulate functions (i.e., reset, turn on/off alarms, system restarts) by touching or tapping. One to five prototype(s) must be delivered. Prototype(s) must have wireless integration allowing for secure data storage and transmission (ie. military grade blockchain encryption and HIPAA/Privacy Act compliant). Detailed initial test and evaluation results and validation process must be delivered with the prototypes. An applicable Food and Drug Administration (FDA) regulatory strategy outlining requirements for testing from safety and effectiveness must be included as well as how FDA approval will be received.

PHASE III DUAL USE APPLICATIONS: Phase III will involve deployment, testing, evaluation, and improvement of the mobile platform for military (and civilian) use. FDA approval is the end goal of this Phase. The inherent dual-use capability of this technology presents opportunities for integration across the military and within civilian healthcare systems. Specifically, target applications include DAF Deployed Personnel: enhancing force health protection for at-risk personnel vulnerable to biological threats, with potential application across the DoW. Commercial applications include civilian healthcare facilities specifically in rural communities: enabling civilian hospitals to remotely monitor and render initial treatment for at-risk patients prior to further patient management.

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KEYWORDS: Wearable device; biosensors; viruses and bacteria detection; biomarkers, drug delivery system; algorithms; monitoring; warfighters, deployers, medical countermeasures innovation

DHA26BZ01-DV005 TITLE: Development of Small Molecules with Antifungal Properties for Safe Use in Human Patients (Direct to Phase II)

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Biomanufacturing (BIO)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Operational Medicine

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: This topic is intended for technology proven ready to move directly into Phase II and accepts Direct to Phase II proposals only. The proposed research will focus on identifying compounds with broad-spectrum activity against clinically relevant fungal pathogens while minimizing toxicity to humans. The primary objective is to identify a small molecule with fungicidal properties that are safe for human use, with FDA clearance.

DESCRIPTION: Fungal infections represent a growing global health challenge, particularly among immuno-compromised individuals. Invasive fungal infections caused by pathogens such as Candida species, Aspergillus species, Fusarium species, and Mucor species are associated with high morbidity and mortality rates. Fungal infections are associated with 130k hospitalizations, 13 million outpatient visits, and result in a financial burden of $19 billion on the civilian health care sector. Fungal wound infections in particular are also growing challenge for the military. Despite the availability of antifungal agents, current treatments are often limited by toxicity, drug resistance, and narrow-spectrum activity. The emergence of multidrug-resistant fungal strains, such as Candida auris, has further exacerbated the need for novel antifungal therapies. Small molecules with antifungal properties offer a promising avenue for addressing these challenges. Their ability to target specific fungal pathways, combined with the potential for oral bioavailability and low manufacturing costs, makes them ideal candidates for therapeutic development. However, significant scientific and technical hurdles remain with the discovery and optimization of small molecules that are both effective against fungal pathogens and safe for human use. Qualified proposals should identify small molecules with antifungal properties from an existing library. These small molecules should be active against all of the following fungi: Fusarium species, Aspergillus species, Candida auris, or Mucorales species. Qualified molecules will have antifungal activity at nanomolar concentrations. Further, these small molecules must have a cytotoxicity profile similar, or better than Amphotericin B.

PHASE I: This topic is intended for technology proven ready to move directly into Phase II. Therefore, the offeror must demonstrate and provide documentation to substantiate that the scientific and technical merit and feasibility described in Phase I has been met and describes the potential commercial applications. Documentation must include all relevant information including, but not limited to technical reports, test data, prototype designs/models, and performance goals/results.
Demonstrate screening of existing chemical libraries or other sources to identify a single or multiple candidate compounds with antifungal properties in vitro. The chemical candidate or candidates must demonstrate antifungal activity at nanomolar concentrations. Acceptable routes of administration are topical and systemic. Further, in vitro cell culture conditions, the chemical candidate or candidates must be as or less cytotoxic than Amphotericin B, as measured by a lactate dehydrogenase assay or similar test. Multiple candidates can be included in Phase II.

PHASE II: During this phase, the lead antifungal candidate or candidates should be optimized into a viable treatment option in vivo. The cytotoxicity must be screened against a skin organoid model and compared to Amphotericin B. The efficacy of the candidate antifungal or antifungals must be tested in these organoid models to determine if the fungal burden is reduced by at least 100-fold. Further, testing on small animals should be completed to ensure that the candidate antifungal or antifungals can reduce

fungal burden in vivo. In vivo studies using a wound model are encouraged. Operational effectiveness must be demonstrated in this phase, with evaluations of the product's stability in austere conditions (temperatures ranging from -55 to 125 ℃). An optimal shelf life in these conditions is one year; a minimum acceptable shelf life is 6 months. The regulatory strategy for this product or products should be clearly defined with a detailed plan to obtain FDA clearance.

PHASE III DUAL USE APPLICATIONS: This phase is critical as it transitions the prototype from an advanced development stage to a fully viable concept ready for real-world military and civilian use. The company should develop partnerships that adhere to Quality Management System (QMS) requirements to demonstrate and commercialize the technology in civilian relevant settings, such as hospitals and clinics. The company should also investigate other funding opportunities with other government agencies, such as Congressionally Directed Medical Research Programs (CDMRP) and Biomedical Advanced Research and Development Authority (BARDA), that may be interested in supporting the development and commercialization of this technology. The goal of this phase is to secure FDA approval. This phase should include testing in large animal models and randomized clinical trials under formal institutional protocol approval. The candidate antifungals must be optimized for a shelf life of one year in austere environments. Once fully developed the antifungal should be positioned for use in both civilian and military settings. The selected offeror is responsible for making the product accessible to potential military and civilian users. Coordinate with military customers to establish a National Stock Number.

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KEYWORDS: Wounds, infections, fungi, antifungals, antimicrobials, biochemistry, microbiology, mycology

OSW/STRATEGIC CAPABILITIES OFFICE (SCO)
Small Business Innovation Research (SBIR) Program
DoW 2026 SBIR BAA Release 1
Direct to Phase II Proposal Submission Instructions

The Strategic Capabilities Office (SCO) seeks small businesses with strong research and development capabilities to pursue and commercialize specific technologies to meet SCO objectives.

Proposers responding to a topic in this solicitation must follow all general instructions provided in the Department of War (DoW) 2026 SBIR Program solicitation. SCO requirements in addition to or deviating from the DoW Program solicitation are provided in the instructions below.

Proposers are encouraged to thoroughly review the solicitation and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The full solicitation is available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate solicitation cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Specific questions pertaining to the SCO SBIR Program should be submitted to the SCO SBIR Program office at: sbir@sco.mil

Offeror Small Business Eligibility Requirements

Each offeror must qualify as a small business at time of award per the Small Business Administration's (SBA) regulations at 13 CFR 121.701-121.705 and certify to this in the Cover Sheet section of the proposal. Small businesses that are selected for award will also be required to submit a Funding Agreement Certification document and be registered with Supplier Performance Risk System https://www.sprs.csd.disa.mil/ prior to award.

Ownership Eligibility

Prior to award, SCO may request business/corporate documentation to assess ownership eligibility as related to the requirements of SBIR/STTR Program Eligibility. These documents include, but may not be limited to, the Business License; Articles of Incorporation or Organization; By-Laws/Operating Agreement; Stock Certificates (Voting Stock); Board Meeting Minutes for the previous year; and a list of all board members and officers. If requested by SCO, the offeror shall provide all necessary documentation for evaluation prior to SBIR award. Failure to submit the requested documentation in a timely manner as indicated by SCO may result in the offeror's ineligibility for further consideration for award.

Organization Conflicts of Interest (OCI)

The basic OCI rules for Contractors that support development and oversight of SBIR topics are covered in 9.505-1 through FAR 9.505-4 as the means of avoiding, neutralizing, or mitigating organizational conflicts of interest. All applicable rules under FAR 9.5 apply.
If you, or another employee in your company, developed or assisted in the development of any SBIR requirement or topic, please be advised that your company may have an OCI. Your company could be precluded from an award under this BAA if your proposal contains anything directly relating to the development of the requirement or topic. Before submitting your proposal, please examine any potential OCI issues that may exist with your company, to include subcontractors and understand that if any exist, your company may be required to submit an acceptable OCI mitigation plan prior to award.
In addition, FAR 3.101-1 states that Government business shall be conducted in a manner above reproach and, except as authorized by statute or regulation, with complete impartiality and with preferential treatment for none. The general rule is to avoid strictly any conflict of interest or even the appearance of a conflict of interest in Government-contractor relationships. An appearance of impropriety may arise where

an offeror may have gained an unfair competitive advantage through its hiring of, or association with, a former Government official if there are facts indicating the former Government official, through their former Government employment, had access to non-public, competitively useful information. (See Health Net Fed. Svcs, B-401652.3; Obsidian Solutions Group, LLC, B-417134, 417134.2). The existence of an unfair competitive advantage may result in an offeror being disqualified and this restriction cannot be waived.

It is SCO policy to ensure all appropriate measures are taken to resolve OCIs arising under FAR 9.5 and unfair competitive advantages arising under FAR 3.101-1 to prevent the existence of conflicting roles that might bias a contractor's judgment and deprive SCO of objective advice or assistance, and to prevent contractors from gaining an unfair competitive advantage.

Use of Foreign Nationals (also known as Foreign Persons), Green Card Holders, and Dual Citizens
See the "Foreign Nationals" section of the DoW SBIR Program announcement for the definition of a Foreign National (also known as Foreign Persons).

ALL offerors proposing to use foreign nationals, green-card holders, or dual citizens, MUST disclose this information regardless of whether the topic is subject to export control restrictions. Identify any foreign nationals or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. You may be asked to provide additional information during negotiations to verify the foreign citizen's eligibility to participate on a SBIR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)). Proposals submitted to export control-restricted topics and/or those with foreign nationals, dual citizens, or green card holders listed will be subject to security review during the contract negotiation process (if selected for award). SCO reserves the right to vet all un-cleared individuals involved in the project, regardless of citizenship, who will have access to Controlled Unclassified Information (CUI) such as export controlled information. If the security review disqualifies a person from participating in the proposed work, the contractor may propose a suitable replacement. In the event a proposed person and/or firm is found ineligible by the Government to perform the proposed work, the Contracting Officer will advise the offeror of any disqualifications but is not required to disclose the underlying rationale. SCO may require offerors to address follow-up questions to determine eligibility.

Export Control Restrictions
The technology within most SCO topics is restricted under export control regulations including the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR). ITAR controls the export and import of listed defense-related material, technical data and services that provide the United States with a critical military advantage. EAR controls military, dual-use and commercial items not listed on the United States Munitions List or any other export control lists. EAR regulates export-controlled items based on user, country, and purpose. The offeror must ensure that their firm complies with all applicable export control regulations. Please refer to the following URLs for additional information: https://www.pmddtc.state.gov/ and
https://www.bis.doc.gov/index.php/regulations/export-administration-regulations-ear.
If the SCO SBIR Direct to Phase II topic is subject to ITAR and/or EAR and if selected for award negotiations, your company will be required to submit a Technology Control Plan (TCP) during the contracting negotiation process.

Rights in Other Than Commercial Technical Data and Computer Software - Small Business Innovation Research Program and Small Business Technology Transfer Program (DFARS 252.227-7018)

Use this link for full description of Data Rights:
https://www.acquisition.gov/dfars/part-252-solicitation-provisions-and-contract-clauses#DFARS_252.227-7018

Direct to Phase II Proposal Submission Guidelines and Requirements

Offerors must create a Cover Sheet using the DoW Proposal submission system. Offerors must provide documentation that satisfies the Phase I feasibility requirement* that will be included in the Phase II proposal. Offerors must demonstrate that they have completed research and development through means other than the SBIR/STTR Program to establish the feasibility of the proposed Phase II effort based on the criteria outlined in the topic description.

Offerors must submit a Phase II proposal using the SCO Phase II proposal instructions below.

* NOTE: Offerors are required to provide information demonstrating that the scientific and technical merit and feasibility has been established. SCO will not evaluate the offeror's related Phase II proposal if it determines that the offeror has failed to demonstrate that technical merit and feasibility has been established or the offeror has failed to demonstrate that work submitted in the feasibility documentation was substantially performed by the offeror and/or the Principal Investigator (PI).

PROPOSAL SUBMISSION

Classified Proposals

Classified proposals ARE NOT accepted. The inclusion of classified data in an unclassified proposal MAY BE grounds for the Agency to determine the proposal as non-responsive and the proposal not to be evaluated. In some instances, work being performed on Phase II contracts will require security clearances. If a Phase II contract requires classified work, the offeror must have a facility clearance and appropriate personnel clearances to perform the classified work. For more information on facility and personnel clearance procedures and requirements, please visit the Defense Counterintelligence and Security Agency Web site at: https://www.dcsa.mil.

Proposal Status

Proposing firms will be notified of selection or non-selection status for a DP2 award within 90 days of the closing date of the BAA. The email will be distributed to the "Corporate Official" and "Principal Investigator" listed on the proposal coversheet and will originate from the sco.baa@sco.mil email address. SCO cannot be responsible for notification to a company that provides incorrect information or changes such information after proposal submission.

Direct to Phase II PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS

Complete proposals must include the following:

- Volume 1: Proposal Cover Sheet
- Volume 2: Technical Volume (15 page Maximum)
    - Part 1: Phase I Justification (5 Pages Maximum)
    - Part 2: Phase II Technical Proposal (10 Pages Maximum)
- Volume 3: Cost Volume
- Volume 4: Company Commercialization Report (2 page Maximum)
- Volume 5: Supporting Documents (5 Pages Maximum)
- Volume 6: Fraud Waste and Abuse Certification

- Volume 7: Disclosure of Foreign Affiliations or Relationships to Foreign Countries

Phase II proposals require a comprehensive, detailed submission of the proposed effort. SCO SBIR Direct to Phase II period of performance is 12 months. SCO may award SBIR Direct to Phase II efforts up to a maximum value of $2,000,000 per award. Commercial and military potential of the technology under development is extremely important. Proposals emphasizing dual-use applications and commercial exploitation of resulting technologies are sought.

Volume 1 - Cover Sheet

The proposal cover sheet is prepared on DSIP. The cover sheet must include a brief technical abstract that describes the proposed R&D project and an anticipated benefits and potential commercial applications discussion. Each section should be no more than 3,000 characters. **Do not include proprietary or classified information in the proposal cover sheet**. If your proposal is selected for negotiation and possible award, the technical abstract and anticipated benefits discussion may be publicly released online. DSIP will assign a proposal number once the cover sheet is saved. You may modify the cover sheet as needed until the topic closes.

Volume 2 – Technical Volume (15 pages)

1. **Phase I Feasibility Documentation (5 Pages Maximum)**. Offerors are required to provide information demonstrating the establishment of the scientific and technical merit and feasibility. **Feasibility documentation MUST NOT be solely based on work performed under prior or ongoing Federally funded SBIR or STTR work.**

2. **Phase II Technical Objectives and Approach (10 Pages Maximum).** List the specific technical objectives of the Phase II research and provide a detailed technical approach to meet these objectives.

   - **Phase II Work Plan.** Provide an explicit, detailed description of the Phase II approach. The plan should indicate what is planned, how and where the work will be carried out, a schedule of major events, and the final product to be developed. A Phase II effort should attempt to accomplish the technical feasibility demonstrated in Phase I, including potential commercialization of results. Phase II is the principal research and development effort and is expected to produce a well-defined deliverable product or process.

   - **Related Work.** Describe significant activities directly related to the proposed effort, including those conducted by the Principal Investigator, the proposing firm, consultants, or others. Report how the activities interface with the proposed project and discuss any planned coordination with outside sources. The proposers' awareness of the state-of-the-art technology and associated science must be demonstrated.

   - **Relationship with Future Research or Research and Development**. State the anticipated results of the proposed approach if the project is successful. Discuss the significance of the Phase II effort in providing a foundation for a Phase III research or research and development effort.

   - **Technology Transition and Commercialization Strategy (NTE 2 pages, counts toward 10 page limit of Phase II technical objective section).** Describe your company's strategy for converting the proposed SBIR research, resulting from your proposed Phase II contract, into a product or non-R&D service with widespread commercial use -- including private sector, military markets, and transition to DoW operational systems. Note that the commercialization strategy is separate from the Commercialization Report described in Section 4.L below. The strategy addresses how you propose to commercialize this research, while the Company Commercialization Report covers what you have done to commercialize the results of past Phase II awards.

Historically, a well-conceived commercialization strategy is an excellent indicator of ultimate Phase III success. The commercialization strategy must address the following questions:

- What is the first product that this technology will go into?
- Who will be your customers, and what is your estimate of the market size?
- How much funding will you need to bring the technology to market, and how will you raise those funds?
- Does your company contain marketing expertise and, if not, how do you intend to bring that expertise into the company?
- Who are your competitors, and what is your price and/or quality advantage over your competitors?

- **Key Personnel**. Identify key personnel, including the Principal Investigator, who will be involved in the Phase II effort. List directly related education and experience and relevant publications (if any) of key personnel. Include a concise resume of the Principal Investigator(s).

- **Facilities/Equipment**. Describe available instrumentation and physical facilities necessary to carry out the Phase II effort. Justify items of equipment to be purchased (as detailed in the cost proposal) including Government Furnished Equipment (GFE). All requirements for government furnished equipment or other assets, as well as associated costs, must be determined and agreed to during Phase II contract negotiations. State whether or not the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name) and local governments. This includes, but is not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.

- **Consultants.** Involvement of university, academic institution, or other consultants in the project may be appropriate. If such involvement is intended, it should be described in detail and identified in the Cost Volume.

Volume 3 - Cost Volume ($2,000,000 Maximum).
A detailed, Phase II Cost Volume must be submitted online and in the proper format shown in the Cost Breakdown Guidance in DoW 2026 SBIR BAA. Some items in the cost volume template may not apply to the proposed project. If such is the case, there is no need to provide information for every item. Provide enough information to allow the SCO evaluators to assess the proposer's plans to use the requested funds if the contract is awarded. Period of performance is 12 months.

Volume 4 – Company Commercialization Report (CCR) (Required)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW 2026 SBIR BAA for full details on this requirement. The information contained in the CCR will not be considered by SCO during proposal evaluations.

Volume 5 – Company Commercialization Report (CCR) (Required)
SCO will accept the following documents under Volume 5:
1. Letter of Support (Optional – use "Letter of Support" upload category)

Volume 6 – Fraud Waste and Abuse Certification (Required)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Volume 7 – Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Required)

Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoW SBIR Program BAA.

METHOD OF SELECTION AND EVALUATION CRITERIA

All proposals will be reviewed for overall merit based on the evaluation criteria published in the DoW 2026 SBIR Program BAA.

CONTRACTUAL CONSIDERATIONS

A. **Awards**. The number of Direct to Phase II awards will depend upon the quality of the Phase II proposals and the availability of funds. Each Phase II proposal selected for award under a negotiated contract requires a signature by both parties before work begins. SCO awards Phase II contracts to Small Businesses based on results of the agency priorities, scientific, technical, and commercial merit of the Phase II proposal.
B. **Reports.** For incrementally funded Phase II projects an interim, midterm written report maybe required (at the discretion of the awarding agency).
C. **Payment Schedule**. SCO Phase II Awards Level of Effort Firm Fixed Price contracts. Monthly invoices are based on the labor hours recorded and the monthly costs associated with the project.
D. **Markings of Proprietary Information**. Per DoW SBIR Program BAA, section 5.3.
E. **Copyrights, Patents and Technical Data Rights**. Per DoW SBIR Program BAA.
F. **Security Information**. SCO anticipates work under Phase II will be classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by DoD 5220.32-M, National Industrial Security Program Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA). The selected company will be required to safeguard classified material IAW DoD 5220.32-M during this contract.

TECHNICAL AND BUSINESS ASSISTANCE (TABA)

The SCO SBIR Program will not participate in the Technical and Business Assistance.

REPORTING OF PHASE III OR ANY OTHER COMMERCIALIZATION EFFORTS

Each small business receiving a Phase II award is required to report all Phase III activities on their Company Commercialization Report (Volume 4 of proposal submission). In addition, please send any corresponding Phase III documents in PDF format to: sbir@sco.mil

Reportable activities include: sales revenue from new products and non-R&D services resulting from the Phase II project; additional investment from sources other than the Federal SBIR program in activities that further the development and/or the commercialization of the Phase II technology; the portion of additional investment representing clear and verifiable investment in the future commercialization of the technology (i.e. "hard investment"); whether the Phase II technology has been used in a fielded DoW system or acquisition program and, if so, which system or program; the number of patents resulting from the contractor's participation in the SBIR/STTR program; growth in number of firm employees, and; whether the firm completed an initial public offering (IPO) of stock resulting in part from the Phase II project.

OSW - SCO
DoW 2026 SBIR BAA
Release 1
Topic Index

OSW26BZ01-DV001 Establishing Ad-Hoc Distributed Network Across Heterogeneous RF Systems

OSW26BZ01-DV002 Scalable Tracking of pLEO Constellations and Debris

OSW26BZ01-DV001 TITLE: Establishing Ad-Hoc Distributed Network Across Heterogeneous RF Systems

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software; Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Successful Joint Operations executed by the U.S. Department of War (DoW) rely on tight coordination, synchronization, and tactical communication across multiple service components and platforms. The Joint Force faces real-time communication and coordination challenges between modern, more flexible systems and the much larger inventory of older legacy platforms that were independently acquired by each service. Virtually all platforms have one or more types of Radio Frequency (RF) apertures and backend electronics that could be used to coordinate effects, but they lack the appropriate firmware (FW) or software (SW) to enable cross platform synchronization of those effects due to compatibility or proprietary software interface constraints. Multiple land, sea, air, and space assets would benefit from software and firmware enhancements to increase communication and synchronization effectiveness across the Joint Force.

The objective of this effort is to assess and implement advanced SW/FW enhancements onto existing platform(s) to enable heterogeneous multifunction RF systems to communicate and synchronize activities to increase effectiveness of Joint Force operations. A large defense contractor that produces high volume, (hundreds or thousands of production units) may be hesitant to change their baseline SW/FW to incorporate new capabilities. The Government is interested in an experienced agile, small business software developer to study and implement communication applications onto a large defense contractor's target software defined radio (SDR) to enable greater Joint interoperability. Key aspects of the study are to assess SW/FW compatibility with the target SDR; identify hardware and software constraints; assess cyber vulnerabilities; and culminate in a proof-of-concept lab demonstration to establish an ad-hoc network between heterogeneous RF systems. Additionally, the study seeks to generate a roadmap and identify risk reduction activities that should be performed in order to fully integrate these new capabilities into operational systems. The proposed solution should support integration with DoW's existing RF systems, payloads, and operations to improve mission agility, reduce mission risk, and enhance Joint Operations.

Competitive proposals must originate from performers that have previously demonstrated SW/FW integration of multi-function operations, in a laboratory environment or in open-air testing, between heterogeneous DoW RF systems. The Government is particularly interested in enabling diverse ad-hoc data network node establishment between dissimilar RF mission systems. These DoW payloads or platforms of interest for this application are Communication, Command and Control (C2), Electronic Warfare (EW), or Synthetic Aperture Radar (SAR) systems. This use of diverse RF mission platforms,

payloads, and leveraging multiplexed signals to establish non-traditional data distribution nodes while still performing the primary mission would greatly increase Joint interoperability.

DESCRIPTION: The proposer will need to work closely with a DoW-DIRECTED defense contractor to implement SW/FW modifications to the target SDR to enable heterogeneous multifunction RF systems to communicate and synchronize activities. The specific defense contractor will be identified to the proposer upon notification of selection for the D2P2 award.

FEASIBILITY DOCUMENTATION:

Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results.

PHASE I: This topic is accepting Direct to Phase II proposals only. Strong proposals should document prior experience collaborating and working with large defense contractors in many or all the following ways:

- Demonstrated ability to transition software enhancements to operational DoW platforms
- Ability to modify / enhance existing legacy tactical platform payloads with additional capabilities
- Real-world or open-air experiments to validate hypotheses and concepts
- Experience implementing and testing Low Probability of Intercept (LPI) & Low Probability of Detection (LPD) waveforms

Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results.

PHASE II: The DoW requires synchronization of effects at tactical timelines, which could be achieved through primary or secondary data pathways. Dynamic, distributed C2 enabled in an active conflict allows for alternate C2 methods should adversary jamming deny specific frequencies in the environment. The purpose of this study is to assess feasibility and implement an alternate data pathway through SW/FW modifications to target hardware, identify constraints, and implement an ad-hoc data network via heterogeneous RF systems. A crawl, walk, run approach is envisioned for this Phase II effort where the contractor would initially assess SW/FW compatibility with the target SDR, identify hardware and software constraints, test code initial on the target hardware, and culminate in a proof-of-concept lab demo with representative HW. A roadmap for future full system integration with the operational system is also desired. An ideal proposal will be able to demonstrate these new SW/FW capabilities on the development hardware kit that will be made available through the large defense contractor to winning proposal(s).

Proposers should have access to an existing SAPF accredited facility and access to classified IT (JWICS, SIC, CORE, or SAV); or provide a credible plan for having access to accredited SAPF facilities and IT systems by the projected contract award date to collaborate with the large defense contractor. Close collaboration via remote telecons or in person meetings with the defense contractor will be required in order to integrate and test the SW/FW updates on the target hardware.

The Phase II Objectives and Tasks:

- Work with the defense contractor to:
  - Evaluate target SRD and assess feasibility of SW/FW updates to perform multiplexed dual mission roles.
  - Establish Ad-Hoc data network via reprogrammable RF systems.
  - Automatically coordinate RF between compliant systems.
- Assess cybersecurity of payload and system.
- Develop a roadmap for future full-system integration.
- Assess the system's existing System Requirement Document (SRD) and derived system specifications for software compatibility. Provide draft requirement language for a SRD and

derived specifications for potential transition to operational systems.

Expected deliverables include:

- Develop and deliver a Security Plan for program protection and operational security during project execution.
- Deliver periodic SW/FW drops (versions) to the Government and defense contractor for integration with the target hardware development kit on a quarterly basis at a minimum, or more frequently if significant advancements are made.
- Deliver software design documentation to the Government regarding SW implementation methodology and constraints.
- Written findings following Technical Exchange Meetings, Design Reviews, or significant interactions with external partners.
- A Midterm Report to assess SW/FW compatibility with target hardware, cyber resiliency of payload and target system, initial lab results from Ad-Hoc data network tests, and mission utility assessment.
- Submit Quarterly status reports.
- A Final Report documenting lab test results, system performance metrics, and integration outcomes achieved over the course of the entire effort.
- A transition and commercialization roadmap outlining next steps for operational use adoption and risk reduction activities to facilitate implementation on operational systems.

The completed documentation, integration, and lab testing results will provide SCO with the opportunity to enhance the agility and operational efficiency across Joint DoW Operations.

PHASE III DUAL USE APPLICATIONS: SCO believes that there is significant potential for this technology to be incorporated across multiple DoW programs and operational platforms to enhance Joint Operations.

REFERENCES:

1. ARCYBER chief's advice to industry: Make interoperable tech and start at the edge, Breaking Defense, https://breakingdefense.com/2025/08/arcyber-chiefs-advice-to-industry-make-interoperable-tech-and-start-at-the-edge/ Accessed 18 Aug. 2025.
2. A Vision for Better, Faster C2 Decision-Making Across All Domains, Breaking Defense, https://breakingdefense.com/2021/08/a-vision-for-better-faster-c2-decision-making-across-all-domains/ Accessed 18 Aug. 2025.

KEYWORDS: Software Integration; Multi-RF Function; Multiplexing ; Ad-Hoc Network ; Joint Warfighting; Integrated Network Systems-of-System; Synthetic Aperture Radar (SAR); Electronic Warfare (EW); Communications; Distributed Command & Control; Software Defined Radio; Low Probability of Intercept (LPI); Low Probability of Detection (LPD)

OSW26BZ01-DV002 TITLE: Scalable Tracking of pLEO Constellations and Debris

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Space Technology

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Develop and deploy scalable, cost-effective, and rapidly iterated wide-field optical tracking solutions to supplement the Space Surveillance Network (SSN) for high-density proliferated Low Earth Orbit (pLEO) constellations and debris. The solution must meet stringent tracking, latency, and dissemination requirements while leveraging Commercial Off-The-Shelf (COTS) components and software-defined sensors.

DESCRIPTION: The increasing density of pLEO constellations and orbital debris presents a significant challenge to the current SSN infrastructure, including radar fences with limited beam count and power. This effort seeks to develop wide-field optical tracking systems capable of addressing these challenges. The solution must integrate COTS components on a COTS bus to ensure low-cost deployment, rapid iteration, and replenishment. Software-defined sensors adaptable to multiple missions and on-board processing capabilities are encouraged. The system should also support extensibility to other missions via side-car payloads or on-orbit firmware/software upgrades.
The system, if deployed at a scale identified by the proposer, must meet the following requirements:
Threshold Requirements:

- Track 95% of objects larger than 1U (10x10 cm^2 optical cross-section, Lambertian surface at 20% reflectivity) at least once per orbit.
- Track 95% of unknown objects as small as 1 cm across once every two weeks.

Goal Requirements:

- Track 99% of unknown objects larger than 1U at least twice per orbit.
- Track 95% of all objects as small as 1 cm once every two weeks.
- Track a cued object down to 1 cm with a maximum latency of 40 minutes, with no requirements for simultaneous tracks.

Dissemination:

- All detections must be injected into the Unified Data Library (UDL) database within 5 minutes of passing through the last Field of Regard of the sensor platform.

FEASIBILITY DOCUMENTATION:

- Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results.

PHASE I: This topic is accepting Direct to Phase II proposals only. Strong proposals should document prior experience:

- Modeling and simulation of wide-field optical systems to validate tracking performance
- Prototyping and testing of COTS-based optical sensors and software-defined systems.
- Integration and testing of on-board processing capabilities for real-time data handling and dissemination.
- Validation of UDL database injection within the required 5-minute latency.
- Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results.

PHASE II: Phase II will focus on:

1. Developing and demonstrating a fully integrated prototype system in a relevant orbital environment.
2. Conducting end-to-end testing to validate tracking performance, latency, and dissemination

requirements.
3. Iterating on the design to incorporate lessons learned and optimize for cost, scalability, and extensibility.
4. Preparing for low-rate initial production (LRIP) and deployment.

PHASE III DUAL USE APPLICATIONS: The solution has significant commercial potential in the growing space industry, including:

1. Supporting commercial satellite operators in collision avoidance and space traffic management.
2. Providing data services to insurance companies for risk assessment and mitigation.
3. Enabling scalable tracking solutions for emerging mega-constellations.
4. Extending capabilities to other missions, such as Earth observation, space domain awareness, and scientific research.

REFERENCES:

1. U.S. Space Force. Spacepower Doctrine: Space Domain Awareness. U.S. Space Force, https://www.spaceforce.mil. Accessed 18 Aug. 2025.
2. NASA. Orbital Debris Quarterly News. Orbital Debris Program Office, https://orbitaldebris.jsc.nasa.gov. Accessed 18 Aug. 2025.
3. U.S. Space Command. Partnerships with Commercial Entities for SDA. U.S. Space Command, https://www.spacecom.mil. Accessed 18 Aug. 2025.
4. LeoLabs. Tracking Satellites and Debris in Low Earth Orbit. LeoLabs, https://www.leolabs.space. Accessed 18 Aug. 2025.
5. Secure World Foundation. Global Space Traffic Management and Debris Mitigation Policies. Secure World Foundation, https://swfound.org. Accessed 18 Aug. 2025.
6. MIT Lincoln Laboratory. Space Surveillance Radar Systems. MIT Lincoln Laboratory, https://www.ll.mit.edu. Accessed 18 Aug. 2025.
7. U.S. Department of Commerce. Open Data Sharing for Space Situational Awareness. U.S. Department of Commerce, https://www.commerce.gov. Accessed 18 Aug. 2025.

KEYWORDS: pLEO; Space Surveillance; Wide-Field Optics, Orbital Debris; COTS, Software-Defined Sensors; UDL, Space Traffic Management

UNITED STATES SPECIAL OPERATIONS COMMAND
Small Business Innovation Research (SBIR)
DoW 2026 SBIR BAA Release 1
Proposal Submission Instructions

INTRODUCTION

The United States Special Operations Command (USSOCOM) Direct to Phase II (DPII) proposal submission instructions cover DPII proposals only and change/append the Department of War (DoW) instructions for Phase II submissions as they apply to USSOCOM Direct to Phase II requirements. The Government will only evaluate responsive proposals.

USSOCOM seeks small businesses with strong research and development capabilities to pursue and commercialize technologies needed by Special Operations Forces (SOF) through the DoW SBIR Program Broad Agency Announcement (BAA).

Offerors responding to a topic in this BAA must follow all general instructions provided in the DoW SBIR Program solicitation. USSOCOM requirements in addition to or deviating from the DoW Program solicitation are provided in the instructions below.

Offerors are encouraged to thoroughly review the DoW Program BAA and register for the DSIP Listserv to stay informed of important programmatic and contractual changes.

- The full DoW solicitation is available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate solicitation cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login

Specific questions pertaining to the administration of the USSOCOM SBIR/STTR programs and these proposal preparation instructions, should be directed to: SBIR@SOCOM.mil.

The USSOCOM SBIR/STTR Program Office will be hosting a virtual USSOCOM Industry Day to specify requirements and stimulate small business/research institute partnership-building. Please visit https://events.sofwerx.org/ for the date and time information.

DIRECT TO PHASE II PROPOSAL GUIDELINES

The topics below are accepting Direct to Phase II (DPII) proposals only. Offerors shall provide the information cited below in the proposal response. Failure to provide requested information may result in a determination that the proposal is non-responsive.

Offerors interested in submitting a DPII proposal must provide documentation to substantiate that the scientific and technical merit and feasibility of the objectives described have been met and provide the potential commercial applications. Documentation should include all relevant information including, but not limited to, technical reports, test data, prototype designs/models, and performance goals/results. Work submitted within the feasibility documentation must have been performed and be owned (data rights) by the offeror and/or the Principal Investigator (PI).

USSOCOM will not evaluate the offeror's related DPII proposal if the offeror fails to demonstrate 1) the technical merit and feasibility of the proposed solution has been established, and/or 2) the work submitted in the feasibility documentation was performed by the offeror and/or the PI.

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoW SBIR/STTR proposal submission. Proposals must be submitted via DSIP; submission by any other means will be disregarded.

Detailed instructions regarding registration and proposal submission via DSIP are provided in the DoW SBIR Program BAA.

Please Note:

1. It is the Offeror's responsibility to ensure compliance with all DoW and USSOCOM instructions to submit all required documents. The DSIP (the DoW SBIR/STTR proposal submission website) will only confirm that the upload process is complete with the notice "100% submitted"; it does not verify compliance with instruction or successful upload of all required documents.
2. USSOCOM does not assist with proposal preparation nor review proposals for completeness prior to formal submission. It is recommended that offerors utilize local and state resources for assistance (refer to DoW Instructions for resources information).
3. Identifying foreign national involvement in a USSOCOM SBIR topic crucial for determining eligibility, particularly for topics within the parameters of the United States Munitions List, Part 121 in the International Traffic in Arms Regulation (ITAR). Firms employing a foreign national(s) (as defined section in the titled "Foreign Nationals" of the DoW SBIR Program BAA) to work on USSOCOM ITAR topics must possess an export license to receive a USSOCOM SBIR contract.
4. The Government may withdraw from negotiations at any time for any reason, including national security concerns (e.g., foreign persons, foreign influence or ownership, inability to clear the firm or personnel for security clearances, or other related issues).
5. **USSOCOM prohibits the inclusion of a Government Letter of Support (LoS). Any proposal that includes a Government LoS will be deemed non-responsive (disqualified).**

Cover Page (Volume 1)

Volume 1 is created as part of the DoW Proposal Submissions process. Follow all instructions provided in the DoW SBIR Program BAA and DSIP.

Technical Volume (Volume 2)

The technical volume is **Not to exceed (NTE) ten (10) pages** and must adhere to the formatting requirements outlined in the "SBIR 25.4 Annual Program BAA", "Format of Technical Volume" and "Content of Technical Volume". USSOCOM will only evaluate the first (10) pages of the Technical Volume; additional pages will not be considered or evaluated.

Content of the Technical Volume

DPII Technical Volume (Volume 2) instructions are specified in the DoW SBIR Program BAA Appendix A, also located on DSIP: https://www.dodsbirsttr.mil/submissions/solicitation-documents/supporting-documents.

The Statement of Objective (SOO) and the list of Contract Data Requirement List (CDRL)s are provided and can be downloaded from https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. The technical proposal shall include a non-proprietary Statement of Work (SOW) with the planned tasks and descriptions to meet the Statement of Objectives (SOO) goals detailed. Do not upload the whole SOO as your SOW with your proposal. The SOO and CDRL are provided to help the offerors consider the required goals, scope, and deliverables when developing the proposal. It is the offeror's responsibility to provide fully responsive, complete, and clear submissions. Exceptions to the requirements need to be identified/explained.

If an offeror is selected for award, the offeror will be required to submit a separate non-proprietary SOW with the planned tasks and descriptions from the proposal and all other applicable sections of the SOO and it shall include no proprietary information, data, or marking. The provided SOW will become Attachment 3

of the resulting Other Transaction Agreement (OTA) Award, incorporating any agreed upon changes if necessary.

Note: The Phase I Feasibility Appendix (Appendix A), documenting the results of the offeror's internal Feasibility Study, is required for the Direct to Phase II proposal and is specified in Volume 5 of these instructions.

Cost Volume (Volume 3)
Offerors must read the instructions before completing the cost volume. The USSOCOM cost volume template (Volume 3 template) is located on DSIP under Supporting Documents and Templates at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations.

For the Direct to Phase II topics in this announcement, the total price limit to provide a testable prototype is listed in Table 1 titled "Consolidated SBIR Topic Information". **Any proposal submitted with a total price above the provided limit will not be evaluated or considered for award.**

The final price of a USSOCOM Phase II SBIR contract/OTA will be negotiated as necessary to reach a determination of price fairness and reasonableness commensurate with the magnitude and complexity of the required research and development effort. The resulting agreement will be a firm priced agreement, OTA contract award.

Proposal information should include the itemized listing (a-h) specified below. The proposal information must include a level of detail that would enable the Government personnel to determine the purpose, necessity, and reasonableness of the proposal and show an understanding of the scope of the work. It is requested that a breakdown of labor hours per labor category and other associated costs be provided **by task.** The Agreements Officer may request additional information to support price analysis or understand the approach if needed. Fully burdened rates should be provided in proposal responses.

a) Special Tooling and Test Equipment and Material: The inclusion of equipment and materials will be carefully reviewed relative to need and appropriateness of the work proposed. The purchase of special tooling and test equipment must, in the opinion of the Government, be advantageous to the Government and relate directly to the specific effort. They may include such items as innovative instrumentation and/or automatic test equipment. The reason for the requirement and the intention of offeror on disposition of the special material/equipment shall be documented in the proposal as well as the reason on why said equipment is charged directly to the effort rather than in the indirect cost of the business.
b) Direct Cost Materials: Justify costs for materials, parts, and supplies with an itemized list that includes item description, part number, quantities, and price.
c) Other Direct Costs: This category of costs includes specialized services such as machining or milling, special testing or analysis, and costs incurred in obtaining temporary use of specialized equipment. Proposals that include leased hardware must provide an adequate lease vs. purchase justification or rationale.
d) Direct Labor: For each individual, include the number of hours, and loaded rate to include all indirect costs. Identify key personnel by name if possible and labor category.
e) Travel: Travel costs must relate to the needs of the project. Proposed travel cost must be in accordance with the Federal Travel Regulation (FTR).

    1. Per Diem Rates can be obtained at: http://www.gsa.gov/perdiem
    2. The following information shall be documented –
        (i) Date (estimated), length and place (city, town, or other similar designation) of the trip;
        (ii) Purpose of the trip; and
        (iii) Number of personnel included in the estimate.

f) Cost Sharing: Cost sharing is permitted. However, cost sharing is not required, nor will it be an evaluation factor in the consideration of a proposal. Please note that cost share contracts do not allow fees/profit.
g) Subcontracts: Involvement of consultants in the planning and/or research stages of the project may be appropriate. If the offeror intends such involvement, describe in detail and include information in the cost proposal. The proposed total of all consultant fees, facility leases or usage fees, and other subcontract or purchase agreements may not exceed one-half of the total contract price or cost, unless otherwise approved in writing by the Agreements Officer.
h) Support subcontract costs with copies of the subcontract agreements. The supporting agreement documents must adequately describe the work to be performed (i.e., cost proposal) and provide a statement of work with a corresponding detailed proposal for each planned subcontract. **Cost proposals that include subcontractors but do not include subcontract agreements will be** determined to be non-responsive.
i) Consultants: Provide a separate agreement letter for each consultant. The letter should briefly state what service or assistance will be provided, the number of hours required and hourly rate. **Failure to provide agreement letters for any proposed consultants will be considered non-responsive.**

SBIR program requires the offerors must do at least 50% of the PHASE II SBIR work. To determine eligibility for award based on this requirement, USSOCOM will divide the overall price submitted/negotiated minus the total cost of subcontractors/consultants amount (with applied indirects), by the total price of the proposal. To qualify for award, the resulting offeror percentage of work shall be 50% or higher. If the percentage is lower, the proposal will not be evaluated.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW SBIR Program BAA for full details on this requirement. Information contained in the CCR will be considered by USSOCOM during proposal evaluations.

Small businesses must complete the CCR by logging into their account at https://www.sbir.gov. Please refer to the "Instructions" and "Guide" documents contained in the DSIP Dashboard for more detail on completing and updating the CCR.

Once the CCR is certified and submitted on SBIR.gov, it must be uploaded to Volume 4: Company Commercialization Report in the Firm Information section of DSIP by the Firm Admin.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoW Program BAA for more information.

In addition to the documentation outlined in the DoW SBIR Program BAA, the following USSOCOM documents must also be included in Volume 5: (1) Feasibility Study (Appendix A), (2) section K and (3) resumes for all key personnel.

1. **<u>Feasibility Study: Offerors must provide documentation to satisfy the feasibility requirement explaining the previously done research and how it applies to the topic as specified in the Phase I topic write-up.</u>** Offerors are required to provide sufficient information to determine, to the extent possible, the scientific, technical, and commercial merit and feasibility of ideas submitted, and that the feasibility assessment was performed by the offeror and/or the Principal Investigator. **If the offeror fails to demonstrate the scientific and technical merit, feasibility, and/or the source of the work, USSOCOM will not continue to evaluate the offeror's proposal.** Refer to

the topic description associated with the Direct to Phase II topic to review the minimum requirements needed to demonstrate feasibility. There is no minimum or maximum page limitation for the Feasibility Appendix (Appendix A).

2. Section K - Titled "Representations, Certifications, and other statements of Offerors": The proposal must also include a completed Section K which does not count toward the page limit and should be uploaded with this volume. The identification of foreign national involvement in a USSOCOM SBIR topic is required to determine if a firm is ineligible for award on a USSOCOM topic that falls within the parameters of the United States Munitions List, Part 121 of the International Traffic in Arms Regulation (ITAR). A firm employing a foreign national(s) (as defined in paragraph 3.7 entitled "Foreign Nationals" of the DoW SBIR 22.4 Announcement) to work on a USSOCOM ITAR topic must possess an export license to receive a SBIR Phase II contract.
3. Resumes: Include Key Personnel resumes.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse (FWA) training is required for Direct to Phase II proposals. Training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, all proposals submitted in response to this BAA will be reviewed for security risks related to foreign affiliations.

- Complete the Disclosures of Foreign Affiliations or Relationships to foreign Countries webform in Volume 7 of the DSIP proposal submission. **PDF uploads will not be accepted.**
- Full proposal submissions cannot be certified and submitted by the Corporate Official until Volume 7 is fully completed and the webform is submitted.
- The Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a Supporting Document in Volume 5. Do not upload any previous versions of this form to Volume 5.
- For additional details, refer to the DoW Program BAA.

INQUIRIES
USSOCOM does not allow direct communication with the topic authors, which differs from the DoW SBIR/STTR Program BAA instructions.

- During the Pre-Release and Open Periods of the DoW SBIR Program BAA, all technical questions that enhance the Offerors' understanding of the topic's requirements must be submitted via the Defense SBIR/STTR Innovation Portal (DSIP) Topic Q&A. All questions and answers submitted to DSIP Topic Q&A will be released to the general public.
- Only questions pertaining to the proposal preparation instructions should be directed to: sbir@socom.mil. All inquiries must include the topic number in the subject line of the e-mail.
- Consistent with DoW SBIR instructions, USSOCOM will not answer programmatic questions such as providing the technical point of contact, the number of contracts to be awarded, the source of funding, or transition strategy.
- Physical site visits will not be permitted during the Pre-release and Open Periods of the DoW SBIR Program BAA.

EVALUATION AND SELECTION
All Offerors will be evaluated in accordance with the evaluation criteria listed in the DoW SBIR Program BAA.

The Government will evaluate only responsive proposals. Feedback is based on strengths and weaknesses to the proposal as compared to Government need. Award selections will be made based on solutions that best meet Government need and available funding.

1. Proposals missing Technical Volume (Volume 2), Cost Volume (Volume 3), Feasibility Appendix (Volume 5), will not be evaluated or those that exceed the maximum price allowed as per Table 1 of this instructions and will be considered non-responsive.

2. Feasibility determination. The Feasibility Appendix (Appendix A) to the Phase II proposal will be evaluated first to determine that the offerors demonstrated they have completed research and development to establish the feasibility of the proposed Phase II effort based on the criteria outlined in the topic description of Phase I**. USSOCOM will not continue evaluating the offeror's related Direct to Phase II proposal if it determines that the offeror failed to demonstrate that feasibility has been established or the offeror failed to demonstrate work submitted in the feasibility documentation was substantially performed by the offeror and/or the Principal Investigator.**

   Refer to the Phase I Topic description associated with the Direct to Phase II topic Statement of Objectives to review the minimum requirements that need to be demonstrated in the feasibility documentation.

3. The technical evaluation will utilize the Evaluation Criteria provided in the DoW SBIR Program BAA instructions. The technical evaluation is performed in two parts:

   Part I: The evaluation of the Technical Volume will utilize the Evaluation Criteria provided in the DoW SBIR Program BAA. Once the evaluations are complete, all offerors will be notified in a timely manner.

   Part II: The Cost Volume award amount is provided as a not to exceed (NTE) amount and a technical evaluation of the proposal cost will be completed to assess price fair and reasonableness. Proposals above the established NTE for the Phase I effort will not be considered for award. The Government will assess the technical approach presented for the effort based on the number of labor hours by labor categories, the key personnel level of involvement, materials, subcontractors and consultants (scope of work, expertise, participation and proposed effort), and other direct cost as proposed.

4. The Cost Volume (Volume 3) evaluation: For these Direct to Phase II efforts, the award amount is set with not to exceed (NTE) amount. Technical evaluation of the proposal cost will be completed to assess the probability of success to obtain a working prototype that meets Government need. Proposals above the set NTE for the effort **will not** be considered for award. The technical team will assess number of labor hours, labor categories, key personnel expertise and level of involvement, materials, equipment, subcontractors and consultants (scope of work, expertise, participation and proposed effort), travel and other direct cost to successfully complete the effort as proposed.

This is a SOCOM SBIR/STTR request for proposals that allows for multiple awards. The Government considers each award to be a separate procurement. Each submitted proposal will be evaluated based on

individual merit and unique technical solutions. Individual proposals may lead to multiple awards, as set forth in the Proposal Submission Instructions. Federal Acquisition Regulation 3.104-1 defines "federal agency procurement" in the context of broad agency announcements and small business innovation research programs as individual and separate procurements although resulting from the same solicitation, because "each proposal received by an agency constitutes a separate procurement…" 48 C.F.R. § 3-104.1 (emphasis added).

The resulting award/s will be a fixed price prototyping agreement and a successful prototype may lead to follow on production. Follow on production awards may be FAR based, Fixed Price or Cost-Plus Fixed Fee contracts. A Defense Contracts Audit Agency approved accounting system will be required to issue a Cost-Plus Fixed Fee contract.

Additionally, input on technical aspects of the proposals may be solicited by USSOCOM from non-Government consultants and advisors who are bound by appropriate non-disclosure requirements. When appropriate, non-Government advisors may have access to Offeror's proposals and may be utilized to objectively review a proposal in a particular functional area and provide comments and recommendations to the Government's decision makers. They may not establish final assessments of risk, rate or rank Offerors' proposals. All advisors shall comply with procurement Integrity Laws and shall sign Non-Disclosure and Rules of Conduct/ Conflict of Interest statements. The Government shall take into consideration requirements for avoiding conflicts of interest. Submission of a proposal in response to this request constitutes approval to release the proposal to Government support contractors.

Offerors will be notified of selection or non-selection status via DSIP within 90 calendar days of the closing date of the BAA. This notification will come by e-mail to the Corporate Official identified by the Offeror during proposal submission. The Government will also notify the Offerors if their proposal is considered non-responsive (disqualified).

A non-selected Offeror can make a written request to the Contracting Officer, within 30 calendar days of receipt of notification of non-selection, for informal feedback. The Contracting Officer will provide informal feedback after receipt of an Offeror's written request rather than a debriefing as specified in the DoW SBIR Program BAA instructions.

Refer to the DoW SBIR Program BAA for procedures to protest the Announcement. As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: sbir@socom.mil.

AWARD AND CONTRACT INFORMATION

Table 1: Consolidated SBIR Topic Information

Topic	Technical Volume (Vol 2)	Period of Performance	Award Amount	Contract Type
SOC26BZ01-DV001	Not to exceed 10 pages	Not to exceed 12 months	NTE $1,000,000.00	Firm-Fixed-Price
SOC26BZ01-DV002	Not to exceed 10 pages	Not to exceed 12 months	NTE $3,500,000.00	Firm-Fixed-Price

SBIR awards for the Direct to Phase II topics will be awarded as a fixed price (level of effort type), Other Transactions Agreements (OTA). Successful completion of the prototype under an OTA may result in a follow-on production OTA or contract. Successful completion of the prototype is defined as meeting one or more threshold requirements.

The OTA template needs to be completed by only those offerors selected for award and will be submitted directly to the Agreements Officer identified in the notification.

Those selected for award would be required to enter their company information, expected milestones (Attachment 1), and provide a non-proprietary Statement of Work (SOW) following the format of the Statement of Objectives (SOO) (Attachment 3).

ADDITIONAL INFORMATION

Direct to Phase II proposals shall NOT include:

1. "Basic Research" (or "Fundamental Research") defined as a "Systematic study directed toward greater knowledge or understanding of the fundamental aspects of phenomena and/or observable facts without specific applications toward processes or products in mind."
2. Discretionary Technical and Business Assistance (TABA).

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Release 1

SOC26BZ01-DV001	Title: Compact UUV Borne LiDAR System
SOC26BZ01-DV002	Title: IRONWALKER

SOC26BZ01-DV001 TITLE: Compact UUV Borne LiDAR System

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Microelectronics; Integrated Network Systems-of-Systems

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: The objective of this topic is to develop applied research towards a low cost and compact Laser interferometric Detection and Ranging (LiDAR) system for rapid clearance of undersea terrain and checkout by divers and vessel operators. To this end, this effort is developing a compact, low-cost LiDAR package for available Government Furnished Equipment (GFE) Unmanned Undersea Vehicle (UUV) systems that can be utilized for offboard collection and inspection of the entire water column, particularly in ports, harbors and confined waterspaces.

DESCRIPTION: As a part of this feasibility study, the proposers shall address all viable overall system design options with respective specifications is needed to both understand the safe operation, maintenance and general use of LiDAR based payloads for man portable UUV systems from Fast Attack Craft (FAC). Further, standardized collection methods and accessible UUV frame integration is needed. This effort seeks to secure designs and rapidly procurable, safe to operate and relatively low cost & compact systems to this end, particularly for shallow water and littoral operational depths (nominally optimal for surf zone to VSW depths, but functional to 200m).

PHASE I (FEASIBILITY STUDY): As a requirement of this Direct to Phase II (DPII) proposers must include a feasibility study that assess what is in the art of the possible that satisfies the requirements specified in the above paragraphs entitled “Objective” and “Description.”

The objective is to document the results of a thorough feasibility study (“Technology Readiness Level 3”) to investigate what is in the art of the possible within the given trade space that will satisfy a needed technology. The feasibility study should investigate all options that meet or exceed the minimum performance parameters specified in this write up. It should also address the risks and potential payoffs of the innovative technology options that are investigated and recommend the option that best achieves the objective of this technology pursuit.

PHASE II: Development of a low cost, man portable (ideally 5” maximum Outer Diameter (OD)) 360-degree scanning LiDAR Systems that can integrate available sonar data into the overall image generation are preferred. The entire kit should be capable of being integrated onto modular UUV frames with open architecture design elements, be fully self-contained and capable of being submitted for laser safety reviews during the execution of the planned effort.

The developed prototype will be installed, and demonstrate the most feasible solution for diver borne or UUV borne LiDAR scanning. Emphasis on fused solutions that incorporate additional sensing, such as side scan sonar (SSS) will be given. This effort will generate two (2) end user prototype systems for modular use in unmanned and diver (e.g. diver propulsion vehicle) applications for survey beneath the waterline. Systems utilized under test and demonstration will not be counted towards the deliverable of two (2) end user evaluation systems.

PHASE III DUAL USE APPLICATIONS: This system could be used in a broad range of military applications where undersea survey is required, to include environmental monitoring, salvage and with the oil and natural gas business.

REFERENCES:

1. A deep-sea corporation is listed that provides seabed oil and natural gas services to describe LiDAR imaging in the commercial sector: https://3datdepth.com

KEYWORDS: LiDAR, UUV, Undersea Warfare

SOC26BZ01-DV002 TITLE: IRONWALKER

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI); Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy; Advanced Computing and Software; Integrated Network Systems-of-Systems; Advanced Materials; Human-Machine Interfaces

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), which controls the export and import of defense-related material and services. Offerors must disclose any proposed use of foreign nationals, their country of origin, and what tasks each would accomplish in the statement of work in accordance with section 5.4.c.(8) of the solicitation. Additionally, Offerors will describe compliance mechanisms offerors have in place or will put in place to address any ITAR issues that arise during the course of agreement administration.

OBJECTIVE: The objective of this topic is to conduct applied research to an innovative capability for a deployable, and user-friendly manufacturing system that integrates Artificial Intelligence (AI) and Augmented Reality (AR) to enhance advanced additive and subtractive machining capabilities. This system will provide each operator with AI-driven advanced manufacturing expertise and AR-based work instructions to train, certify, and guide them in operating complex machinery for the production of air, ground, and maritime components.

DESCRIPTION: This effort will explore, design, and evaluate an innovative manufacturing capability that combines Artificial Intelligence (AI) and Augmented Reality (AR) to enhance advanced additive and subtractive machining operations. As a part of this feasibility study, the proposers shall address all viable overall system design options with respective specifications that enable rapid deployment, ease of use, and minimal training burden while maintaining the precision and repeatability required for air, ground, and maritime component production.

The research will focus on integrating AI-driven advanced manufacturing expertise with AR-based work instructions to guide operators through the full lifecycle of production—from setup and calibration to machining, quality assurance, and certification. The system shall leverage modular, transportable platforms suitable for forward-deployed or austere environments, with consideration for integration into both manned and autonomous (e.g., bipedal robotic) operations in future phases.

Proposers shall detail specification for key system attributes, including but not limited to:

- **AI Capability:** Real-time adaptive guidance, error detection, and optimization based on operator input and environmental factors.
- **AR Work Instructions:** Interactive overlays for step-by-step tasks, safety checks, and certification pathways.
- **Machining Integration:** Compatibility with advanced additive and subtractive manufacturing processes using multiple metal and composite materials.
- **Deployment and Sustainment:** Footprint, power requirements, portability, and environmental resilience.
- **Cybersecurity and Data Management:** Secure handling of operational data, digital twin integration, and compliance with DoD cybersecurity requirements.

PHASE I (FEASIBILITY STUDY): As a requirement of this Direct to Phase II (DPII) proposers must include a feasibility study that assess what is in the art of the possible that satisfies the requirements specified in the above paragraphs entitled "Objective" and "Description."

The objective is to document the results of a thorough feasibility study ("Technology Readiness Level 3") to investigate what is in the art of the possible within the given trade space that will satisfy a needed technology. The feasibility study should investigate all options that meet or exceed the minimum performance parameters specified in this write-up. It should also address the risks and potential payoffs of the innovative technology options that are investigated and recommend the option that best achieves the objective of this technology pursuit.

PHASE II: Develop, integrate, and demonstrate a fully functional IRONWALKER prototype system. based on the optimal solution identified in the completed Phase I feasibility study. This effort will focus on merging mature additive and subtractive machining systems, commercial off-the-shelf (COTS) augmented reality platforms, and advanced AI-driven manufacturing guidance engines into a cohesive, operator-centric solution.

PHASE III DUAL USE APPLICATIONS: This system could be used in a broad range of military applications where rapid, deployable, and precise manufacturing capabilities are needed to sustain operations in contested, remote, or resource-limited environments. In the commercial sector, IRONWALKER's portable, AI/AR-guided manufacturing capability could support industries such as aerospace, maritime shipping, oil and gas, construction, heavy equipment repair, and disaster recovery operations. Its ability to integrate with both human operators and future autonomous robotic platforms allows for flexible deployment in locations where skilled machinists are unavailable or where traditional manufacturing facilities are inaccessible.

REFERENCES:

1. Department of Defense Additive Manufacturing Strategy, Office of the Under Secretary of Defense for Research and Engineering, January 2021. https://www.cto.mil/wp-content/uploads/2021/01/dod-additive-manufacturing-strategy.pdf
2. National Institute of Standards and Technology (NIST) – Additive Manufacturing Standards Roadmap, Version 3.0, July 2023. https://www.nist.gov/additive-manufacturing
3. Massachusetts Institute of Technology Lincoln Laboratory – Human-Machine Teaming for Manufacturing and Maintenance, Technical Report, September 2022. https://www.ll.mit.edu/sites/default/files/page/doc/2023-06/MITLL_2022_Annual_Report.pdf
4. Defense Innovation Board – AI Principles: Recommendations on the Ethical Use of Artificial Intelligence by the Department of Defense, October 2019. https://insidecybersecurity.com/sites/insidecybersecurity.com/files/documents/2019/oct/cs2019_0292.pdf

KEYWORDS: Artificial Intelligence; Augmented Reality; Additive Manufacturing; Subtractive Machining; Advanced Manufacturing; Human-Machine Teaming; Operator Training; Expeditionary Manufacturing; Deployable Manufacturing; Digital Work Instructions; CNC Machining; 3D Printing; Autonomous Manufacturing; Field Repair; Point-of-Need Production

Appendix A
TECHNICAL PROPOSAL TEMPLATE (VOLUME 2)

INSTRUCTIONS

These instructions and template apply to DoW SBIR/STTR Phase I topics and provide general guidelines for completing the Phase I Technical Volume. Information provided in the Service/Component-specific instructions for the topic of interest take precedence over any instructions listed below.

The template (beginning on the following page) is the format model that may be used to prepare the Phase I Technical Volume. Do not include the instructions provided on this page or any bracketed [ ] guidance in the template.

Disclosure

Offerors that include in their proposals data which they do not want disclosed to the public for any purpose, or used by the U.S. Government except for evaluation purposes, must:

(1) Mark the first page of each Volume of the Submission with the following legend:
"This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of-or in connection with-the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]";

(2) Mark each sheet of data it wishes to restrict with the following legend:
"Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

Format

The Technical Volume shall meet the following requirements:

- Please refer to Service/Component-specific topic instructions for the page limit and how a technical volume is handled if the stated page count is exceeded. It is the proposing firm's responsibility to verify that the Technical Volume does not exceed the page limit after upload to DSIP. Unless otherwise noted, all content in the Technical Volume will count toward the limit.
- Single column format single-spaced typed lines.
- Standard 8 ½" x 11" paper format.
- Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
- **The header on each page of the Technical Volume should contain your company name, topic number, and DSIP-assigned proposal number.**
- No font smaller than 10-point. For headers, footers, imbedded tables, figures, images, or graphics that include text, a font size of smaller than 10-point is allowable, though proposers are cautioned that the text may be unreadable by evaluators.

Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

Delete this instruction page and begin the Technical Volume starting with the following page.

[Title]

Volume 2: Technical Volume

[Note: Remove the disclosure statement below if not applicable to your proposal. Refer to Instructions.]

This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed – in whole or in part – for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of – or in connection with – the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages <insert numbers or other identification of sheets>.

1. Identification and Significance of the Problem or Opportunity.

[Define the specific technical problem or opportunity addressed and its importance.]

2. Phase I Technical Objectives.

[Enumerate the specific objectives of the Phase I work, including the questions the research and development effort will try to answer to determine the feasibility of the proposed approach.]

3. Phase I Statement of Work (include Subcontractors and/or Research Institutions).

(a) [Provide an explicit, detailed description of the Phase I approach. If a Phase I option is required or allowed by the Component (refer to Component-specific instructions for topic of interest), describe appropriate research activities which would commence at the end of Phase I base period should the Component elect to exercise the option. The Statement of Work should indicate what tasks are planned, how and where the work will be conducted, a schedule of major events, and the final product(s) to be delivered. The Phase I effort should attempt to determine the technical feasibility of the proposed concept. The methods planned to achieve each objective or task should be discussed explicitly and in detail. This section should be a substantial portion of the Technical Volume.

(b) The solicitation may contain topics that have been identified by the Program Manager as research or activities involving Human/Animal Subjects and/or Recombinant DNA. If Phase I performance includes performance of these kinds of research or activities, please identify the applicable protocols and how those protocols will be followed during Phase I. Please note that funds cannot be released or used on any portion of the project involving human/animal subjects or recombinant DNA research or activities until all of the proper approvals have been obtained. **SBCs proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal in order to avoid potential delay of contract award.**]

4. Related Work.

[Describe significant activities directly related to the proposed effort, including any conducted by the principal investigator, the proposing firm, consultants, or others. Describe how these activities interface with the proposed project and discuss any planned coordination with outside sources. The Technical Volume must persuade evaluators of the proposer's awareness of the state of the art in the topic. Describe any previous work not directly related but similar to the proposed effort. Provide the following: (1) a short description, (2) the client for which work was performed (including the Government Point of Contact to be contacted including e-mail address and phone number), and (3) date of performance including project completion.]

5. <u>Relationship with Future Research or Research and Development</u>.

(a) [State the anticipated results of the proposed approach if the project is successful.

(b) Discuss the significance of the Phase I effort in providing a foundation for a Phase II research or research and development effort.

(c) Identify the applicable clearances, certifications and approvals required to conduct Phase II testing. Outline the plan for ensuring timely completion of stated authorizations in support of a Phase II research or research and development effort.]

6. <u>Commercialization Strategy.</u>

[Describe in approximately one page the SBC's strategy for commercializing this technology in DoW, other Federal Agencies, and/or private sector markets. Provide specific information on the market need the technology will address and the size of the market. Also include a schedule showing the quantitative commercialization results from the project that your company expects to achieve.]

7. <u>Key Personnel</u>.

[Identify key personnel who will be involved in the Phase I effort including information on directly related education and experience. A concise technical resume of the principal investigator, including a list of relevant publications (if any), must be included (Please do not include Privacy Act Information). All resumes will count toward the page limit for Volume 2, as specified in the Component-specific instructions.]

[Principal Investigator Name]
[School, Degree, Year]

Relevant Experience
[A concise description of the principal investigator's relevant technical experience and its application to this topic.]

Relevant Awards or Patents
[List any awards received or patents granted or applications submitted for work related to this topic.]

Relevant Publications
[List any publications relevant to this topic.]

[Repeat this format as necessary to address the qualifications of all key personnel.]

8. <u>Foreign Citizens</u>.

[Identify any foreign citizens or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. The proposal may be deemed nonresponsive if the requested information is not provided. You may be asked to provide additional information during negotiations in order to verify the foreign citizen's eligibility to participate on a SBIR/STTR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)). Note: If no foreign nationals will be involved in proposed work, the word "None" can be substituted for the table.]

Name [include direct employees, subcontractors, and consultants]	Foreign National (Yes/No)	Country of Origin	Type of Visa or Work Permit	Level of Involvement (Role)

9. Facilities/Equipment.

[Describe available instrumentation and physical facilities necessary to carry out the Phase I effort. Justify equipment purchases in this section and include detailed pricing information in the Cost Volume. State whether or not the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name), and local Governments for, but not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.]

10. Subcontractors/Consultants.

[Propose efforts as applicable to either SBIR or STTR as follows:

SBIR. Involvement of a university or other subcontractors or consultants in the project may be appropriate. A minimum of two-thirds of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be carried out by the proposing small business firm, unless otherwise approved in writing by the Contracting Officer. SBIR efforts may include subcontracts with Federal Laboratories and Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories and FFRDCs; however, proposers must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoW SBIR/STTR solicitation for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.)

STTR. Involvement of a Research Institution in the project is required. A minimum of 40 percent of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be conducted by the proposing small business firm, and a minimum of 30 percent of the research and/or tasks in Phase I, as measured by direct and indirect costs, must be conducted by a single Research Institution. STTR efforts may include subcontracts with Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories, but they do not qualify as a Research Partner; proposers may only subcontract to Federal Laboratories within the remaining 30 percent and must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoW SBIR/STTR solicitation for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.]

11. Prior, Current or Pending Support of Similar Proposals or Awards.

[If a proposal submitted in response to this solicitation is substantially the same as another proposal that was funded, is now being funded, or is pending with another Federal Agency, another or the same DoW Service/Component, you must disclose this on the proposal cover sheet and provide the following information:

a) Name and address of the Federal Agency(s) or DoW Component to which a proposal was submitted, will be submitted, or from which an award is expected or has been received.

b) Date of proposal submission or date of award.

c) Title of proposal.

d) Name and title of principal investigator for each proposal submitted or award received.

e) Title, number, and date of solicitation(s) under which the proposal was submitted, will be submitted, or under which award is expected or has been received.

f) If award was received, provide contract number.

g) Specify the applicable topics for each proposal submitted or award received.

Note: If this does not apply, state in the proposal "No prior, current, or pending support has been provided for proposed work."]

12. <u>Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software.</u>

The Offeror asserts for itself, or the persons identified below, that the Government's rights to use, release, or disclose the following technical data or computer software should be restricted:

Technical Data or Computer Software to be Furnished with Restrictions	Basis for Assertion	Asserted Rights Category	Name of Person or Organization Asserting Restrictions
[(LIST)]	[(LIST)]	[(LIST)]	[(LIST)]

[Completion of this table and submission of the proposal constitutes signature for the information listed in the table above.]

[**ADDITIONAL INFORMATION/INSTRUCTION:** Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this solicitation generally remain with the contractor, except that the Government obtains a royalty-free license to use such technical data only for Government purposes during the period commencing with contract award and ending twenty years after completion of the project under which the data were generated. This data <u>must</u> be marked with the restrictive legend specified in DFARS 252.227-7018 (JAN 2025). Upon expiration of the twenty-year restrictive license, the Government has unlimited rights in the SBIR data. During the license period, the Government may not release or disclose SBIR data to any person other than its support services contractors except: (1) For evaluation purposes; (2) As expressly permitted by the contractor; or (3) A use, release, or disclosure that is necessary for emergency repair or overhaul of items operated by the Government. See DFARS clause 252.227-7018 (JAN 2025).

If a proposer plans to submit assertions in accordance with DFARS 252.227-7017 (JAN 2025), those assertions must be identified and assertion of use, release, or disclosure restriction MUST be included with your proposal submission. The contract cannot be awarded until assertions have been approved. Please note that only the table is included in the page limitation; any supporting data concerning the contract/grant number and awarding agency, as well as planned use or need of the data asserted, can be provided in Volume 5, Supporting Documents.

The following instructions apply to the fields in the table above (Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software).

1) For technical data (other than computer software documentation) pertaining to items, components, or processes developed at private expense, identify both the deliverable technical data and each such item, component, or process. For computer software or computer software documentation identify the software or documentation.

2) Generally, development at private expense, either exclusively or partially, is the only basis for asserting restrictions. For technical data, other than computer software documentation, development refers to development of the item, component, or process to which the data pertain. The Government's rights in computer software documentation generally may not be restricted. For computer software, development refers to the software. Indicate whether development was accomplished exclusively or partially at private expense. If development was not accomplished at private expense, or for computer software documentation, enter the specific basis for asserting restrictions.

3) Enter asserted rights category (e.g., Government purpose license rights from a prior contract, rights in SBIR/STTR data generated under another contract, limited, restricted, or government purpose rights under this or a prior contract, or specially negotiated licenses).

4) Corporation, individual, or other person, as appropriate.

Enter “none” when all data or software will be submitted without restrictions.]

Appendix B
DEFINITIONS

The following definitions from the SBA SBIR/STTR Policy Directive, the Federal Acquisition Regulation (FAR) and other cited regulations apply to this solicitation.

Commercialization

The process of developing products, processes, technologies, or services, and the production and delivery (whether by the originating party or others) of the products, processes, technologies, or services for Federal Government or commercial markets purchase or use.

Cooperative Research and Development

An SBC and a research institution jointly conduct R&D. For purposes of the STTR Program, the SBC performs 40 percent of the work, and the single research institution performs not less than 30 percent of the work. For purposes of the SBIR Program, this refers to work a research institution conducts as the SBC's subcontractor. The proposing SBC must conduct at least two-thirds of the research and/or analytical work in Phase I.

Covered Individual

An individual who contributes in a substantive, meaningful way to the scientific development or execution of a R&D project proposed to be carried out with a DoW-funded award. DoW has further designated covered individuals as including all proposed key personnel.

Essentially Equivalent Work

Work that is substantially the same research, which is proposed for funding in more than one contract proposal or grant application submitted to the same federal agency or submitted to two or more different federal agencies for review and funding consideration; or work where a specific research objective and the research design for accomplishing the objective are the same or closely related to another proposal or award, regardless of the funding source.

Export Control

The International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, and the Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, apply to all projects with military or dual-use applications that develop beyond fundamental research, which is basic and applied research ordinarily published and shared broadly within the scientific community. Details are available here.

NOTE: Export control compliance statements found in the individual Service/Component-specific proposal instructions are not meant to be all inclusive. They do not remove any liability from the submitter to comply with applicable ITAR or EAR export control restrictions or from informing the U.S. Government of any potential export restriction as fundamental R&D efforts proceed.

Federal Laboratory

In 15 U.S.C. §3703, it means any laboratory, any federally funded R&D center (FFRDC), or any center established under 15 U.S.C. §§ 3705 & 3707 that a federal agency owns, leases, or otherwise uses and the Federal Government funds, whether the U.S. Government or the contractor operates.

Federally Funded Award

A Phase I, Phase II (including Direct to Phase II, sequential Phase II/subsequent Phase II and cross-agency Phase II), or Phase III SBIR or STTR award made using a funding agreement.

Foreign Affiliation

Under 15 U.S.C. § 638(e)(16), foreign affiliation means a funded or unfunded academic, professional, or institutional appointment or position with a foreign government or government-owned entity, whether full-time, part-time, or voluntary (including adjunct, visiting, or honorary). This includes appointments or positions deemed adjunct, visiting, or honorary with research institutions located in a foreign country of concern.

Foreign Country of Concern

In 15 U.S.C. § 638(e)(17), foreign country of concern means the People's Republic of China, the Democratic People's Republic of Korea, the Russian Federation, the Islamic Republic of Iran, or any other country the Secretary of State determines as a country of concern.

Foreign Entity

Foreign entity means any branch, partnership, group or sub-group, association, estate, trust, corporation or division of a corporation, non-profit, academic institution, research center, or organization that foreign owners, foreign investors, foreign management, or a foreign government establish, direct, or control.

Foreign Government

Foreign government means any government or governmental body, organization, or instrumentality, including government owned-corporations, other than the U.S. Government or U.S. state, territorial, tribal, or jurisdictional governments or governmental bodies. The term includes, but is not limited to, non-U.S. national and subnational governments, including their respective departments, agencies, and instrumentalities.

Foreign National

Foreign National (also known as Foreign Person) under 22 CFR 120.16 mean any natural person who is not a lawful permanent resident as defined by 8 U.S.C. § 1101(a)(20) or who is not a protected individual as defined by 8 U.S.C. § 1324b(a)(3). It also means any foreign corporation, business association, partnership, trust, society, or any other entity or group that is not incorporated or organized to do business in the United States, as well as international organizations, foreign governments and any agency or subdivision of foreign governments (e.g., diplomatic missions).

"Lawfully admitted for permanent residence" means the status of having been lawfully accorded the privilege of residing permanently in the United States as an immigrant in accordance with the immigration laws, such status not having changed.

"Protected individual'' means an individual who (A) is a citizen or national of the United States, or (B) is an alien who is lawfully admitted for permanent residence, is granted the status of an alien lawfully admitted for temporary residence under 8 U.S.C. § 1160(a) or 8 U.S.C. § 1255a(a)(1), is admitted as a refugee under 8 U.S.C. § 1157, or is granted asylum under Section 8 U.S.C. § 1158; but does not include

(i) an alien who fails to apply for naturalization within six months of the date the alien first becomes eligible (by virtue of period of lawful permanent residence) to apply for naturalization or, if later, within six months after November 6, 1986, and (ii) an alien who has applied on a timely basis, but has not been naturalized as a citizen within two years after the date of the application, unless the alien can establish that the alien is actively pursuing naturalization, except that time consumed in the Service's processing the application shall not be counted toward the two-year period.

Fraud, Waste and Abuse

a. Fraud includes any false
b. about a material fact or any intentional deception designed to deprive the United States unlawfully of something of value or to secure from the United States a benefit, privilege, allowance, or consideration to which an individual or business is not entitled.
c. Waste includes extravagant, careless or needless expenditure of government funds, or the consumption of government property, that results from deficient practices, systems, controls, or decisions.
d. Abuse includes any intentional or improper use of government resources, such as misuse of rank, position, or authority or resources.
e. The SBIR/STTR Program training related to Fraud, Waste and Abuse is available here. See Section 1.13 for reporting fraud, waste, and abuse.

Funding Agreement

Any contract, grant, or cooperative agreement entered between any Federal Agency and any SBC for the performance of experimental, developmental, or research work, including products or services, Federal Government-funded in whole or in part. DoW Services/Components will only use contracts and other transaction authority (OTA) agreements for all SBIR awards.

Historically Black Colleges and Universities, and Minority-Serving Institutions

Department of Education list for historically Black colleges and universities and minority-serving institutions.

HUBZone Certified Small Business Concern

An SBC with SBA certification under the Historically Underutilized Business Zones (HUBZone) Program (13 C.F.R. § 126) as a HUBZone firm listed in the dynamic small business search (DSBS).

Innovation

As defined and as used in SBA Policy Directive 3(t), the term innovation means-
Something new or improved, having marketable potential, that includes the development of new technology, the refinement of existing technology, or the development of new applications for existing technology.

Malign Foreign Talent Recruitment Program

As defined in 42 U.S.C § 19237, the term “malign foreign talent recruitment program” means-

(A) any program, position, or activity that includes compensation in the form of cash, in-kind compensation, including research funding, promised future compensation, complimentary foreign travel, things of non de minimis value, honorific titles, career advancement opportunities, or other

types of remuneration or consideration directly provided by a foreign country at any level (national, provincial, or local) or their designee, or an entity based in, funded by, or affiliated with a foreign country, whether or not directly sponsored by the foreign country, to the targeted individual, whether directly or indirectly stated in the arrangement, contract, or other documentation at issue, in exchange for the individual-

(i) engaging in the unauthorized transfer of intellectual property, materials, data products, or other nonpublic information owned by a United States entity or developed with a Federal R&D award to the government of a foreign country or an entity based in, funded by, or affiliated with a foreign country regardless of whether that government or entity provided support for the development of the IP, materials, or data products;

(ii) being required to recruit trainees or researchers to enroll in such program, position, or activity;

(iii) establishing a laboratory or company, accepting a faculty position, or undertaking any other employment or appointment in a foreign country or with an entity based in, funded by, or affiliated with a foreign country if such activities are in violation of the standard terms and conditions of a Federal R&D award;

(iv) being unable to terminate the foreign talent recruitment program contract or agreement except in extraordinary circumstances;

(v) through funding or effort related to the foreign talent recruitment program, being limited in the capacity to carry out a R&D award or required to engage in work that would result in substantial overlap or duplication with a Federal R&D award;

(vi) being required to apply for and successfully receive funding from the sponsoring foreign government's funding agencies with the sponsoring foreign organization as the recipient;

(vii) being required to omit acknowledgment of the recipient institution with which the individual is affiliated, or the Federal research agency sponsoring the R&D award, contrary to the institutional policies or standard terms and conditions of the Federal R&D award;

(viii) being required to not disclose to the Federal research agency or employing institution the participation of such individual in such program, position, or activity; or

(ix) having a conflict of interest or conflict of commitment contrary to the standard terms and conditions of the Federal R&D award; and

(B) a program that is sponsored by-

(i) a foreign country of concern or an entity based in a foreign country of concern, whether or not directly sponsored by the foreign country of concern;

(ii) an academic institution on the list developed under section 1286(c)(8) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232) ; or

(iii) a foreign talent recruitment program on the list developed under section 1286(c)(9) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232).

Performance Benchmark Requirements

SBCs with multiple SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The purpose of these requirements is to ensure that Phase I SBCs with multiple prior SBIR/STTR awards are making progress towards commercializing the work done under those awards. The Phase I to Phase II transition rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The commercialization benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' applications for more experienced firms. Detailed information on benchmark calculations and increased performance standards for more experienced firms can be found here.

Personal Conflict of Interest

A situation in which an individual has a financial interest, personal activity, or relationship that could impair the employee's ability to act impartially and in the best interest of the government when performing under the contract. (A de minimis interest that would not "impair the employee's ability to act impartially and in the best interest of the government" is not covered under this definition.)

Among the sources of personal conflicts of interest are:

a. The covered employee's, their close family members' or other members of their household's financial interests;
b. Other employment or financial relationships (including seeking or negotiating for prospective employment or business); and
c. Gifts, including travel.

Financial interests referred to in this definition's first paragraph may arise from:

a. Compensation, including wages, salaries, commissions, professional fees, or fees for business referrals;
b. Consulting relationships (including commercial and professional consulting and service arrangements, scientific and technical advisory board memberships, or serving as an expert witness in litigation);
c. Services provided in exchange for honorariums or travel expense reimbursements;
d. Research funding or other forms of research support;
e. Investment in the form of stock or bond ownership or partnership interest (excluding diversified mutual fund investments);
f. Real estate investments;
g. Patents, copyrights, and other IP interests; or
h. Business ownership and investment interests.

Principal Investigator/Program Manager

The principal investigator/project manager is the proposing SBC-designated individual who provides the scientific and technical direction to a funding agreement-supported project.

Proprietary Information

Proprietary information is any information that an SBC considers to be non-public information the SBC owns and is marked accordingly.

Research Institution

Any organization located in the United States that is:

a. A university.
b. A nonprofit institution as defined in Section 4(5) of the Stevenson-Wydler Technology Innovation Act of 1980.
c. A contractor-operated federally funded R&D center, as identified by the National Science Foundation in accordance with the government-wide FAR issued in accordance with the Office of Federal Procurement Policy Act Section 35(c)(1). A list of eligible FFRDCs is here.

Research or Research and Development

Any activity that is:

a. A systematic, intensive study directed toward greater knowledge or understanding of the subject studied.
b. A systematic study directed specifically toward applying new knowledge to meet a recognized need; or
c. A systematic knowledge application toward the production of useful materials, devices, systems, or methods, including design, development, and improvement of prototypes and new processes to meet specific requirements.

Research Involving Animal Subjects

All activities involving animal subjects shall be conducted in accordance with DoDI 3216.01 "Use of Animals in DoD Programs," 9 C.F.R. parts 1-4 "Animal Welfare Regulations," National Academy of Sciences Publication "Guide for the Care & Use of Laboratory Animals," as amended, and the Department of Agriculture rules implementing the Animal Welfare Act (7 U.S.C. §§ 2131-2159), as well as other applicable federal and state law and regulation and DoD instructions.

"Animal use" protocols apply to all activities that meet any of the following criteria:

a. Any research, development, test, evaluation or training, (including experimentation) involving an animal or animals.
b. An animal is defined as any living or dead, vertebrate organism (non-human) that is being used or is intended for use in research, development, test, evaluation or training.
c. A vertebrate is a member of the subphylum Vertebrata (within the phylum Chordata), including birds and cold-blooded animals.

See DoDI 3216.01 for definitions of these terms and more information about the applicability of DoDI 3216.01 to work involving animals.

Research Involving Human Subjects

All research involving human subjects shall be conducted in accordance with 32 C.F.R. § 219 "The Common Rule," 10 U.S.C. § 980 "Limitation on Use of Humans as Experimental Subjects," and DoDI 3216.02 "Protection of Human Subjects and Adherence to Ethical Standards in DoD-Supported Research," as well as other applicable federal and state law and regulations, and DoW Services/Component guidance. Proposing SBCs must be cognizant of and abide by the additional restrictions and limitations imposed on the DoW regarding research involving human subjects, specifically as they regard vulnerable populations (DoDI 3216.02), recruitment of military research subjects (DoDI 3216.02), and informed consent and surrogate consent (10 U.S.C. § 980) and chemical and biological agent research (DoDI 3216.02). Food and Drug Administration regulation and policies may also apply.

"Human use" protocols apply to all research that meets any of the following criteria:

a. Any research involving an intervention or an interaction with a living person that would not be occurring or would be occurring in some other fashion but for this research.
b. Any research involving identifiable private information. This may include data/information/specimens collected originally from living individuals (broadcast video, web-use logs, tissue, blood, medical or personnel records, health data repositories, etc.) in which the

identity of the subject is known, or the identity may be readily ascertained by the investigator or associated with the data/information/specimens.

See DoDI 3216.02 for definitions of these terms and more information about the applicability of DoDI 3216.02 to research involving human subjects.

Research Involving Recombinant DNA Molecules

Any recipient performing research involving recombinant DNA molecules and/or organisms and viruses containing recombinant DNA molecules shall comply with the National Institutes of Health Guidelines for Research Involving Recombinant DNA Molecules, dated January 2011, as amended. The guidelines can be found at: https://osp.od.nih.gov/wp-content/uploads/2016/05/NIH_Guidelines.pdf. Recombinant DNA is defined as (i) molecules that are constructed outside living cells by joining natural or synthetic DNA segments to DNA molecules that can replicate in living cells or (ii) molecules that result from the replication of those described in (i) above.

Service-Disabled Veteran-Owned Small Business (SDVOSB)

A service-disabled veteran or service-disabled veterans-owned and controlled SBC defined in Small Business Act 15 USC § 632(q)(2) and SBA's implementing SDVOSB regulations (13 CFR 125).

Small Business Concern (SBC)

A concern that meets the requirements set forth in 13 C.F.R. § 121.702 (available here).

An SBC must satisfy the following conditions on the date of award:

a. Is organized for profit, with a place of business located in the United States, which operates primarily within the United States, or which makes a significant contribution to the United States economy through payment of taxes or use of American products, materials or labor;
b. Is in the legal form of an individual proprietorship, partnership, limited liability company, corporation, joint venture, association, trust or cooperative, except that if the concern is a joint venture, each entity to the venture must meet the requirements set forth in paragraph (c) below;
c. Is more than 50 percent directly owned and controlled by one or more individuals (who are citizens or permanent resident aliens of the United States), other SBCs (each of which is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States), or any combination of these; and
d. Has, including its affiliates, not more than 500 employees. (See here for definition of an affiliate.)

Subcontract

A subcontract is any agreement, other than one involving an employer-employee relationship, including consultants, the funding agreement awardee enters calling for supplies or services for the performance of the original funding agreement.

Subcontractor

Subcontractor means any supplier, distributor, vendor, firm, academic institution, research center, or other person or entity that furnishes supplies or services pursuant to a subcontract, at any tier.

United States

United States means the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

Women-Owned Small Business Concern

An SBC where one or more women own at least 51 percent, or in the case of any publicly owned business, women own at least 51 percent of the stock, and women control the management and daily business operations.

Appendix C
POTENTIAL APPLICABLE FEDERAL ACQUISITION REGULATION, DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT CLAUSES

Note: An independent review and determination of required clauses should be complete for each award. Green cells are potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Blue cells are potential required FAR and DFARS clauses, when applicable.

Clause Number	Title	Date	When Applicable
52.203-17	Contractor Employee Whistleblower Rights	Nov-23	
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements	Jan-17	
52.204-10	Reporting Executive Compensation and First-Tier Subcontract Awards	Jun-20	
52.204-13	System for Award Management Maintenance	Oct-18	
52.204-18	Commercial and Government Entity Code Maintenance	Aug-20	
52.204-19	Incorporation by Reference of Representations and Certifications	Dec-14	
52.204-21	Basic Safeguarding of Covered Contractor Information Systems	Nov-21	
52.204-23	Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab and Other Covered Entities	Dec-23	
52.204-25	Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment	Nov-21	
52.204-27	Prohibition on a ByteDance Covered Application	Jun-23	
52.204-30	Federal Acquisition Supply Chain Security Act Orders—Prohibition	Dec-23	
52.209-06	Protecting the Government's Interest When Subcontracting with Contractors Debarred, Suspended, or Proposed for Debarment	Jan-25	
52.209-10	Prohibition on Contracting with Inverted Domestic Corporations	Nov-15	
52.219-06	Notice of Total Small Business Set-Aside	Nov-20	
52.219-08	Utilization of Small Business Concerns	Jan-25	
52.219-28	Post-Award Small Business Program Representation	Jan-25	
52.222-03	Convict Labor	Jun-03	
52.222-19	Child Labor-Cooperation with Authorities and Remedies	Jan-25	
52.222-21	Prohibition of Segregated Facilities	Apr-15	
52.222-25	Affirmative Action Compliance	Apr-84	
52.222-26	Equal Opportunity	Sep-16	
52.222-36	Equal Opportunity for Workers with Disabilities	Jun-20	

Clause Number	Title	Date	When Applicable
52.222-50	Combating Trafficking in Persons	Nov-21	
52.225-01	Buy American-Supplies	Oct-22	
52.225-13	Restrictions on Certain Foreign Purchases	Feb-21	
52.226-07	Drug-Free Workplace	May-24	
52.226-08	Encouraging Contractor Policies to Ban Text Messaging While Driving	May-24	
52.227-01 Alt I	Authorization and Consent - Alternate I (Apr-84)	Jun-20	
52.227-02	Notice and Assistance Regarding Patent and Copyright Infringement	Jun-20	
52.227-11	Patent Rights-Ownership by the Contractor	May-14	
52.227-20	Rights in Data-SBIR Program	May-14	
52.232-01	Payments	Apr-84	
52.232-11	Extras	Apr-84	
52.232-23	Assignment of Claims	May-14	
52.232-25	Prompt Payment	Jan-17	
52.232-33	Payment by Electronic Funds Transfer - System for Award Management	Oct-18	
52.232-39	Unenforceability of Unauthorized Obligations	Jun-13	
52.232-40	Providing Accelerated Payments to Small Business Subcontractors	Mar-23	
52.233-01	Disputes	May-14	
52.233-04	Applicable Law for Breach of Contract Claim	Oct-04	
52.242-15	Stop-Work Order	Aug-89	
52.243-01 Alt V	Changes-Fixed-Price Alternate V (Apr-84)	Aug-87	
52.244-06	Subcontracts for Commercial Products and Commercial Services	Jan-25	
52.246-09	Inspection of Research and Development (Short Form)	Apr-84	
52.252-02	Clauses Incorporated by Reference	Feb-98	
52.252-06	Authorized Deviations in Clauses	Nov-20	
52.253-01	Computer Generated Forms	Jan-91	
252.203-7000	Requirements Relating to Compensation of Former DoD Officials	Sep-11	
252.203-7002	Requirement to Inform Employees of Whistleblower Rights	Dec-22	
252.204-7000	Disclosure of Information	Oct-16	
252.204-7003	Control of Government Personnel Work Product	Apr-92	
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls	Oct-16	
252.204-7009	Limitations on the Use or Disclosure of Third-Party Contractor Reported Cyber Incident Information	Jan-23	
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting (DEVIATION 2024-O0013)	May-24	

Clause Number	Title	Date	When Applicable
252.204-7016	Covered Defense Telecommunications Equipment or Services—Representation	Dec-19	
252.204-7017	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services—Representation	May-21	
252.204-7018	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services	Jan-23	
252.204-7019	Notice of NISTSP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7020	NIST SP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7022	Expediting Contract Closeout	May-21	
252.204-7024	Notice on the use of the Supplier Performance Risk System	Mar-23	
252.227-7016	Rights in Bid or Proposal Information	Jan-23	
252.227-7018	Rights in Other Than Commercial Technical Data and Computer Software—Small Business Innovation Research Program and Small Business Technology Transfer Program	Jan-25	
252.227-7019	Validation of Asserted Restrictions--Computer Software	Jan-25	
252.227-7025	Limitations on the Use or Disclosure of Government-Furnished Information Marked with Restrictive Legends	Jan-25	
252.227-7030	Technical Data--Withholding of Payment	Mar-00	
252.227-7037	Validation of Restrictive Markings on Technical Data	Jan-25	
252.227-7039	Patents--Reporting of Subject Inventions	Apr-90	
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports	Dec-18	
252.232-7006	Wide Area WorkFlow Payment Instructions	Jan-23	
252.232-7010	Levies on Contract Payments	Dec-06	
252.235-7010	Acknowledgment of Support and Disclaimer	May-95	
252.235-7011	Final Scientific or Technical Report	Dec-19	
252.243-7001	Pricing of Contract Modifications	Dec-91	
252.244-7000	Subcontracts for Commercial Items	Nov-23	
52.203-03	Gratuities	Apr-84	Exceeding the simplified acquisition threshold.
52.203-05	Covenant Against Contingent Fees	May-14	Exceeding the simplified acquisition threshold.
52.203-06 or Alt I	Restrictions on Subcontractor Sales to the Government or ALT I	Nov-21	Exceeding the simplified acquisition threshold, Alt I commercial products or commercial services.
52.203-07	Anti-Kickback Procedures	Jun-20	Contracts exceeding $150,000.
52.203-08	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-12	Limitation on Payments to Influence Certain Federal Transactions	Jun-20	Contracts exceeding $150,000.

Clause Number	Title	Date	When Applicable
52.204-02	Security Requirements	Mar-21	May require access to classified information; cost contract (see 16.302) for research and development with an educational institution is contemplated.
52.212-04	Contract Terms and Conditions—Commercial Products and Commercial Services	Nov-23	Commercial products or commercial services.
52.212-05	Contract Terms and Conditions Required to Implement Statutes or Executive Orders-Commercial Items	May-24	Commercial products or commercial services.
52.219-14	Limitations on Subcontracting	Oct-22	Set aside for small business and the contract amount is expected to exceed the simplified acquisition threshold.
52.222-35	Equal Opportunity for Veterans	Jun-20	Contracts exceed $150,000.
52.222-37	Employment Reports on Veterans	Jun-20	If contract contains 52.222-35.
52.222-40	Notification of Employee Rights Under the National Labor Relations Act	Dec-10	Exceeding the simplified acquisition threshold.
52.222-54	Employment Eligibility Verification	Jan-25	Contracts exceed $150,000.
52.223-03	Hazardous Material Identification and Material Safety Data	Feb-21	Requires the delivery of hazardous materials.
52.229-03	Federal, State, and Local Taxes	Feb-13	A fixed-price contract is contemplated; and the contract is expected to exceed the simplified acquisition threshold.
52.232-02	Payment under Fixed-Price Research and Development Contracts	Apr-84	Fixed Price R&D.
52.233-03	Protest After Award	Aug-96	Exceed the simplified acquisition threshold.
52.242-13	Bankruptcy	Jul-95	Exceed the simplified acquisition threshold.
52.242-17	Government Delay of Work	Apr-84	Supplies other than commercial or modified-commercial products.
52.245-01	Government Property	Sep-21	When property is expected to be furnished.
52.245-09	Use and Charges	Apr-12	When the clause at 52.245-1 is included.
52.246-04	Inspection of Services-Fixed Price	Aug-96	Services, or supplies that involve the furnishing of services, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-16	Responsibility for Supplies	Apr-84	Supplies, services involving the furnishing of supplies, or research and development, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-23	Limitation of Liability	Feb-97	Exceed the simplified acquisition threshold.
52.247-34	F.o.b. Destination	Nov-91	When the delivery term is f.o.b. destination.
252.204-7015	Notice of Authorized Disclosure of Information for Litigation Support	Jan-23	Commercial products and commercial services.
252.209-7004	Subcontracting with Firms that are Owned or Controlled by the Government of a Country that is a State Sponsor of Terrorism	May-19	Contracts value of $150,000 or more.
252.211-7003	Item Identification and Valuation	Jan-23	For supplies, and for services involving the furnishing of supplies.
252.223-7001	Hazard Warning Labels	Dec-91	Requires submission of hazardous material data sheets.
252.223-7008	Prohibition of Hexavalent Chromium	Jan-23	For supplies, maintenance and repair services, or construction.

Clause Number	Title	Date	When Applicable
252.225-7001	Buy American and Balance of Payments Program	Feb-24	Acquisition of commercial products and commercial services.
252.225-7002	Qualifying Country Sources as Subcontractors	Mar-22	(i) 252.225-7001, Buy American and Balance of Payments Program. Use if one or more is included: (ii) 252.225-7021, Trade Agreements. (iii) 252.225-7036, Buy American - Free Trade Agreements - Balance of Payments Program.
252.225-7012	Preference for Certain Domestic Commodities	Apr-22	Commercial products and commercial services.
252.225-7052	Restriction on Acquisition of Certain Magnets, Tantalum, and Tungsten	May-24	Products and commercial services, that exceed the simplified acquisition threshold.
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime	Jan-23	Commercial products and commercial services.
252.225-7060	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region	Jun-23	Products utilizing funds appropriated or otherwise made available for any fiscal year.
252.225-7972	Prohibition on the Procurement of Foreign-Made Unmanned Aircraft Systems (DEVIATION 2024-O0014)	Aug-24	Commercial products and commercial services.
252.225-7967	Prohibition Regarding Russian Fossil Fuel Business Operations (DEVIATION 2024-O0006, Revision 1)	Feb-24	Exceeding the simplified acquisition threshold and for commercial products and commercial services.
252.228-7001	Ground and Flight Risk	Mar-23	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.228-7005	Mishap Reporting and Investigation Involving Aircraft, Missiles, and Space Launch Vehicles	Nov-19	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.235-7002	Animal Welfare	Dec-14	Uses live vertebrate animals.
252.235-7004	Protection of Human Subjects	Jul-09	Involving human subjects.
252.243-7002	Requests for Equitable Adjustment	Dec-22	Exceeding the simplified acquisition threshold.
252.245-7003	Contractor Property Management System Administration	Jan-25	Containing the clause at FAR 52.245-1, Government Property.
252.245-7005	Management and Reporting of Government Property	Jan-24	Containing the clause at FAR 52.245-1, Government Property.
252.247-7023	Transportation of Supplies by Sea	Oct-24	Except - those with an anticipated value at or below the simplified acquisition threshold.